



Scientific Committee on Consumer Safety

SCCS

OPINION

ON PROSTAGLANDIN ANALOGUES:

**Methylamido-Dihydro-Noralfaprostal (MDN)
(CAS No. 155206-01-2)**

**Isopropyl Cloprostenate (IPCP)
(CAS No. 157283-66-4)**

**Dechloro Dihydroxy Difluoro Ethylcloprostenolamide (DDDE)
(CAS No. 1185851-52-8)**



The SCCS adopted this document by written procedure on 2 February 2026

ACKNOWLEDGMENTS

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This Opinion has been subject to a commenting period of minimum eight weeks after its initial publication (from 6 June to 18 August due to holiday period). Comments received during this period were considered by the SCCS. Main changes occurred in all three preambles, 3.1.1.5 incl. SCCS comment, 3.1.5 incl. SCCS comment, 3.16 incl. SCCS comment, SCCS comment under 3.1.8, SCCS comment under 3.2.4, overall SCCS comment under 4.5.3.2, SCCS comment under 5.1.2.4 was deleted, SCCS comment under study 2 Applicant 2 of section 5.3.8, discussion 5.5, and chapter 8. SCCS concerns. Conclusions remain unchanged but have been slightly reworded and restructured.

All Declarations of Working Group members are available on the following webpage:
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1. ABSTRACT

The SCCS concludes the following:

1. *In light of the data provided and taking under consideration the concerns raised by the SCCS in Opinion SCCS/1635/21,*
 - a) *does the SCCS consider Ethyl Tafluprostamide safe when used in eyelash and eyebrow products up to a maximum concentration of 0.018 %?*
 - b) *does the SCCS consider Methylamido-Dihydro-Noralfaprostal safe when used in eyelash and eyebrow products up to a maximum concentration of 0.03 %?*
 - c) *does the SCCS consider Isopropyl Cloprostenate safe when used in eyelash and eyebrow products up to a maximum concentration of 0.005 %?*

Having evaluated all the evidence provided by the Applicants to support the safe use of the three prostaglandin analogues (PGAs) - Isopropyl Cloprostenate, *Methylamido-Dihydro-Noralfaprostal* and Dechloro Dihydroxy Difluoro Ethylcloprostenolamide - the SCCS is of the opinion that none of them can be considered safe for use in cosmetic products intended for promoting the growth of eyelashes and eyebrows.

The SCCS conclusion is based on the potent pharmacological activity of the PGAs, even at low concentrations, and the lack of acceptable data that would make it possible to exclude any potential adverse effects resulting from reproductive/developmental toxicity. The latter is crucially important for PGAs in view that users of the PGA-containing cosmetic products will most likely be women of child-bearing age.

More details on the SCCS concerns regarding safety of the PGAs are given in the section 8.

2. *Alternatively, what is according to the SCCS the maximum concentration considered safe for use of Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal and Isopropyl Cloprostenate, respectively, in eyelash and eyebrow products?*

/

3. *Does the SCCS have any further scientific concerns regarding the use of Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal and Isopropyl Cloprostenate in cosmetic products?*

The available evidence has shown that neither of the 3 PGAs assessed in this Opinion has a potential for genotoxicity, and hence no concern over genotoxic carcinogenicity. However, as indicated in the previous SCCS Opinion (SCCS/1635/21), in the absence of experimental data on carcinogenicity, the possibility of non-genotoxic carcinogenicity cannot be excluded.

Keywords: SCCS, scientific opinion, prostaglandin analogues, PGAs, Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal, Isopropyl Cloprostenate, Regulation 1223/2009

Opinion to be cited as: SCCS (Scientific Committee on Consumer Safety), Opinion on prostaglandin analogues used in cosmetic products, preliminary version of 28 May 2025, final version of 2 February, SCCS/1680/25

About the Scientific Committees

Two independent non-food Scientific Committees provide the Commission with the scientific advice it needs when preparing policy and proposals relating to consumer safety, public health and the environment. The Committees also draw the Commission's attention to the new or emerging problems which may pose an actual or potential threat.

These Committees are: the Scientific Committee on Consumer Safety (SCCS) and the Scientific Committee on Health, Environmental and Emerging Risks (SCHEER) and they are made up of scientists appointed in their personal capacity.

In addition, the Commission relies upon the work of the European Food Safety Authority (EFSA), the European Medicines Agency (EMA), the European Centre for Disease prevention and Control (ECDC) and the European Chemicals Agency (ECHA).

SCCS

The Committee shall provide Opinions on questions concerning health and safety risks (notably chemical, biological, mechanical and other physical risks) of non-food consumer products (for example cosmetic products and their ingredients, toys, textiles, clothing, personal care and household products such as detergents, etc.) and services (for example: tattooing, artificial sun tanning, etc.).

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2. MANDATE FROM THE EUROPEAN COMMISSION

Background

In 2018, the German Federal Institute for Risk Assessment (BfR) informed the European Commission that cosmetic products intended to promote the growth of eyelashes were increasingly available on the market. These products contained substances such as prostaglandins or their analogues that may pose risks for consumers. Prostaglandins and their analogues were not listed in the Annexes to the Cosmetic Regulation (EC) No. 1223/2009 and their use was not otherwise restricted in cosmetic products.

In 2019, Member States were invited to participate in a survey on cosmetic products for eyelash growth containing prostaglandins (and analogues) and their potential undesirable health effects, which led to launching a Call for data in 2020 (from June to October 2020). During the call for data, stakeholders submitted evidence on the safety of prostaglandins and two dossiers on the safety of the analogues 'Ethyl Tafluprostamide'(DDDE) (CAS No. 1185851-52-8) and 'Isopropyl Cloprostenate' (CAS No. 157283-66-4), respectively. In 2021, the Commission services requested the SCCS to carry out a safety assessment on the safety of prostaglandins and their analogues in cosmetic products in view of the information provided.

In the Opinion SCCS/1635/21¹, the SCCS was not able to conclude on the safety of 'Isopropyl Cloprostenate' and 'Ethyl Tafluprostamide' in cosmetic products in view of the limited data provided and the available information in published literature. Nevertheless, the SCCS noted concerns about the safety prostaglandin analogues in cosmetic products intended for use in the proximity of the eye.

By January 2024, industry submitted additional evidence to support the safe use of 'Ethyl Tafluprostamide' (DDDE), 'Methylamido-Dihydro-Noralfaprostal' (MDN) and 'Isopropyl Cloprostenate' (IPCP) in eyelash and eyebrow products. The Commission requests the SCCS to carry out a safety assessment on these three substances in view of the new information provided.

Terms of reference

- In light of the data provided and taking under consideration the concerns raised by the SCCS in Opinion SCCS/1635/21,*
 - does the SCCS consider Ethyl Tafluprostamide safe when used in eyelash and eyebrow products up to a maximum concentration of 0.018 %?*
 - does the SCCS consider Methylamido-Dihydro-Noralfaprostal safe when used in eyelash and eyebrow products up to a maximum concentration of 0.03 %?*
 - does the SCCS consider Isopropyl Cloprostenate safe when used in eyelash and eyebrow products up to a maximum concentration of 0.005 %?*
- Alternatively, what is according to the SCCS the maximum concentration considered safe for use of Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal and Isopropyl Cloprostenate, respectively, in eyelash and eyebrow products?*
- Does the SCCS have any further scientific concerns regarding the use of Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal and Isopropyl Cloprostenate in cosmetic products?*

¹ SCCS (Scientific Committee on Consumer Safety), Opinion on Prostaglandins and prostaglandin-analogues used in cosmetic products, preliminary version of 27 September 2021, final version of 3 February 2022, SCCS/1635/21.

3. OPINION ON METHYLAMIDO-DIHYDRO-NORALFAPROSTAL (MDN)

Preamble

In the submitted safety dossier, the Applicant provided experimental data for Methylamido-Dihydro-Noralfaprostal (MDN) from studies on skin and eye irritation, skin sensitisation and genotoxicity. A dermal absorption study for MDN was also provided. No data were provided on either toxicokinetics or systemic toxicological endpoints, e.g. repeated dose toxicity and developmental and reproductive toxicity. To fill this gap, the Applicant proposed a read-across approach using data from Bimatoprost - a structural analogue of MDN. After careful evaluation, the SCCS is of the opinion that the proposed read-across between MDN and Bimatoprost does not fulfil the critical requirements for an unbiased and transparent read-across. In this regard, no details have been provided on how the analogues were searched, how many were identified, and on what basis they were deselected or selected for use in the read-across.

Furthermore, although predictions derived from a non-testing NAM like read-across provide an important line of evidence, they are not considered sufficient as a standalone evidence to allow forming a conclusion on the lack of toxicity of a chemical substance against a given endpoint. Read-across outcomes therefore need to be supported by other lines of evidence, where the overall weight of evidence can provide a consistent and more reliable basis for decision making in regulatory risk assessments. For example, see EFSA (2025) Guidance on Read-across.

In view of this, the proposed read-across from a single analogue (Bimatoprost) can only be considered if it is not the only line of evidence for a given endpoint – *i.e.* if there is further information from other line(s) of evidence to support the read-across outcome. This means that the SCCS could only rely on the use of read-across from Bimatoprost to MDN for any given endpoint where it formed part of a weight of evidence.

However, since no other lines of evidence were provided for MDN, the read-across as proposed by the Applicant has not been used in the current Opinion.

3.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

3.1.1 Chemical identity

3.1.1.1 Primary name and/or INCI name

INCI name: Norbimatoprost

Ref: Cayman, 2023a; SCCS, 2022

3.1.1.2 Chemical names

IUPAC name: Methylamido-Dihydro-Noralfaprostal (Abbreviation: MDN)

Formal name: (5Z)-7-[(1R,2R,3R,5S)-3,5-dihydroxy-2-[(1E,3S)-3-hydroxy-5-phenyl-1-penten-1-yl]cyclopentyl]-N-methyl-5-heptenamide

Ref: Cayman, 2023a; SCCS, 2022

SCCS comment

The IUPAC name of MDN is ((Z)-7-[(1R,2R,3R,5S)-3,5-dihydroxy-2-[(E,3S)-3-hydroxy-5-phenylpent-1-enyl]cyclopentyl]-N-methylhept-5-enamide).

3.1.1.3 Trade names and abbreviations

Bimatoprost methyl amide
17-phenyl trinor Prostaglandin F2a methyl amide
17-phenyl trinor PGF2a methyl amide

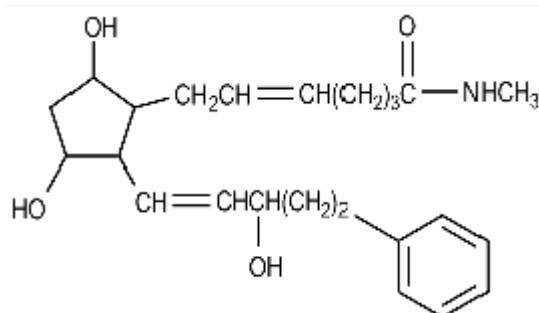
Ref: Cayman, 2023a, b

3.1.1.4 CAS / EC number

CAS: 155206-01-2; EC / List No.: none

Ref: Cayman, 2023a; SCCS, 2022

3.1.1.5 Structural formula



Ref: Cayman, 2023a; SCCS, 2022; CosIng, 2023

The substance as supplied is a single pure isomer (**5Z**)-7-[(**1R,2R,3R,5S**)-3,5-dihydroxy-2-[(**1E,3S**)-3-hydroxy-5-phenyl-1-penten-1-yl]cyclopentyl]-N-methyl-5-heptenamide. The alphanumeric characters in **bold** specify the single optical isomer present at each of 7 possible chiral centers.

SCCS comment

The SCCS has noted that MDN has chiral carbons in its structure and hence can exist in different stereoisomeric forms. The Applicant's claim that the substance is supplied as a pure specific single isomer needs to be substantiated by details on the synthesis route, the method used for purification of the isomer, and the analytical technique used to verify the purity and the isomeric form of the substance.

3.1.1.6 Empirical formula

Empirical (molecular) formula: C₂₄H₃₅NO₄

Ref: Cayman, 2023a

3.1.2 Physical form

Pure (pristine) material: oily viscous liquid, pale
Generally, the material is supplied as a 25% solution in ethanol.

Ref: Cayman, 2018c, 2023a; Laus, 2023a

3.1.3 Molecular weight

Molecular weight: 401.5 g/mol

Ref: Cayman, 2023a

3.1.4 Purity, composition and substance codes

Purity specification: $\geq 98\%$ (HPLC)
Batch 0682542: 98.0% (HPLC, toxicity testing)
Batch 0696557: 98.8% (HPLC, toxicity testing)
Batch 0667472: 98.3% (HPLC, physico-chemical data)

Ref: Cayman, 2018a, 2018b, 2018c

3.1.5 Impurities / accompanying contaminants

Regarding impurities, the following information was provided.

Water content: Negligible but not regularly analysed.
Isomers: HPLC Purity demonstrates regularly that the C-5,6 trans double bond isomer of the C-5,6 cis double bond in the alpha chain is $\leq 2\%$.
Residual solvents: Negligible, prior to formulation in ethanol, but not regularly analysed.
Elemental Impurities: Elemental impurities (e.g., heavy metals) are not analyzed as the risk for presence of these impurities (including Cadmium, Lead, Arsenic, and Mercury) in the MDN to be higher than the ICH Permitted Cutaneous Concentration has been assessed.
. The primary source of these elements is potable water. The water employed in the synthesis at Cayman Chemical and its wholly owned subsidiary Cayman Pharma is regulated by their respective national environmental departments, where these impurities are controlled to very low levels. Further, the ICH limits assume a daily intake of 10 g of the concerned material, the use of MDN in cosmetics is at least over 1000-fold less. Therefore, the probability of occurrence for these impurities in a final product at or near the allowable limit is very low.

Ref: Cayman, 2024a

SCCS comments on purity and impurities

The study report on analytical methodology and information on the validation of the HPLC method used for purity/impurity analysis was not provided.

Chemical identification of the test substance along with the analytical method and detailed data on the chemical identification and quantitation of the organic impurities (isomers), have not been provided.

Also, analysis of elemental impurities was not provided.

The quantity of any remaining solvents has not been reported.

3.1.6 Solubility

Water solubility: Measured according to OECD TG105/ Regulation (EC) No 440/2008 A.6:
3.80 g/L (at 20 °C, RSD: 2.4%)

Water solubility of MDN at 20°C was determined by the Applicant using the flask method. The mass concentration of the test item in the aqueous solution was determined by HPLC at 205 nm. According to the study report, in all measurements a second signal was observed as shoulder at the signal of the test item. This second signal was neither present in the calibration measurements nor in the validation and standards measurements, which were performed without contact to water. Thus, according to the water solubility report, it seems that the test item is not stable, since it changes during stirring in water.

During the commenting period the Applicant provided SCCS with new information after contacting the performing laboratory (Consilab). The lab confirmed to the Applicant that MDN is only sparingly soluble in water and usually requires a co-solvent like ethanol, isopropanol or DMSO in order to form dilute aqueous solutions more rapidly. However, once formed such aqueous solutions are exceedingly stable with no decomposition noted in several years.

Further solubilities (measured):

Ethanol:	about 50 mg/mL	Acetonitrile:	about 3 mg/mL
DMSO:	about 25 mg/mL		
Dimethyl formamide:	about 25 mg/mL		
PBS (pH 7.2)	about 0.3 mg/mL		

Ref.: Consilab, 2024c

SCCS comments

The SCCS notes that MDN may not be stable in water alone. Data on the stability of the test substance in water with co-solvents have not been provided by the Applicant.

3.1.7 Partition coefficient (Log P_{ow})

Partition coefficient (Log P_{ow}): Measured:
2.07 ± 0.01 (HPLC, OECD 117, Regulation EC No. 440/2008 method A.24, GLP)

Ref: Consilab, 2024d

3.1.8 Additional physical and chemical specifications

UV/VIS Absorption Spectra:

Measured:

MDN showed two absorption maxima/shoulders according to the OECD-Guideline 101 (1981) and Test Guideline OPPTS 830.7050 (1996) tested under GLP conditions:

λ (nm)	pH	log ϵ	λ (nm)	pH	log ϵ
262	6.6	2.27	269	6.6	2.15
	1.2	2.27		1.2	2.15
	13.9	2.28		13.9	2.17

Therefore, there is no trigger to investigate this endpoint as MDN showed consistently two absorption maxima in the UV/VIS spectrum at 262 nm and at 269 nm under acidic (pH 1.2), around neutral (pH 6.6) and alkalic (pH 13.9) conditions, respectively. At the wavelength of 300 nm and above no absorption was noted. According to EFSA and SCCS criteria no phototoxicity testing should be performed in cases where a test material only absorbs at wavelengths lower than 313 nm, which is clearly the case with MDN.

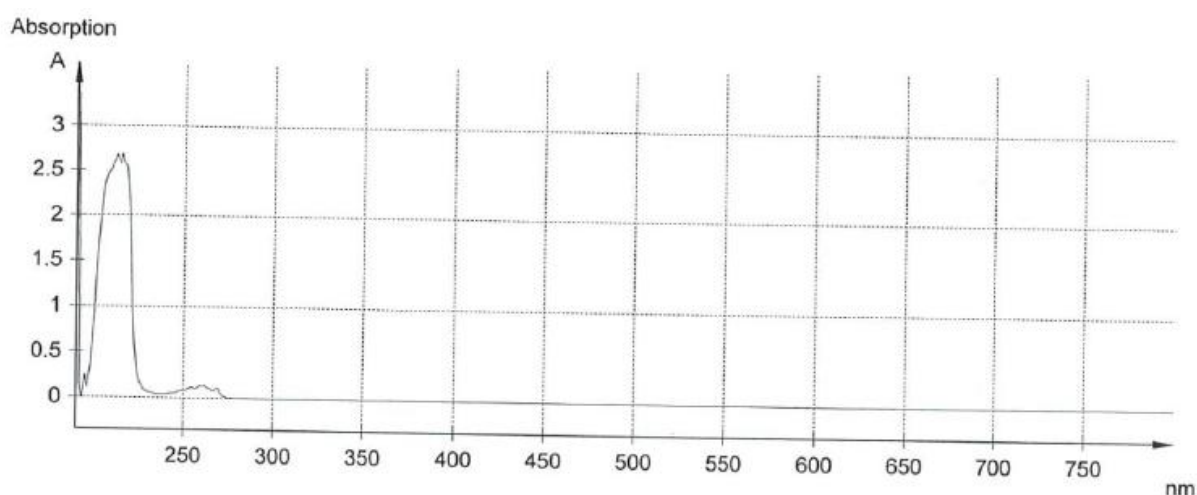


Figure 1: UV-spectra from the method development (190 – 800 nm) ($c = 312.5$ mg/L)

Since the absorption at approx. 220 nm did not decrease significantly by diluting the solution, this absorption was not evaluated.

7 Figures

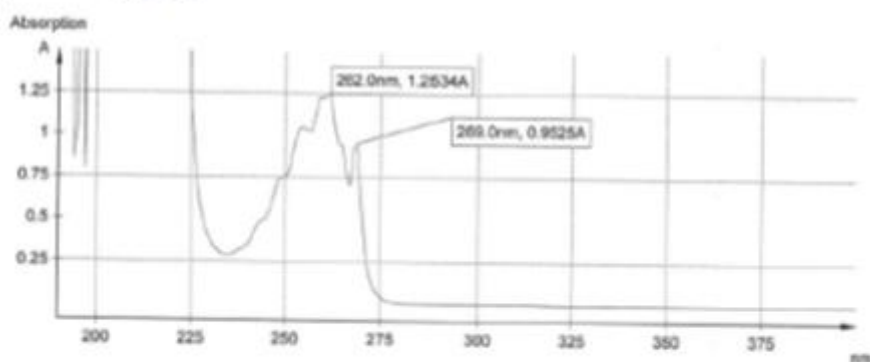


Figure 2: First test series: UV-spectra of the neutral solution (c = 2.7 g/L)

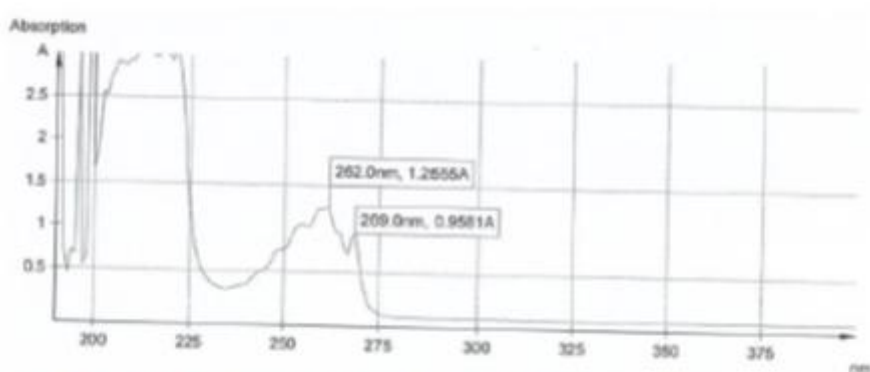


Figure 3: First test series: UV-spectra of the acidic solution (c = 2.7 g/L)

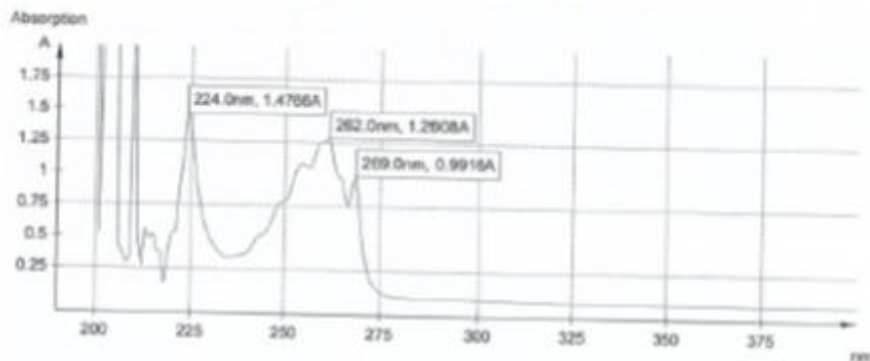


Figure 4: First test series: UV-spectra of the basic solution (c = 2.7 g/L)

Melting Point:

Measured:

MDN had no melting point between -90° C and the decomposition temperature of 230°C at atmospheric pressure (1009 to 1010 hPa) as determined by Differential Scanning Calorimetry (OECD 102, Regulation (EC) No 440/2008 method A.1., GLP).

Boiling Point:

Measured:

MDN showed no boiling point up to the decomposition temperature of 230°C at atmospheric pressure (1009 to 1010 hPa) as determined by Differential Scanning Calorimetry (OECD 103, Regulation (EC) No 440/2008 method A.2., GLP).

Vapour Pressure (Pa, 25°C): **Predicted:**
2.04E-015 (Prediction software: MPBPWIN v.1.44)

Ref: Consilab, 2024b; Epi Suite, 2023; SCCS, 2023; ToxMinds, 2024a

SCCS comments

No experimental data were provided on:

- organoleptic properties (odour and taste, if relevant)
- flash point
- density
- viscosity
- pKa
- pH
- refractive index.

For predicted values, information about the reliability of the predictions compared to experimental data have not been provided by the Applicant. In particular, the applicability domain of the specific models was not properly addressed, considering if there were similar substances in the training set of the model and if their predictions were accurate.

3.1.9 Homogeneity and Stability

Stability: ≥ 2 years (at – 20°C)
In the product at ambient temperature at least 6 months

Ref: Cayman, 2023a; M2 Beauté, undated a

SCCS comment

The Applicant has not provided experimental data on homogeneity and stability of MDN.

SCCS general comments on physicochemical characterisation

No data were provided on chemical characterisation by IR, ^{13}C -NMR or ^1H -NMR for any of the batches used in the studies.

A full report of chemical characterisation of MDN in terms of purity, identity and impurities in batches 0682542, 0696557, 0667472 has not been provided, and the validity of the analytical methodologies used was not shown. Information on identity and concentration of impurities have not been provided by the Applicant.

Data on the homogeneity of the test solutions, with respect to the content of the test substance under experimental conditions, have not been provided.

Data on the stability of the test substance under experimental conditions of the reported studies and under conditions of use have not been provided. Validated analytical procedures need to be used to determine stability of the test substance.

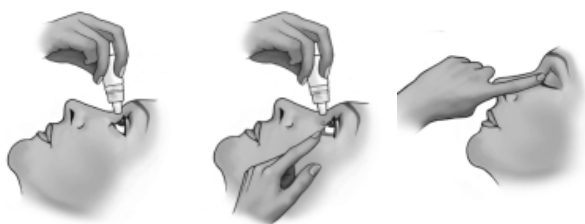
3.2 EXPOSURE ASSESSMENT & TOXICOKINETICS

3.2.1 Function and uses

According to the Applicant, MDN is chemically a fully synthetic substance in the form of a 17-phenyl trinor prostaglandin F2alpha (PGF2α) methyl amide analogue (Reference: Cayman 2023a). It is a substance with high structural similarity to the drug Bimatoprost (CAS No. 155206-00-1). See section 3.3.10 for more details on the read across justification. In cosmetics MDN is used to conditioning eyelashes and promote natural eyelash growth.

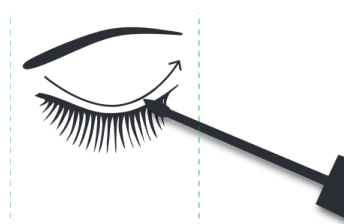
There are clear instructions for application (product packaging and instruction leaflet). The usage instructions on the label/package for the eyelash serum is unambiguously formulated. The intended purpose clearly emerges from the product description. Misuse is not to be expected from the usage instructions. The product is for external use only and are designed not to come into contact with the eyes (References: M2 Beauté, undated a and b).

As can be seen from the images below, there is a clear difference between the use and application of an eyedrop medication drug and eyelash serum (References: Santen, 2020; M2 Beauté, undated b).



Eyedrop medication:

Position the head horizontally in the neck, pull downward lower eyelid, one drop between lower eyelid and eye, eyes to be closed and press the finger about 1 minute on the inner edge of the eye to inhibit drainage via the lacrimal duct.



Eyelash serum:

Apply the serum once a day after cleansing the eye area, preferably at night. Your eye area should be oil- and grease-free. The serum is applied to the upper lash line like a normal eyeliner. One brush stroke is sufficient for both eyes.

Importantly, further differences in use between eyedrop medication and eyelash serum use have to be considered. All protective mechanisms of the eye for protection against and removal of foreign substances are intact, when using an eyelash serum. These mechanisms are the eyelid reflex and an open lacrimal duct. In accordance with the literature "*Greater than 80% of a solution dropped into the human eye is excreted within one to two minutes*" (Reference: OECD, 2020; Mikkelsen *et al.*, 1973). In contrast, when using eye drops, these protective mechanisms must be specifically suppressed (keeping the eye open and squeezing the tear duct shut) so that the drug can act at all.

SCCS comment

It is notable that the intended application of MDN containing formulation(s) will be in close proximity to the eye.

3.2.2 Dermal / percutaneous absorption**Study design**

Reference:	Eurofins, 2024
Date of report:	30.05.2024
Study period:	20.12.2023 – 18.03.2024
Guidelines/Methods:	OECD 428, SCCS Notes of Guidance (12 revision), SCCS basic criteria for the <i>in vitro</i> assessment of dermal absorption of cosmetic ingredient, OECD Guidance No. 28
Test system:	Human split-thickness skin membranes (300 – 400 µm)
Group size:	3 female donors (age range: 30 – 54 years), 1 male donor (age: 48 years)
Number of cells:	12 cells in total (3 replicates from each of the 4 donors)
Test substances:	1) MDN 2) [³ H]-MDN 3) Eyelash serum product without MDN
Batch (Purity):	1. 0667472 (Purity: 98.3 (HPLC)) 2. 23-1127-93 (Purity: around 98% (HPLC), Specific activity: 1.1 Ci/mmol (2.74 mCi/mg)) 3. 824386001
Application area	1 cm ² (flow of receptor fluid = 1.5 mL/h)
Active ingredient concentration (w/w)	0.0275%
Dose	10 µL mg/cm ²
Amount of reference item applied on skin	2.75 µg
Receptor fluid:	PBS (0.01M pH 7.4)
Diffusion cells.	Dynamic cells (application area = 1 cm ² , flow of receptor fluid = 1.5 mL/h)
Skin temperature:	32 ± 1°C
Skin integrity measurement:	Transepidermal water loss (TEWL)
Check of the test item stability at 32°C during 24 hours	Yes (n=1)
Sampling of Receptor Fluid	30 min, 2h, 4h, 6h, 12h and 24h after application
Stratum corneum tape stripping:	A maximum of 11 strips was performed and were pooled as follows: 1-2, 3-7, 8-11
GLP:	Yes
Published:	No

Material and methods:

The aim of this study was to investigate the rate and extent of the *in vitro* dermal absorption of MDN (0.0275%) from a cosmetic formulation as test item by using the [³H]-MDN as a tracer according to OECD 428. This test item was applied to the surface of healthy human skin mounted on dynamic cells. The test item containing the reference item in the form of an isotopic solution of MDN and the radiolabelled compound [³H]-MDN was prepared at the test site. Approximately 0.5 µCi was applied on each cell (10 µL/cm²). The homogeneity of the radiolabelled test item was checked on 10 µL of the radiolabelled preparation. The receptor fluid consisted of PBS (0.01 M, pH 7.4) and the flow of receptor fluid was 1.5 mL/h. The radiochemical purity analysed by HPLC was 98.23% at the day of formulation, 97.71% at the day of preparation (T0) and 97.02% after storage for 24 hours at 32 °C indicating stability during the experiment.

The skin samples were placed on the receptor compartment and the donor compartment were placed onto the skin samples. The skin surface temperature was maintained at $32^{\circ}\text{C} \pm 1^{\circ}\text{C}$, with a fixed water bath integrated in the dynamic system. After at least 30 minutes of stabilization, the integrity of the skin was verified using the TEWL method. The donor compartment was not closed (non-occlusive system) and the preparation for the test item was applied homogeneously at $10 \mu\text{L}/\text{cm}^2$ ($10 \mu\text{L}/\text{cell}$) and was left on the skin for 24 hours. The receptor fluid was collected continuously passing through the dynamic system at a flow rate of 1.5 mL/h. At each interval time, 30 min, 2h, 4h, 6h, 12h and 24h after application, the receptor fluid was placed into one or several vials. Finally, the skin was washed at 24h post-application and thereafter the cells were dismantled and the Rinsing Donor Compartment (RCD) was obtained.

The stratum corneum was taken off using adhesive scotch by stripping. The number of strips for all cells ranged from 2 to 11 strips considering the fragility of the skin samples after 24 h in contact with the prepared test item. The strips were pooled as follow for analysis: 1-2, 3-7, 8-11. Thereafter, the epidermis and dermis were separated. Samples were analysed for radiolabel content by scintillation counting (Tricarb 2910TR, Perkin) and calculations were performed using Excel directly from the raw data obtained with the scintillation counter. Conversion of the counts per minute (cpm) to disintegrations per minute (dpm) was performed directly.

Results and conclusion:

The skin samples were considered as suitable for the use in the test as the TEWL revealed acceptable values in the range between 0.5 - 13 g/m²/h. The homogeneity of the test item was in the acceptable range and the radiolabelled test item was shown to be stable throughout the experimental period. The summary of the mean absorption of MDN through human split thickness skin is presented in following Tables 1 and 2:

Table 1: Distribution of ³H-MDN after application of the eyelash serum product to human skin (% applied dose)

Reference item	MDN (% applied dose)	
Test item	Eyelash serum product	
No. of cells	n=12	
	Mean	SD
Total <i>stratum corneum</i> strips	2.18	1.33
Skin Excess*	93.28	5.62
Epidermis	4.20	2.67
Dermis	0.08	0.07
Receptor fluid + RCR**	0.67	0.35
Absorbed part (Epidermis + Dermis + receptor fluid + RCR)	4.95	2.81
TOTAL RECOVERY	100.41	5.16

*Skin excess corresponds to: Washing + Donor compartment rinsing + Remaining skin

**RCR = Rinsed receptor compartment

Table 2: Distribution of ^3H -MDN after application of the eyelash serum product to human skin ($\mu\text{g}_{\text{eq}}/\text{cm}^2$)

Reference item Test item No. of cells	MDN (% applied dose) Eyelash serum product n=12	
	Mean	SD
Test item amount applied (mg/cm^2)	9.26	0.52
Test substance applied ($\mu\text{Ci}/\text{cm}^2$)	0.45	0.03
Total stratum corneum strips	0.06	0.04
Skin Excess*	2.38	0.11
Epidermis	0.11	0.07
Dermis	0.002	0.002
Receptor fluid + RCR**	0.02	0.01
Absorbed part (Epidermis + Dermis + receptor fluid + RCR)	0.13	0.07

*Skin excess corresponds to: Washing + Donor compartment rinsing + Remaining skin

**RCR = Rinsed receptor compartment

Conclusion:

Following volatility tests and the pilot study, MDN was shown to be slightly volatile. In order to keep the compound and obtain complete mass balances, trapping systems were placed on top of the cells. A total of 4 donors were used in triplicates. The mean total recovery was $100.41\% \pm 5.16\%$ and therefore within the established acceptance criteria of SCCS guidance (85-115%), validating the results obtained.

The number of strips for all cells ranged from 2 to 11 strips considering the fragility of the skin samples after 24h in contact with the prepared test item.

The bioavailable portion consisting of receptor fluid plus rinse receptor compartment plus epidermis plus dermis after 24-hour was $4.95\% \pm 2.81\%$ of the applied dose corresponding to $0.13 \mu\text{g}_{\text{eq}}/\text{cm}^2 \pm 0.07 \mu\text{g}_{\text{eq}}/\text{cm}^2$.

Therefore, the value of 7.76% (mean plus standard deviation) will be used as dermal absorption value for the Margin of safety (MoS) calculation. The bioavailable portion as measure for skin absorption was $4.95\% \pm 2.81\%$ [$0.13 \pm 0.07 \mu\text{g}/\text{cm}^2$].

SCCS comment

The SCCS has noted that dermal absorption estimated by the Applicant at $4.95\% \pm 2.81\%$.

3.2.3 Other studies on toxicokinetics

There are no experimental data on toxicokinetics of MDN. According to the Applicant, Bimatoprost is the structural analogue of MDN and there are strong indications that both substances share the same main metabolite Bimatoprost acid, key and relevant kinetics studies derived from the pharmacology/toxicology review and evaluation performed by the Center for Drug Evaluation and Research (CDER) of the US Food and Drug Administration (FDA) on Bimatoprost (Reference: CDER, 2001) are presented in Annex 1.

SCCS comments

As expressed in the Preamble, the proposed read-across between MDN and Bimatoprost does not fulfil the requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence, it has not been considered relevant by the SCCS for use in the safety assessment of MDN.

3.2.4 Calculation of SED/LED

According to the Applicant the calculation for the systemic exposure dose (SED) is expressed mathematically as follows (Reference: SCCS, 2023a):

$$\text{SED} = E_{\text{product}} \times \frac{C}{100} \times \frac{DA_p}{100}$$

Where:

SED (mg/kg bw/day) - Systemic Exposure Dose

E_{product} (mg/kg bw/day) - Estimated daily exposure to a cosmetic product per kg body weight, based on the amount applied and the frequency of application (for calculated relative daily exposure levels for different cosmetic product types (SCCS/1628/21).

C (%) - Concentration of the substance under study in the finished cosmetic product on the application site.

DA_p (%) - Dermal Absorption expressed as a percentage of the test dose assumed to be applied in real-life conditions.

There are no daily exposure levels provided for eyelash sera in Europe in the most recent SCCS guidance document (SCCS, 2023a). The eyelash serum is applied as a thin layer to the eyelid above the eyelash line. This is comparable to the use of liquid eyeliner. However, there are clear differences in exposure as eyeliner will generally be applied above and below the eyelash lines of the upper and lower eye lids, respectively. In addition, and in contrast to the single use of the eyelash sera, eyeliner will be applied twice a day (Reference: SCCS, 2023a). To close the existing data gap, the Applicant has very recently performed analyses on the exact amount of the daily uses of his eyelash serum according to his application instructions (products' labels and information leaflets) including HPLC analytic on MDN. HPLC analysis on two eyelash sera showed that M2Beaute Eyelash Serum contained 0.030 % (w/w) bimatoprost methyl amide. The Applicant also analysed how much eyelash serum is applied on the eyelids. The average applied weight was 1.7 ± 0.7 mg. This corresponds to an amount of 0.51 ± 0.21 µg bimatoprost methyl amide.

Ref: Cayman, 2024b, c

SCCS comments

The Applicant has measured the daily amount of eyelash serum on the basis that the value in the SCCS Notes of Guidance (SCCS/1647/22) is based on a single application of eyeliner and not on a twice per day application. However, the SCCS has noted that the measured value for two applications instead of one is 3-fold lower than 0.005 g/day, which according to the SCCS Notes of Guidance (SCCS/1647/22), results from a single application of eyeliner. An explanation for this discrepancy is not given.

3.3 TOXICOLOGICAL EVALUATION

3.3.1. Irritation and corrosivity

3.3.1.1 Skin irritation

***In vitro* EpiDerm™ Skin Irritation Test**

Guideline: OECD 439

Test System:	EpiDerm™ SIT, Reconstructed human Epidermis (RhE) test model (three-dimensional human epidermis tissue model)
Replicates:	Triplicate
Test substance:	MDN
Batch:	0696557
Purity:	98.8% (HPLC)
Test item:	Oily viscous liquid, pale
Dose level:	Neat
Dose applied:	30µl
Treatment period:	In total 60 minutes: 35 minutes at 37°C, followed by 25 minutes at room temperature
Post-treatment incubation time:	42 ± 1 hour
Positive control:	5% aqueous solution of Sodium Dodecyl Sulfate
Negative control:	Dulbecco's Phosphate Buffered Saline (DPBS)
Solvent control:	Not applicable
Dead dermis negative control:	Not done
Interaction with MTT:	Negative
Colouring of tissue:	Negative
GLP:	In compliance
Study period:	January 2024

Material and methods:

The possible *in vitro* skin irritation potential of neat MDN (batch: 0696557 (purity: 98.8%)) was investigated according to OECD 439 under GLP conditions considering also current SCCS guidance criteria. The Reconstructed human Epidermis (RhE) test model (EpiDerm® tissue kit) was used to evaluate the skin irritation potential of the test substance. One valid experiment was performed.

The neat liquid test item was applied to a three-dimensional human epidermis tissue model in triplicate for an exposure time of 60 minutes. DPBS buffer was used as negative control and 5 % aqueous SDS solution was used as positive control. After treatment, the respective substance was rinsed from the tissue. Thereafter, the cell viability of the tissues was evaluated by addition of MTT (3-(4,5-dimethyl thiazole 2-yl) 2,5-diphenyltetrazolium bromide), which can be reduced to formazan. The formazan production was evaluated by measuring the optical density (OD) of the resulting solution.

Results:

The obtained mean optical density (OD) measured at 570 nm of the two replicates per tissue, the mean OD of the three relating tissues and the standard deviation (SD) are provided in the following table.

Table 3: Mean OD and SD Negative Control, Positive Control and Test Item

Designation	Negative Control (DPBS)	Test Item	Positive Control (5% aqueous SDS)
Mean OD Tissue 1 (OD – Mean Blank Control)	2.397	0.122	0.028
Mean OD Tissue 2 (OD – Mean Blank Control)	2.140	0.118	0.047
Mean OD Tissue 3 (OD – Mean Blank Control)	1.698	0.123	0.039
Mean OD of the three relating tissues	2.078	0.121	0.038
SD	17.0%	0.1%	0.5%

For the test item and the positive control, the following percentage values of tissue viability were calculated in comparison to the negative control and shown in the table below:

Table 4: % Viability

Designation	Test Item	Positive Control
% Tissue viability (tissue 1)	5.8%	1.3%
% Tissue viability (tissue 2)	5.7%	2.3%
% Tissue viability (tissue 3)	5.9%	1.9%
% Tissue viability (mean)	5.8%	1.8%

The mean value of relative tissue viability of the test item was reduced to 5.8 % after the treatment. This value is below the threshold for skin irritation (50 %) and therefore, the test item has to be considered as irritant to skin.

Positive and negative controls showed the expected OD and cell viability values within acceptable limits and indicated the suitability and sensitivity of the test system.

Conclusion:

Under the conditions specified in this *in vitro* study, neat MDN is considered irritant to the skin in the Reconstructed human Epidermis (RhE) test method according to the criteria of the OECD Guideline 439.

Ref. Laus, 2024a

SCCS comments

The skin irritation study was performed in accordance with the test guideline. The SCCS agrees with the Applicants' conclusion that MDN is a skin irritant. However, the test method used does not differentiate whether MDN is only a skin irritant or also a corrosive substance.

3.3.1.2 Mucous membrane irritation / eye irritation

EpiOcular™, Reconstructed human Cornea-like Epithelium (RhCE) model

Guideline:	OECD 492
Test system:	EpiOcular™ tissue
Replicates:	2 cornea tissue models per test condition
Test substance:	MDN
Batch:	0696557
Test item:	Oily viscous liquid, pale
Purity:	98.8% (HPLC)
Vehicle:	None
Dose:	Neat
Dose applied:	50µl
Treatment period:	28 ±2 minutes
Post-treatment incubation time:	120 ±2 minutes
Positive control:	Methyl acetate
Negative control:	Sterile demineralised water
Dead dermis negative control:	Not done
Interaction with MTT:	Negative
Colouring of tissue:	Negative
GLP:	In compliance
Study period:	January 2024

Material and methods:

The possible *in vitro* eye irritation potential of neat MDN (0696557 (purity: 98.8%) was investigated according to OECD 492 under GLP conditions considering also current SCCS guidance criteria. The EpiOcular™ reconstructed human cornea-like epithelium (RhCE) was used to evaluate the test substance. One valid experiment was performed.

The neat test item was applied to a three-dimensional human cornea tissue model in duplicate for an exposure time of 28 minutes. Sterile demineralised water was used as the negative control and methyl acetate was used as the positive control. After treatment, the respective substance was rinsed from the tissue. Thereafter, the cell viability of the tissues was evaluated by addition of MTT, which can be reduced to formazan. The formazan production was evaluated by measuring the optical density (OD) of the resulting solution.

Results:

After treatment with the negative control, the absorbance values were within the required acceptability criterion of $0.8 < \text{mean OD} < 2.8$, OD was 1.100.

The positive control showed clear eye irritating effects, and the mean value of the relative tissue viability was 1.5 % (required: < 50 %). The variation within tissue replicates of the controls and the test item was acceptable (< 20 %). Thus, the sensitivity and validity of the test system was given.

The obtained mean optical density (OD) measured at 570 nm of the two replicates per tissue, the mean OD of the three relating tissues and difference are provided in Table 3.

For the test item and the positive control, the following percentage values of tissue viability were calculated in comparison to the negative control and presented in Table 5.

Table 5: Mean OD and SD of the Negative Control, Positive Control and Test Item

Designation	Negative Control (Sterile demineralised water)	Positive Control (Methyl acetate)	MDN
Mean OD Tissue 1 (OD – Mean Blank Control)	1.120	0.012	1.066
Mean OD Tissue 2 (OD – Mean Blank Control)	1.080	0.023	0.988
Mean OD of the two relating tissues	1.100	0.017	1.027
Difference	3.6 %	1.0 %	7.1 %

Table 6: % Viability Positive Control and Test Item

Designation	Positive Control (Methyl acetate)	MDN
% Viability (Tissue 1)	1.0 %	96.9 %
% Viability (Tissue 2)	2.0 %	89.8 %
% Viability Mean	1.5 %	93.4 %

After treatment with the test item, the mean value of relative tissue viability was 93.4 %. As this value is above the threshold for eye irritation potential (≤ 60 %) test items that induce these values are considered non-eye irritant.

Conclusion:

Under the conditions of this *in vitro* study, neat MDN is considered as non-eye irritant in the *in vitro* EpiOcular™ Eye Irritation Test according to the criteria of the OECD Guideline 492.

Ref: Laus, 2024b

In addition, the structural analogue Bimatoprost did not cause irritation to the ocular structures in monkeys when administered once or two-times a day for at least 29 or 52 weeks followed by a 24 or 20 week recovery period after repeated application of concentrations at 0.03% and 0.1% directly into the eye.

Clinical observations and ocular examinations showed increased iridal pigmentation and periocular changes characterized by a prominent upper and/or lower sulcus and/or widening of the palpebral fissure in the treated eyes. No functional or anatomic ocular abnormalities were noted. This can also be interpreted that no indication of irritation was observed. The periocular findings were completely reversed by the end of the recovery period, while the increased iridal pigmentation was not reversible. However, iridal pigmentation per se is not considered as toxicological effect. These findings were also observed in monkeys with other PG analogues and were considered to be related to this pharmacological class. No systemic toxicity was observed at any dose

Ref: CDER, 2001

SCCS comments

The *in vitro* study showed that neat MDN is not an eye irritant. The study was well performed; however, this outcome is contrary to the fact that MDN was shown to be a skin irritant. Substances identified as skin irritants often exhibit (severe) eye irritation, particularly when tested as neat substance. Furthermore, according to the Cayman safety data sheet (2023b), MDN is classified as Eye Irritation 2A (causes serious eye irritation). As an example, in a clinical setting where the product was directly applied in the eye, it was also observed that the structural analogue Bimatoprost induced adverse eye effects in a subset of patients. The clinical manifestations included eye itching, inflammation, dryness, burning sensation and redness (conjunctival hyperaemia) (BfR, 2018).

3.3.2 Skin sensitisation

The possible *in vitro* skin sensitizing potential of MDN was investigated in the Direct Peptide Reactivity Assay (DPRA) and in the KeratinoSens™ method (ARE-Nrf2 Luciferase test method) following the two out of three (2o3) prediction strategy as laid down in the guideline on defined approaches (DA) for skin sensitization (OECD 497) and are presented below.

Direct Peptide Reactivity Assay (DPRA)

Guideline:	OECD 442C: <i>In Chemico</i> Skin Sensitisation Assays addressing the AOP key event on covalent binding to proteins
Replicates:	3
Test substance:	Methylamido-Dihydro-Noralfaprostal
Batch:	0696557
Purity:	98.8%
Test item:	oily viscous liquid, pale
Solvent:	Acetonitrile
Dose level:	100 mM
Treatment period:	24± 2 hours
Positive control:	Cinnamaldehyde 100 mM and acetonitrile
Negative control:	/

Co-elution control samples:	Test item stock solution and acetonitrile combined with phosphate buffer
Reference control samples:	A: Peptide stock solution with acetonitrile for system suitability check B: Peptide stock solution with acetonitrile for peptide stability check. C: Peptide stock solution with acetonitrile for the test item and the positive control
GLP:	In compliance
Study period:	February 2024

Material and methods:

The possible sensitizing potential of MDN (batch: 0696557, purity: 98.8%) was investigated by means of *in chemico* determination with the Direct Peptide Reactivity Assay (DPRA) with high-performance liquid chromatography (HPLC) analysis according to OECD 442C and considering SCCS criteria under GLP conditions. In order to derive a prediction for the test item, six independent valid tests were evaluated, three with cysteine and three with lysine peptides. Executing three full runs was necessary according to the decision tree of OECD 497 guideline, because the overall peptide depletion was not between 0-3% in the first full run (which would indicate a clear negative outcome) and the overall peptide depletion was borderline in the second full run. Accordingly, the reactivity of the test item and synthetic Cysteine as well as Lysine containing peptides was evaluated by combining the test chemical with a solution of the peptides and monitoring the remaining concentration of the peptide following 24 hours of interaction time at room temperature. Relative concentrations of the peptides following the 24-hour incubation were determined by HPLC with gradient elution and UV detection at 220 nm. The samples were also evaluated with regards to possible occurrence of precipitation. Reaction samples, reference controls A, B and C, co-elution controls and positive control s were prepared and analysed in triplicates. The total HPLC analysis time was less than 30 hours.

Results:

No precipitation was observed as acetonitrile dissolved the test item completely and homogenous and clear solution was achieved.

Means of the peak areas versus the concentrations of cysteine and lysine peptide showed good linearity, covering the concentration range from 0.0167 mM to 0.534 mM.

- The system suitability was verified as the mean peptide concentration of reference control A sample replicates was 0.53 mM, 0.55 mM and 0.53 mM for the cysteine peptide; 0.51 mM, 0.50 mM and 0.50 mM for the lysine peptide.
- the mean peptide concentration of reference control B sample replicates were 0.50 mM, 0.51 mM and 0.49 mM for cysteine peptide; 0.51 mM, 0.50 mM and 0.50 mM for lysine peptide.
- the mean peptide concentration of reference control C (acn) sample replicates were 0.49 mM, 0.51 mM and 0.49 mM for cysteine peptide; 0.51 mM, 0.50 mM and 0.50 mM for lysine peptide.

The acceptance criteria were met for the positive control with cysteine peptide depletion values of 71.4 % $\pm$ 1 %, 72.6 % $\pm$ 1 % and 73.3 % $\pm$ 1%; with lysine peptide depletion values of 51.4 % $\pm$ 1 %, 49.9 % $\pm$ 0.2 % and 51.7 % $\pm$ 1 %. The mean cysteine peptide depletion values obtained with the test item were 6.7 % $\pm$ 2 %, 9.2 % $\pm$ 6 % and 8.4 % $\pm$ 3 % while the lysine peptide depletion values of the test item were 1.2 % $\pm$ 1 %, 1.6 % $\pm$ 0.3 % and 0.7 % $\pm$ 0.5 %. The SD for the test item were within the acceptance range.

No co-elution was observed with the peptides.

The mean cysteine peptide depletion values of the test item were 6.7 % $\pm$ 2 % for the first run, 9.2 % $\pm$ 6 % for the second run and 8.4 % $\pm$ 3 % for the third run; while the lysine

peptide depletion values of the test item were $1.2 \% \pm 1 \%$ for the first run, $1.6 \% \pm 0.3 \%$ for the second run and $0.7 \% \pm 0.5 \%$ for the third run. The overall mean peptide depletion of the test item was 4 % in the case of the first run, 5.4 % in the case of the second run and 4.6 % in the case of the third run and therefore finally indicative for negative DPRA prediction. Peptide depletion resulting from the positive control cinnamaldehyde was within the expected percentage range and confirmed the suitability and sensitivity of the test system. All validity criteria were met, confirming the validity of the study.

Conclusion:

Under the conditions of this *in chemico* Direct Peptide Reactivity Assay and based on the obtained results and the cysteine 1:10 / lysine 1:50 prediction model, MDN showed no or minimal reactivity and was therefore concluded to be negative according to the prediction criteria.

Ref. ToxiCoop, 2024b

SCCS comment

The DPRA was well-conducted according to the OECD TG. Under the conditions of the test, MDN was negative in the DPRA.

KeratinoSens™

Guideline:	OECD 442D <i>In vitro</i> skin sensitisation assays addressing the AOP key event on keratinocyte activation.
Test System:	KeratoSens™ cell line
Replicates:	triplicate
Test substance:	MDN
Batch:	0696557
Purity:	98.8% (HPLC)
Test item:	oily viscous liquid, pale
Solvent:	DMSO
Dose level:	1st and 2nd test: twelve concentrations: 0.98 µM - 2000.00 µM, 2-fold dilution factor
Treatment period:	48 ± 1 hours (37 ± 1 °C in the presence of 5 % CO ₂)
Positive control:	Trans-Cinnamaldehyde
Negative control:	Vehicle (DMSO)
GLP:	In compliance
Study period:	January 2024

Material and methods:

The possible sensitizing potential of MDN (batch: 0696557 (purity: 98.8%)) was studied using the KeratoSens™ method (ARE-Nrf2 Luciferase test method) according to OECD 442D and considering SCCS criteria under GLP conditions. In total, two independent tests with each three replicates were performed with the test item. In both tests, twelve concentrations ranging from 0.98 µM - 2000.00 µM using a 2-fold dilution factor were used. DMSO was used as suitable solvent and negative control, while Trans-Cinnamaldehyde was the positive control. The KeratoSens™ cells were derived from human keratinocytes (HaCaT) and transfected with selectable plasmids containing luciferase gene under the transcriptional control of the AKR1C2 ARE gene sequence. After 48 hours of incubation, Luciferase gene induction was measured in the cell lysates by luminescence detection using a light producing luciferase substrate. Cytotoxicity and the relative luminescence intensity of luciferase substrate in the lysates were measured and luciferase induction compared to solvent/vehicle control was calculated.

Results:

Precipitation was not observed at any concentration in any of the two tests.

The test item induced cytotoxicity in KeratinoSens™ cells compared to the solvent/vehicle control in both tests at the two highest tested concentrations (2000 and 1000 µM).

IC30 and IC50 values could be determined in both tests. The overall IC30 was determined as 794.35 µM, while the overall IC50 was calculated as 879.26 µM.

The induction values for luciferase activity did exceed the 1.5-fold threshold only in the first test in two out of three replicates at the second highest and cytotoxic concentration of 1000.00 µM with regards to the OECD Test Guideline 442D prediction criteria. However, there was no statistical significance when the mean value was compared to the respective negative control. All the other induction values did not exceed the 1.5-fold threshold in both tests.

When evaluated according to the OECD Test Guideline 497 prediction model, the non-statistically significant mean induction value of the test item was between the borderline threshold only in the first test at the second highest and cytotoxic concentration of 1000 µM. All the other induction values were below the borderline threshold range (1.351.67-fold) at tested concentrations for the first and second tests. The overall maximal fold induction (Imax) was 1.27-fold. Furthermore, no dose response was observed in any of the tests.

Based on the obtained results and on the OECD 442D and OECD 497 prediction models, both tests were concluded negative, *i.e.* not indicative for skin sensitising potential.

Conclusion:

Overall, under the conditions of this study and based on the obtained results, MDN showed no *in vitro* skin sensitisation potential under the experimental conditions of the KeratinoSens™ method, when tested up to clear cytotoxic concentrations and considering the OECD 442D and OECD 497 prediction model criteria.

Ref. ToxiCoop, 2024a

SCCS comments

The Keratinosens assay was well-conducted according to the OECD TG. Under the conditions of the test, MDN was negative in the Keratinosens assay.

According to the Applicant, no *in vivo* studies were available for MDN. A Guinea Pig Maximisation test was performed for the structural analogue Bimatoprost. Under the conditions of this study, Bimatoprost was also shown to have no skin sensitising potential (Reference: CDER, 2001).

Applicants' overall summary on skin sensitisation:

The possible *in vitro* skin sensitising potential of MDN was investigated in the Direct Peptide Reactivity Assay (DPRA) and in the KeratinoSens™ method (ARE-Nrf2 Luciferase test method) following the two out of three (2o3) prediction strategy as laid down in the guideline on defined approaches (DA) for skin sensitization (OECD 497).

Both *in vitro* tests revealed no skin sensitizing potential considering the OECD 497 prediction model criteria. In the DPRA, MDN showed no or minimal reactivity and was therefore concluded to be negative according to the prediction criteria. In the KeratinoSens™ test, no *in vitro* skin sensitization potential was detected, when tested up to clear cytotoxic concentrations.

In addition, also the structural analogue Bimatoprost revealed no sensitizing potential *in vivo* when tested in the Guinea pig Maximization test.

SCCS comments

Taking the data from the DPRA and Keratinosens assay together and applying the two out of three (2o3) Defined Approach as described in OECD TG497, it can be concluded that MDN is not a skin sensitiser.

3.3.3 Acute toxicity

According to the Applicant, there are no acute toxicity studies on MDN available. The Applicant provided the SCCS with data of the structural analogue Bimatoprost (Reference: CDER, 2001) as surrogate based on the read-across justification provided in section 3.3.10. These data on acute toxicity, although limited and with uncommon exposure routes, are provided for sake of completeness and to provide a complete overview on the toxicological profile with regards to acute toxicity in Annex 2. Acute toxicity studies using the common oral, dermal or inhalation route are not available.

3.3.3.1 Acute oral toxicity

/

3.3.3.2 Acute dermal toxicity

/

3.3.3.3 Acute inhalation toxicity

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3.3.3.4 Acute toxicity – other routes**SCCS comment**

Since the product applied above the eyelid only contains small amounts of MDN, there is no concern over acute toxicity of MDN.

3.3.4 Repeated dose toxicity

According to the Applicant, there are no repeated-dose toxicity studies on MDN available. The Applicant provided data on longer-term oral administration studies (13 weeks and 52 weeks) of the structural analogue Bimatoprost (Reference: CDER, 2001) as surrogate based on the read-across justification provided in section 3.3.10. Studies using repeated ocular or intravenous application were not taken into consideration. A summary of these studies can be found in Annex 3.

3.3.4.1 Repeated dose (28 days) oral / dermal / subcutaneous / inhalation toxicity

/

3.3.4.2 Sub-chronic (90 days) oral / dermal / inhalation toxicity

/

3.3.4.3 Chronic (> 12 months) toxicity

/

3.3.5 Reproductive and developmental toxicity

There are no reproductive nor developmental toxicity studies on MDN available. Therefore, studies of the structural analogue Bimatoprost (Reference: CDER, 2001; CDER, 2010; CDER, 2020) were taken as surrogate based on the read-across justification provided in section 3.3.10. A summary of these studies can be found in Annex 4.

SCCS comments

As expressed in the Preamble, the SCCS is of the view that the proposed read-across between MDN and Bimatoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Also, the SCCS has concerns about the developmental and reprotoxic effects of prostaglandin analogues. Besides Bimatoprost, Tafluprost was also identified as a structural analogue of MDN by the Applicant. Concerns about the developmental toxicity of this analogue are described in more detail in section 5.3.6.2. on DDDE. Similar concerns were raised by the BfR (2018) for Bimatoprost, the structural analogue used by the Applicant to derive a NOAEL. In their report, BfR has described that Bimatoprost led to miscarriages in embryonal and foetal development studies in rodents. These occurred at doses of at least 860 times or 1700 times higher than the human dose, while no developmental impairments were observed. In peri-/postnatal studies on rats with ≥ 0.3 mg/kg/day (at least 41 times the systemic exposure of a therapeutic application to the eye for humans) toxic effects for female animals led to shortened gestation periods, foetus death and reduced body weight of young animals. The read-across performed by the Applicant for Bimatoprost shows that in a developmental toxicity study, a NOAEL of 0.1 mg/kg bw/day was identified. The data of Bimatoprost are in line with data for other prostaglandin analogues, showing that this class has adverse developmental and reprotoxic effects. Although the read-across data has not been used in the MoS calculation by the SCCS, the potential reproductive effects of MDN become evident when all the available data are taken together. This raises a serious safety concern because the majority of those using these products will most likely be women of child-bearing age. Therefore, it is not possible to evaluate the safety of MDN in the intended cosmetic applications in the absence of data on reproductive/ developmental toxicity.

3.3.6 Mutagenicity / genotoxicity

3.3.6.1 Mutagenicity / genotoxicity *in vitro*

Study #1: Bacterial Reverse Mutation Test

Guideline:	OECD TG 471
Species/Strain:	Salmonella typhimurium strains TA98, TA100, TA1535, TA1537, Escherichia coli WP2 uvrA
Replicates:	Triplicates
Test substance:	Methylamido Dihydro Noralfaprostal (MDN)
Batch:	0682542
Purity:	98% (HPLC)
Solvent:	DMSO
Positive controls:	Without S9 mix: Sodium azide (NaN ₃): 10 µg/plate for TA100, TA1535 4-Nitro-o-phenylenediamine (4-NOPD): 10 µg/plate for TA98; 50 µg/plate for TA 1537 Methyl methanesulfonate (MMS): 2.0 µL/plate for WP2 uvr A

Without S9-mix:	2-aminoanthracene (2-AA): 2.5 µg/plate for TA98, TA100, TA1535, TA1537, 10 µg/plate for WP2 uvr A
Negative control:	Untreated and solvent (DMSO)
Treatment:	Plate incorporation (Experiment I): with and without S9-mix: 0; 3; 10; 33; 100; 333; 1000; 2500; 5000 µg/plate Pre-incubation test (Experiment II): with and without S9-mix: 0, 33; 100; 333; 1000; 2500; 5000 µg/plate
GLP:	In compliance
Study period:	October – November 2023

Material and methods:

MDN (batch: 0682542, purity: 98%) was investigated for its possible mutagenic potential in *Salmonella typhimurium* (*S. typhimurium*) strains TA98, TA100, TA1535, TA1537 and *Escherichia coli* (*E. coli*) strain WP2 uvrA according to OECD 471 under GLP conditions. The tested concentrations of the test item dissolved in DMSO were 0; 3; 10; 33; 100; 333; 1000; 2500 and 5000 µg/plate for the plate incorporation test (experiment I) and 0, 33; 100; 333; 1000; 2500 and 5000 µg/plate for the pre-incubation assay (experiment II) for all strains with and without the addition of a rat liver homogenate metabolizing system (S9 fraction of phenobarbital/β-naphthoflavone-induced rat liver). Three replicates for each concentration level and controls were used. Concurrent untreated plates, solvent control and appropriate reference mutagens (sodium azide, 4-nitro-o-phenylene-diamine, methyl methane sulfonate, 2-aminoanthracene) were investigated in parallel.

Results:

The test item was adequately soluble in DMSO. No precipitation of the test item was observed on plates in the tested bacterial strains at any examined concentration level with and without metabolic activation. The plates incubated with the test item showed normal background growth up to 5000 µg/plate with and without S9 mix in all strains used. No toxic effects, evident as reduction in the number of revertants (below the indication factor of 0.5), occurred in all strains with and without metabolic activation. No substantial increase in revertant colony numbers of any of the five tester strains was observed following treatment with MDN at any dose level, neither in the presence nor absence of metabolic activation (S9 mix). There was also no tendency of higher mutation rates with increasing concentrations in the range below the generally acknowledged border of biological relevance. Appropriate reference mutagens were used as positive controls and showed a distinct increase of induced revertant colonies, which were clearly within the historical laboratory control data range. Thus, the sensitivity and suitability of the test system was demonstrated.

Conclusion:

Under the conditions of this test, MDN did not induce gene mutations by frameshift or base-pair substitution in the genome of the tester strains, when tested with and without metabolic activation up to cytotoxic and/or regulatory compliant concentrations. Thus, MDN is considered to reveal no mutagenic potential in this bacterial reverse mutation assay.

Ref: ICCR, 2023a

SCCS comment

The Ames test is considered valid by the SCCS as it was performed according to the OECD TG. The data show that MDN does not induce gene mutations.

Study #2: In vitro micronucleus test in cultured human lymphocytes

Guideline:	OECD TG 487
Cells:	Human lymphocytes
Replicates:	Duplicates
Test substance:	Methylamido Dihydro Noralfaprostal (MDN),

	CAS 55206-01-2
Batch:	0682542
Purity:	98% (HPLC)
Solvent:	DMSO
Positive controls:	Without S9-mix: Mitomycin C (MMC, pulse treatment): 1.0 µg/mL; Vinblastine (continuous treatment): 12.5 µg/mL With S9-mix: Cyclophosphamide (CPA): 17.5 µg/mL
Negative control:	Solvent control (culture medium with 1.0 % DMSO)
Treatment:	3/28 h exposure/preparation time without S9-mix: Pre-test: 0; 13.3; 23.2; 40.6; 71.1; 124; 218; 381; 666; 1166; 2041 µg/mL Experiment I: 0; 215; 376; 658; 1152; 1267; 1394; 1533; 1687; 1855; 2041 µg/mL 28/28 h exposure/preparation time without S9-mix: Experiment II: 0; 13.3; 23.2; 40.6; 71.1; 124; 218; 381; 666; 1166; 2041 µg/mL 3/28 h exposure/preparation time with S9-mix: Pre-test: 0; 13.3; 23.2; 40.6; 71.1; 124; 218; 381; 666; 1166; 2041 µg/mL Experiment I: 0; 215; 376; 658; 1152; 1267; 1394; 1533; 1687; 1855; 2041 µg/mL
GLP:	In compliance
Study period:	July – September 2023

Material and methods:

The possible clastogenic and aneugenic potential of the test item MDN (0682542 (purity: 98% (HPLC) dissolved in DMSO was evaluated according to OECD 487 under GLP conditions in Human lymphocytes in three independent experiments. The induction of cytogenetic damage in human lymphocytes was assessed in three independent experiments with one preparation interval (28 hours). The cytokinesis blocker, cytochalasin B was added to the cell culture to ensure that there are binucleate cells to be evaluated for micronuclei. Treatments started after a 48-hour stimulation period with phytohaemagglutinin (PHA) when cells were actively proliferating and the cells were prepared at approx. 1.5 – 2.0 fold of the normal cell cycle time (start of exposure until preparation of cells). All cell cultures were set up in duplicates.

In each experimental group, two parallel cultures were analysed. Per culture 1000 binucleated cells were evaluated for cytogenetic damage. The dose selection of the cytogenetic experiment was performed considering the toxicity data in accordance with OECD Guideline 487. With regard to the purity (98 %) of the test item, 2041 µg/mL were applied as top concentration for treatment of the cultures in the pre-test. Test item concentrations ranging from 13.3 to 2041 µg/mL (with and without S9 mix) were chosen for the evaluation of cytotoxicity. In Experiment I, 2041 µg/mL of the test item were applied as top concentration. The test item concentrations ranging from 215 to 2041 µg/mL (with and without S9 mix) were chosen. Considering observed toxicity, 2041 µg/mL were chosen as top concentration in continuous treatment in Experiment II

Phenobarbital/β-naphthoflavone-induced rat liver S9 was used as the metabolic activation system. Concurrent solvent control (DMSO) and appropriate reference mutagens (Mitomycin C, Vinblastine, Cyclophosphamide) were investigated in parallel.

Results:

In Experiment I, neither precipitation nor phase separation of the test item in the culture medium was observed in the absence and in the presence of S9 mix at the end of treatment. In addition, no precipitation occurred in Experiment II in the absence of S9 at the end of treatment. No relevant influence on osmolarity or pH was observed.
Short time treatment without metabolic activation

In the pre-experiment in the absence of S9 mix, no cytotoxicity was observed after treatment with 1166 µg/mL. The next higher tested concentration (2041 µg/mL), which was separated by a factor smaller than requested by the guideline (factor $\times 1.75$) was outside the recommended range of cytotoxicity and therefore not evaluated for cytogenetic damage. To meet the requested range for cytotoxicity, the experiment was repeated as Experiment I with smaller spacing (factor $\times 1.1$) in the concentration range of interest. In Experiment I, no relevant cytotoxicity (14.1% cytostasis) was observed at the highest evaluable concentration (1687 µg/mL). The next higher tested concentration (1855 µg/mL) was not evaluable for the genotoxicity parameter due to strong cytotoxic effects. This indicates that the test item induces a very steep cytotoxic gradient, even though the small meshed spacing between the concentrations were tested. For this reason, further testing was not considered expedient.

In Experiment I in the absence of S9 mix, no relevant increase in the numbers of micronucleated cells were observed after treatment with the test item. The mean percentage of the micronuclei in all treated conditions was within the 95% control limit and none of the values were statistically significantly increased, when compared to the solvent control. There also was no concentration related increase in micronucleus formation, as judged by an appropriate trend test.

Continuous treatment without metabolic activation

In Experiment II in the absence of S9 mix after continuous treatment, clear cytotoxicity (66.8% cytostasis) was observed at the highest evaluated concentration.

In the continuous treatment experiment, no relevant increase in the numbers of micronucleated cells were observed after treatment with the test item. The mean percentage of the micronuclei in all treated conditions was within the 95% control limit and none of the values were statistically significantly increased, when compared to the solvent control. There also was no concentration related increase in micronucleus formation, as judged by an appropriate trend test.

Short time treatment with metabolic activation

In the pre-experiment in the presence of S9 mix, no cytotoxicity (0.8% cytostasis) was observed after treatment with 1166 µg/mL. The next higher tested concentration (2041 µg/mL), which was separated by a factor smaller than requested by the guideline (factor $\times 1.75$) was outside the recommended range of cytotoxicity and therefore not evaluated for cytogenetic damage. To meet the requested range for cytotoxicity the experiment was repeated as Experiment I with smaller spacing (factor $\times 1.1$) in the concentration range of interest. In Experiment I, moderate cytotoxicity (33.7% cytostasis) was observed at the highest evaluable concentration (1855 µg/mL). The next higher tested concentration (2041 µg/mL) was not evaluable for the genotoxicity parameter, due to strong cytotoxic effects. This indicates that the test item induces a very steep cytotoxic gradient, even though the small meshed spacing between the concentrations were tested. For this reason, it was considered to provide sufficient cytotoxicity data for the assessment and that further testing is therefore not expedient.

In Experiment I in the presence of S9 mix, no relevant increase in the numbers of micronucleated cells were observed after treatment with the test item. None of the values were statistically significantly increased, when compared to the solvent control. There also was no concentration related increase in micronucleus formation, as judged by an appropriate trend test. The mean percentage of the micronuclei, however, in the solvent control (1.00%) and in the treatment concentrations 1267 µg/mL (1.05%) and 1855 µg/mL (1.15%) slightly exceeded the 95% control limit of the historical data of the solvent control (0.00 – 0.94% micronucleated cells). Because all values were within the min-max range (0.05 – 1.18% micronucleated cells), these findings can be considered as biologically irrelevant. The outcome of the experiment is therefore negative and valid.

Appropriate mutagens were used as positive control s. They induced statistically significant increases in cells with micronuclei which were clearly within the historical laboratory control data range. Thus, the sensitivity and suitability of the test system was demonstrated.

Conclusion:

Under the conditions of this study, MDN did not induce micronuclei as determined by the *in vitro* micronucleus test in human lymphocytes. Therefore, MDN is considered to be non-mutagenic in this *in vitro* micronucleus test, when tested with and/or without metabolic activation up to cytotoxic or to the highest evaluable concentrations investigating potential structural (clastogenicity) and numerical (aneugenicity) chromosome aberrations.

Ref.: ICCR, 2023b

SCCS comments

The micronucleus test was performed according to the OECD TG and is therefore considered valid. The results show that MDN did not induce aneugenic or clastogenic effect *in vitro*.

Additional information from the Applicant:

Mutagenicity studies with the structural analogue Bimatoprost:

The structural analogue Bimatoprost was tested for its mutagenic potential in three standard and GLP compliant tests (for details see reference: CDER, 2001). The summaries of these studies are in Annex 5.

3.3.6.2 Mutagenicity / genotoxicity *in vivo*

According to the Applicant, no *in vivo* studies were conducted with MDN.

The Applicant did provide descriptive information on an *in vivo* clastogenicity assay in mice (Micronucleus assay) for Bimatoprost. A summary of this study can be found in Annex 5.

Applicants' overall summary on mutagenicity /genotoxicity:

MDN was investigated for its possible mutagenic potential in standard *in vitro* studies compliant to OECD and SCCS requirements under GLP conditions.

No mutagenic/genotoxic properties of MDN were observed. Namely, not in the Ames Test pursuant to OECD 471 investigating potential gene mutations and not in the *in vitro* mammalian micronucleus test pursuant to OECD 487, investigating potential structural (clastogenicity) and numerical (aneugenicity) chromosome aberrations.

The obtained results are in line with results obtained for the structural analogue Bimatoprost, which indicated also neither a mutagenic potential *in vitro* (Ames, L5178Y mouse lymphoma cells) nor *in vivo* (mice micronucleus test).

Finally, the mutagenic / genotoxic potential was sufficiently covered and also in line with SCCS criteria and the ECHA requirements (Reference: ECHA, 2017a). Therefore, there is no need for additional testing.

SCCS overall comment on mutagenicity

Two valid *in vitro* tests were provided by the Applicant, showing no induction of gene mutations in bacteria or chromosomal damage in human lymphocytes by MDN. Therefore, the SCCS considers that there are no concerns in relation to genotoxicity for MDN.

3.3.7 Carcinogenicity**From Opinion SCCS/1635/21**

No relevant experimental data on the carcinogenicity of the prostaglandin analogues assessed in this Opinion have been provided.

A review of the data from available literature has indicated the involvement of prostaglandin E2 in tumorigenesis, mainly through interaction with E Prostaglandin receptors (EP) (Piotrowski *et al.*, 2020). EP2 receptor subtype is known to act as tumour promoter

predominantly through activation of angiogenesis related to induction of Vascular Endothelial Growth Factor (VEGF). EP4 is another receptor with well-described connection to tumourigenesis. The PGE2-induced activation of EP4 receptor results in the development of pro-tumourigenic immune response. EP4 receptor pathway has been implicated in the activation of Treg lymphocytes, which inhibit the inflammatory response to the tumour or can promote growth and invasiveness of tumour cells.

Considering the limited available data, the SCCS cannot exclude the potential carcinogenicity of the prostaglandin analogues.

Ref.: Piotrowski *et al.*, 2020

According to the Applicant, there are no experimental data regarding the possible carcinogenicity of MDN available. The structural analogue Bimatoprost revealed no such potential in rats and mice after daily oral administration of Bimatoprost for 2 years at doses up to 2 mg/kg bw/d in rats and up to 1 mg/kg bw/d in mice. As a consequence, the concern that MDN may be a non-genotoxic carcinogen can be clearly excluded.

SCCS comments

As expressed in the Preamble, the SCCS is of the view that the proposed read-across between MDN and Bimatoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. However, since MDN is not a genotoxic compound, the SCCS has no concern for genotoxic carcinogenicity.

3.3.8 Photo-induced toxicity

3.3.8.1 Phototoxicity / photo-irritation and photosensitisation

No data are available with regard to phototoxicity of MDN.

However, there is no trigger to investigate this endpoint as MDN showed consistently two absorption maxima in the UV/VIS spectrum at 262 nm and at 269 nm under acidic (pH 1.2), around neutral (pH 6.6) and alkalic (pH 13,9) conditions, respectively. At the wavelength of 300 nm and above no absorption was noted. According to EFSA and SCCS criteria no phototoxicity testing should be performed in cases where a test material only absorbs at wavelengths lower than 313 nm, which is clearly the case with MDN.

Ref: Consilab, 2024a; SCCS, 2023

SCCS comment

The SCCS agrees with the Applicant that no phototoxicity testing is needed.

3.3.8.2 Photomutagenicity / photoclastogenicity

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3.3.9 Human data

According to the Applicant, there are no human data regarding MDN per se available.

However, in this respect it is important to mention and in line with deviations of the regular SCCS procedure in its opinion referring also to product data, that recently a clinical study was performed with an eyelash serum product containing MDN as active ingredient in 34 volunteers over a period of twelve weeks.

The objective of this study was to precisely investigate the skin tolerability and efficacy of the product according to clinical-dermatological and ophthalmological test criteria. The primary study objectives consisted in testing skin, eye mucosa and eye tolerance, possible changes in intraocular pressure (IOP) and, if applicable, the sensitization potential. From the start of the study to the end of the study, all effects were to be recorded within the application period after four, eight and twelve weeks of in-home use. The product was applied under real conditions once daily above the lash line according to the instructions on the packaging over the entire application period. The subjects were instructed not to use any equivalent product in the test area during the test period.

From a clinical-dermatological and ophthalmological point of view, there were no relevant reactions in the test area and the product was very well tolerated within the period of twelve weeks. Neither intolerance reactions in terms of skin irritation nor allergic reactions (contact dermatitis) were detected but in two cases were subjective perception of burning eyes reported. Therefore, from a dermatological viewpoint it was concluded that the tested product might lead to skin irritations in particularly sensitive persons.

All test subjects showed healthy ocular mucous membranes before, during and after the application study. There were no pathological changes to mucous membrane of the eyes or the eye itself, despite unintentional misapplication in the eye.

The use of the eyelash serum product clearly showed no findings that indicate or suggest serious or severe side effects in the eye tissue or periocular tissue of the users. This included the absence of serious or irreversible histological changes. There was no evidence from the investigations to suggest the occurrence of potentially serious or irreversible histological changes. Consequently, further histological examinations were not indicated.

Especially no changes in intraocular pressure were detected. The measured IOP values were all within the range of normal values and showed a normal fluctuation range with no obvious connection to the use of the product over the period of twelve weeks. No test interruption, even less ophthalmological treatment was performed in any case.

Finally, the product was very well tolerated and did not lead to any ophthalmologically relevant changes in the ocular mucous membranes or eyes in any of the test subjects. The adverse effects reported by the users are within a range that was usual and to be expected for the incorrect application of eye cosmetics. Very sporadically observed changes in visual acuity were within the normal fluctuation range and could not be observed consistently.

The singularly observed blepharitis was not persistent and was therefore classified as irrelevant. Likewise, the "subjective perceived eye burning" that occurred at the end had no clinical relevance.

In this respect it has to be emphasized again (see section 5.1 Function and uses) that all protective mechanisms of the eye for protection against and removal of foreign substances are intact, when using an eyelash serum. These mechanisms are the eyelid reflex and an open lacrimal duct.

SCCS comments

The Dermatest study was performed in 34 volunteers. This is a small study and it has therefore limited precision to estimate the frequency of adverse effects (0% has an upper 95% confidence limit of 10.3%). Of note, after 8 weeks of use, seven participants (20.6%, 95% CI: 8.7--37.9%) showed mild or moderate blepharitis, which resolved in 6 participants, with one persistent case of moderate blepharitis and one additional case of severely burning eyes at week 12. It is not clear why these reactions were considered by the Applicant as "clinically irrelevant". The SCCS considers that the frequency of side effects is relatively high in such a small group. Besides the studies submitted by the Applicant, there is solid clinical evidence regarding side-effects that are seen with several other prostaglandin analogues, for example the loss of periorbital tissue leading to a "sunken eye" appearance has been shown

not only in the Dermatest study but also in other studies with eyelash lengtheners with prostaglandin analogues.

Ref. Dermatest, 2024

3.3.10 Special investigations

***In silico* assessment of toxicity**

From Opinion SCCS/1635/21

SCCS comments

As noted above, the available information on prostaglandin analogues was either very patchy or missing altogether. In view of this, the SCCS used an *in silico* modelling approach to gain some insights into the potential toxicities of prostaglandin analogues using QSAR models and read-across. It needs to be stressed that the SCCS carried out this exercise only to identify the first level alerts for any potential harmful effects that might impact consumers' health from the use of cosmetic products containing prostaglandin analogues. The endpoints assessed included skin sensitisation, carcinogenicity, mutagenicity and reproductive toxicity. The *in silico* systems used included QSAR-based systems (VEGA-QSAR and US EPA-TEST) and read-across (TOXREAD).

The exercise indicated that Norbimatorprost (MDN) is:

1. not predicted to be mutagenic or genotoxic.
2. flagged for potential carcinogenicity with a reasonable model certainty. This raises the concern that Norbimatorprost may be a non-genotoxic carcinogen.
3. flagged for reproductive and developmental toxicity with a reasonable model certainty, raising the concern that it is potentially a reproductive/ developmental toxicant.

From the Applicant (current submission):

Due to the lack of data on MDN within its safety assessment at that time, the SCCS performed an *in silico* assessment to gain some insights into the potential toxicities of prostaglandin analogues using QSAR models and read-across (Reference: SCCS, 2022). The SCCS raised the concern that PG analogues in general may be non-genotoxic carcinogens. However, the weakness of this information was that there was no evidence whether MDN was flagged specifically. Due to insistence and follow-up, the SCCS had to admit that MDN was outside the prediction space of some of the models used in its *in silico* analysis for the endpoint carcinogenicity (Reference: SCCS, 2024).

An independently conducted *in silico* analysis by an accredited institute (ToxMinds), which was even more comprehensive in terms of models than that performed by the SCCS, confirmed not only that MDN should not be considered a non-genotoxic carcinogen, but also that there was some weakness as SCCS did not use SAR tools generally recommended by EU regulatory bodies for this purpose and that some models also used by the SCCS were not considered as reliable as predictions were outside the applicability domain. In addition, the current QSAR models for carcinogenicity are considered as not powered enough to provide reliable results based on the experience acquired so far by the accredited institute (Reference: ToxMinds, 2024a).

The models used by the accredited institute comprised expert-rules based SAR tools (OECD QSAR Toolbox v.4.6, Derek Nexus v.6.3.0) and statistical based QSAR models (VEGA v.1.2.3, Danish (Q)SAR models) for the *in silico*-based analysis of the carcinogenicity concerns raised. In addition, *in silico* analysis of the potential bacterial mutagenicity by QSAR tools (Derek Nexus v.6.3.0, Sarah Nexus v.3.3.0) were carried out.

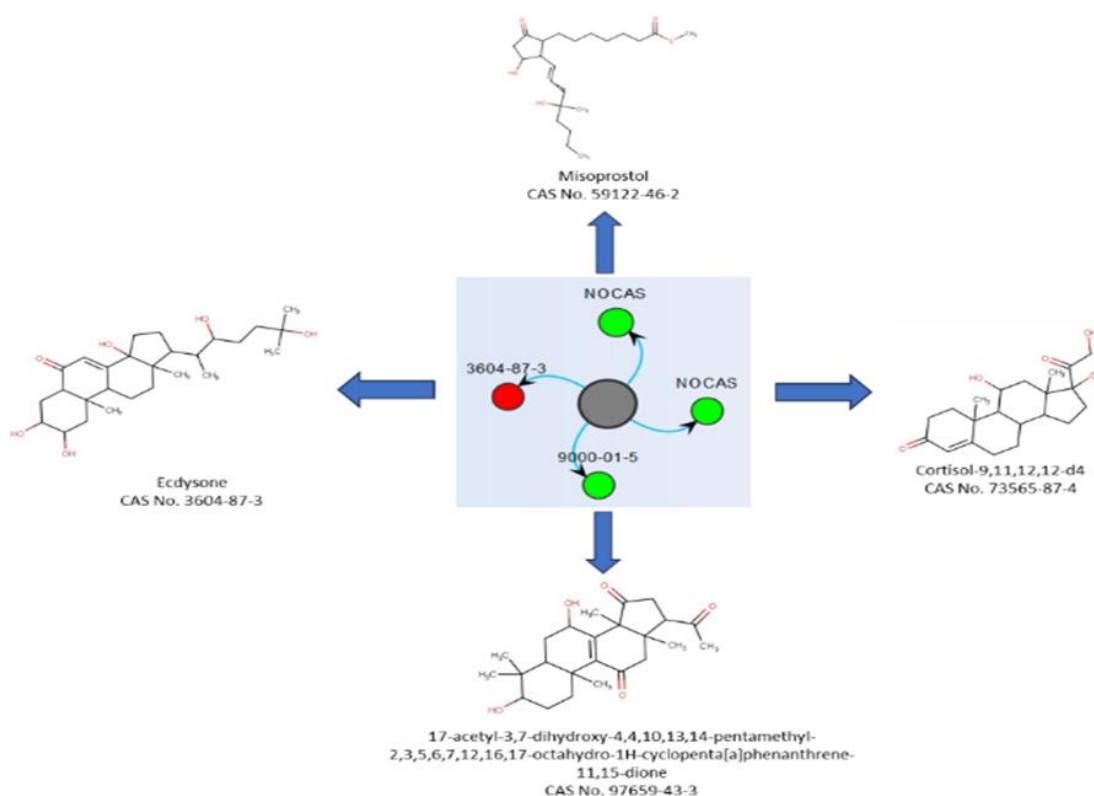
In addition, the ToxRead tool v.0.25 was evaluated to determine whether this model could provide an answer with a reasonable model certainty. The analogues identified by ToxRead

and thus assumed to have been considered by SCCS as suitable analogues are shown in the figure below.

However, it was emphasized that e.g., according to the analogue justification guidance under the ECHA read across assessment framework these ToxRead analogues cannot be considered as good read across substances for MDN and even more so for the analogue that is carcinogenic, Ecdysone as they have a low Dice similarity index of below < 0.29 . In this respect it has to be considered that the OECD QSAR toolbox recommends a minimum Dice index of 0.5 for the presumption of chemical similarity (Reference ToxMinds, 2024a).

Finally, the SCCS concern on the non-genotoxic carcinogenicity potential of MDN can be ruled out for the following reasons:

- 1) The SAR analysis did not indicate any carcinogenicity concern related to MDN;
- 2) The predictions of the QSAR analysis are not reliable, as they are always outside the applicability domain. Also, it should be pointed out that the current QSAR tools are not powered enough to address the carcinogenicity endpoint;
- 3) The *in silico* analysis using the ToxRead tool did not identify suitable analogues for the carcinogenicity endpoint;
- 4) On the basis of the lack on carcinogenicity in two species shown for the close analogue Bimatoprost, MDN is not considered to be carcinogenic, neither regarding a direct nor a non-genotoxic mechanism.



Moreover, based on a scientifically recognized structure-analogy analysis for Bimatoprost, there is no plausible evidence that MDN possesses the property of a non-genotoxic carcinogen.

As already mentioned in sections 6. MDN revealed no mutagenic potential, when tested in the OECD/SCCS compliant Ames test. This is a confirmation of the prediction made by the QSAR analysis on bacterial mutagenicity as both tools predicted MDN to be negative in the Ames test, with the Sarah prediction being within the applicability domain. The lack of possible mutagenicity/genotoxicity was also confirmed by the *in vitro* micronucleus test. No carcinogenic potential is likely considering the fact that the structural analogue Bimatoprost

revealed no such potential in rats and mice after daily oral administration of Bimatoprost for 2 years at doses up to 2 mg/kg bw/d and 1 mg/kg bw/d, respectively. According to the Applicant, as a consequence, the concern that MDN may be a non-genotoxic carcinogen can be clearly excluded.

SCCS comments

As mentioned in section 3.3.6, the SCCS has regarded MDN of no concern regarding genotoxicity, and thus by implication of no concern over genotoxic carcinogenicity.

In regard to non-genotoxic carcinogenicity, the SCCS has noted the Applicant's detailed explanation and the use of QSAR models and read-across approach to exclude non-genotoxic carcinogenicity. The SCCS shares the Applicant's views that the ability of the currently available *in silico* models and tools to predict non-genotoxic carcinogenicity may be limited because they do not cover large enough chemical space relevant to prostaglandin analogues, and also do not cover all the different mechanistic pathways for non-genotoxic carcinogenicity. It, however, needs to be reminded that the SCCS had clearly indicated in the previous Opinion on prostaglandin analogues (SCCS /1635/21) that the *in silico* exercise was carried out 'only to identify the first level alerts for any potential harmful effects' not to obtain definitive evidence to base the overall safety conclusions on.

In this regard, it also needs to be noted that the SAR tools particularly pointed out by the Applicant (e.g. QSAR Toolbox profilers based on structural alerts) are meant for category formation and not for use as predictive tools. The SCCS considers that the profilers are not reliable enough to allow acceptance of a negative prediction (e.g. see a recent analysis of the profilers by AlJallal *et al.* (2024: <https://doi.org/10.1038/s41598-024-69157-1>). Also, as mentioned in section 3.3.5 of this Opinion, there are serious concerns in regard to potential reproductive and developmental toxicity of MDN from the collective view of all the available information. Discounting the concerns over non-genotoxic carcinogenicity and reproductive/developmental toxicity needs experimental data, not *in silico* predictions alone without any supporting data from other lines of evidence.

Read across justification from the Applicant:

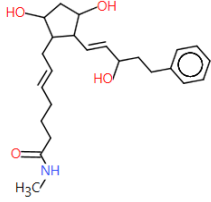
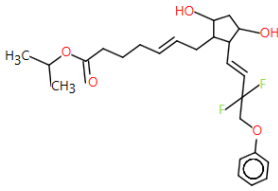
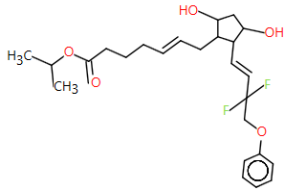
In principle, the read across justification was performed according to the principles of the analogue approach as laid down in the Read-Across Assessment Framework (RAAF) guidance document of ECHA (Reference: ECHA, 2017b). The RAAF provides a framework and principles for scientifically examining a read-across case. In this context of the RAAF, the simplest case of an analogue approach is considered: read-across from a single source substance to a single target substance. However, in this case Bimatoprost and Tafluprost have been evaluated for their suitability as analogues for the target substance MDN (syn. Methylamido Dihydro Noralfaprostal).

In principle, the analogue approach considers the following criteria.

- Common functional groups and structure
- Common structural alerts or reactivity
- Common physicochemical properties
- Likelihood of common breakdown products via biological/degradation processes

The hypothesis was that Bimatoprost (CAS No. 155206-00-1) and Tafluprost (CAS No. 209860-87-7) are suitable read across substances for MDN. Details of the analogue evaluation are presented in Table 7.

Table 7: Analogue evaluation

	Target substance	Analogue 1	Analogue 2
CAS number	155206-01-2	155206-00-1	209860-87-7
Name	MDN	Bimatoprost	Tafluprost
2D Structure			
SMILES from PubChem	<chem>CNC(=O)CCCC=CCC1C(CC(C1C=CC(CCC2=CC=CC=C2)O)O)O</chem>	<chem>CCNC(=O)CCCC=CCC1C(C(C1C=CC(CCC2=CC=CC=C2)O)O)O</chem>	<chem>CC(C)OC(=O)CCCC=CC1C(CC(C1C=CC(COC2=CC=CC=C2)(F)F)O)O</chem>
Analogue ranking	-	Suitable	Suitable with Interpretation
Structural similarity			
Dice index from OECD QSAR Toolbox v.4.6	-	0.91	0.66
Physico-chemical properties			
Molecular weight (Da)	401.55	415.58	452.54
Melting Point (deg C), Prediction software: MPBPWIN v.1.44	261.59	267.01	204.56
Boiling Point (deg C), Prediction software: MPBPWIN v.1.44	604.87	616.48	509.44
Vapour Pressure (Pa, 25°C), Prediction software: MPBPWIN v.1.44	2.04E-015	7.03E-016	5.06E-011
Log Know, Prediction software: KOWWIN v1.69	3.48	3.97	6.51
Water solubility (mg/L), Prediction software: WSKOW v1.43	3.194	0.9948	0.003928

	Target substance	Analogue 1	Analogue 2
	Functional groups and profiling		
Organic functional groups from OECD QSAR Toolbox v.4.6	Alcohol Alkene moiety Allyl Aryl Cycloalkane Organic amide and thioamide	Alcohol Alkene moiety Allyl Aryl Cycloalkane Organic amide and thioamide	Alcohol Alkane, branched with secondary carbon Alkene moiety Alkyl halide Allyl Aryl Carboxylic acid ester Cycloalkane Dihydroxyl derivatives Ether moiety Isopropyl
Structural alerts from OECD QSAR Toolbox v.4.6	DNA binding by OECD: Michael addition >> P450 Mediated Activation to Quinones and Quinone-type Chemicals >> Arenes Estrogen Receptor Binding: Strong binder, OH group Toxic hazard classification by Cramer: High (Class III)	DNA binding by OECD: Michael addition >> P450 Mediated Activation to Quinones and Quinone-type Chemicals >> Arenes Estrogen Receptor Binding: Strong binder, OH group Toxic hazard classification by Cramer: High (Class III)	Estrogen Receptor Binding: Strong binder, OH group Toxic hazard classification by Cramer: High (Class III)
	Metabolism		
Prediction of the first metabolic reaction using Meteor Nexus v.3.2.0 and literature data	Meteor v.3.2.0: Hydrolysis of Acyclic Carboxylic Amides with the formation of Bimatoprost	Literature: Bimatoprost is rapidly hydrolyzed to form its biologically active acid metabolite (Al-Ghananeem and Crooks, 2007)	Literature: Tafluprost is an ester prodrug which is rapidly hydrolyzed by corneal esterases to form its biologically active acid metabolite. Tafluprost acid is further metabolized via fatty acid β -oxidation and phase II conjugation (CDER, 2011)

Thus, the main outcome of the analogue assessment can be summarized as follows:

- Bimatoprost presents a high Dice index of 0.91, indicative of structural similarity, while Tafluprost presents a lower similarity index of 0.66.
- Bimatoprost presents the same key functional groups (e.g., alcohol, cycloalkane, alkene moiety and organic amide and thioamide) as compared to the MDN, while Tafluprost presents additional functional groups like 'alkyl halide' and 'ether moiety'.
- With respect to the structural alerts, the (Q)SAR analysis (OECD QSAR Toolbox v.4.6) revealed that Bimatoprost and Tafluprost are classified as Cramer Class III (high toxicity), similar to MDN.
- Bimatoprost also presents the same structural alerts as MDN, which have been identified by the 'DNA binding by OECD' and 'Estrogen Receptor Binding' and 'Oncologic Primary

Classification' profilers. Tafluprost does not present the alert identified by the 'DNA binding by OECD' profiler.

- Bimatoprost presents physicochemical properties in the same range as compared to the target substance, while Tafluprost presents differences with respect to water solubility and log Kow values. These differences suggest that Tafluprost is less bioavailable under oral exposure conditions compared to the target substance.
- Data available for Bimatoprost indicate that the substance is rapidly hydrolyzed to form Bimatoprost acid; Tafluprost also undergoes hydrolysis giving rise to Tafluprost acid. MDN is predicted by MeteorTM Nexus v.3.2 to undergo predominantly hydrolysis of the acyclic carboxylic amide moiety to form Bimatoprost acid and to a lesser extent aromatic hydroxylation.

Bimatoprost is ranked as suitable analogue, while Tafluprost is considered only as 'suitable with interpretation' analogue due to differences in physicochemical properties and functional groups. The key arguments that Bimatoprost can be considered as a reliable structural and functional analogue to MDN are the obtained facts that Bimatoprost presents a high similarity index, shares the same structural alerts and functional groups and gives rise to the same metabolite, Bimatoprost acid as MDN. Finally, considering all of the above presented data, the profile of Bimatoprost is considered as sufficiently valid to justify the structural analogue read across approach for the target MDN.

SCCS comments

As expressed in the Preamble, the SCCS is of the view that the proposed read-across between MDN and Bimatoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence.

3.4 SAFETY EVALUATION (including calculation of the MoS)

According to the Applicant:

The obtained NOAEL of 0.1 mg/kg bw/d (100 µg/kg bw/d) of the oral (gavage) developmental and prenatal/postnatal developmental toxicity study in rats of the structural analogue Bimatoprost is used as PoD for the calculation of the Margin of Safety (MoS) compliant to the guidance of the SCCS. This will be related to the value of 7.76% (mean: 4.95% plus 2.81% SD) for dermal absorption for the calculation of the Systemic Exposure Dose (SED, mg/kg bw/day).

Eyelash serum with 0.0275% MDN		
Amount of product applied daily (qx)	0.0017	g/day
Default human body weight	60	kg
Relative daily exposure (Eproduct = qx/60 kg)	0.0283	mg/kg bw/d
Dermal absorption as percentage (Dap (%), mean + 1 SD)	7.76	%
Concentration in the product (C (%))	0.0275	%
Systemic exposure dose (SED, mg/kg bw/d) = Eproduct x C (%)/100 x Dap (%)/100)	0.0006046	µg/kg bw/d
Point of departure (PoD, former NOAEL from the oral (gavage) developmental and perinatal/postnatal reproduction toxicity study with the structural analogue Bimatoprost in rats)	100	µg/kg bw/d
Route-to-Route Faktor	0.5	
PoDsys = PoD x Route-to Route Faktor	50	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	82695	

With respect to the standard procedure, an ingredient is considered safe for use in humans at a margin of safety (MoS) at and above 100. It is important to consider that the factor 100 is a composite value. It consists of 4 individual factors (4.0 x 2.5 x 3.2 x 3.2: animal data: Factor 4 for cross-species kinetics, Factor 2.5 for cross-species dynamics; Human data: Factor 3.2 for variability in kinetics, factor 3.2 for variability in dynamics).

However, this approach is not fixed and in case of valid and reliable scientific data, especially from toxicokinetics and toxicodynamics, it is justified to be modified on a case-by-case consideration.

In the extensive Center for Drug Evaluation and Research (CDER) of the US Food and Drug Administration (FDA) registration document for Bimatoprost, there is clear evidence that Bimatoprost was metabolized very rapidly to the active metabolite Bimatoprost acid in mice, rats, rabbits, monkeys, and humans. (CDER, 2001). The comparative *in vitro* studies indicated clearly that there were no relevant species differences in the metabolic profile. Thus, species differences can clearly be excluded. Therefore, the factor 4 for interspecies variability in kinetics is omitted in the calculation. Based on this fact, a margin of safety of 25 clearly satisfies the criteria of a valid and robust safety assessment.

Consequently, MDN can be regarded as safe for humans, when using eyelash sera at a concentration of up to 0.0275%.

SCCS comments

The MoS calculation of MDN was based on a NOAEL derived from a study with Bimatoprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between MDN and Bimatoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a PoD for MDN the SCCS cannot derive a MoS for MDN.

3.5 DISCUSSION

A general remark is that nearly no toxicological data were provided on the substance under consideration, MDN. Experimental data were provided on skin and eye irritation, skin sensitisation, genotoxicity and dermal absorption. To fill this gap, the Applicant proposed a read-across approach using data from Bimatoprost - a structural analogue of MDN. As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across from a single analogue (Bimatoprost) can only be considered if it is not the only evidence for a given endpoint – *i.e.* if there is further information from other line(s) of evidence to support the read-across outcome. However, since no other lines of evidence were provided for MDN, the read-across as proposed by the Applicant has not been used in the current Opinion. Only the results obtained by using the compound MDN are taken into consideration.

Physicochemical properties

A full report of the chemical characterization of MDN in terms of purity, identity and impurities in the different batches has not been provided by the Applicant.

No data were provided on chemical characterisation regarding IR, ¹³C-NMR and ¹H-NMR for all the batches used in the studies.

It is noted that MDN is not stable in water.

Data on homogeneity and stability of the test substance under experimental conditions has not been provided by the Applicant.

Function and use

MDN is used in eyelash serum that is intended to be applied once a day to the upper lash line like normal eyeliner.

Exposure assessment and Toxicokinetics

A dermal absorption value of 7.76% (4.95% ± 2.81%) was determined in an *in vitro* skin absorption study.

There are no experimental data on toxicokinetics of MDN.

Toxicological Evaluation

Irritation and corrosivity

Skin irritation was assessed *in vitro* according to OECD TG439. MDN was irritant to the skin. The test method used does not allow to differentiate whether MDN is only a skin irritant or also a corrosive substance.

Eye irritation was assessed *in vitro* according to 492. Neat MDN was not an eye irritant in this assay. The study was well-performed; however, this outcome is contrary to the fact that MDN was shown to be a skin irritant. Substances identified as skin irritants often exhibit (severe) eye irritation, particularly when tested as neat substance. Furthermore, according to the Cayman safety data sheet (2023b), MDN is classified as Eye Irritation 2A (causes serious eye irritation). As an example, in a clinical setting, where the product was directly applied in the eye, it was also observed that the structural analogue Bimatoprost induced adverse eye effects in a subset of patients.

Skin sensitisation

Skin sensitisation potential of MDN was tested in two *in vitro* assays. MDN was negative in the DPRA and Keratinosens assay. Taking the data together by applying the 2o3 Defined Approach, as described in OECD TG497, it was concluded that MDN is not a skin sensitizer.

Acute toxicity

Since the product applied above the eyelid only contains small amounts of MDN, there is no concern over acute toxicity of MDN.

Repeated dose toxicity

There are no repeated dose toxicity studies on MDN available.

Reproductive and developmental toxicity

There are no reproductive or developmental toxicity studies on MDN available. The SCCS has concerns on the developmental and reprotoxic effects of prostaglandin analogues. Therefore, it is not possible to evaluate the safety of MDN in the intended cosmetic applications in the absence of data on reproductive /developmental toxicity.

Mutagenicity / genotoxicity

Mutagenicity and genotoxicity of MDN was studied in the Ames test and the micronucleus test. Both tests were considered valid and were negative. Hence, MDN did not induce gene mutations in bacteria or chromosomal damage in human lymphocytes. Therefore, there are no concerns in relation to genotoxicity for MDN.

Carcinogenicity

There are no experimental data on carcinogenicity of MDN provided. The SCCS has regarded MDN of no concern regarding genotoxicity, and thus by implication of no concern over genotoxic carcinogenicity.

Photo-induced toxicity

No data are available with regard to phototoxicity of MDN. However, there is no trigger to investigate this endpoint as MDN showed consistently two absorption maxima in the UV/VIS spectrum at 262 nm and at 269 nm under acidic (pH 1.2), around neutral (pH 6.6) and alkaline (pH 13.9) conditions, respectively. At the wavelength of 300 nm and above no absorption was noted. According to EFSA and SCCS criteria no phototoxicity testing should be performed in cases where a test material only absorbs at wavelengths lower than 313 nm, which is clearly the case with MDN.

Human data

A clinical study was available in which skin tolerability and efficacy of an eyelash with MDN as active ingredient was tested in 34 volunteers. This is a small study with limited precision. Of note, after 8 weeks of use, seven participants showed mild or moderate blepharitis, which resolved in 6, with one persistent case of moderate blepharitis and one additional case of severely burning eyes at week 12. It is not clear why these reactions were considered by the Applicant as "clinically irrelevant". The frequency of side effects is relatively high in such a small group. Besides the studies submitted by the Applicant, there is solid clinical evidence regarding side-effects that are seen with several other prostaglandin analogues. E.g. the loss of periorbital tissue leading to a "sunken eye" appearance has been shown not only in the Dermatest study but also in other studies with eye lash lengtheners with prostaglandin analogues.

Special investigation

The Applicant performed used QSAR models and read-across approach to exclude nongenotoxic carcinogenicity. The SCCS shares the Applicant's views that the ability of the currently available *in silico* models and tools to predict non-genotoxic carcinogenicity may be limited because they do not cover large enough chemical space relevant to prostaglandin analogues and also do not cover all the different mechanistic pathways for non-genotoxic carcinogenicity. It, however, needs to be reminded that the SCCS had clearly indicated in the previous Opinion on prostaglandin analogues (SCCS/1635/21) that the *in silico* exercise was carried out 'only to identify the first level alerts for any potential harmful effects' not to obtain definitive evidence to base the overall safety conclusions on.

Safety evaluation

The MoS calculation of MDN as performed by the Applicant was based on a NOAEL derived from a study with Bimatoprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between MDN and Bimatoprost does not fulfil the critical

requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a PoD for MDN the SCCS cannot derive a MoS for MDN.

4. OPINION ON ISOPROPYL CLOPROSTENATE (IPCP)

Preamble

In the submitted safety dossier, the Applicants provided experimental data for Isopropyl Cloprostenate (IPCP) from studies on skin and eye irritation, skin sensitisation and genotoxicity. Also, a dermal absorption study for IPCP was provided. No data were provided either on toxicokinetics, or systemic toxicological endpoints, e.g. repeated dose toxicity and developmental and reproductive toxicity. To fill this gap, the Applicants proposed a read-across approach using data from Cloprostenol (Applicant-1) or Travoprost (Applicant-2).

In this regard, it is not clear on what basis the selection of Cloprostenol as the only analogue for read-across was made by Applicant-1.

Whilst the SCCS appreciates Applicant-2 for discussing the read-across process, and explaining how Tanimoto similarity scores were calculated, it is not clear on what basis the three analogues (Latanoprost, Travoprost, Bimatoprost) were selected out of several other prostaglandin analogues. Furthermore, despite the Tanimoto similarity index being >0.8, the finally selected analogue (Travoprost) differs from the target substance (IPCP) in aspects that are important determinants of toxicokinetic behaviour. There is a large difference in the solubilities of the two compounds (Travoprost being 161 times more soluble than IPCP as provided by the Applicant-2), and Travoprost has a -CF₃ moiety instead of chlorine on the phenyl ring in IPCP. These differences are likely to make the two compounds different in terms of toxicokinetic and other toxicological aspects, and therefore Travoprost cannot be regarded as 'similar' to IPCP for use in read-across.

In this regard, no details have been provided on how the analogues were searched, how many were identified, and on what basis they were deselected or selected for use in the read-across. Furthermore, although predictions derived from a non-testing NAM like read-across provide an important line of evidence, they are not considered sufficient as a standalone evidence to allow forming a conclusion on the lack of toxicity of a chemical substance against a given endpoint. Read-across outcomes therefore need to be supported by other lines of evidence, where the overall weight of evidence can provide a consistent and more reliable basis for decision making in regulatory risk assessments. For example, see EFSA (2025) Guidance on Read-across.

In view of this, the proposed read-across from a single analogue (Cloprostenol or Travoprost) can only be considered if it is not the only line of evidence for any given endpoint – *i.e.* if there is further information from other line(s) of evidence to support the read-across outcome. This means that the SCCS could only rely on the use of read-across from Cloprostenol or Travoprost to IPCP for a given endpoint where it formed part of a weight of evidence.

However, since no other lines of evidence were provided for IPCP, the read-across as proposed by the Applicant has not been used in the current Opinion.

4.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

4.1.1 Chemical identity

4.1.1.1 Primary name and/or INCI name

Isopropyl Cloprostenate

4.1.1.2 Chemical names**Isopropyl cloprostenate (PubChem, Cayman):**

IUPAC: propan-2-yl (Z)-7-[(1R,2R,3R,5S)-2-[(E,3R)-4-(3-chlorophenoxy)-3-hydroxybut-1-enyl]-3,5 dihydroxycyclopentyl]hept-5-enoate

(+)-Cloprostenol isopropyl ester

(+)-Cloprostenol isopropyl ester, ethanol solution

Cloprostenol isopropyl ester

Isopropyl (Z)-7-((1R,2R,3R,5S)-2-((R,E)-4-(3-chlorophenoxy)-3-hydroxybut-1-en-1-yl)-3,5-dihydroxycyclopentyl)hept-5-enoate

Propan-2-yl (Z)-7-[(1R,2R,3R,5S)-2-[(E,3R)-4-(3-chlorophenoxy)-3-hydroxybut-1-enyl]-3,5-dihydroxycyclopentyl]hept-5-enoate

5-Heptenoic acid, 7-((1R,2R,3R,5S)-2-((1E,3R)-4-(3-chlorophenoxy)-3-hydroxy-1-buten-1-yl)-3,5-dihydroxycyclopentyl)-, 1-methylethyl ester, (5Z)-

5-Heptenoic acid, 7-((1R,2R,3R,5S)-2-((1E,3R)-4-(3-chlorophenoxy)-3-hydroxy-1-butenyl)-3,5-dihydroxycyclopentyl)-, 1-methylethyl ester, (5Z)-

5-Heptenoic acid, 7-(2-(4-(3-chlorophenoxy)-3-hydroxy-1-butenyl)-3,5-dihydroxycyclopentyl)-, 1-methylethyl ester, (1R-(1 α (Z),2 β (1E,3R*),3 α ,5 α))

Propan-2-yl (5Z)-7-[(1R,2R,3R,5S)-2-[(1E,3R)-4-(3-chlorophenoxy)-3-hydroxybut-1-en-1-yl]-3,5-dihydroxycyclopentyl]hept-5-enoate

(+)-5-cis Cloprostenol isopropyl ester

(+)-16-m-Chlorophenoxy tetranor Prostaglandin F2 α isopropyl ester

4.1.1.3 Trade names and abbreviations

Isopropyl Cloprostenate: /

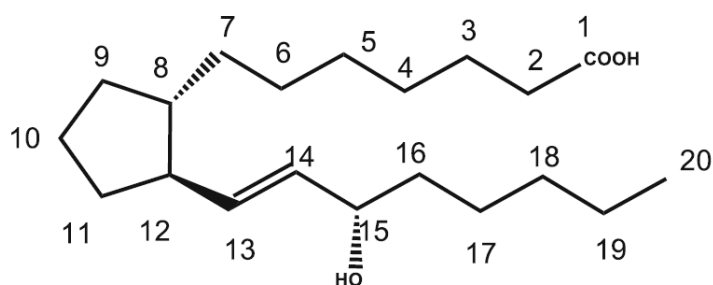
4.1.1.4 CAS / EC number

Isopropyl Cloprostenate: CAS 157283-66-4 / EC 682-025-2

4.1.1.5 Structural formula

The basic skeleton of all the naturally occurring prostaglandins consists of 20 carbon chain in which C-8 to C-12 is a cyclopentyl ring (Fig. 1). A carboxy group is present at C-1 terminal and an α -hydroxy group is present at C-15 position along with a double bond at C-13 position. This basic pharmacophore moiety is essential for its binding with the receptor. The chain containing carboxy group (C-1 to C-7) is known as the α -chain, and the chain containing hydroxyl group (C-13 to C-20) is termed as the β -chain (Piplani *et al.*, 2016).

Figure 1. The structural formula of a PG prototype. The basic structure is shown without the type-specific modifications (oxo-, hydroxy group(s)) of the cyclopentane ring and without the second double bond in the α -side chain (C5-C6) (Piplani *et al.* 2016).



INCI Name/Substance Name	CAS No.	Structural formula
ISOPROPYL CLOPROSTENATE	157283-66-4	

SCCS comment

The SCCS has noted that Isopropyl Cloprostenate has chiral carbons in its structure and hence can exist in different stereoisomeric forms.

4.1.1.6 Empirical formula

Isopropyl Cloprostenate: $C_{25}H_{35}ClO_6$

4.1.2 Physical form

Isopropyl Cloprostenate:

Pure (pristine) material: oily viscous liquid, pale.
Generally, the material is supplied as a 10% solution in ethanol.

Ref.: Cayman, 2018e

4.1.3 Molecular weight

Isopropyl Cloprostenate: 467 g/mol

4.1.4 Purity, composition and substance codes

Applicant 1:

Isopropyl Cloprostenate:

Purity specification: $\geq 98\%$ (HPLC) Batch 0684851: 100% (HPLC, toxicity testing)
Batch 0696699: 98% (HLPC, toxicity testing)
Batch 0695607: 100% (HLPC, toxicity testing)
Batch 0697058: 98% (HPLC, physicochemical data)

Ref.: Cayman, 2018a, 2018b, 2018c, 2018e

Applicant 2: /

4.1.5 Impurities / accompanying contaminants

Applicant 1:

Isopropyl Cloprostenate:

Water content: Negligible but not regularly analysed.
Isomers: HPLC Purity demonstrates regularly that the C-5,6 trans double bond isomer of the C-5,6 cis double bond in the alpha chain is $\leq 2\%$.
Residual solvents: Negligible, prior to formulation in ethanol, but not regularly analysed.
Elemental Impurities: Elemental impurities are not assessed as the risk for presence of these impurities (including Cadmium, Lead, Arsenic, and Mercury) in the Isopropyl Cloprostenate to be higher than the ICH Permitted Cutaneous Concentration has been assessed.
The primary source of these elements is potable water. The water employed in the synthesis at Cayman Chemical and its wholly owned subsidiary Cayman Pharma is regulated by their respective national environmental departments, where these impurities are controlled to very low levels.
Further, the ICH limits assume a daily intake of 10 g of the concerned material, the use of Isopropyl Cloprostenate in cosmetics is at least over 1000-fold less. Therefore, the probability of occurrence for these impurities in a final product at or near the allowable limit is very low.

Ref.: Cayman, 2024a

Applicant 2:

/

SCCS comments on purity and impurities

Given the presence of multiple chiral centers and double bonds in Isopropyl Cloprostenate, the potential for numerous stereo-isomers exists.

Details on the analytical method or its validation have not been shown, nor confirmed whether a stereoselective analytical method was used to verify the purity.

Chemical identification of the test substance along with the analytical method and detailed data on the chemical identification and quantitation of the organic impurities (isomers), have not been provided.

Residual solvents have not been reported in detail nor analysed regularly from batch to batch. The heavy metal content (including Cadmium, Lead, Arsenic, and Mercury) has not been reported.

4.1.6 SolubilityApplicant 1:**Isopropyl Cloprostenate:**

Water solubility: **Measured according to OECD TG105:**
99.1 mg/L (at 20 °C, RSD: 1.0%, OECD 105 and Regulation (EC) No 440/2008 method A.6, GLP)

Further solubilities:

Ethanol: completely
PBS (pH): about 10 mg/mL
DMSO: about 100 mg/mL
Dimethyl formamide: about 100 mg/mL

Ref.: Consilab, 2024c

Applicant 2: /**SCCS comment**

The available information suggests that Isopropyl Cloprostenate is (practically) insoluble in water.

4.1.7 Partition coefficient (Log P_{ow})Applicant 1:**Isopropyl Cloprostenate:**

Partition coefficient (Log P_{ow}): **Measured according to OECD 117:**
3.86 ± < 0.01 (HPLC, OECD 117, Regulation EC No. 440/2008 method A.24, GLP)

Ref.: Consilab, 2024d

Applicant 2: /**SCCS comment**

The available information suggests that Isopropyl Cloprostenate is hydrophobic.

4.1.8 Additional physical and chemical specificationsApplicant 1:

UV/VIS Absorption Spectra: **Measured:**
Isopropyl Cloprostenate (showed two absorption maxima/shoulders according to the OECD-Guideline 101 (1981) and Test Guideline OPPTS 830.7050 (1996) tested under GLP conditions:

λ (nm)	pH	log ϵ	λ (nm)	pH	log ϵ
275	6.8	3.15	282	6.8	3.11
	1.3	3.15		1.3	3.11
	13.9	3.16		13.9	3.12

Therefore, there is no trigger to investigate this endpoint as IPCP showed consistently two absorption maxima in the UV/VIS spectrum at 275 nm and at 282 nm under acidic (pH 1.3), neutral (pH 6.8) and alkalic (pH 13,9) conditions, respectively. At the wavelength of 300 nm and above no absorption was noted. According to EFSA and SCCS criteria no phototoxicity testing should be performed in cases where a test material only absorbs at wavelengths lower than 313 nm, which is clearly the case with IPCP.

Ref.: Consilab, 2024a

Melting Point:

Measured:

IPCP has no melting point between -90°C and the decomposition temperature of 340°C at atmospheric pressure (984 to 987 hPa) as determined by Differential Scanning Calorimetry (OECD 102, Regulation (EC) No 440/2008 method A.1., GLP).

Ref.: Consilab, 2024b

Boiling Point:

Measured:

IPCP shows no boiling point but is evaporating at 230°C under concurrent decomposition at atmospheric pressure (984 to 987 hPa) as determined by Differential Scanning Calorimetry (OECD 103, Regulation (EC) No 440/2008 method A.2., GLP).

Ref.: Consilab, 2024b

Vapour Pressure (Pa, 25°C):

Predicted:

4.83E-014 (Prediction software: MPBPWIN v.1.44)

Ref.: Epi Suite, 2023; ToxMinds, 2024b

SCCS comments

The SCCS notes that no data was provided on:

- organoleptic properties (odour, taste if relevant)
- flash point
- density
- viscosity
- pKa
- pH
- refractive index.

For the predicted value of vapour pressure, information about the reliability of the predictions have not been provided by the Applicant.

Applicant 2: /

4.1.9 Homogeneity and Stability

Applicant 1:

Isopropyl Cloprostenate:

Stability: ≥ 2 years (at -20°C)
In the product at ambient temperature at least 6 months.

Ref.: Cayman, 2021

Applicant 2: /

SCCS comment

The Applicant has not provided experimental data on homogeneity and stability of Isopropyl Cloprostenate under use conditions.

SCCS general comments to physicochemical characterisation

No data were provided on the chemical identification of the test substance by IR, ^{13}C -NMR or ^1H -NMR for any of the batches used in the studies.

A full report on the chemical characterisation of ICPC, including its identity, purity, and impurity profile (e.g. isomeric impurities), has not been provided. Detailed analytical data and validation of the methods used for identification and quantitation of the test substance and its impurities are missing.

Data on the homogeneity of the test solutions, with respect to the content of the test substance under experimental conditions, have not been provided.

Data on the stability of the test substance under experimental conditions of the reported studies and under conditions of use have not been provided. Validated analytical procedures need to be used to determine stability of the test substance.

4.2 EXPOSURE ASSESSMENT & TOXICOKINETICS

4.2.1 Function and uses

Isopropyl Cloprostenate:

Applicant 1:

Isopropyl Cloprostenate is chemically a fully synthetic substance in the form of the Isopropyl ester of Cloprostenate, which is converted under physiological conditions into Cloprostenol, a synthetic prostaglandin analogue (Reference: SCCS, 2022).

In cosmetics Isopropyl Cloprostenate is used to conditioning eyelashes and eyebrows and promote natural growth by the use of eyelash and eyebrow serums products.

There are clear instructions for application for both products (products packages). The usage instructions on the packaging for both, the eyelash and the eyebrow serums are unambiguously formulated. The intended purpose clearly emerges from the product description. Misuse is not to be expected from the usage instructions. Both products are for external use only and are designed not to come into contact with the eyes (References: Nutracosmetic, 2024a, b).

As can be seen from the images below, there is a clear difference between the use and application of an eyedrop medication drug and eyelash serums (References: Santen, 2020; Nutracosmetic, 2024a).

			
Eyedrop medication: Position the head horizontally in the neck, pull downward lower eyelid, one drop between lower eyelid and eye, eyes to be closed and press the finger about 1 minute on the inner edge of the eye to inhibit drainage via the lacrimal duct.			Eyelash serum: Only once a day, preferably before bedtime, a thin layer of product (approx. 1 – 2 mm) to the base of upper eyelashes, only a small amount of the fluid, no need for redip the brush, applied fluid drying by itself, rubbing of eyes to be avoided until the fluid is totally absorbed, application only on dry skin surface.

Importantly, further differences in use between eyedrop medication and eyelash serum use have to be considered. All protective mechanisms of the eye for protection against and removal of foreign substances are intact, when using a, eyelash serum. These mechanisms are the eyelid reflex and an open lacrimal duct. In accordance with the literature "*Greater than 80% of a solution dropped into the human eye is excreted within one to two minutes*" (Reference: OECD, 2020; Mikkelsen *et al.*, 1973). In contrast, when using eye drops, these protective mechanisms must be specifically suppressed (keeping the eye open and squeezing the tear duct shut) so that the drug can act at all.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate
by the Applicant 1. 2024-09-16

SCCS comments

It is notable that the intended application of Isopropyl Cloprostenate containing formulation(s) will be in close proximity to the eye.

4.2.2 Dermal / percutaneous absorption

A full dermal absorption study of radiolabelled Isopropyl Cloprostenate from a cosmetic formulation through healthy human skin *in vitro* was provided by **Applicant 1**.

Applicant 1:

In vitro dermal penetration study

Study Design:

Reference:	Eurofins, 2024
Date of report:	30.07.2024
Study period:	15.03.2024 – 16.05.2024
Guidelines/Methods:	OECD 428, SCCS Notes of Guidance (12 revision), SCCS basic criteria for the <i>in vitro</i> assessment of dermal absorption of cosmetic ingredient, OECD Guidance No. 28
Test system:	Human split-thickness skin membranes (300 – 400 µm)
Group size:	3 female donors (age range: 31 – 50 years), 1 male donor (age: 31 years)
Number of cells:	12 cells in total (3 replicates from each of the 4 donors)

Opinion on Prostaglandin analogues used in cosmetic products

Test substances:	1) Isopropyl Cloprostenate 2) [³ H]-Isopropyl Cloprostenate 3) Eyelash serum product without Isopropyl Cloprostenate
Batch (Purity):	1) 0695607 (Purity: 100% (HPLC, Cayman, 2018e) 2) 240216 (Purity: >98%) 3) NL5103 / 1414NBN23
Application area	1 cm ² (low of receptor fluid = 1.5 mL/h)
Active ingredient concentration (w/w)	0.005%
Dose	10 µL mg/cm ²
Amount of reference item applied on skin	2.75 µg
Receptor fluid:	3% polyethylene glycol 20 oleyl ether in PBS (0.01 M, pH 7.4)
Diffusion cells:	Dynamic cells (application area = 1 cm ² , flow of receptor fluid = 1.5 mL/h)
Skin temperature:	32 ± 1°C
Skin integrity measurement:	Transepidermal water loss (TEWL)
Check of the test item stability at 32°C during 24 hours	Yes (n=1)
Sampling of Receptor Fluid	30 min, 2h, 4h, 6h, 12h and 24h after application
Stratum corneum tape stripping:	A maximum of 11 strips was performed and were pooled as follows: 1-2, 3-7, 8-11
GLP:	Yes
Published:	No

Material and methods:

The aim of this study was to investigate the rate and extent of the *in vitro* dermal absorption of Isopropyl Cloprostenate (0.005%) from a cosmetic formulation as test item by using the [³H]-Isopropyl Cloprostenate as a tracer according to OECD 428 including specific requirements of the Scientific Committee for Consumer Safety (SCCS, Notes of guidance, 12th revision, 2023) under GLP conditions. This test item was applied to the surface of healthy human skin of 4 donors mounted on dynamic cells. The test item contained the reference item in the form of an isotopic solution of Isopropyl Cloprostenate and the radiolabelled compound [³H]-Isopropyl Cloprostenate was prepared at the test site. Approximately 0.5 µCi was applied on each cell (10 µL/cm²). The homogeneity of the radiolabelled test item was checked on 10 µL of the radiolabelled preparation. The receptor fluid consisted of 3% polyethylene glycol 20 oleyl ether in PBS (0.01 M, pH 7.4) and the flow of receptor fluid was 1.5 mL/h. The radiochemical purity analysed by HPLC was 98.20% at the day of formulation, 97.95% at the day of preparation (T0) and 98.57% after storage for 24 hours at 32 °C indicating stability during the experiment.

The skin samples were placed on the receptor compartment and the donor compartment were placed onto the skin samples. The skin surface temperature was maintained at 32°C ± 1°C, with a fixed water bath integrated in the dynamic system. After at least 30 minutes of stabilization, the integrity of the skin was verified using the TEWL method. The donor compartment was not closed (non-occlusive system) and the preparation for the test item was applied homogeneously at 10 µL/cm² (10 µL/cell) and was left on the skin for 24 hours. The receptor fluid was collected continuously passing through the dynamic system at a flow rate of 1.5 mL/h. At each interval time, 30 min, 2h, 4h, 6h, 12h and 24h after application, the receptor fluid was placed into one or several vials. Finally, the skin was washed at 24h post-application and thereafter, the cells were dismantled and the Rinsing Donor Compartment (RCD) was obtained.

The stratum corneum was taken off using adhesive scotch by stripping. The number of strips for all cells ranged from 2 to 11 strips considering the fragility of the skin samples after 24h in contact with the prepared test item. The strips were pooled as follow for analysis: 1-2, 3-7, 8-11. Thereafter, the epidermis and dermis were separated. Samples were analysed for radiolabel content by scintillation counting (Tricarb 2910TR, Perkin) and calculations were

performed using Excel directly from the raw data obtained with the scintillation counter. Conversion of the counts per minute (cpm) to disintegrations per minute (dpm) was performed directly.

Results and conclusion:

The skin samples were considered as suitable for the use in the test as the TEWL revealed acceptable values in the range between 0.5 - 13 g/m²/h. The homogeneity of the test item was in the acceptable range and the radiolabelled test item was shown to be stable throughout the experimental period.

As the cells J and K did neither had a recovery rate of 100 ± 20.00% according to the guideline OECD Number 28 (2004) and OECD 156 (2011) for volatile test substances like IPCP, the results of this study were calculated without the J and K cells.

The summary of the mean absorption of Isopropyl Cloprostenate through human split thickness skin is presented in following Tables 1 and 2:

Table 1: Distribution of ³H-Isopropyl Cloprostenate after application of the eyelash serum product to human skin (% applied dose)

Reference item	Isopropyl Cloprostenate (% applied dose)	
Test item	Eyelash serum product	
No. of cells	n=10 (without J and K cells)	
	Mean	SD
Total stratum corneum strips	2.53	1.12
Skin Excess*	94.39	6.63
Epidermis	2.56	1.36
Dermis	0.15	0.13
Receptor fluid + RCR**	1.17	0.59
Absorbed part (Epidermis + Dermis + receptor fluid + RCR)	3.87	1.74
TOTAL RECOVERY	100.79	7.34
*Skin excess corresponds to: Washing + Donor compartment rinsing + Remaining skin		
**RCR = Rinsed receptor compartment		

Table 2: Distribution of ³H-Isopropyl Cloprostenate after application of the eyelash serum product to human skin (µg_{eq}/cm²)

Reference item	Isopropyl Cloprostenate (% applied dose)	
Test item	Eyelash serum product	
No. of cells	n=10 (without J and K cells)	
	Mean	SD
Test item amount applied (mg/cm ²)	9.48	0.70
Test substance applied (µCi/cm ²)	0.46	0.03
Total stratum corneum strips	0.0011	0.0005
Skin Excess*	0.0424	0.0004
Epidermis	0.0012	0.0006
Dermis	0.00006	0.00006
Receptor fluid + RCR**	0.0005	0.0003
Absorbed part (Epidermis + Dermis + receptor fluid + RCR)	0.0017	0.0008
*Skin excess corresponds to: Washing + Donor compartment rinsing + Remaining skin		
**RCR = Rinsed receptor compartment		

Conclusion:

Following volatility tests, Isopropyl Cloprostenate was shown to be volatile up to 36%. In order to keep the compound and obtain complete mass balances, trapping systems were placed on top of the cells. A total of 4 donors were used in triplicates. The mean total recovery was within the established acceptance criteria of SCCS guidance (85-115%), validating the results obtained.

The number of strips for all cells ranged from 2 to 11 strips considering the fragility of the skin samples after 24h in contact with the prepared test item.

As the recovery of cell J (78.88%) and K (128.62%) were not $100\% \pm 20.00\%$, the results of this study are calculated and presented without these two cells.

The mean total recovery was $100.79\% \pm 7.34\%$ (without the cells J and K, $n=10$).

The bioavailable portion consisting of receptor fluid plus rinse receptor compartment plus epidermis plus dermis after 24-hour was $3.87\% \pm 1.74\%$ of the applied dose corresponding to $0.0017 \mu\text{g}_{\text{eq}}/\text{cm}^2 \pm 0.0008 \mu\text{g}_{\text{eq}}/\text{cm}^2$.

Therefore, in accordance with the SCCS guidance principles (SCCS/1647/22, Reference: SCCS, 2023) the value of 5.61% (mean plus standard deviation) will be used as dermal absorption value for the Margin of safety (MoS) calculation. This value considers already a high variation according to the basic SCCS criteria for the *in vitro* assessment of dermal absorption of cosmetic ingredient (Reference: SCCS, 2010).

Ref.: Eurofins, 2024.

SCCS comments

The SCCS considers the study valid and agrees that the value of 5.61% (mean plus standard deviation) can be used for dermal absorption.

In addition to the dermal absorption study, a validation of the LC-MS/MS method has been conducted to quantitate both Isopropyl Cloprostenate and the metabolite – Cloprostenol in different matrices including washing (1-100 ng/mL for Isopropyl Cloprostenate; 0.5-50 ng/mL for Cloprostenol); Stratum Corneum – tape strips (0.1-50 ng/mL for Isopropyl Cloprostenate; 0.25-125 ng/mL for Cloprostenol); human dermatomed skin samples 1 cm² (0.05-25 ng/mL for Isopropyl Cloprostenate; 1-500 ng/mL for Cloprostenol) and receptor fluid (1-500 ng/mL for both compounds).

4.2.3 Other studies on toxicokinetics

No new data on toxicokinetics of Isopropyl Cloprostenate have been provided with the current submission. The Applicant 1 provided the information from the previous Opinion SCCS/1635/21.

Applicant 1:

There are no studies on toxicokinetics of Isopropyl Cloprostenate available.

However, Isopropyl Cloprostenate is chemically a fully synthetic substance in the form of the isopropyl ester of Cloprostenate, which is converted under physiological conditions into Cloprostenol, a synthetic prostaglandin analogue (Reference: SCCS, 2022).

Moreover, as esters can metabolically be cleaved by esterases, it is considered as appropriate to take into account data on Cloprostenol and R-Cloprostenol with regards to toxicokinetics of Isopropyl Cloprostenate. These data on toxicokinetics data of Cloprostenol and R-Cloprostenol are available from the summary evaluation of the European Medicines Agency (EMA, references, EMA 2004), which were also presented in the SCCS Opinion on prostaglandins and prostaglandin analogues (Reference: SCCS, 2022).

In addition, it is scientifically accepted that the epidermis of the skin has a great metabolic capacity, which is around one third of the metabolic capacity of the liver. This fact can also

be considered as another legitimate [reason] to consider toxicokinetics data of Cloprostenol and R-Cloprostenol.

Therefore, the available data, taken out of the SCCS opinion, are presented below.

- *In rats, after subcutaneous administration of doses ranging from 20 to 200 µg/kg bw of cloprostenol, 60% of the administered radioactivity was recovered in urine and 14% in faeces within 48 hours. Excretion was complete in 7 days. Major urinary metabolites were identified by GC-MS: tetranor acid of 9-keto-cloprostenol, δ-lactone of the tetranor acid of cloprostenol. Unchanged cloprostenol (less than 10%) and an acid labile conjugate were minor components. No precise information was given about the relative percentage of metabolites.*
- *In marmosets, 55% of the administered radioactivity was recovered in urine and 16% in faeces within 72 hours after subcutaneous administration of 100 µg of ¹⁴C-cloprostenol per kg bw. Unchanged cloprostenol (approximately 40%) was identified in urine together with its dinor acid metabolite (approximately 57%) formed by one stage of β-oxidation.*
- *In pigs, after a single intramuscular administration of ¹⁴C-cloprostenol (acid in the form of sodium salt) at a dose of 200 µg, the highest plasma level of radioactivity (0.70 ± 0.14 µg/L) was measured at 1 hour after dosing. At 24 hours post dosing, the levels were close to 0.04 µg/L. Fifty percent of the dose administered was recovered either via urine or faeces. In urine, the following compounds were identified: the parent cloprostenol (approximately 10-14%), the tetranor acid metabolite (approximately 37%) and polar compounds (26-32%).*
- *After intramuscular administration of 75 µg of R-cloprostenol to sows, the maximum concentration of R-cloprostenol in plasma was close to 2 µg/L and appeared between 30 and 80 minutes after injection. The half-life of elimination $T_{1/2\beta}$ was estimated to be 3 h 10 min.*
- *In dairy cows, after a single intramuscular injection of 500 µg of free acid ¹⁴C-cloprostenol (specific activity 122 µCi/mg free acid), the highest plasma level (0.43 ± 0.043 µg free acid equivalent/L) was reached within 30 minutes after dosing. The concentrations were lower than 0.01 µg free acid equivalents/L at 24 hours post dosing. The $T_{1/2\beta}$ was 3 hours. After 16 hours, 52.5 ± 4.8% of radioactivity was recovered from urine.*
- *Cloprostenol was extensively metabolised in the cow by β-oxidation to give the tetranor acid of cloprostenol, isolated as δ-lactone and as glucuronide conjugates (44%). The parent compound represents 18% of the radioactivity excreted.*
- *After intramuscular administration of 150 µg of R-cloprostenol to cows, the highest plasma concentration of R-cloprostenol was found at 90 minutes after injection (approximately 1.4 µg/L). The half-life of elimination $T_{1/2\beta}$ was estimated to be 1 h 37 min.*
- *Residue depletion studies were performed in pigs and cattle. In pigs, 30 minutes after intramuscular administration of 200 µg ¹⁴C-cloprostenol (acid in the form of sodium salt), the highest concentrations of radioactivity were detected at the injection site and in kidneys (43.98 ± 6.90 and 19.00 ± 4.13 µg cloprostenol equivalent/kg respectively). At 24 hours post injection, the amounts were lower than the limit of quantification (0.04 µg cloprostenol equivalent/kg) in muscle and in fat. The residues in liver and kidney were of the same magnitude (close to 0.10 µg cloprostenol equivalent/kg). At the injection site the levels were still relatively high (0.83 ± 0.53 µg cloprostenol equivalent/kg).*
- *A first tissue depletion study was carried out in cows dosed with 500 µg of ¹⁴C-cloprostenol (122 µCi/mg free acid) by single intramuscular injection. At 30 minutes post injection, residues of ¹⁴C-related cloprostenol radioactivity could only be detected in kidney (19.10 ± 2.73 µg cloprostenol equivalent/kg), in liver (7.25 ± 0.641 µg cloprostenol equivalent/kg) and at the injection site (162 ± 19.10 µg cloprostenol equivalent/kg). At 24 hours post dose, the amounts were much lower: 0.123 ± 0.019 µg cloprostenol equivalent/kg in kidney, 0.036 ± 0.010 µg cloprostenol equivalent/kg*

in liver and $0.493 \pm 0.198 \mu\text{g}$ cloprostenol equivalent/kg at the injection site. No residues were detectable after 48 hours post administration.

- A second depletion study, carried out with unlabelled compound at the same dose, showed that all the residues were below the limit of quantification of the radioimmunoassay method used at 24 hours post administration (0.3 or $0.5 \mu\text{g}$ cloprostenol/kg according to the tissue). In a third non-radiometric study, at 16 hours after a single administration of $150 \mu\text{g}$ of R-cloprostenol to cows, $0.092 \pm 0.063 \mu\text{g}$ R-cloprostenol/kg was measured at the injection site, $0.051 \pm 0.003 \mu\text{g}$ R-cloprostenol/kg in liver and $0.120 \pm 0.021 \mu\text{g}$ R-cloprostenol/kg in kidney. Only traces of R-cloprostenol were noted in muscle.
- Less than 1% of radioactivity administered was eliminated via cow's milk. After an intramuscular dose of $500 \mu\text{g}$ ^{14}C -cloprostenol (specific activity $86.35 \mu\text{Ci}/\text{mg}$ acid in the form of sodium salt - $22 \mu\text{Ci}$ per animal) to cows, the highest levels, corresponding to $4 \mu\text{g}$ free acid equivalents/L, were found in samples collected over the 0-4 hour period. By 24 hours after dosing, levels had fallen below $0.012 \mu\text{g}/\text{L}$.
- In a second depletion study carried out with unlabelled compound no detectable cloprostenol residues (less than $0.1 \mu\text{g}$ cloprostenol/L) were found nine hours after dosing or in subsequent samples after single intramuscular dose of $500 \mu\text{g}$ cloprostenol.
- In cows, the ratios of parent compound to total residues were estimated from tissue samples collected at 30 minutes after intramuscular administration of $500 \mu\text{g}$ ^{14}C -cloprostenol. Ninety percent of the radioactivity could be extracted. The parent compound represented 85% of total residues in muscle, 35.5% in kidney and liver. In liver and kidney, the percentage of cloprostenol metabolites, lactone and its tetranor acid, was close to 20%. In milk, the ratio of cloprostenol to total radioactivity ranged from 65% (at 0 to 4 hours) to 16% (24 hours post dosing).
- Twenty-four hours after intramuscular treatment tissue residues in cattle and pigs are generally only present at the injection site and to a lesser extent in the liver and kidney. Total radioactivity in cow's milk was less than 1% of the administered dose. The maximum intake of total residues that might be ingested from animals slaughtered 24 hours after treatment would be 6.7% of the ADI for pig meat (including injection site) plus cow's milk and 4.8% for cattle meat and milk. These values decrease to below 1% without the injection site. The available pharmacokinetic and residues depletion data do not indicate any significant variability between the mammalian species that have been investigated, therefore, any possible difference in pharmacokinetics in goats, including goat's milk, would not be expected to have a significant impact on this percentage.

Overall, these data showed that both R-Cloprostenol and Cloprostenol are eliminated rapidly and that the extensive metabolism resulted in metabolites without pharmacological activity. Based on the rational presented above, the same or at least similar toxicokinetics effects can be assumed for Isopropyl Cloprostenate.

Overall summary of toxicokinetics by the Applicant 1:

The dermal absorption *in vitro* was investigated in the SCCS compliant *in vitro* dermal absorption on Isopropyl Cloprostenate (0.005%) from a cosmetic formulation by using the $[^3\text{H}]$ -Isopropyl Cloprostenate as a tracer and applied for 24 hours onto the surface of healthy human skin. This resulted in a mean total recovery of $100.79\% \pm 7.34\%$. The bioavailable portion as measured for skin absorption was $3.87\% \pm 1.74\%$ [$0.017 \pm 0.0008 \mu\text{g}/\text{cm}^2$].

Therefore, in accordance with the SCCS guidance principles (SCCS/1647/22, Reference: SCCS, 2023) the value of 5.61 % (mean plus standard deviation) will be used as dermal absorption value for the Margin of safety (MoS) calculation.

Despite this recent study there are no experimental data on toxicokinetics of Isopropyl Cloprostenate available.

However, Isopropyl Cloprostenate is chemically a fully synthetic substance in the form of the isopropyl ester of Cloprostenate, which is converted under physiological conditions into Cloprostenol, a synthetic prostaglandin analogue.

Moreover, as esters can metabolically be cleaved by esterases, also in the metabolically active skin epidermis, it is considered as appropriate to take into account data on Cloprostenol and R-Cloprostenol with regards to toxicokinetics of Isopropyl Cloprostenate. This is also supported by data and gained knowledge of the read across justification (for details see section 7 Read across justification).

The data on Cloprostenol showed that elimination of both, R-Cloprostenol and Cloprostenol from the body is rapid and that the extensive metabolism resulted in metabolites without pharmacological activity. Based on the above presented rational, the same or at least similar toxicokinetics effects can be assumed for Isopropyl Cloprostenate.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16 and in SCCS/1635/21

SCCS overall comment on toxicokinetics

No new studies on toxicokinetics of Isopropyl Cloprostenate were provided, except a valid study on dermal penetration, based on which the value of 5.61% (mean plus standard deviation) can be used for dermal absorption.

As expressed in the Preamble, the proposed read-across between Cloprostenol and Isopropyl Cloprostenate does not fulfil the requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence, it has not been considered relevant by the SCCS for use in the safety assessment of Isopropyl Cloprostenate.

4.2.4 Calculation of SED/LED

Applicant 1:

Following the SCCS Notes of Guidance (12th Revision, SCCS/1647/22) the calculation for the systemic exposure dose (SED) is expressed mathematically as follows (Reference: SCCS, 2023a):

$$SED = E_{\text{product}} \times \frac{C}{100} \times \frac{DA_p}{100}$$

Where:

SED (mg/kg bw/day) - Systemic Exposure Dose

E_{product} (mg/kg bw/day) - Estimated daily exposure to a cosmetic product per kg body weight, based on the amount applied and the frequency of application (for calculated relative daily exposure levels for different cosmetic product types (SCCS/1628/21).

C (%) - Concentration of the substance under study in the finished cosmetic product on the application site

DA_p (%) - Dermal Absorption expressed as a percentage of the test dose assumed to be applied in real-life conditions.

There are no daily exposure levels provided for eyelash serums and eyebrow serums in Europe in the most recent SCCS guidance document (Reference: Reference: SCCS, 2023). The eyelash serum is applied as a thin layer to the eyelid above the eyelash line. This is comparable to the use of a liquid eyeliner. However, there are clear differences in exposure as eyeliner will generally be applied above and below the eyelash lines of the upper and lower

eye lids, respectively. And in contrast to the single use of the eyelash sera, eyeliner will be applied twice a day (Reference: SCCS, 2023).

To close the existing data gap, the applicant has very recently performed analyses on the exact amount of the daily uses of his eyelash serum and eyebrow serum according to his application instructions (products packages) including HPLC analytic on Isopropyl Cloprostenate. The following results were obtained (References: Cayman, 2024 b, 2024c and 2024d) and will be used for the subsequent calculations:

Eyelash serum: 0.58 ± 0.34 mg (corresponding to 0.03 ± 0.018 µg Isopropyl Cloprostenate)

Eyebrow serum: 4.0 ± 1.7 mg (corresponding to 0.10 ± 0.04 µg Isopropyl Cloprostenate).

Ref.: Cayman, 2024b, 2024c, 2024d

SCCS comments

The Applicant has measured the daily amount of eyelash serum considering that the value in the SCCS Notes of Guidance (SCCS/1647/22) is based on a single application and not on a twice per day application. However, the SCCS has noted that the measured value for two applications instead of one is 9-fold lower (*i.e.* 0.58 ± 0.34 mg vs. 5 mg) than 0.005 g/day, which according to the SCCS Notes of Guidance, results from a single application of eyeliner. An explanation for this discrepancy is not given.

Applicant 2:

Application to the upper lash line of both eyes of lash serum products containing IPCP from two producers' results, on average, in exposure to 0.0000084 mg or 0.000013 mg of IPCP.

Ref.: Safety assessment dossier on Isopropyl Cloprostenate by Applicant 2. 2023-10-17.

4.3 TOXICOLOGICAL EVALUATION

4.3.1 Irritation and corrosivity

4.3.1.1 Skin irritation

Applicant 1:

Study Design:

Reference:	Laus, 2024a
Date of report:	29.01.2024
Study period:	10.01.2024 – 12.01.2024
Guidelines/Methods:	OECD 439, Commission Regulation (EU) 2023/464, EU Method B.46, SCCS Notes of Guidance, 12 th revision, 2023
Test system:	EpiDerm®, Reconstructed human Epidermis (RhE) test model (three-dimensional human epidermis tissue model)
Replicates:	Triplicate
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0696699 (purity: 98.0% (HPLC))
Vehicle:	None
Test item:	Neat
Dose applied:	30µl
Exposure:	In total 60 minutes: 35 minutes at 37°C, followed by 25 minutes at room temperature

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Post-treatment incubation time	42 ± 1 hour
Negative control:	Dulbecco's Phosphate Buffered Saline (DPBS)
Positive control:	Sodium Dodecyl Sulfate (SDS, 5% aqueous solution)
Dead dermis negative control	Not done
Interaction with MTT	Negative
Colouring of tissue	Negative
GLP:	Yes
Published:	No

Material and methods:

1. The possible *in vitro* skin irritation potential of neat Isopropyl Cloprostenate (batch: 0696699, purity: 98.0%) was investigated according to OECD 439 under GLP conditions considering also current SCCS guidance criteria. The Reconstructed human Epidermis (RhE) test model (EpiDerm® tissue kit) was used to evaluate the skin irritation potential of the test substance. One valid experiment was performed.
2. The neat liquid test item was applied to a three-dimensional human epidermis tissue model in triplicate for an exposure time of 60 minutes. DPBS buffer was used as negative control and 5 % aqueous SDS solution was used as positive control. After treatment, the respective substance was rinsed from the tissue. Thereafter, the cell viability of the tissues was evaluated by addition of MTT (3-(4,5-dimethyl thiazole 2-yl) 2,5-diphenyltetrazolium bromide), which can be reduced to formazan. The formazan production was evaluated by measuring the optical density (OD) of the resulting solution.

Results:

The obtained mean optical density (OD) measured at 570 nm of the two replicates per tissue, the mean OD of the three relating tissues and the standard deviation (SD) are provided in the following table.

Table 3: Mean OD and SD Negative Control, Positive Control and Test Item

Designation	Negative Control (DPBS)	Test Item	Positive Control (5% aqueous SDS)
Mean OD Tissue 1 (OD – Mean Blank Control)	2.397	0.064	0.028
Mean OD Tissue 2 (OD – Mean Blank Control)	2.140	0.043	0.047
Mean OD Tissue 3 (OD – Mean Blank Control)	1.698	0.049	0.039
Mean OD of the three relating tissues	2.078	0.052	0.038
SD	17.0%	0.5%	0.5%

For the test item and the positive control, the following percentage values of tissue viability were calculated in comparison to the negative control and shown in the table below:

Table 4: % Viability

Designation	Test Item	Positive Control
-------------	-----------	------------------

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% Tissue viability (tissue 1)	3.1 %	1.3%
% Tissue viability (tissue 2)	2.0 %	2.3%
% Tissue viability (tissue 3)	2.3 %	1.9%
% Tissue viability (mean)	2.5%	1.8%

The mean value of relative tissue viability of the test item was reduced to 2.5 % after the treatment. This value is below the threshold for skin irritation (50 %) and therefore, the test item has to be considered as irritant to skin.

Positive and negative controls showed the expected OD and cell viability values within acceptable limits and indicated the suitability and sensitivity of the test system.

Conclusion:

Under the conditions specified in this *in vitro* study, neat Isopropyl Cloprostenate is considered irritant to the skin in the Reconstructed human Epidermis (RhE) test method according to the criteria of the OECD Guideline 439.

Ref.: Laus, 2024a

SCCS comment

The SCCS notes that the test does not allow to differentiate between irritant and potential corrosive properties of the neat substance.

4.3.1.2 Mucous membrane irritation / eye irritation

Applicant 1:**Study Design:**

Reference:	Laus, 2024b
Date of report:	30.01.2024
Study period:	10.01.2024 - 11.01.2024
Guidelines/Methods:	OECD 492, Commission Regulation (EU) 2023/464, EU Method B.69, SCCS Notes of Guidance, 12 th revision, 2023
Test system:	EpiOcular®, Reconstructed human Cornea-like Epithelium (RhCE) model (three-dimensional RhCE tissue construct)
Replicates	2 cornea tissue models per test condition
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0696699 (purity: 98.0% (HPLC))
Vehicle:	None
Dose:	Neat test item
Dose applied:	50 µL
Exposure:	30 min.
Post-treatment incubation time:	120 min.
Negative control:	Sterile demineralised water
Positive control:	Methyl acetate (MA)
Dead dermis negative control:	Not done
Interaction with MTT:	Negative
Colouring of tissue:	Negative
GLP:	Yes
Published:	No

Material and methods:

The possible *in vitro* eye irritation potential of neat Isopropyl Cloprostenate (0696699, purity: 98.0%) was investigated according to OECD 492 under GLP conditions considering also current SCCS guidance criteria. The EpiOcular® reconstructed human cornea-like epithelium (RhCE) was used to evaluate the test substance. Two valid experiments were performed as the first experiment was invalid and had to be repeated.

The neat test item was applied to a three-dimensional human cornea tissue model in duplicate for an exposure time of 30 minutes. Sterile demineralised water was used as the negative control and methyl acetate was used as the positive control.

After treatment, the respective substance was rinsed from the tissue. Thereafter, the cell viability of the tissues was evaluated by addition of MTT, which can be reduced to formazan. The formazan production was evaluated by measuring the optical density (OD) of the resulting solution.

Results:

After treatment with the negative control, the absorbance values were within the required acceptability criterion of $0.8 < \text{mean OD} < 2.8$, the OD value was 1.676.

The positive control showed clear eye irritating effects and the mean value of the relative tissue viability was 17.7 % (required: < 50 %). The variation within tissue replicates of the controls and the test item was acceptable (< 20 %). Thus, the sensitivity and validity of the test system was given.

The obtained mean optical density (OD) measured at 570 nm of the two replicates per tissue, the mean OD of the three relating tissues and difference are provided in the following table.

Table 5: Mean OD and SD of the Negative Control, Positive Control and Isopropyl Cloprostenate

Designation	Negative Control (Sterile demineralized water)	Positive Control (Methyl acetate)	Isopropyl Cloprostenate
Mean OD Tissue 1 (OD – Mean Blank Control)	1.743	0.318	0.811
Mean OD Tissue 2 (OD – Mean Blank Control)	1.609	0.276	0.834
Mean OD of the two relating tissues	1.676	0.297	0.822
Difference	8.0%	2.5%	1.4%

For the test item and the positive control, the following percentage values of tissue viability were calculated in comparison to the negative control and presented in the table below:

Table 6: % Viability Positive Control and Test Item

Designation	Positive Control (Methyl acetate)	Isopropyl Cloprostenate
% Viability (Tissue 1)	18.9%	48.4%
% Viability (Tissue 2)	16.4%	49.8%
% Viability Mean	17.7%	49.1%

After treatment with the test item, the mean value of relative tissue viability was 49.1 %. As this value is above the threshold for eye irritation potential (≤ 60 %) test items that induce these values are considered as eye irritant.

Conclusion:

Under the conditions of this *in vitro* study, neat Isopropyl Cloprostenate is considered as eye irritant in the *in vitro* EpiOcular™ Eye Irritation Test according to the criteria of the OECD Guideline 492.

Overall summary on irritation and corrosivity by the Applicant:

The skin and eye irritational potential of the undiluted Isopropyl Cloprostenate was tested in *in vitro* systems according to internationally recognized standards in accredited testing laboratories in accordance with Good Laboratory Practice. Namely, the *in vitro* EpiOcular® Reconstructed human Cornea-like Epithelium (RhCE) Test Method (OECD 492) and the *in vitro* EpiDerm® Reconstructed human Epidermis (RhE) Test Method (OECD 439).

The results of these tests indicated that the pure Isopropyl Cloprostenate possesses both, a skin and eye irritation potential. However, with regards to the observed skin and eye irritation potential, this intrinsic hazard property is considered to represent no risk for the consumer with regards to the very low concentration of Isopropyl Cloprostenate in cosmetic products at a maximum use concentration of 0.005%.

Therefore, according to the general understanding it can be considered as highly unlikely that the low use concentration of 0.005% of Isopropyl Cloprostenate in eyelash serums and eyebrow serums may lead to skin or eye irritation.

Thus, the endpoint of the skin and eye irritation potential *in vitro* was sufficiently covered. Therefore, there is no need for additional testing in line with SCCS criteria (Reference: SCCS, 2023) and the ECHA requirements (Reference: ECHA, 2017a).

Ref.: Laus, 2024b

SCCS comments

The SCCS does not agree with the Applicant's conclusion. The prediction model states that if the mean tissue viability $\leq 60\%$ 'no prediction can be made' for classification purposes. This is because in case of a true positive, the method cannot resolve between UN GHS Categories 1 and 2. Furthermore, RhCE test methods show a high percentage of false positive results.

Applicant 2:**The Hen's Egg Test** - Utilizing the Chorioallantoic Membrane (HET-CAM)**Method:**

Fresh, fertile, White Leghorn eggs were obtained from a suitably licensed/approved vendor. The eggs were stored at this facility for two days, at approximately 12°-16°C, after arrival at this facility. For incubation Days 0 - 10, the eggs were placed in a Kuhl, humidified incubator so that the acutely angled ends face down. The incubator is such that the eggs are automatically rotated approximately once every hour. The temperature was maintained at 37° C ($\pm 2^\circ$ C) for the incubation period.

On day ten (10) each egg was removed from the incubator and placed in a Plexiglas work enclosure. This enclosure had been preheated and humidified so that its environment approached that of the incubator. A cut was made in the larger end of each egg, where the air sack is located. A dedicated tool was used to make each cut. Forceps were then used to remove the shell down to the shell-membrane junction. The inner egg membrane was then hydrated with a warm, physiological saline solution. The saline was removed after a two (2) to five (5) minute exposure. Utilizing pointed forceps, the inner egg membrane was then carefully removed to reveal the CAM.

The test or reference article, at a dosage of three-tenths of one milliliter (0.3 ml) of a liquid or three-tenths of one gram (0.3 g) of a solid, was then administered to each of four (4) CAM's. Twenty seconds later, the test or reference article was rinsed from each CAM with five (5) milliliters of physiological saline. All CAM's were observed immediately prior to test article administration and at 30 seconds, two (2) and five (5) minutes after exposure to the test

article. The reactions of the CAM, the blood vessels, including the capillaries, and the albumin were examined and scored for irritant effects according to a routine scoring criteria.

The numerical, time dependent scores were totaled for each CAM. Each reaction type can be recorded only once for each CAM, therefore the maximum score per CAM is 32. The mean score was determined for all CAM's similarly tested.

Discussion:

Previous studies have shown that the CAM of the hen's egg is more sensitive to liquid irritants than is the rabbit eye. Therefore, 50% dilutions of the liquid reference articles were used to approximate their 100% irritation potential. The Sponsor requested the irritation potential of their article at 0.1%. Therefore, a 0.05% dilution of the liquid test article was used to approximate its 0.1% irritation potential.

Historical *In vivo* Results:

The reference products have historically been categorized as being practically non-irritating, eliciting scores approaching 0, at 24 hours, when dosed at 100% and tested using the Draize ocular irritation methodologies (Draize Scale: 0 - 110).

Conclusion:

Under the conditions of this test, the results indicate that the sponsor-submitted product, isopropyl ester at 0.1%, would have practically no ocular irritation potential *in vivo*.

Ref.: Consumer Product Testing Company. The Hen's Egg Test Utilizing the Chorioallantoic Membrane (HET-CAM). 2022-10-13

SCCS comment

In the study the test article (0.1% IPCP) showed total score of 1.5, while two reference articles showed total score of 0.5 and 0.75.

The SCCS is of the opinion that the HET-CAM test is not sufficient to state that a chemical has no eye irritation potential, as it can only discriminate between strong and non-irritating chemicals.

4.3.2 Skin sensitisation

Applicant 1:

The possible *in vitro* skin sensitizing potential of Isopropyl Cloprostenate was investigated in the Direct Peptide Reactivity Assay (DPRA) and in the KeratinoSens™ method (ARE-Nrf2 Luciferase test method) following the two out of three (2o3) prediction strategy as laid down in the guideline on defined approaches (DA) for skin sensitization (OECD 497) and are presented below.

Study Design: Direct Peptide Reactivity Assay (DPRA)

Reference:	ToxiCoop, 2024b
Date of report:	25.01.2024
Study period:	19.12.2023 – 20.12.2023
Guidelines/Methods:	OECD 442C, OECD 497, SCCS Notes of Guidance, 12 th revision, 2023
Test system:	Direct Peptide Reactivity Assay (DPRA) with high-performance liquid chromatography (HPLC) analysis
Replicates:	Triplicate
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0696699 (purity: 98.0% (HPLC))

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Vehicle:	Acetonitrile
Dose:	100 mM test solution
Exposure:	<i>In chemico</i> by combining the test chemical with a solution of the synthetic Cysteine peptide
Exposure duration:	24 ± 2 hours (25°C)
Controls:	Reference control A: Peptide stock solution with acetonitrile for system suitability check. Reference control B: Peptide stock solution with acetonitrile for peptide stability check. Reference control C: Peptide stock solution with acetonitrile for the test item and the positive control Co-elution controls Test item stock solution and acetonitrile combined with phosphate buffer. Positive control: Peptide stock solution combined with Cinnamaldehyde as positive control stock solution (100 mM) and acetonitrile (acn)
GLP:	Yes
Published:	No

Material and methods:

The possible sensitizing potential of Isopropyl Cloprostenate (batch: 0696699, purity: 98.0%) was investigated by means of *in chemico* determination with the Direct Peptide Reactivity Assay (DPRA) with high-performance liquid chromatography (HPLC) analysis according to OECD 442C and considering SCCS criteria under GLP conditions.

Accordingly, the reactivity of the test item and synthetic Cysteine containing peptides was evaluated by combining the test chemical with a solution of the peptide and monitoring the remaining concentration of the peptide following 24 hours of interaction time at room temperature. Relative concentrations of the peptides following the 24 hours incubation were determined by HPLC with gradient elution and UV detection at 220 nm. The samples were also evaluated with regards to possible occurrence of precipitation. Reaction samples, reference controls A, B and C, co-elution controls and positive controls were prepared and analysed in triplicates. The total HPLC analysis time was less than 30 hours. The Cysteine 1:10 prediction model was used for the discrimination between sensitiser and non-sensitiser.

Results:

Precipitation was observed in the case of cysteine peptides.

Means of the peak areas versus the concentrations of cysteine peptide showed good linearity, covering the concentration range from 0.0167 mM to 0.534 mM.

Peptide depletion resulting from the positive control cinnamaldehyde was within the expected percentage range and confirmed the suitability and sensitivity of the test system.

The mean back-calculated peptide concentrations of the reference control replicates were within the expected molarity concentration range and the CV % value for the nine reference controls B and C in acetonitrile were acceptable.

The mean cysteine peptide depletion value of the test item was 2 % ± 3 % indicative for negative DPRA prediction. However, precipitation was noted following incubation.

Table 7 below showed that the acceptance criteria was met for the positive control with a cysteine peptide depletion value of 74.3 % ± 1 % and that the mean cysteine peptide depletion value obtained with the test item was 2 % ± 3 %.

Table 7: Cysteine peptide depletion values for the positive control, reference control c (dms+acn) and the test item

Sample	Peptide conc. calculated (mM)	Mean peptide conc. (mM)	Peptide depletion (%)	Mean Peptide depletion (%)	Standard deviation (SD) for peptide depletion (%)
Positive control, rep I	0.12	0.12	74.9	74.3	1

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Positive control, rep II	0.12		74.4		
Positive control, rep III	0.12		73.7		
Ref C dmso+acn, rep I	0.45	0.45	-		
Ref C dmso+acn, rep II	0.45		-	-	-
Ref C dmso+acn, rep III	0.45		-		
Test item, rep I	0.45		-0.4*		
Test item, rep II	0.44	0.44	0.8	2	3
Test item, rep III	0.42		6.3		
*Negative depletion value was submitted with zero for calculation of the mean.					

No co-elution was observed with the peptide. The mean cysteine peptide depletion of the test item was $2 \% \pm 3 \%$, which was clearly below the threshold of 13.89 % indicating a negative DPRA prediction.

Peptide depletion resulting from the positive control cinnamaldehyde was within the expected percentage range and confirmed the suitability and sensitivity of the test system.

All validity criteria were met, confirming the validity of the study.

Conclusion:

The cysteine 1:10 prediction model results obtained with Isopropyl Cloprostenate under the conditions of the study suggested a negative DPRA prediction, since showing no or minimal reactivity under the experimental conditions of the *in chemico* DPRA method indicative for a non-sensitizer.

However, following incubation precipitation was observed in the case of cysteine peptides and therefore a final conclusion on the lack of reactivity cannot be drawn with sufficient confidence as peptide depletion may be underestimated according to the OECD 442C guideline criteria.

Ref.: TOXI-COOP, 2024b

SCCS comment on the DPRA study

It is the SCCS's opinion, based on the OECD 442C guideline criteria and due to precipitation following incubation, that the result of the DPRA is inconclusive.

Applicant 1:

Study Design: KeratinoSens™ method (ARE-Nrf2 Luciferase test method)

Reference:	ToxiCoop, 2024b
Date of report:	22.01.2024
Study period:	18.12.2023 – 11.01.2024
Guidelines/Methods:	OECD 442C, OECD 497, SCCS Notes of Guidance, 12 th revision, 2023
Test system:	Direct Peptide Reactivity Assay (DPRA) with high-performance liquid chromatography (HPLC) analysis
Replicates:	Triplicate
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0696699 (purity: 98.0% (HPLC))
Vehicle:	DMSO
Dose:	1 st test: twelve concentrations: 0.98 µM - 2000.00 µM, 2-fold dilution factor 2 nd test and 3 rd test: twelve concentrations: 5.58 µM - 100 µM, 1.3-fold dilution

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Exposure duration:	48 ± 1 hours (37 ± 1 °C in the presence of 5 % CO ₂)
Negative control:	Vehicle (DMSO)
Positive control:	Trans-Cinnamaldehyde
GLP:	Yes
Published:	No

Material and methods:

The possible sensitizing potential of Isopropyl Cloprostenate (batch: 0652603-2, purity: 98.5%) was studied using the KeratinoSens™ method (ARE-Nrf2 Luciferase test method) according to OECD 442D and considering SCCS criteria under GLP conditions. In total, three independent tests with each three replicates were performed with the test item. In the first test, twelve concentration ranging from 0.98 µM - 2000.00 µM using a 2-fold dilution factor were used. However, since in the first test there was only one concentration where the induction value of the test item exceeded the 1.5-fold threshold; and strong cytotoxicity was observed at the higher seven tested concentrations, a lower top concentration and narrower dilution factor was used in the second and third tests in order to be able to determine EC1.5 and IC30 and IC50 values more precisely. Therefore, twelve doses ranging from 100.00 µM to 5.58 µM and a 1.3-fold dilution factor were investigated. DMSO was used as suitable solvent and negative control, while Trans-Cinnamaldehyde was the positive control. The KeratinoSens™ cells were derived from human keratinocytes (HaCaT) and transfected with selectable plasmids containing luciferase gene under the transcriptional control of the AKR1C2 ARE gene sequence. After 48 hours of incubation, Luciferase gene induction was measured in the cell lysates by luminescence detection using a light producing luciferase substrate. Cytotoxicity and the relative luminescence intensity of luciferase substrate in the lysates were measured and luciferase induction compared to solvent/vehicle control was calculated.

Results:

Precipitation was not observed at any concentration in any of the three tests.

First test:

The test item induced strong cytotoxicity (or viability below 70 %) at the highest seven tested concentrations (2000.00 – 31.25 µM) compared to the solvent/vehicle control in KeratinoSens™ cells. Thus, IC30 as 25.17 µM and IC50 as 27.93 µM was determined. The induction value of the test item was higher than the 1.5-fold threshold compared to the respective negative control at the concentration of 15.63 µM with induction value that was statistically significant over solvent control. According to the OECD Test Guideline 497 prediction model, the induction value of the test item was within, at 15.63 µM, the borderline threshold range (1.35-1.67-fold) compared to the respective negative control. The maximal fold induction (Imax) was 1.65-fold. Therefore, the EC1.5 value could be determined as 12.28 µM.

Second test:

The tested concentrations were decreased in order to achieve more accurate results. Therefore, the tested concentrations were varied from 100.00 – 5.58 µM.

The test item induced strong cytotoxicity (viability below 70 %) at the highest six tested concentrations (100.00 – 26.93 µM) compared to the solvent/vehicle control in KeratinoSens™ cells. Thus, IC30 as 26.57 µM was determined. IC50 could be determined as 31.02 µM.

The induction values of the test item were higher than the 1.5-fold threshold compared to the respective negative control at the concentrations of 20.72 and 26.93 µM, with induction values that were statistically significant over solvent control. Moreover, according to the OECD Test Guideline 497 prediction model, the induction values of the test item were above at 20.72 and 26.93 µM, the borderline threshold range (1.35-1.67-fold) compared to the respective negative control and statistically significant compared with solvent control. The maximal fold induction (Imax) was 5.21-fold. Therefore, the EC1.5 value could be determined as 16.50 µM. No precipitation was observed at any concentration during the second test.

Third test:

The test item induced strong cytotoxicity (or viability below 70 %) at the highest five tested concentrations (100.00 – 35.01 µM) compared to the solvent/vehicle control in KeratinoSens™ cells. Thus, IC30 as 30.43 µM and IC50 as 32.84 µM was determined.

The induction value of the test item was higher than the 1.5-fold threshold compared to the respective negative control at the concentrations of 20.72, 26.93 and 35.01 µM with induction value that was statistically significant over solvent control. According to the OECD Test Guideline 497 prediction model, the induction value of the test item was above or within at the concentrations of 20.72, 26.93 and 35.01 µM, the borderline threshold range (1.35-1.67-fold) compared to the respective negative control. The maximal fold induction (Imax) was 5.67-fold. Therefore, the EC1.5 value could be determined as 16.61 µM.

The obtained results can be summarized as follows.

The test item induced strong cytotoxicity in KeratinoSens™ cells compared to the solvent/vehicle control in all the three tests at seven highest tested concentrations (2000.00 – 31.25 µM) in case of the first test, at six highest tested concentrations (100.00 – 26.96 µM) in the second test and at five highest tested concentrations (100.00 – 35.01 µM) in the third test. Thus, IC30 and IC50 values could be determined in all the three tests. The overall IC30 was determined as 27.39 µM, while the overall IC50 was calculated as 30.60 µM.

The induction values of the test item exceeded the 1.5-fold threshold compared to the respective negative control and were statistically significant compared with solvent control at one tested concentration for the first test (15.63 µM), two tested concentrations for the second test (20.72 and 26.93 µM) and three tested concentrations for the third test (20.72, 26.93 and 35.01 µM). Moreover, according to the OECD Test Guideline 497 prediction model, there was one tested concentration for the first test (15.63 µM), two tested concentrations for the second test (20.72 and 26.93 µM) and three tested concentrations for the third test (20.72, 26.93 and 35.01 µM) where the induction values of the test item were above or within the borderline threshold range (1.35-1.67-fold) compared to the respective negative control and statistically significant compared with solvent control.

The suitability and sensitivity of the test system was confirmed as all validity criteria were met and the negative control reacted as expected without indicating a skin sensitizing potential, while the luciferase activity induction obtained with the positive control (Trans-Cinnamaldehyde) was dose-dependently and statistically significantly above the threshold of 1.5-fold in all tests.

With regards to the obtained results and based on the OECD 442D and OECD 497 prediction models, the test item was concluded positive, since all criteria for a positive result were met in the three tests. EC1.5 values were determined for each test, and the overall EC1.5 value was 15.13 µM. Furthermore, the overall Imax value was determined as 4.18-fold. A clear dose response could be observed throughout the study.

Conclusion:

Overall, under the conditions of this study and based on the obtained results, Isopropyl Cloprostenate showed *in vitro* a skin sensitization potential under the experimental conditions of the KeratinoSens™ method (ARE-Nrf2 Luciferase Test Method), when tested up to clear cytotoxic concentrations and considering the OECD 442D and OECD 497 prediction model criteria.

Ref.: TOXI-COOP, 2024a

SCCS comment on the KeratinoSens study

The SCCS considers Isopropyl Cloprostenate positive in the KeratinoSens test.

Overall summary on skin sensitisation by the Applicant 1:

The possible *in vitro* skin sensitising potential of Isopropyl Cloprostenate was investigated in the Direct Peptide Reactivity Assay (DPRA) and in the KeratinoSens™ method (ARE-Nrf2 Luciferase test method).

In the DPRA test, the cysteine 1:10 prediction model results suggested a negative DPRA prediction and thus, no sensitizing potential. However, as precipitation following incubation was observed a conclusion on the lack of reactivity, *i.e.* lack of sensitization was finally not possible.

In the KeratinoSens™ test, Isopropyl Cloprostenate showed a skin sensitization potential according to the OECD 442D and OECD 497 criteria, when tested up to clear cytotoxic concentrations.

Overall, although no consistent results were obtained in these *in vitro* test systems, a skin sensitizing potential for Isopropyl Cloprostenate cannot be excluded with certainty.

However, even when a skin sensitising potential is assumed, this intrinsic hazard property is considered to represent no risk for the consumer with regards to the very low concentration of Isopropyl Cloprostenate in cosmetic products at a maximum use concentration of 0.005%.

Applicant 2:

The Applicant 2 provided analysis of several HRIPTs, in which cosmetic lash serums containing up to 0.005% IPC were used.

Overall SCCS comments on skin sensitisation

The SCCS has noted for the studies provided by Applicant 1 that two *in silico* models (OECD Toolbox, Derek Nexus) classified Isopropyl Cloprostenate as positive, and one of the *in vitro* tests performed was positive. Following the two out of three (2o3) Defined Approach, an additional test should have been 2o3 potential.

The SCCS considers HRIPTs unethical, additionally the cosmetic product results are not evaluated by the SCCS.

4.3.3 Acute toxicity

4.3.3.1 Acute oral toxicity

Applicant 1:

There are no acute toxicity studies on Isopropyl Cloprostenate available.

However, Isopropyl Cloprostenate is chemically a fully synthetic substance in the form of the isopropyl ester of Cloprostenate, which is converted under physiological conditions into Cloprostenol (3.2.3. Other studies on toxicokinetics).

Therefore, and with regards to supportive data and gained knowledge of the read across justification (for details see section 3.3.10 Special investigations. Read across justification), the available data, taken out of the SCCS opinion, are presented below.

Acute oral toxicity:

The acute toxicity of Cloprostenol and R-Cloprostenol is considered low. In rats the oral LD50 values were higher than 25 mg/kg bw.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16, SCCS/1635/21

4.3.3.2 Acute dermal toxicity

Applicant 1:

Acute dermal toxicity:

Three male and three female rats received a topical dose of 250 µg of Cloprostenol Sodium in one mL sterile water on day 1 and day 4 (1.25 mg/kg b.w.) on the shaved area of skin. No particular adverse effects were shown.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16, SCCS/1635/21

4.3.3.3 Acute inhalation toxicity

SCCS comment

No relevant experimental data on acute inhalation toxicity of Isopropyl Cloprostenate have been provided.

4.3.3.4 Acute toxicity – other routes

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

Overall comment on acute toxicity of Isopropyl Cloprostenate by the Applicant 1:
There are no acute toxicity studies available on Isopropyl Cloprostenate on the structural analogue Cloprostenol.

Overall, as Isopropyl Cloprostenate is chemically the form of the isopropyl ester of Cloprostenate, which is converted into Cloprostenol and with regards to supportive data and gained knowledge of the read across justification, it is considered as appropriate to assume that the acute toxicity of Isopropyl Cloprostenate can also be regarded as low.

SCCS overall comment on acute toxicity

Since the product intended to be applied above the eyelid only contains small amounts of Isopropyl Cloprostenate, there is no concern over acute toxicity of Isopropyl Cloprostenate.

4.3.4 Repeated dose toxicity

4.3.4.1 Repeated dose (28 days) oral / dermal / subcutaneous / inhalation toxicity

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

SCCS comments

Some experimental data presented in SCCS/1653/21 on subcutaneous administration are available for Cloprostenol; the data show that even low doses of this analogue can exert a systemic effect. This raises concern about the potential for similar effects by Isopropyl Cloprostenate.

4.3.4.2 Sub-chronic (90 days) oral / dermal / inhalation toxicity

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

SCCS comments

Some experimental data presented in SCCS/1653/21 on sub-chronic oral administration are available for Cloprostenol. Although not directly relevant, the data indicate systemic effects

caused by very low doses of Cloprostenol. This raises concern about the potential for similar effects by Isopropyl Cloprostenate.

4.3.4.3 Chronic (> 12 months) toxicity

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

Conclusion on repeated dose toxicity of Isopropyl Cloprostenate by the Applicant 1:

With respect to repeated dose toxicity there are no data on Isopropyl Cloprostenate available. The available data on repeated dose toxicity of Cloprostenol and R-Cloprostenol indicated that systemic toxic effects can be induced at low doses but also that clear NOELs can be achieved. After subchronic repeated oral exposure in rats and marmosets with these substances, in both species the same NOEL of 50 µg/kg bw/d was obtained.

Overall and due to the rationale presented above, with regards to systemic toxicity after repeated oral exposure of Cloprostenol, it can be considered as likely that also low but clear NOAELs in the same order of magnitude for Isopropyl Cloprostenate can be expected.

SCCS overall comment on repeated dose toxicity taken from SCCS/1635/21

Some experimental data on sub-chronic oral administration are available for Cloprostenol.

No relevant experimental data on repeated dose toxicity of Isopropyl Cloprostenate via relevant routes (oral, dermal) have been provided. Although not directly relevant, the available data on Cloprostenol indicate that even very low doses of Isopropyl Cloprostenate may have the potential to exert systemic effects.

4.3.5 Reproductive and developmental toxicity

4.3.5.1 Fertility and reproduction toxicity

No new experimental data have been provided on reproductive toxicity of Isopropyl Cloprostenate in the current submission dossier.

Applicant 1:

A specific investigation carried out to evaluate the impact of Isopropyl Cloprostenate on the male reproductive system suggests significant histopathologic modifications in the testis of rats and mice following intraperitoneal exposure to 0.1 mg/kg bw/day for 28 day.

In a 3-generation study carried out in rats, the oral administration of doses of 0, 10, 15, 20 and 40 µg of Cloprostenol/kg bw/d did not induce effects on reproductive performance of the animals. The only effects seen were the slight reduction in neonatal viability attributable to the prematurity of the offspring. A NOEL of 15 µg/kg bw/d for Cloprostenol was retained.

In a series of reproductive studies performed with Cloprostenol, it was shown that the sensitivity of the rat to termination of pregnancy resulting from luteolysis varies depending on the point in pregnancy, when the compound is administered. The oral dose 25 µg/kg bw/d of Cloprostenol did not terminate pregnancy; the most sensitive period to luteolytic action of Cloprostenol was just prior to the parturition.

Applicant 2:

CD-1 mice exposed to Travoprost at up to 1 mg/kg/day via intravenous and, subsequently, i.p. injection had no evidence of treatment-related microscopic lesions in their reproductive tissues (U.S. FDA 2000).

There were no microscopic findings in male and female reproductive tissues in CD rats following 6 months' s.c. exposure to Travoprost at up to 0.1 mg/kg/day. The critical effect in the 6-month study was altered bone histopathology (NOAEL = 0.01 mg/kg/day, LOAEL = 0.1 mg/kg/day) (U.S. FDA 2000).

In a 3-generation study, Sprague-Dawley rats were dosed by s.c. injection at 0, 0.12, 0.36, and 0.72 µg/kg/day on GD 7 to postnatal day (PND) 21. Reported effects included a decrease in gestation length and an increase in litters with stillborn pups at all dose levels in the F0 generation and an increase in litter loss and a decrease in litters with viable pups and numbers of pups per litter were reported at 0.72 µg/kg/day. Effects reported in the F1 generation included decreases in pup survival on PND 1 to 4, physical development, and motor activity at all dose levels in the F1 generation. No treatment-related effects were reported in the F2 generation (U.S. FDA 2000). The LOAEL for this study was 0.12 µg/kg/day, which was the lowest dose tested. A similar developmental toxicity study was available in which one generation of Sprague-Dawley rats received slightly lower doses of 0, 0.01, 0.03, and 0.1 µg/kg Travoprost on GD 7 to PND 21 by s.c. injection. There were no effects on pup development. The NOAEL for this study was 0.1 µg/kg/day (U.S. FDA 2000).

Based on the weight of available evidence, the Applicant identified the subcutaneous LOAEL of 0.12 µg/kg/day (0.00012 mg/kg/day) from the 3-generation rat study with Travoprost as the point of departure for margin of safety calculations (U.S. FDA 2000).

This value is >100-fold lower than LOAELs for adverse effects on reproductive tissues following parenteral Cloprostenol exposure. Even though Travoprost is not a cosmetic ingredient, use of Cloprostenol as the sole surrogate for IPC would result in a less conservative risk assessment because the identified N(L)OAELs for Cloprostenol are higher than those for Travoprost for the critical reproductive endpoints.

SCCS comments on evaluation of reproductive studies

The Applicants provided evaluation of some studies on reproductive toxicity with Cloprostenol or Travoprost, however as explained in the Preamble the SCCS is of the view that the proposed read-across between Isopropyl Cloprostenate and Cloprostenol or Travoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence.

Although the read-across data have not been used in the MoS calculation by the SCCS, the potential reproductive effects of Isopropyl Cloprostenate become evident when all the available data are taken together. This raises a serious safety concern because the majority of those using these products will most likely be women of child-bearing age.

4.3.5.2 Developmental Toxicity

No new experimental data have been provided on developmental toxicity of Isopropyl Cloprostenate in the current submission dossier.

Applicant 1:

No teratogenic properties of Cloprostenol were reported in the two teratogenicity studies performed either in rats after oral administration of 0, 10, 25, 50 and 100 µg/kg bw/d or in rabbits after subcutaneous administration of 0, 0.025, 0.075 and 0.250 µg/kg bw/d.

Thus, the available data on reproduction and prenatal developmental toxicity of Cloprostenol and R-Cloprostenol indicated that effects can be caused by low doses, with the exception of teratogenicity. However, it was also demonstrated that clear NOELs were obtained.

With regards to risk assessment, the oral data, both, with regards to reproductive as well as to prenatal developmental toxicity, are considered as most relevant due to the more reliable route of exposure.

Thus, in rats no teratogenic properties were detected in the prenatal developmental toxicity study up to the highest dose of 100 µg/kg bw/d. In the 3-generation reproduction toxicity study in this species, a NOEL of 15 µg/kg bw/d for Cloprostenol was achieved.

Finally, with regards to reproductive and developmental toxicity after repeated oral exposure of Cloprostenol, it can be considered as likely that also low but clear NOAELs for Isopropyl Cloprostenate can be expected, which may be in the same order of magnitude.

Overall conclusions on reproductive dose toxicity of Isopropyl Cloprostenate by the Applicant 1:

There are no reproductive toxicity data on Isopropyl Cloprostenate available. The available data on Cloprostenol and R-Cloprostenol indicated that effects can be caused by low doses, with the exception of teratogenicity, but that there were also clear NOELs. Thus, in rats no teratogenic properties were detected in the prenatal developmental toxicity study up to the highest dose of 100 µg/kg bw/d. In the 3-generation reproduction toxicity study in this species, a NOEL of 15 µg/kg bw/d for Cloprostenol was achieved. As a consequence, and with regards to its sensitivity and physiological relevance, the NOEL of 15 µg/kg bw/d obtained in the 3-generation reproduction toxicity study in rat will be used for risk assessment and margin of safety calculation.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16

Applicant 2:

Sprague-Dawley rats, animals were dosed by s.c. injection at 0, 1, 3, and 10 µg/kg/day Travoprost for 4 weeks prior to and through GD 13 (females) or 2 weeks prior to mating through GD 7 (males). Corpora lutea, implantations, fetal viability, estrous cyclicity, and sperm parameters were not affected by treatment. Authors reported an increase in early fetal resorptions at 10 µg/kg/day. The NOAEL for this study was 3 µg/kg/day (U.S. FDA 2000). Female CD-1(ICR)BR mice were given s.c. injections of 0, 0.1, 0.3, and 1 µg/kg Travoprost on GD 6-16 and were sacrificed on GD 18. The incidences of early deliveries, litter loss, and total litter resorption as well as the number of early resorptions were increased, while the number of viable fetuses was reduced. No teratogenic effects were observed in the fetuses. Authors identified a NOAEL of 0.3 µg/kg/day for this study (U.S. FDA 2000).

Ref.: Safety assessment dossier on Isopropyl Cloprostenate by Applicant 2. 2023-10-17

SCCS comments on evaluation of developmental studies

The Applicants provided evaluation of some studies on developmental toxicity with Cloprostenol or Travoprost, however as explained in the Preamble the SCCS is of the view that the proposed read-across between Isopropyl Cloprostenate and Cloprostenol or Travoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence.

SCCS overall comment on reproductive and developmental toxicity

No new data have been provided on the reproductive and developmental toxicity of Isopropyl Cloprostenate in the current dossiers.

The available data on Cloprostenol effects on reproduction and on developmental toxicity in rats presented in SCCS/1635/21 indicate its activity at very low doses. This raises concern about the potential for similar effects by Isopropyl Cloprostenate.

The Applicants provided evaluation of studies on reproductive and developmental toxicity with Cloprostenol or Travoprost, however as explained in the Preamble the SCCS is of the view that the proposed read-across approaches between Isopropyl Cloprostenate and Cloprostenol or Travoprost do not fulfil the critical requirements for an unbiased and transparent read-across and are not supported by other line(s) of evidence.

Therefore, it is not possible to regard Isopropyl Cloprostenate safe in the intended cosmetic applications in the absence of data that can exclude reproductive/developmental toxicity.

4.3.6 Mutagenicity / genotoxicity**4.3.6.1 Mutagenicity / genotoxicity *in vitro***

Four new mutagenicity studies on Isopropyl Cloprostenate were available and evaluated below by the SCCS:

- two studies on gene mutation in bacteria (Ames test) – study #1 and #2
- two mammalian cell micronucleus tests on human lymphocytes (study #3) and Chinese hamster ovary cells (Study #4).

Applicant 1:

Study #1: Gene mutation test in bacteria (Ames test)**Study Design:**

Reference:	ICCR, 2024
Date of report:	04.01.2024
Study period:	17.11.2023 – 27.11.2023
Guidelines/Methods:	OECD 471, Council Regulation (EU) No 440/2008 B13/14 Mutagenicity, SCCS Notes of Guidance, 12 th revision, 2023
Test system:	<i>Salmonella typhimurium</i> strains TA98, TA100, TA1535, TA1537, <i>Escherichia coli</i> WP2 uvrA
Replicates:	Triplicates
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0684851 (purity: 100% (HPLC))
Vehicle:	DMSO
Concentrations:	Plate incorporation (Experiment I): ±S9-mix: 0; 3; 10; 33; 100; 333; 1000; 2500; 5000 µg/plate Pre-incubation test (Experiment II): ±S9-mix: 0, 33; 100; 333; 1000; 2500; 5000 µg/plate
Positive controls:	-S9-mix: Sodium azide (NaN ₃): 10 µg/plate for TA100, TA1535 4-Nitro-o-phenylenediamine (4-NOPD): 10 µg/plate for TA98; 50 µg/plate for TA 1537 Methyl methanesulfonate (MMS): 2.0 µL/plate for WP2 uvr A +S9-mix: 2-aminoanthracene (2-AA): 2.5 µg/plate for TA98, TA100. TA1535, TA1537, 10 µg/plate for WP2 uvr A
Negative control:	Vehicle and DMSO
GLP:	Yes
Published:	No

Material and methods:

Isopropyl Cloprostenate (batch: 0684851, purity: 100%) was investigated for its possible mutagenic potential in *Salmonella typhimurium* (*S. typhimurium*) strains TA98, TA100, TA1535, TA1537 and *Escherichia coli* (*E. coli*) strain WP2 uvrA according to OECD 471 under GLP conditions. The tested concentrations of the test item dissolved in DMSO were 0; 3; 10; 33; 100; 333; 1000; 2500 and 5000 µg/plate for the plate incorporation test (experiment I) and 0, 33; 100; 333; 1000; 2500 and 5000 µg/plate for the pre-incubation assay (experiment II) for all strains with and without the addition of a rat liver homogenate metabolizing system (S9 fraction of phenobarbital/β-naphthoflavone-induced rat liver). Three replicates for each concentration level and controls were used. Concurrent untreated plates, solvent control and appropriate reference mutagens (sodium azide, 4-nitro-o-phenylene-diamine, methyl methane sulfonate, 2-aminoanthracen) were investigated in parallel.

Results:

The test item precipitated in the overlay agar in the test tubes from 1000 to 5000 µg/plate. Precipitation of the test item in the overlay agar on the incubated agar plates was observed in Experiment I at 5000 µg/plate without S9 mix and from 2500 to 5000 µg/plate with S9 mix and in Experiment II at 5000 µg/plate with and without S9 mix.

The plates incubated with the test item showed normal background growth up to 5000 µg/plate with and without S9 mix in all strains used.

Toxic effects, evident as a reduction in the number of revertants (below the indication factor of 0.5), occurred in strain TA 100 (without S9-mix at 2500 µg/plate and/or 5000 µg/plate) and TA 1537 (without S9-mix at 5000 µg/plate).

No substantial increase in revertant colony numbers of any of the five tester strains was observed following treatment with Isopropyl Cloprostenate at any dose level, neither in the presence nor absence of metabolic activation (S9 mix). There was also no tendency of higher mutation rates with increasing concentrations in the range below the generally acknowledged border of biological relevance.

Appropriate reference mutagens were used as positive controls and showed a distinct increase of induced revertant colonies, which were clearly within the historical laboratory control data range. Thus, the sensitivity and suitability of the test system was demonstrated.

Conclusion:

Under the conditions of this test, Isopropyl Cloprostenate did not induce gene mutations by base pair changes or frameshifts in the genome of the bacterial strains used.

Therefore, Isopropyl Cloprostenate is considered to reveal no mutagenic potential in this *Salmonella typhimurium* and *Escherichia coli* reverse mutation assay, when tested with and without metabolic activation up to toxic and/or regulatory required doses.

Ref.: ICCR, 2024

SCCS comments on *in vitro* mutagenicity study #1

The results indicate no mutagenic effect of Isopropyl Cloprostenate in the Ames test.

The SCCS noted a cytotoxic effect in TA100 and TA1537 strains at the two highest concentrations tested, which was more pronounced for the TA100 strain without metabolic activation.

Applicant 2:**Study #2: Gene mutation test in bacteria (Ames test)****Study Design:**

Reference:	CIR, 2024
Date of report:	22.02.2024
Study period:	07.02.2024 – 21.02.2024
Guidelines/Methods:	OECD 471
Test system:	<i>Salmonella typhimurium</i> strains TA97a, TA98, TA100, TA102, TA1535
Replicates:	Triplicates
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0695607-4 (purity: not provided)
Vehicle:	Sterile deionized water
Concentrations:	Plate incorporation (Experiment I): ±S9-mix: 0; 50; 100; 500; 1000; 5000 µg/plate
Positive controls:	-S9-mix: Sodium azide: TA100, TA1535 Daunomycin: TA98 Mitomycin C: TA102 ICR 191 Acridine: TA97a

	+S9-mix: 2-aminoanthracene for all strains
Negative control:	Vehicle
GLP:	Yes
Published:	As attachment to CIR, 2024

Materials and methods:

Detailed study protocol is available as Appendix A to the study report.

Solubility:

The test article was found to be completely soluble in Sterile Deionized Water. This was the solvent used to dissolve the test article in this study.

Bacterial reverse mutation (Ames mutagenicity) assay:

The bacterial reverse mutation assay was used to evaluate the mutagenic potential of the test article at five (5) concentrations of the test article per plate: .5.0, 1.0, 0.5, 0.1, and 0.05 milligrams.

Testing was done with the appropriate solvent control and positive cultures were plated with overnight cultures of the test systems (TA97a, TA98, TA100, TA102, and TA1535) on selective minimal glucose agar in the presence and absence of PB/BNF-induced rat liver S9. All dose levels of the test article, solvent control and positive controls were plated in triplicate.

Conclusion / discussion:

The results in Tables 1 and 2 show that the test strains are sensitive to the positive control mutagens and showed the appropriate mutagenic response (*i.e.* positive control counts were greater than 2.5 times the negative solvent control) The spontaneous reversion rate was well within the accepted values of each strain, indicating that under the test conditions, the strains were sensitive to the detection of potentially genotoxic agents. The data in Tables 1 and 2 shows that the test article was not cytotoxic to the test system at all five (5) dose levels. There was no precipitation of the test article noted at any test concentration either with or without S-9 for the test system.

The metabolic activation using the S9 activation mixture shows an active microsomal preparation.

Using the same test conditions, there was no detectable genotoxic activity at the concentration shown above (*i.e.* the test article did not show a 2.5 fold increase in counts over the negative solvent control) associated with M24-0224.01) neither in the absence (Table 1) or presence (Table 2) of the S9 enzyme activation.

Ref.: CIR, 2024, study report by Consumer Product Testing Company (cptclab.com, USA)

SCCS comments on *in vitro* mutagenicity study #2

The SCCS considers the study of limited reliability. Although the study was claimed to be performed according to GLP principles, the SCCS noted the following inconsistencies/lack of data:

- distilled water was used as a vehicle, and it was stated by the authors of the report that no precipitation was observed in either the stock solution (50 mg/mL) or in plates with the highest concentration tested, *i.e.* 5000 µg/plate. This observation is difficult to explain considering the low solubility of Isopropyl Cloprostanate in water (99.1 µg/mL as reported by the applicant) and is in contradiction to the observation made in the *in vitro* study #1 where precipitation was clearly indicated at the concentration of 5000 and 2500 µg/plate.
- no historical control values were provided, hence it is not possible to verify if the almost equal number of revertants for the tester strains TA97a, TA98, TA100 and TA102 after using different positive control mutagens (both ±S9 mix) is reliable. Clear information on the concentrations of the mutagens used was not provided.
- purity of the test item was not provided.

Applicant 1:**Study #3: Mammalian Cell Micronucleus Test on human lymphocytes****Study Design:**

Reference:	ICCR, 2023
Date of report:	09.01.2024
Study period:	23.08.2023 – 02.10.2023
Guidelines/Methods:	OECD 487, SCCS Notes of Guidance, 12 th revision, 2023
Test system:	Human lymphocytes
Replicates:	Duplicates
Test substance:	Isopropyl Cloprostenate
Batch (Purity):	0684851 (purity: 100% (HPLC))
Vehicle:	DMSO
Concentrations and exposures:	3/28 h exposure/preparation time without S9-mix: Experiment I: 0; 13.3; 23.2; 40.6; 71.1; 124; 218; 381; 666; 1166; 2041 µg/mL Experiment II: 0, 14.4, 25.2, 44.1, 77.2, 88.8, 102, 117, 135, 155, 179 µg/mL 28/28 h exposure/preparation time without S9-mix: Experiment II: 3.7, 6.5, 11.4, 19.9, 34.8, 60.9, 107, 187, 327 µg/mL 3/28 h exposure/preparation time with S9-mix: Experiment I: 0; 13.3; 23.2; 40.6; 71.1; 124; 218; 381; 666; 1166; 2041 µg/mL
Positive controls:	-S9-mix: Mitomycin C (MMC, pulse treatment): 0.6 µg/mL Vinblastine (continuous treatment): 17.5 µg/mL +S9-mix: Cyclophosphamide (CPA): 15.0 µg/mL
Negative control:	Vehicles
GLP:	Yes
Published:	No

Material and methods:

The possible clastogenic and aneugenic potential of the test item Isopropyl Cloprostenate (0684851 (purity: 100%)) dissolved in DMSO was evaluated according to OECD 487 under GLP conditions in Human lymphocytes in three independent experiments. The induction of cytogenetic damage in human lymphocytes was assessed in two independent experiments with one preparation interval (28 hours). The cytokinesis blocker, cytochalasin B was added to the cell culture to ensure that there are binucleate cells to be evaluated for micronuclei. Treatments started after a 48-hour stimulation period with phytohaemagglutinin (PHA) when cells were actively proliferating and the cells were prepared at approx. 1.5 – 2.0 fold of the normal cell cycle time (start of exposure until preparation of cells). All cell cultures were set up in duplicates.

The study was performed according to the following exposure procedure:

	Without S9 mix		With S9 mix
	Experiment I** & II	Experiment II	Experiment I
Stimulation period (h)	48	48	48
Exposure period (h)	3	28*	3
Cytochalasin B exposure (h)	25	28*	25
Total culture period (h)	76	76	76
* Co-treatment of test item and Cytochalasin B			
** Was repeated since the cytotoxicity range was not met			

In each experimental group, two parallel cultures were analysed. Per culture 1000 binucleated cells were evaluated for cytogenetic damage.

The highest applied concentration in this study (2041 µg/mL of the test item) was chosen with regard to the initially provided purity (98 %) of the test item and with respect to the OECD Guideline 487.

The selected concentrations were as follows:

Exp.#	Prep. interval	Exposure period	Concentrations in µg/mL									
Without S9 mix												
I*	28 h	3 h	13.3	23.2	40.6	71.1	124 ^{PS}	218 ^{PS}	381 ^{PS}	666 ^{PS}	1166 ^{PS}	2041 ^{PS}
II	28 h	28 h	14.4	25.2	44.1	77.2	88.8^{PS}	102 ^{PS}	117 ^{PS}	135 ^{PS}	155 ^{PS}	179 ^{PS}
												250 ^{PS}
II	28 h	28 h	3.7	6.5	11.4	19.9	34.8	60.9	107 ^{PS}	187 ^{PS}	327 ^{PS}	571 ^{PS}
												1000 ^{PS}
With S9 mix												
I	28 h	3 h	13.3	23.2	40.6	71.1	124^{PS}	218 ^{PS}	381 ^{PS}	666 ^{PS}	1166 ^{PS}	2041 ^{PS}
Evaluated experimental points are shown in bold characters												
Exp. = experiment												
^{PS} Phase separation was observed at the end of treatment												
* = was repeated since the cytotoxicity range was not met												

Phenobarbital/β-naphthoflavone induced rat liver S9 was used as the metabolic activation system. Concurrent solvent control (DMSO) and appropriate reference mutagens (Mitomycin C, Vinblastine, Cyclophosphamide) were investigated in parallel.

Results:

In Experiment I, neither precipitation nor phase separation of the test item in the culture medium was observed in the absence and in the presence of S9 mix at the end of treatment. In addition, no precipitation occurred in Experiment II in the absence of S9 at the end of treatment. No relevant influence on osmolarity or pH was observed.

Short time treatment with and without metabolic activation

In Experiment I and II in the presence and absence of S9 mix after short time treatment, no relevant cytotoxicity was observed up to the highest evaluated concentration, which showed phase separation.

In the presence and absence of S9 mix, no relevant increases in the numbers of micronucleated cells were observed after treatment with the test item.

In the presence of S9 mix, however, the mean percentage of the micronuclei in two of three test item concentrations (0.95% and 1.05% micronucleated cells after treatment with 40.6 and 124 µg/mL) as well as in the solvent control (0.95% micronucleated cells), slightly exceeded the 95% control limit of the historical control data (0.00 – 0.94% micronucleated cells). Since all values were clearly within the min-max range (0.05 -1.18% micronucleated cells), the single cultures showed highly homogeneous values and none of the values were statistically significantly increased, when compared to the solvent control, these findings were considered as biologically irrelevant. This includes that there also was no concentration related increase in micronucleus formation (as judged by an appropriate trend test). Therefore, the outcome of this experimental part is clearly negative and valid.

Continuous treatment without metabolic activation

In Experiment II in the absence of S9 mix after continuous treatment, clear cytotoxicity (59.7% cytostasis) was observed at the highest evaluated concentration (34.8 µg/mL).

After continuous treatment, no relevant increases in the numbers of micronucleated cells were observed after treatment with the test item. The mean percentage of the micronuclei in all treated conditions was within the 95% control limit and none of the values were statistically

significantly increased, when compared to the solvent control. There also was no concentration related increase in micronucleus formation, as judged by an appropriate trend test.

The value of 1.25% micronucleated cells of the solvent control slightly exceeded the 95% control limit of the historical control data (0.06 – 1.21% micronucleated cells). Since the value is within the min-max range of the historical control data (0.10 – 1.35% micronucleated cells) and all values of the evaluated test item concentrations are clearly within the 95% control limit, this finding was considered as biologically irrelevant, and the experiment assessed as valid.

Overall, the outcome of the whole study is considered as negative without induction of micronuclei in the investigated human lymphocytes.

Appropriate mutagens were used as positive controls. They induced statistically significant increases in cells with micronuclei which slightly exceeded the historical laboratory control data range but were clearly in the min-max range of the historical control data (HCD). Thus, the sensitivity and suitability of the test system was demonstrated.

Conclusion:

Under the conditions of this study, Isopropyl Cloprostenate did not induce micronuclei as determined by the *in vitro* micronucleus test in human lymphocytes. Therefore, Isopropyl Cloprostenate is considered to be non-mutagenic in this *in vitro* micronucleus test, when tested with and/or without metabolic activation up to cytotoxic or to the highest evaluable concentrations investigating potential structural (clastogenicity) and numerical (aneugenicity) chromosome aberrations.

Ref: ICCR, 2023

SCCS comments on *in vitro* mutagenicity study #3

The SCCS considers the study as valid, and the results as negative.

The SCCS noted that in Experiment II, in which 28 h continuous exposure without S9-mix was used, a borderline cytostasis of 59.7% was observed for the highest concentration of 34.8 µg/mL.

The following inconsistency was noted for the concentration of Vinblastine used for continuous treatment: in the Table included in the dossier, the concentration reported was 17.5 µg/mL, while in the original study report it was 7.5 ng/mL. The SCCS assumes the latter value is correct.

Applicant 2:

Study #4: Mammalian Cell Micronucleus Test on Chinese hamster ovary cells

Study Design:

Reference:	Nucro-Technics Study No 403393
Date of report:	09/05/2024
Study period:	12/02/2024 – 09/05/2024
Guidelines/Methods:	OECD 487
Test system:	Chinese hamster ovary cells
Replicates:	Duplicates
Test substance:	Isopropyl Cloprostenate; clear colorless liquid, 10% w/w in ethanol (~75 mg/mL)
Batch (Purity):	0695607-8 (purity: 100% (HPLC))
Vehicle:	Ethanol
Concentrations and exposures:	4/23 h exposure/preparation time ± S9-mix: Experiment I: 0, 2.9, 5.9, 12, 23, 47, 94, 190, 380, 750 µg/mL 23 h exposure/preparation time without S9-mix: Experiment II: 0, 0.23, 0.47, 0.94, 1.9, 3.8, 7.5, 15, 30 µg/mL

Positive controls:	-S9-mix: Mitomycin C (MMC) (4h): 0.20 and 0.25 µg/mL Colchicine (COL) (23 h): 0.15 and 0.18 µg/mL +S9-mix: Cyclophosphamide (CP) (4 h): 2.0 and 4.0 µg/mL
Negative control:	Vehicle - ethanol
GLP:	Yes
Published:	No

Materials and methods:

The test item, (+)-Cloprostenol Isopropyl Ester (Clo-IPE), was evaluated for its potential to induce micronuclei (clastogenic response) or hypodiploidy (aneugenic response) in cultured Chinese Hamster Ovary (CHO) cells in the absence and presence of metabolic activation (S9). The experimental design followed the OECD Guideline for the Testing of Chemicals – 487, *In vitro* Mammalian Cell Micronucleus Test.

As the test item was provided as a solution in ethanol (10% w/w; 75 mg/mL), a solubility test was performed to evaluate its solubility when added to the test system.

Dose-dependent test item precipitate was observed for the three highest concentrations in Growth Medium. Based on the results of the solubility test, ethanol was selected as the vehicle for the main study experiments.

Micronuclei frequency assessment was performed using a commercially available micronucleus analysis kit based on flow cytometry measurements.

Results:

In Experiment 1, the test item was tested at 750, 380, 190, 94, 47, 23, 12, 5.9, and 2.9 µg/mL using three treatment conditions, short-term treatments (4 hours) in the absence and presence of S9 and extended exposure (23 hours) in the absence of S9. Concurrent negative/vehicle and positive controls were included in each treatment condition to verify the test system.

In Experiment 1, dose-dependent test item precipitate was observed at 750, 380, and 190 µg/mL in the absence of S9, and at 750 and 380 µg/mL in the presence of S9. Relative Increase in Cell (Nuclear) Counts (RICC), which is based on the number of nuclei present in the treated samples versus the negative control samples, was used as a measure of cytotoxicity. Following treatment with Clo-IPE, RICC values ranged from 101 to -41% for all treatment conditions indicating there was dose-dependent test item-induced cytotoxicity. When the maximum test item concentration is based on cytotoxicity, the highest concentration for evaluation should aim to achieve an RICC of $45 \pm 5\%$. At least three non-cytotoxic (RICC $\geq 40\%$) concentrations of the test item were available for the evaluation of micronuclei for the short-term treatments, and micronuclei were scored in at least 2000 nuclei test item concentration and control ($\geq 15,806$ per replicate). Therefore, Clo-IPE was tested with a sufficient number of concentrations up to the limit of cytotoxicity in accordance with OECD guidelines. Due to overt cytotoxicity (RICC $< 40\%$) in the long-term treatment, only the two lowest concentrations of Clo-IPE were available for evaluation of micronuclei, therefore, this condition was repeated in Experiment 2 with a modified dose design.

In Experiment 2, the test item was tested at 30, 15, 7.5, 3.8, 1.9, 0.94, 0.47, and 0.23 µg/mL for 23 hours in the absence of S9. Concurrent negative/vehicle and positive controls were included to verify the test system. No test item precipitate was observed for any concentration when treated to the test system. Test item RICC values ranged from 102 to -25% indicating there was dose-dependent test item-induced cytotoxicity. At least three non-cytotoxic concentrations of the test item were available for the evaluation of micronuclei, and micronuclei were scored in at least 2000 nuclei per test item concentration and control ($\geq 18,349$ per replicate). Therefore, Clo-IPE was tested with a sufficient number of concentrations up to the limit of cytotoxicity in accordance with OECD guidelines.

For both experiments, when compared to the concurrent negative control, no change in pH (indicated by a change in treatment media colour and/or measurement by pH meter) and osmolality (measurement by osmometer) was observed following treatment with Clo-IPE, where applicable.

In all treatment conditions, there were no biologically relevant increases in either the frequency of micronuclei (mean %MN) or in the frequency of hypodiploidy (mean %HD) compared to the concurrent negative control for any concentration of Clo-IPE. The mean %MN and %HD values were all < 3-fold or < 7-fold the concurrent negative controls, respectively, and within the laboratory historical negative control range. In the absence of any indication of a biologically relevant response, no statistical analysis was performed for any Clo-IPE concentration for any treatment condition. Clo-IPE was considered to be negative, *i.e.* the test item did not induce chromosome breaks and/or loss in cultured CHO cells.

The replication increases of the vehicle controls met the acceptance criteria of being ≥ 1.8 demonstrating that a substantial proportion of the negative control cells underwent cell division. The frequencies of MN and HD for the concurrent negative controls were within the historical negative control data. At least one concentration of the positive controls for each treatment condition produced increases in %MN that were ≥ 3 -fold when compared with the concurrent negative controls and fell within the normal characteristic range of the historical positive control data. Consistent with its mode of action (aneugen), colchicine induced increases in %HD that were ≥ 7 -fold when compared with the concurrent negative control and that fell within the normal characteristic range of the historical positive control data. Therefore, the study met *all* the acceptance criteria for a valid test.

Conclusions by the Authors of the report:

In conclusion, the test item, Clo-IPE, did not induce micronuclei (via chromosome breaks and/or loss of whole chromosome(s)) or hypodiploidy (chromosome loss, sub-2n nuclei) in cultured CHO cells in the absence (4 and 23 hours treatment) and presence (4 hours treatment) of metabolic activation when tested up to the limit of cytotoxicity under the conditions of the test.

Ref.: CIR, 2024, study report by Nucro-Technics' Study No.: 403393

SCCS comment on *in vitro* mutagenicity study #4

The SCCS considers the study as valid, and the results as negative.

Overall conclusions on mutagenicity / genotoxicity of Isopropyl Cloprostenate by the Applicant 1:

Isopropyl Cloprostenate was investigated for its possible mutagenic potential in standard *in vitro* studies compliant to OECD and SCCS requirements under GLP conditions.

No mutagenic/genotoxic properties of Isopropyl Cloprostenate were observed. Namely, not in the Ames Test pursuant to OECD 471 investigating potential gene mutations in bacteria and not in the *in vitro* mammalian micronucleus test pursuant to OECD 487, investigating potential structural (clastogenicity) and numerical (aneugenicity) chromosome aberrations in human lymphocytes.

The obtained results are in line with results obtained for the structural analogue Cloprostenol, which indicated also neither a mutagenic potential *in vitro* in the Ames and Mouse Lymphoma tests, a more than questionable clastogenicity nor *in vivo* (mice micronucleus test).

According to information provided by the SCCS (Reference: SCCS, 2022) Isopropyl Cloprostenate was predicted to be not mutagenic in Ames test according their models used.

This is in line with the independently performed *in silico* analysis, which concluded that Isopropyl Cloprostenate is considered to be non-genotoxic.

Finally, the mutagenic / genotoxic potential was sufficiently covered and was also in line with SCCS criteria and the ECHA requirements (Reference: ECHA, 2017a). Therefore, there is no need for additional testing.

SCCS overall comment on mutagenicity of Isopropyl Cloprostenate

Isopropyl Cloprostenate was tested in two bacterial gene mutation tests, one valid, and one of limited reliability. Both of the tests showed negative results. Two valid micronucleus studies, on human lymphocytes and CHO cells, were also negative.

Therefore, the SCCS considers that there are no concerns in relation to mutagenicity of Isopropyl Cloprostenate.

4.3.7 Carcinogenicity

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

Applicant 1:

Overall conclusions on carcinogenicity of Isopropyl Cloprostenate by the Applicant 1:

With respect to the possible carcinogenic potential there are no data on Isopropyl Cloprostenate available.

However, no mutagenic potential was detected in the OECD and SCCS compliant standard *in vitro* tests comprising the bacterial gene mutation assay (Ames test) and the *in vitro* mammalian micronucleus test investigating potential structural (clastogenicity) and numerical (aneugenicity) chromosome aberrations. Thus, it was demonstrated that Isopropyl Cloprostenate possess no mutagenic/genotoxic potential *in vitro*.

Based on the *in silico* analysis performed by the SCCS there are concerns that Isopropyl Cloprostenate may be a non-genotoxic carcinogen. Therefore, the *in silico* analysis was repeated by an accredited institute but including additional and more reliable methods.

On the one hand, its analysis confirmed that Isopropyl Cloprostenate does not present a genotoxic potential. On the other hand and with regards to possible carcinogenicity, no conclusion was possible as the QSAR results were inconclusive and in most cases outside the applicability domain. In addition, it was emphasized that based on the experience acquired so far by the institute, the current QSAR models for carcinogenicity are not powered enough to provide reliable results.

In addition, in the read across justification there are also structural alerts presented for possible genotoxic and non-genotoxic carcinogenicity. Unique for both, Isopropyl Cloprostenate and its structural analogue Cloprostenol. However, based on the available experimental data of both substances, there is no evidence for any mutagenicity/genotoxicity. Therefore, it was clearly demonstrated that the concern regarding possible carcinogenicity on a genotoxic mechanism can be excluded.

Whether the observed model alerts can be used as indication for a reliable concern regarding a possible non-genotoxic carcinogenicity potential is considered more than unlikely but cannot be answered with final certainty. However, it is generally accepted by scientific bodies as well as by authorities that even for proved non-genotoxic and carcinogenic compounds clear threshold values are available. As a consequence, this knowledge could also be applied for Isopropyl Cloprostenate. Unfortunately, no experimental carcinogenicity data on the structural analogue Cloprostenol are available.

However, from two other prostaglandin analogues, namely Tafluprost and Bimatoprost there are carcinogenicity studies in rats and mice available, which clearly showed no indication for a carcinogenic potential (CDER, 2010, 2011). As both compounds also revealed no indication for mutagenicity/genotoxicity, a possible non-genotoxic mechanism can be excluded for these compounds.

Finally, there is currently no consistent and reliable picture with regards to a possible carcinogenic potential of Isopropyl Cloprostenate available. However, there is evidence that this could only be due to a non-genotoxic mechanism. A mechanism for which it is generally accepted that a clear threshold exists.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16

Applicant 2:

Travoprost was not carcinogenic in 2-year studies in male and female CD-1 mice and CD rats dosed at 10, 30, and 100 µg/kg/day by s.c. injection. There was no evidence of carcinogenic potential associated with treatment, although the U.S. FDA reviewer noted that the doses administered to mice were too low and were below the maximum tolerated dose (MTD), a dose level that produces mild evidence of systemic toxicity, which serves as an indicator that

negative results are not attributable to factors such as low bioavailability. At the 100 µg/kg/day dose level, male rats were only dosed for 82 weeks (U.S. FDA 2000), which is shorter than the duration of a typical lifetime carcinogenicity bioassay.

Ref.: Safety assessment dossier on Isopropyl Cloprostenate by Applicant 2. 2023-10-17

SCCS overall comment on carcinogenicity of Isopropyl Cloprostenate

As expressed in the Preamble, the SCCS is of the view that the proposed read-across between Isopropyl Cloprostenate and Cloprostenol or Travoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. However, since Isopropyl Cloprostenate is not a genotoxic compound, the SCCS has no concern for genotoxic carcinogenicity.

4.3.8 Photo-induced toxicity

4.3.8.1 Phototoxicity / photo-irritation and photosensitisation

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

Applicant 1:

No data are available with regards to photo-toxicity on Isopropyl Cloprostenate. However, there is no trigger to investigate this endpoint as IPCP showed consistently two absorption maxima in the UV/VIS spectrum at 275 nm and at 282 nm under acidic (pH 1.3), neutral (pH 6.8) and alkalic (pH 13.9) conditions, respectively. At the wavelength of 300 nm and above no absorption was noted. According to EFSA and SCCS criteria no phototoxicity testing should be performed in cases where a test material only absorbs at wavelengths lower than 313 nm, which is clearly the case with IPCP.

Moreover, there is no indication to assume that Isopropyl Cloprostenate may exhibit phototoxic properties as no extended light absorbing chromophore is evident in the chemical structure.

SCCS comment

As UV/VIS absorption maxima ($\lambda_{\max}$) are at wavelength lower than 313 nm, no phototoxicity is expected.

4.3.8.2 Photomutagenicity / photoclastogenicity

No new experimental data have been provided on Isopropyl Cloprostenate in the current submission dossier.

Applicant 1:

Overall conclusions on photo-induced toxicity of Isopropyl Cloprostenate by the Applicant 1: No data are available with regards to photo-toxicity on Isopropyl Cloprostenate. However, there is no indication to assume that Isopropyl Cloprostenate may exhibit phototoxic properties as no extended light absorbing chromophore is evident in the chemical structure. In addition, whether there may be any trigger to investigate this endpoint the unfortunately outstanding investigation on the UV/VIS spectrum according to OECD 101 under GLP will show.

Ref.: Consilab, 2024a

SCCS comments on photo-induced toxicity of Isopropyl Cloprostenate

No relevant experimental data on the photo-induced toxicity of Isopropyl Cloprostenate are available. However, as UV/VIS absorption maxima are at wavelength lower than 313 nm, no phototoxicity or photomutagenicity are expected.

4.3.9 Human data

Applicant 2:

The Applicant provided a summary of results (in the form of PPT presentation) on two new safety clinical studies and one safety assessment document (full versions of all three documents have been provided to the SCCS) presented during CIR Expert Panel Meeting on December 4, 2023 ("New Studies Support Safety of Isopropyl Cloprostenate in Cosmetics"). The PPT presentation was focused on the Study 1. Conclusions from the presentation were formulated as follows:

Study 1. 8-Month /120 Subject Clinical Study run on a **lash serum** containing 0.0044% IC (the "Clinical Study")

- General safety: Safe for use by both contact and non-contact lens wearers.
- Ocular irritation: Slight potential for transient ophthalmological irritation.
- Ocular pigmentation: No statistically significant difference in visible color of iris on RGB scale after 8 months.
- Periorbital Volume: No change in periorbital volume after 8 months.

Ref.: Consumer Product Testing Company. Ophthalmological in-use safety evaluation of an eyelash serum. 2023-03-11

Study 2. Intraocular Pressure Assay run on a **lash serum** containing 0.005% IC (the "IOP Assay")

Intraocular pressure: No statistically significant difference in intraocular pressure after 28 days.

Ref.: Consumer Product Testing Company. Safety evaluation of an eyelash serum using changes in intraocular pressure. 2023-09-28

Study 3. Toxicological Safety Assessment evaluating safety of lash serums containing up to 0.005% IC (the "Tox Report").

Systemic toxicity: low risk of potential systemic toxicity due to de minimis exposure.

Ref.: Safety assessment dossier on Isopropyl Cloprostenate by Applicant 2. 2023-10-17

SCCS comments on the human data provided

Although the SCCS does not take human studies with finished products into consideration, it has noted that the results observed in the "Clinical Study" indicate that consumers have experienced changes in ocular pigmentation and a decrease in periorbital volume after using eyelash products containing Isopropyl Cloprostenate.

4.3.10 Special investigationsApplicant 1:***In silico* assessment of toxicity**

Due to the lack of data on Isopropyl Cloprostenate within its safety assessment at that time, the SCCS performed an *in silico* assessment to gain some insights into the potential toxicities of prostaglandin analogues using QSAR models and read-across (Reference: SCCS, 2022). The SCCS raised the concern that PG analogues in general may be non-genotoxic carcinogens. However, the weakness of this information was that there was no evidence whether Isopropyl Cloprostenate was flagged specifically. Due to insistence and follow-up, the SCCS had to admit that Isopropyl Cloprostenate was outside the prediction space of some of the models used in its *in silico* analysis for the endpoint carcinogenicity (Reference: SCCS, 2024).

An independently conducted *in silico* analysis by an accredited institute (ToxMinds), which was even more comprehensive in terms of models than that performed by the SCCS indicated that, there was some weakness as SCCS did not use SAR tools generally recommended by EU regulatory bodies for this purpose and that some models also used by the SCCS were not considered as reliable as predictions were outside the applicability domain. In addition, the current QSAR models for carcinogenicity are considered as not powered enough to provide reliable results based on the experience acquired so far by the accredited institute (Reference: ToxMinds, 2024a).

The models used by the accredited institute comprised expert-rules based SAR tools (OECD QSAR Toolbox v.4.6, Derek Nexus v.6.3.0) and statistical based QSAR models (VEGA v.1.2.3, Danish (Q)SAR models) for the *in silico*-based analysis of the carcinogenicity concerns raised. In addition, *in silico* analysis of the potential bacterial mutagenicity by QSAR tools (Derek Nexus v.6.3.0, Sarah Nexus v.3.3.0) were carried out.

Based on its comprehensive *in silico* analysis it was finally concluded that Isopropyl Cloprostenate does not present a genotoxic potential. With regards to possible carcinogenicity, no conclusion was possible as the QSAR results were inconclusive and in most cases outside the applicability domain. In addition, it was emphasized that based on the experience acquired so far by the institute, the current QSAR models for carcinogenicity are not powered enough to provide reliable results.

Noteworthy to mention that the prediction for a lack of a mutagenic potential is in line with the negative results obtained by the Ames test and the *in vitro* micronucleus test, the SCCS compliant standard battery (for details see section 6.6. above).

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16

SCCS comments

As mentioned in section 4.3.6, the SCCS has regarded Isopropyl Cloprostenate of no concern regarding genotoxicity, and thus by implication of no concern over genotoxic carcinogenicity.

In regard to non-genotoxic carcinogenicity, the SCCS has noted the Applicant's detailed explanation and the use of QSAR models and read-across approach to exclude non-genotoxic carcinogenicity. The SCCS shares the Applicant's views that the ability of the currently available *in silico* models and tools to predict non-genotoxic carcinogenicity may be limited because they do not cover large enough chemical space relevant to PGAs and also do not cover all the different pathways for non-genotoxic carcinogenicity. It should however be noted that the SCCS clearly indicated in the previous Opinion on PGAs (SCCS/1635/21) that the *in silico* exercise was carried out 'only to identify the first level alerts for any potential harmful effects' and not to obtain definitive evidence on which to base the overall safety conclusions.

In this regard, it also needs to be noted that the SAR tools particularly pointed out by the Applicant (e.g. QSAR Toolbox profilers based on structural alerts) are meant for category formation and not for use as predictive tools. The SCCS considers that the profilers are not

reliable enough to allow acceptance of a negative prediction (e.g. see a recent analysis of the profilers by AlJallal *et al.* 2024 <https://doi.org/10.1038/s41598-024-69157-1>).

Also, as mentioned in section 4.3.5 of this Opinion, there are serious concerns in regard to potential reproductive and developmental toxicity of Isopropyl Cloprostenate from the collective view of all the available information. In order to discount the concerns over non-genotoxic carcinogenicity and reproductive/developmental toxicity, experimental data would be needed, not *in silico* predictions alone without further supporting data from other lines of evidence.

Read across justification from the Applicant 1:

In principle, the read across justification was performed according to the principles of the analogue approach as laid down in the Read-Across Assessment Framework (RAAF) guidance document of ECHA (Reference: ECHA, 2017b). The RAAF provides a framework and principles for scientifically examining a read-across case. In this context of the RAAF, the simplest case of an analogue approach is considered: read-across from a single source substance to a single target substance.

In this case Cloprostenol has been evaluated for its suitability as analogue for the target substance Isopropyl Cloprostenate. Supported by the fact that Isopropyl Cloprostenate is chemically a fully synthetic substance in the form of the isopropyl ester of Cloprostenate, which is converted under physiological conditions into Cloprostenol, a synthetic prostaglandin analogue (Reference: SCCS, 2022).

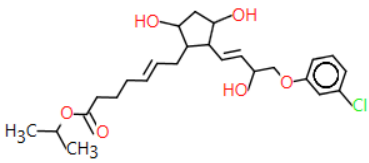
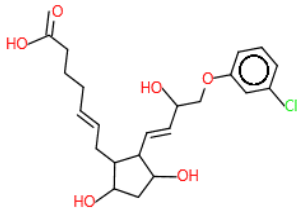
In principle, the analogue approach considers the following criteria.

- Common functional groups and structure
- Common structural alerts or reactivity
- Common physico-chemical properties
- Likelihood of common breakdown products via biological/degradation processes

The hypothesis was that Cloprostenol (CAS No. 54276-21-0) is a suitable read across substance for Isopropyl Cloprostenate.

Details of the analogue evaluation are presented in the table below.

Table 8. Analogue evaluation

	Target substance	Analogue
CAS number	157283-66-4	54276-21-0
Name	Isopropyl cloprostenate	Cloprostenol
2D Structure		
SMILES from PubChem	<chem>CC(C)OC(=O)CCCC=CCC1C(CC(C1)C=CC(COC2=CC(=CC=C2)Cl)O)O</chem>	<chem>C1C(C(C(C1O)C=CC(COC2=CC(=CC=C2)Cl)O)CC=CCCCC(=O)O)O</chem>
Analogue ranking	-	Suitable with interpretation
Structural similarity		
Dice index from OECD QSAR Toolbox v.4.6	-	0.82

Opinion on Prostaglandin analogues used in cosmetic products

	Target substance	Analogue
Physicochemical properties		
Molecular weight (Da)	467.01	424.93
Melting Point (deg C), Prediction software: MPBPWIN v.1.44	245.38	253.24
Boiling Point (deg C), Prediction software: MPBPWIN v.1.44	570.17	587.01
Vapour Pressure (Pa, 25°C), Prediction software: MPBPWIN v.1.44	4.83E-014	1.04E-014
Log Know, Prediction software: KOWWIN v1.69	5.15	3.95
Water solubility (mg/L), Prediction software: WSKOW v1.43	0.04694	2.251
Functional groups and profiling		
Organic functional groups from OECD QSAR Toolbox v.4.6	Alcohol Alkane, branched with secondary carbon Alkene moiety Allyl Aryl Aryl halide Carboxylic acid ester Cycloalkane Ether moiety Isopropyl	Alcohol Alkene moiety Allyl Aryl Aryl halide Carboxylic acid Cycloalkane Ether moiety
Structural alerts from OECD QSAR Toolbox v.4.6	Estrogen Receptor Binding: Strong binder, OH group Toxic hazard classification by Cramer: High (Class III) Carcinogenicity (genotox and nongenotox) alerts by ISS: Halogenated benzene (Non-genotox) <i>In vivo</i> mutagenicity (Micronucleus) alerts by ISS: H-acceptor-path3-H-acceptor Oncologic Primary Classification: Halogenated Aromatic Hydrocarbon Type Compounds Protein Binding Potency h-CLAT: Monohaloarenes	Estrogen Receptor Binding: Strong binder, OH group Toxic hazard classification by Cramer: High (Class III) Carcinogenicity (genotox and non-genotox) alerts by ISS: Halogenated benzene (Nongenotox) <i>In vivo</i> mutagenicity (Micronucleus) alerts by ISS: H-acceptor-path3-H-acceptor Oncologic Primary Classification: Halogenated Aromatic Hydrocarbon Type Compounds Protein Binding Potency h-CLAT: Monohaloarenes
Metabolism		

	Target substance	Analogue
Prediction of the first metabolic reaction using Meteor v.3.2.0 and literature data	Meteor v.3.2.0: Hydrolysis of acyclic carboxylic esters giving rise to isopropanol and Cloprostenol	Literature: Beta-oxidation of carboxylic acids (Al-Ghananeem and Crooks, 2007; Bourne et al., 1979)

Thus, the main outcome of the analogue assessment can be summarized as follows:

- Cloprostenol presents a high Dice index (0.82), indicative of structural similarity.
- Cloprostenol presents the same key functional groups (e.g., alcohol, aryl halide and ether moiety) as compared to Isopropyl Cloprostenate; it does not contain the carboxylic acid ester and isopropyl functional groups.
- With respect to the structural alerts, (Q)SAR analysis, as provided by the OECD QSAR Toolbox v.4.6 (OECD, 2023), revealed that Cloprostenol is classified as Cramer Class III (high toxicity), similar to Isopropyl Cloprostenate. Cloprostenol also presents the same structural alerts as Isopropyl Cloprostenate, which have been identified by the 'estrogen receptor binding', 'carcinogenicity (genotox and non-genotox) alerts by ISS', 'oncologic primary classification', 'in vivo mutagenicity (micronucleus) alerts by ISS' and 'protein binding potency h-CLAT 'profilers.
- Cloprostenol presents physicochemical properties in the same range as compared to the target substance, with lower log Kow and higher water solubility. These differences suggest that Cloprostenol is more bioavailable compared to the target substance.
- Finally, data available for Cloprostenol indicate that the substance undergoes beta-oxidation, while Isopropyl Cloprostenate is predicted by Meteor™ Nexus v.3.2 to undergo hydrolysis of acyclic carboxylic esters giving rise to isopropanol and Cloprostenol.

Thus, Cloprostenol was ranked as 'suitable with interpretation' analogue, as Cloprostenol presents a high similarity index, shares the same structural alerts and functional groups and is more bioavailable as compared to the target substance, thus representing a worst-case. In addition, Isopropyl Cloprostenate as the isopropyl ester of Cloprostenol is predicted to give rise to Cloprostenol upon the first metabolic reaction.

Finally, considering all of the above presented data, the profile of Cloprostenol is considered as sufficiently valid to justify the structural analogue read across approach for the target Isopropyl Cloprostenate.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16

SCCS comments on the Applicant's 1 Read across justification for Isopropyl Cloprostenate

Although it is plausible that Isopropyl Cloprostenate undergoes hydrolysis in the exposed organisms giving rise to Isopropanol and Cloprostenol, the SCCS is of the view (as expressed in the Preamble) that the proposed read-across between Isopropyl Cloprostenate and Cloprostenol does not fulfil the critical requirements for an unbiased and transparent read-across.

Applicant 1:

READ-ACROSS APPROACH

Read-across leverages available scientific data on structurally and/or mechanistically similar substances for the purpose of identifying human health hazards. As described by OECD (2014), the following attributes can be considered when identifying read-across substances:

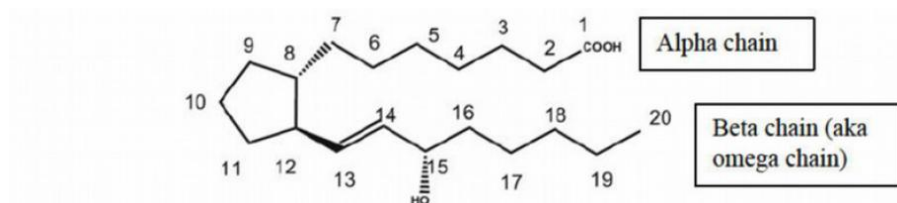
- Common functional groups
- Common mechanism of action: pharmacokinetic and pharmacodynamic properties, key events
- Common chemical class

Importantly, members of the same chemical class are often consistently positive or negative with respect to a specific health effect (OECD 2014).

The Applicant initially identified Bimatoprost, Latanoprost, and Travoprost as potential surrogates because they are all PGF₂a analogues, have robust toxicological datasets, and were identified in the SCCS Opinion on the use of prostaglandin analogues in cosmetics (SCCS 2022).

The Applicant first assessed the structural similarity of the three analogue candidates to the target chemical, IPC. The prostaglandin backbone structure is shown below in Figure 2. The analogues are all variations of this basic structure.

Figure 2. Basic structure of endogenous prostaglandins (SCCS 2022)



Available information on each potential surrogate elemental composition, chemical functional groups, and selected physicochemical properties is shown below in Table 9. The surrogate candidates are similar with respect to size, charge, and functional groups, with one notable exception: IPC, Latanoprost, and Travoprost are esters (RCOOR' group), while Bimatoprost is an amide (RCONHR'). In contrast to the other prostaglandin analogues (collectively termed prostanoids), Bimatoprost is considered a prostamide due to the presence of an amide rather than an ester at the C1 position on the alpha chain (Ishida et al. 2006).

Table 9. Physicochemical properties of IPC and surrogate candidates (PubChem 2023, SCCS 2022)

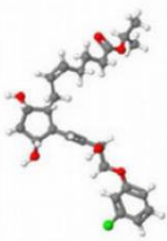
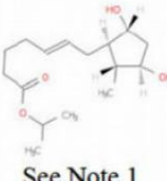
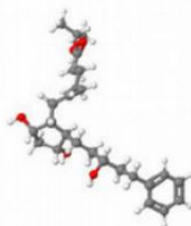
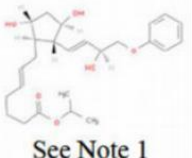
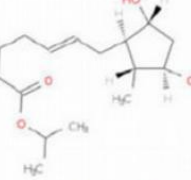
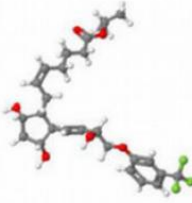
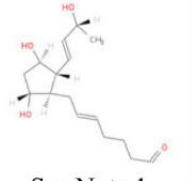
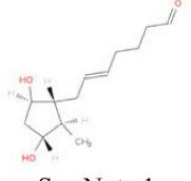
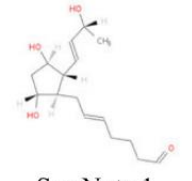
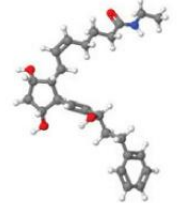
Property	Isopropyl Cloprostenate (IPC)	Latanoprost	Travoprost	Bimatoprost
CAS	157283-66-4	130209-82-4	157283-68-6	155206-00-1
MF	C ₂₅ H ₃₅ ClO ₆	C ₂₆ H ₄₀ O ₅	C ₂₆ H ₃₅ F ₃ O ₆	C ₂₅ H ₃₇ NO ₄
MW	466.9985	432.597	500.5505	415.57
Charges	0	0	0	0
Water solubility	0.047 mg/L @ 25°C (calculated)	12.9 mg/L	7.59 mg/L	18.7 mg/L
Partition coefficient	5.15 (calculated)	3.98	4.6	3.2
Individual Elements				
C	25	26	26	25
H	35	40	35	37
O	6	5	6	4
Ca	0	0	0	0
F	0	0	3	0
Cl	1	0	0	0
N	0	0	0	1
Na	0	0	0	0
S	0	0	0	0
Functional Groups				
RNH ₂	0	0	0	0
R ₂ NH	0	0	0	0
R ₃ N	0	0	0	0
RCONHR'	0	0	0	1
ROPO ₃	0	0	0	0
ROH	3	3	3	3
RCHO	0	0	0	0
RCOR	0	0	0	0
RCOOH	0	0	0	0
RCOOR	1	1	1	0
ROR	1	0	1	0
RCCH	0	0	0	0
RCN	0	0	0	0
RINGS	2	2	2	2
AROMATIC	1	1	1	1

Molecular formula (MF), molecular weight (MW), charges, atom counts, functional group counts, and ring counts. Physicochemical properties evaluation methods described in Dreyer et al. 2018.

To compare the structure of IPC to the three potential surrogates and evaluate whether these molecules are closely related in structure, the Applicant used ChemMine tools (ChemMine 2020) and OECD Toolbox (OECD 2020). Generally, chemical similarity analysis involves structural descriptors and similarity coefficient calculations, with the Tanimoto coefficient being widely used due to its utility and predictive accuracy (Chen and Reynolds 2002). The Tanimoto coefficient ranges from 0 to 1, with 0 being the least similar and 1 being the most similar. A Tanimoto coefficient of 0.8 or greater is often used to group similar chemicals. The Applicant compared chemicals in a pairwise manner using ChemMine's Similarity Workbench. For each pair, ChemMine calculated an atom pair Tanimoto similarity score (AP Tanimoto) and a maximum common substructure Tanimoto similarity score (MCS Tanimoto) (Chen and Reynolds 2002, Cao et al. 2008). Similarly, the Applicant compared chemicals in a pairwise manner using OECD Toolbox to calculate an atom pair Tanimoto (Jaccard) similarity score.

The results of the structural similarity analyses are shown in Table 10. Based on Tanimoto scores from the two software programs, IPC and Travoprost display the greatest structural similarity.

Table 10. Structural similarity analysis of four prostaglandin analogues

	Isopropyl Cloprostenate	Latanoprost	Travoprost	Bimatoprost
SMILES	<chem>CC(C)OC(=O)CCC\C=C/C[C@H]1[C@@H](O)[C@@H](O)[C@@H]1C=C[C@H](O)COC(=O)C(C)C</chem>	<chem>CC(C)OC(=O)CCC\C=C/C[C@H]1[C@@H](O)[C@@H](O)[C@@H]1C=C[C@H](O)COC(=O)C(C)C</chem>	<chem>CC(C)OC(=O)CCC\C=C/C[C@H]1[C@@H](O)[C@@H](O)[C@@H]1C=C[C@H](O)COC(=O)C(C)C</chem>	<chem>CCNC(=O)CCC\C=C/C[C@H]1[C@@H](O)[C@@H](O)[C@@H]1C=C[C@H](O)COC(=O)C(C)C</chem>
Isopropyl Cloprostenate		ChemMine AP Tanimoto: 0.623 MCS Tanimoto: 0.465 MCS Size: 20 MCS Min: 0.645 MCS Max: 0.625 OECD Toolbox Tanimoto (Jaccard): 0.537	ChemMine AP Tanimoto: 0.742 MCS Tanimoto: 0.861 MCS Size: 31 MCS Min: 0.969 MCS Max: 0.886 OECD Toolbox Tanimoto (Jaccard): 0.811	ChemMine AP Tanimoto: 0.549 MCS Tanimoto: 0.476 MCS Size: 20 MCS Min: 0.667 MCS Max: 0.625 OECD Toolbox Tanimoto (Jaccard): 0.512
Latanoprost	 See Note 1		ChemMine AP Tanimoto: 0.534 MCS Tanimoto: 0.435 MCS Size: 20 MCS Min: 0.645 MCS Max: 0.571 OECD Toolbox Tanimoto (Jaccard): 0.500	ChemMine AP Tanimoto: 0.596 MCS Tanimoto: 0.356 MCS Size: 16 MCS Min: 0.533 MCS Max: 0.516 OECD Toolbox Tanimoto (Jaccard): 0.564
Travoprost	 See Note 1	 See Note 1		ChemMine AP Tanimoto: 0.471 MCS Tanimoto: 0.444 MCS Size: 20 MCS Min: 0.667 MCS Max: 0.571 OECD Toolbox Tanimoto (Jaccard): 0.477
Bimatoprost	 See Note 1	 See Note 1	 See Note 1	

Note 1: Image represents the common substructure found in the two compounds being compared, based on an analysis of similarity in ChemMine (ChemMine 2020).

AP Tanimoto and Tanimoto (Jaccard) - similarity between corresponding atom pairs based on Tanimoto coefficient. MCS Tanimoto - similarity of the maximum common substructure (MCS) based on the Tanimoto coefficient. Structural similarity evaluation supported methods described in Dreyer et al. 2018.

MECHANISTIC SIMILARITY AMONG IPC AND POTENTIAL ANALOGUES

In addition to structural similarity, similarity in toxicologic or pharmacologic mechanism of action can serve as the basis for surrogate identification (OECD 2014). Prostaglandins are a group of endogenous bioactive molecules. These molecules play important roles in tissue homeostasis/normal physiology and are also activated during inflammation. IPC, Latanoprost,

Travoprost, and Bimatoprost are synthetic analogues of prostaglandin F_{2a} (PGF_{2a}). PGF_{2a} mediates female reproductive functions, cardiovascular function, and pain, and is upregulated during inflammation. Binding to its cognate receptor activates G-protein coupled intracellular signal transduction pathways, resulting in increased intracellular calcium (Ricciotti and FitzGerald 2011).

The PGF_{2a} analogues used as anti-glaucoma agents function as PGF_{2a} receptor agonists. In the eye, agonist-mediated activation of the PGF_{2a} receptor ultimately causes relaxation of muscles controlling outflow of vitreous fluid and/or remodeling of the extracellular matrix via increased matrix metalloproteinase activity. The analogues may also upregulate endogenous prostaglandin production and/or may increase ocular blood flow (Ishida et al. 2006). Several chemical-specific factors can influence the extent to which an analogue can participate in a mechanism or mode of action (and, hence, its suitability as a surrogate), such as absorption, metabolism to a biologically active form, and receptor affinity and specificity. The PGF_{2a} analogues must first be absorbed from the application site (skin, in the case of IPC used in cosmetics, or cornea, in the case of drug uses of Bimatoprost, Latanoprost, and Travoprost). Available data demonstrate that Latanoprost and Travoprost are rapidly absorbed from the cornea (U.S. FDA 1996, 2000) where they undergo in situ esterase-mediated hydrolysis to the active form. Although Bimatoprost is also readily absorbed, in contrast to other prostanoids, Bimatoprost is pharmacologically active as the intact molecule (Woodward et al. 2003). Once activated, the analogues bind the PGF_{2a} receptor with varying effectiveness (affinity).

The Applicant searched for data on structural features that can impact binding to the PGF_{2a} receptor, also known as the prostaglandin F receptor (FP receptor). Schuster et al. (2000) demonstrated that the carboxylic acid functional group (COOH, see Figure 2, above) is essential for FP receptor binding: modification of the carboxylic acid moiety to an amide reduced FP receptor binding affinity by more than 2 orders of magnitude. While substitution of the hydroxyl group at position 15 did not appreciably impact FP receptor binding affinity, introduction of a phenyl group on the omega chain enhanced FP receptor binding (Shuster et al. 2000).

SUMMARY

The Applicant's assessment of structural and mechanistic properties of the surrogate candidates indicates that Bimatoprost is different from Latanoprost and Travoprost in both aspects. A structural comparison of IPC to Latanoprost, Travoprost, and Bimatoprost indicates that Travoprost displays the greatest structural similarity to IPC. Available mechanistic data suggest that Bimatoprost exerts different mechanism(s) of action than IPC and therefore is not a strong a surrogate for IPC.

The Applicant's analog comparison results are summarized in Table 11. Based on high structural similarity, similar therapeutic potency, and available mechanism of action and metabolism information for prostaglandin analogues, the Applicant identifies Travoprost as the most suitable surrogate for evaluation of IPC's potential health effects, including carcinogenicity, reproductive toxicity, and developmental toxicity.

Table 11. Prostaglandin analog comparison

Potential Analog	Structural Similarity to IPC (Low, Intermediate, High)	Mechanism of Action	Metabolism	Pharmacology	Toxicology
Latanoprost	Intermediate --Based on low Tanimoto coefficient but similar functional groups.	PGF2 α receptor agonist.	Rapidly hydrolyzed to the active form.	Effectively reduces IOP.	Can induce local ocular effects. May also induce developmental effects at high doses.
Travoprost	High --Based on Tanimoto coefficient and functional groups	PGF2 α receptor agonist.	Rapidly hydrolyzed to the active form.	Effectively reduces IOP; similar in potency compared to IPC.	Can induce local ocular effects. May also induce developmental effects at high doses.
Bimatoprost	Low --Based on low Tanimoto coefficient and different functional groups	Less selective for PGF2 α receptor; binds other receptor forms.	Parent is the active form; not readily hydrolyzed.	Greater IOP lowering potency.	Slightly higher likelihood of local ocular effects. May also induce developmental effects at high doses.

Ref.: Safety assessment dossier on Isopropyl Cloprostenate by Applicant 2. 2023-10-17

SCCS comments on the Applicant's 2 Read across justification for Isopropyl Cloprostenate

Whilst the SCCS appreciates Applicant-2 for discussing the read-across process, and explaining how Tanimoto similarity scores were calculated, it is not clear on what basis the three analogues (Latanoprost, Travoprost, Bimatoprost) were selected out of several other prostaglandin analogues. Furthermore, despite the Tanimoto similarity index being >0.8, the finally selected analogue (Travoprost) differs from the target substance (IPCP) in aspects that are important determinants of toxicokinetic behaviour. There is a large difference in the solubilities of the two compounds (Travoprost being 161 times more soluble than IPCP as calculated by the Applicant-2), and Travoprost has a -CF₃ moiety instead of chlorine on the phenyl ring in IPCP. These differences are likely to make the two compounds different in terms of toxicokinetic and other toxicological aspects, and therefore Travoprost cannot be regarded as 'similar' to IPCP for use in read-across.

Therefore, after careful evaluation, the SCCS is of the opinion that the proposed read-across between IPCP and Travoprost does not fulfil the critical requirements for an unbiased and transparent read-across. In particular, no details have been provided on how the analogues were searched, how many were identified, and on what basis they were deselected or selected for use in the read-across. Therefore, the SCCS has considered that the proposed read-across from a single analogue (Travoprost) can only be considered if it is not the only evidence for a given endpoint – *i.e.* there is further information from other line(s) of evidence to support the read-across outcome. This means that the SCCS could only rely on the use of read-across from Travoprost to IPCP for a given endpoint where it formed part of a weight of evidence. Since no other lines of evidence were provided for IPCP, the read-across as proposed by the Applicant has not been used in the current Opinion.

Detailed results of similarity assessment between Isopropyl Cloprostenate, Latanoprost, Bimatoprost, and Travoprost prepared by Applicant 2 are presented in ANNEX 6.

4.4 SAFETY EVALUATION (including calculation of the MoS)

Applicant 1:

The obtained NOAEL of 15 µg/kg bw/d achieved in the three-generation reproductive toxicity study in rats of the structural analogue Cloprostenol will be used as PoD for the calculation of the Margin of Safety (MoS) compliant to the guidance of the SCCS. This will be related to the value of 5.61% (mean: 3.87% plus 1.74% SD) for dermal absorption for the calculation of the Systemic Exposure Dose (SED, mg/kg bw/day, references: EMA, 2004; SCCS, 2022, 2023).

This standard SCCS procedure resulted in very comfortable Margins of Safety as demonstrated in the calculations below:

Eyelash serum with 0.005% Isopropyl Cloprostenate		
Amount of product applied daily (qx)	0.00058	g/day
Default human body weight	60	kg
Relative daily exposure (Eproduct = qx/60 kg)	0.0097	mg/kg bw/d
Dermal absorption as percentage (Dap (%))	5.61	%
Concentration in the product (C (%))	0.005	%
Systemic exposure dose (SED, mg/kg bw/d) = Eproduct x C (%) / 100 x Dap (%) / 100	0.00003	µg/kg bw/d
Point of departure (PoD, 3 generation reproduction toxicity study in rats of the structural analogue Cloprostenol)	15	µg/kg bw/d
Route-to-Route Faktor	0.5	
PoDsys = PoD x Route-to Route Faktor	7.5	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	27660	

Eyebrow serum with 0.005% Isopropyl Cloprostenate		
Amount of product applied daily (qx)	0.004	g/day
Default human body weight	60	kg
Relative daily exposure (Eproduct = qx/60 kg)	0.0667	mg/kg bw/d
Dermal absorption as percentage (Dap (%))	5.61	%
Concentration in the product (C (%))	0.005	%
Systemic exposure dose (SED, mg/kg bw/d) = Eproduct x C (%) / 100 x Dap (%) / 100	0.0016667	µg/kg bw/d
Point of departure (PoD, 3 generation reproduction toxicity study in rats of the structural analogue Cloprostenol)	15	µg/kg bw/d
Route-to-Route Faktor	0.5	
PoDsys = PoD x Route-to Route Faktor	7.5	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	40107	

Aggregate Exposure: Eyelash serum and eyebrow serum each with 0.005% Isopropyl Cloprostenate		
Eyelash serum SED	0.00003	µg/kg bw/d
Eyebrow serum SED	0.00019	µg/kg bw/d
PoDsys = PoD x Route-to Route Faktor	7.5	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	35028	

With respect to the standard procedure, an ingredient is considered safe for use in humans at a MoS at and above 100, one has to consider that the above calculations are very

conservative, represent a worst-case scenario assessment and can be considered as a clear overestimation based on the preliminary estimation value of 10%. Thus, despite the applied worst-case scenarios, there are comfortable margins of safety available, which are clearly above the acceptable safety margin limit of 100.

The respective MOS values are:

- 276600 for the individual exposure calculation regarding eyelash serum.
 - 40107 for the individual exposure calculation regarding eyebrow serum.
 - 35028 for the aggregate exposure calculation regarding eyelash and eyebrow sera.
- Consequently, Isopropyl Cloprostenate can be regarded as safe for humans, when using eyelash and/or eyebrow serums individually or combined at a concentration of up to 0.005% in cosmetic products.

Ref.: Updated safety assessment dossier on Isopropyl Cloprostenate by the Applicant 1.
2024-09-16

SCCS comment

The MoS calculation for Isopropyl Cloprostenate is based on a NOAEL derived from a study performed on Cloprostenol.

Applicant 1:

The margin of safety (MoS) is the ratio of an acceptable intake/exposure level (such as the ADI, RfD, WEEL, or DNEL) to an estimated human exposure level or dose. Larger MoS are generally more health protective (ChemSafetyPro 2019).

For cosmetics ingredients, the MOS is calculated by dividing the highest NO(A)EL value that is lower than the lowest LO(A)EL of the cosmetic substance (identified throughout this safety assessment as a point of departure) under study by its exposure through use of the cosmetic (SCCS 2023). If the point of departure is obtained from a study in which dosing was conducted on fewer than 7 days per week, the NOAEL or LOAEL is adjusted for daily exposure, and the adjusted N(L)OAEL is used in MOS calculations. Similarly, if the point of departure is either a LOAEL or is obtained from a systemic toxicity study less than 90 days' duration, a safety factor of 3 is applied, except if the critical effect in a shorter-term study is on a reproductive or developmental endpoint, in which case the safety factor is not needed (SCCS 2023).

As outlined in SCCS (2023), for most cosmetic ingredients evaluated by the SCCS, the exposure is compared to an oral NOAEL or LOAEL. Generally, the N(L)OAEL identified in a toxicity study corresponds to the dose that has been administered orally (*i.e.*, the external dose). However, for cosmetic ingredients, the MoS is calculated by dividing the internal (systemic) dose (designated N(L)OAEL_{sys}) by the SED. For cosmetic ingredients, SCCS considers not more than 50% of an orally administered dose to be systemically available. Thus, in the absence of data supporting a different value, 50% of the orally administered dose is the default oral absorption value. The N(L)OAEL_{sys} is derived from the N(L)OAEL by dividing by a factor of 2: $N(L)OAEL_{sys} = N(L)OAEL/2$. If there is information to suggest poor oral bioavailability, a default value of 10% oral absorption may be considered (SCCS 2023).

After deriving NOAEL_{sys}, the MoS is calculated as follows:

$$\text{MoS} = \frac{\text{NOAEL}_{\text{sys}} \text{ or } \text{LOAEL}_{\text{sys}}}{\text{SED}}$$
$$\text{SED} = [\text{A (mg/day)} * \text{C (\%)/100} * \text{DA (\%)/100} * \text{DR}] / \text{BW}$$

The SED is a composite value that takes into account specific exposure conditions:

A is the estimated daily exposure to the cosmetic product in mg/day,
C is the concentration of the substance of interest,
DA is dermal absorption,
DR is dermal retention, and
BW is body weight.

In the absence of substance-specific data for dermal absorption or for formulations containing dermal penetration enhancers, a dermal absorption factor of 50% is included in the SED calculation as a conservative approach in accordance with SCCS (2023) guidelines. For leave-on, semi leave-on, and rinse-off products, a dermal retention factor of 1.0, 0.1, and 0.01, respectively, is additionally included in the SED calculations (SCCS 2023).

A sufficiently large MOS demonstrates safety of the raw material when used in the specific product application. For purposes of establishing safety, an acceptable MOS of at least 100 is required (factors of 10, each accounting for extrapolating from animal to human and interhuman variability), with additional factors added to account for items such as use of a lowest observable adverse effect level (LOAEL) (additional factor of 3), or use of a subacute study (additional factor of 3). In these cases, the acceptable MOS is 300. In contrast, if the PoD is derived from a clinical or epidemiological study, the factor of 10 for animal to human extrapolation is no longer necessary, and an MOS of 10 would be acceptable.

Table below identifies the point of departure, the PoD_{sys}, an SED, and subsequent MoS for IPC at a use level of 0.005% in a cosmetic lash serum.

Table E-1: Margin of Safety Calculation for Isopropyl Cloprostenate in a Cosmetic Lash Serum			
Parameter	Explanation	Value	Reference
A	Estimated daily exposure to cosmetic lash serum (mg/day)	0.28	Avomeen (2020)
C	Concentration of IPC (as decimal)	0.00005	Confidential
DA	Dermal absorption (as decimal)	0.5	SCCS (2023)
DR	Dermal retention (as decimal)	1	SCCS (2023)
BW	Body weight (kg)	60	SCCS (2023)
SED	Systemic exposure dose (mg/kg/day) = $C * A * DA * DR / BW$	1.17E-07	Calculated
POD	Point of departure (mg/kg/day), a LOAEL from a 3-generation study with surrogate travoprost.	0.00012	U.S. FDA 2000
PODsys	No adjustment to the POD needed because the study providing the LOAEL used parenteral exposure.	0.00012	U.S. FDA 2000
MOS	$MOS = POD_{sys} / SED$	1,029	Calculated

Ref.: Safety assessment dossier on Isopropyl Cloprostenate by Applicant 2. 2023-10-17

SCCS comments

The MoS calculation of Isopropyl Cloprostenate is based on a LOAEL derived from a study performed with Travoprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between Isopropyl Cloprostenate and Travoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a PoD for Isopropyl Cloprostenate, the SCCS cannot derive a MoS for Isopropyl Cloprostenate.

4.5 DISCUSSION

Prostaglandin analogues (PGAs) are intended to be used in cosmetic products that function as eyelash and/or eyebrow conditioners. These products are comprised of numerous cosmetic ingredients including Isopropyl Cloprostenate that are designed to enhance the appearance of natural eyelashes and eyebrows by nourishing, moisturising and protecting eyelashes and eyebrows from breakage.

Physicochemical properties

From the experimental physicochemical data provided, Isopropyl Cloprostenate can be considered as hydrophobic substance that is practically insoluble in water.

The Applicants have not provided experimental data on homogeneity and stability of Isopropyl Cloprostenate.

Exposure assessment & Toxicokinetics

The experimental data on percutaneous absorption of Isopropyl Cloprostenate indicate absorption at 5.61% (mean plus standard deviation).

The SCCS is of the opinion that the toxicokinetics, including systemic uptake via the dermal or oral route of Isopropyl Cloprostenate can be expected to be different from that of Cloprostenol and R-Cloprostenol. Although metabolic transformation of Isopropyl

Cloprostenate will yield Cloprostenol, it can be assumed that before the metabolism, systemically available Isopropyl Cloprostenate can be distributed differently in the organism compared with Cloprostenol. Hence, drawing conclusions on the toxicokinetics profile of Isopropyl Cloprostenate based on the toxicokinetics data available for Cloprostenol and R-Cloprostenol is not appropriate.

Toxicological Evaluation

Irritation and corrosivity

The skin and eye irritational potential of the undiluted Isopropyl Cloprostenate was tested in *in vitro* systems in accordance with Good Laboratory Practice, in the *in vitro* EpiOcular® Reconstructed human Cornea-like Epithelium (RhCE) Test Method (OECD 492), the *in vitro* EpiDerm® Reconstructed human Epidermis (RhE) Test Method (OECD 439) and Hen's Egg test - utilizing the chorioallantoic membrane (HET-CAM).

The results of these tests indicated that the pure Isopropyl Cloprostenate possesses both, a skin and eye irritation potential, which can be expected to be minimal or none at the low concentration of Isopropyl Cloprostenate at a maximum concentration of 0.005%.

Skin sensitisation

The SCCS has noted that two *in silico* models (OECD Toolbox, Derek Nexus) classified Isopropyl Cloprostenate as positive, and one of the *in vitro* tests performed by the Applicant was positive. A skin sensitising potential for Isopropyl Cloprostenate therefore cannot be excluded with certainty even when considering that the maximum intended concentration of Isopropyl Cloprostenate in eyelash and eyebrow products is very low (0.005%).

The SCCS considers HRIPTs unethical, additionally the cosmetic product results are not evaluated by the SCCS.

Acute toxicity

There are no acute toxicity studies on Isopropyl Cloprostenate available. Since the product is applied in small amounts above the eyelid, there is no concern on acute toxicity for Isopropyl Cloprostenate.

Repeated dose toxicity

There are no repeated dose toxicity studies on Isopropyl Cloprostenate available.

Reproductive and developmental toxicity

There are no reproductive and developmental toxicity studies on Isopropyl Cloprostenate available.

The available data on Cloprostenol effects on reproduction and on developmental toxicity in rats presented in SCCS/1635/21 indicate its activity at very low doses. This raises concern about the potential for similar effects by Isopropyl Cloprostenate.

Therefore, it is not possible to regard Isopropyl Cloprostenate safe in the intended cosmetic applications in the absence of data that can exclude reproductive/developmental toxicity.

Mutagenicity / genotoxicity

Isopropyl Cloprostenate was tested in two bacterial gene mutation tests, one valid, and one with limited reliability, both tests with negative results. Two valid micronucleus studies, on human lymphocytes and CHO cells, were negative.

Based on the available data, the SCCS is of the opinion that genotoxic potential of Isopropyl Cloprostenate could be excluded.

Carcinogenicity

No relevant experimental data on carcinogenicity of Isopropyl Cloprostenate have been provided.

As Isopropyl Cloprostenate is not a genotoxic compound, the SCCS has no concern for genotoxic carcinogenicity. However, the possibility for non-genotoxic carcinogenicity has not been ruled out.

Photo-induced toxicity

No relevant experimental data on photo-induced toxicity of Isopropyl Cloprostenate have been provided.

However, as UV/VIS absorption maxima for Isopropyl Cloprostenate are at wavelength lower than 313 nm no phototoxicity is expected.

Human data

In the submission, the Applicant provided a summary of results on three new safety clinical studies on human subjects.

Although the SCCS is not considering human studies with finished products, it has noted that the results observed in these clinical studies indicate occurrence of changes in ocular pigmentation and decrease in periorbital volume in consumers after use of Isopropyl Cloprostenate in eyelash products.

Special investigation

In the submission the Applicants provided the results of profiling of Isopropyl Cloprostenate and Cloprostenol and Travoprost, and considered the profile of Cloprostenol and Travoprost to be sufficiently valid to justify the structural analogue read across approach for the target Isopropyl Cloprostenate.

As expressed in the Preamble, the SCCS is of the view that the proposed read-across between Cloprostenol or Travoprost and Isopropyl Cloprostenate does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence.

Safety evaluation

The MoS calculations of Isopropyl Cloprostenate by the Applicants were based on a NOAEL derived from a study with Cloprostenol or a LOAEL derived from a study with Travoprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between Isopropyl Cloprostenate and Cloprostenol or Travoprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a PoD for Isopropyl Cloprostenate the SCCS cannot derive a MoS for Isopropyl Cloprostenate.

5. OPINION ON DECHLORO DIHYDROXY DIFLUORO ETHYLCLOPROSTENOLAMIDE (DDDE) or ETHYL TAFLUPROSTAMIDE

Preamble

In the submitted safety dossier, the Applicant provided experimental data for Dechloro Dihydroxy Difluoro Ethylcloprostenolamide (DDDE) or Ethyl Tafluprostamide from studies on skin and eye irritation, skin sensitisation and mutagenicity/genotoxicity. Also, dermal absorption studies and a metabolism study for DDDE were provided. No further data were provided either on toxicokinetics or systemic toxicological endpoints, e.g. repeated dose toxicity and developmental and reproductive toxicity. To fill this gap, the Applicant proposed a read-across approach using data from Tafluprost – taken as a structural analogue of DDDE. After careful evaluation, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across. In this regard, no details have been provided on how the analogues were searched, how many were identified, and on what basis they were deselected or selected for use in the read-across.

Furthermore, although predictions derived from a non-testing NAM like read-across provide an important line of evidence, they are not considered sufficient as a standalone evidence to allow forming a conclusion on the lack of toxicity of a chemical substance against a given endpoint. Read-across outcomes therefore need to be supported by other lines of evidence, where the overall weight of evidence can provide a consistent and more reliable basis for decision making in regulatory risk assessments. For example, see EFSA (2025) Guidance on Read-across.

In view of this, the proposed read-across from a single analogue (Tafluprost) can only be considered if it is not the only line of evidence for any given endpoint – *i.e.* if there is further information from other line(s) of evidence to support the read-across outcome. This means that the SCCS could only rely on the use of read-across from Tafluprost to DDDE for a given endpoint where it formed part of a weight of evidence.

However, since no other lines of evidence were provided for DDDE, the read-across as proposed by the Applicant has not been used in the current Opinion.

5.1 CHEMICAL AND PHYSICAL SPECIFICATIONS

5.1.1 Chemical identity

5.1.2 Primary name and/or INCI name

Dechloro dihydroxy difluoro ethylcloprostenolamide (DDDE)

Ref.: (Athena Cosmetics Inc., 2020)

Ethyl Tafluprostamide

Ref.: (Athena Cosmetics Inc., 2023)

5.1.2.1 Chemical names

IUPAC: (Z)-7-[(1R,2R,3R,5S)-2-[(E)-3,3-difluoro-4-phenoxybut-1-enyl]-3,5-dihydroxycyclopentyl]-N-ethylhept-5-enamide

Tafluprost ethyl amide (TEA)

Tafluprostamide

Dechloro dihydroxy difluoro ethylcloprostenolamide [INCI]

Ref.: (Athena Cosmetics Inc., 2023)

5.1.2.2 Trade names and abbreviations

Not available

5.1.2.3 CAS / EC number

Ethyl Tafluprostamide: CAS 1185851-52-8

Ref.: (Athena Cosmetics Inc., 2023)

Tafluprost: CAS 209860-87-7

Ref.: (BfR 7-3721-00-10144629, 2018)

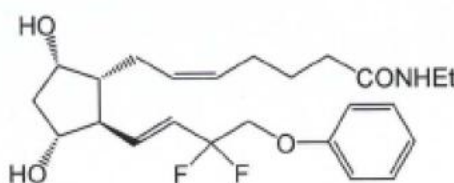
Added by the SCCS:

Ethyl Tafluprostamide: EC 867-521-0

Tafluprost: EC 682-458-7

Ref.: (Pubchem Tafluprost, n.d.)
(Pubchem tafluprost ethyl amide, n.d.)

5.1.2.4 Structural formula



Ref.: (Athena Cosmetics Inc., 2020)

5.1.2.5 Empirical formula

C₂₄H₃₃F₂NO₄

Ref.: (Athena Cosmetics Inc., 2023)

5.1.3 Physical form

Colourless to pale yellow solution.

Ref.: (Athena Cosmetics Inc., 2023)

5.1.4 Molecular weight

437.52 g/mol.

Ref.: (Athena Cosmetics Inc., 2023)

5.1.5 Purity, composition and substance codes**Applicant 1**

Purity: not less than 99% (99.78% by LCMS)

Ref.: (Athena Cosmetics Inc., 2023)

Applicant 2

Purity specification: $\geq 98\%$ (HPLC) Batch 0652603: 98.5% (HPLC, toxicity testing)

Batch 0663687: 99.2% (HPLC, physico-chemical data)

Ref.: (CaymanChemicals, 2023a)

5.1.6 Impurities / accompanying contaminants**Applicant 1**

Impurities: not more than 1% (NMT 0.22%)

Ref.: (Athena Cosmetics Inc., 2023)

Applicant 2

Water content: Negligible but not regularly analysed.

Isomers: HPLC Purity demonstrates regularly that the C-5,6 trans double bond isomer of the C-5,6 cis double bond in the alpha chain is $\leq 2\%$.

Residual solvents: Negligible, prior to formulation in ethanol, but not regularly analysed.

Elemental Impurities: Elemental impurities are not assessed as the risk for presence of these impurities (including Cadmium, Lead, Arsenic, and Mercury) in the Ethyl Tafluprostamide to be higher than the ICH Permitted Cutaneous Concentration, has been assessed.

The primary source of these elements is potable water. The water employed in the synthesis at Cayman Chemical and its wholly owned subsidiary Cayman Pharma is regulated by their respective national environmental departments, where these impurities are controlled to very low levels. Further, the ICH limits assume a daily intake of 10 g of the concerned material, the use of Ethyl Tafluprostamide in cosmetics is at least over 1000-fold less. Therefore, the probability of occurrence for these impurities in a final product at or near the allowable limit is very low.

Ref.: (CaymanChemicals, 2023a)

SCCS comments

The impurity(ies) have not been identified and quantified; batch numbers have not been provided; the first sentence under 'Elemental impurities' is unclear.

5.1.7 Solubility**Applicant 1**

Water solubility at 20°C: 1.05 g/L (Experimental)

Ref.: (Athena Cosmetics Inc., 2023)

Applicant 2

Water solubility: Measured: 1.05 g/L $\pm$ 3.6% RSD at 20 °C (OECD 105)

Further solubilities:

Ethanol: completely
DMSO: about 30 mg/mL
Dimethyl formamide: about 30 mg/mL

Ref.: (Walberg, 2023)

5.1.8 Partition coefficient (Log Pow)

Applicant 1

$2.74 \pm < 0.01$ at 25°C

Applicant 2

Partition coefficient (Log Pow):

Measured: $2.74 \pm < 0.01$ at 25 °C (HPLC, OECD 117)

Ref.: (Walberg, 2023)

Ref.: (Athena Cosmetics Inc., 2023)

5.1.9 Additional physicochemical specifications

Applicant 1

Where relevant:

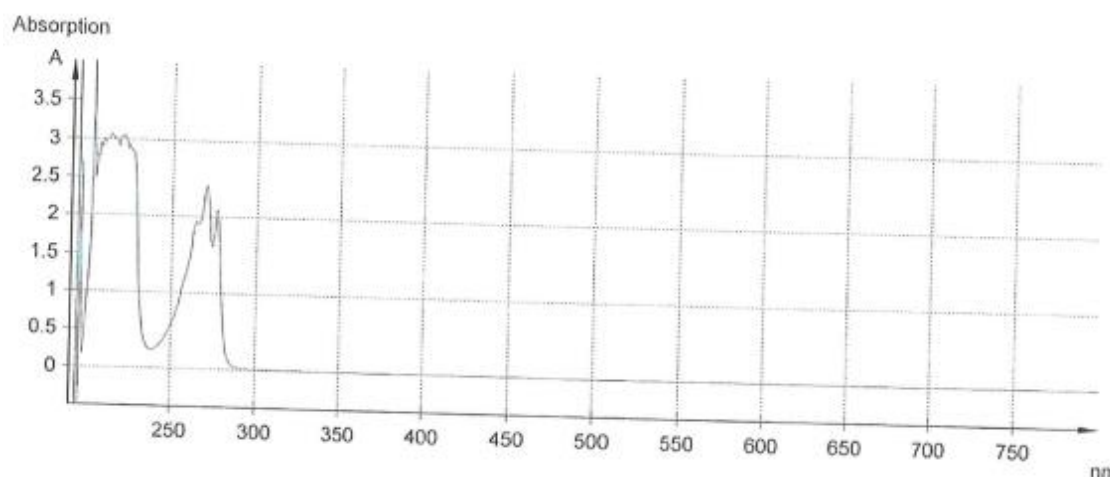
- organoleptic properties (colour, odour, taste if relevant): pale yellow
- melting point: 98.08°C
- boiling point: 503.76°C
- flash point: 14°C (57.2°F) – closed cup
- vapour pressure: 1.2E-13 Pa at 25°C
- density: 1.21 g/cm³
- pKa: no data
- pH: not relevant
- UV/visible light: 2 absorption peaks: 226-276 nm

Ref.: (Athena Cosmetics Inc., 2023)

Applicant 2

UV/VIS Absorption Spectra were analyzed according to OECD-Guideline 101 (1981) and Test Guideline OPPTS 830,7050 (1996).

Initially, spectrophotometric measurements were carried out over the whole wavelength range of 190 to 800 nm.



UV-spectra from the method development (190 - 800 nm, $c = 1$ g/L)

Since no absorption took place above 300 nm, further measurements were performed in the wavelength range of 190 to 400 nm. The "neutral", acidic and basic solutions showed three absorption maxima/shoulders at wavelengths of about 220, 270 and 276 nm.

Ref.: (Walberg, 2023)

5.1.10 Homogeneity and Stability

No data provided.

SCCS comment

DDDE is stable for > 2 years if stored at -20°C

Ref.: (CaymanChemicals, 2023)

5.2 EXPOSURE & TOXICOKINETICS

5.2.1 Function and uses

DDDE is used in product X as an eyelash conditioner product that protects against breakage and brittleness while improving flexibility. It is meant to be applied once daily with a fine brush applicator as a thin line directly to the eyelashes. An average of 2.4mg was used with each brush stroke or 0.432 μ g DDDE (product X with 0.018% DDDE). With a maximum of 0.72 μ g DDDE per application (with a maximum of 4 mg product X per brush stroke).

Ref.: (Athena Cosmetics Inc., 2023)

SCCS comments

Although exposure data of the ingredient under consideration is important, here a product is used, not the pure DDDE substance. The amount of eyeliner used according to the Notes of Guidance (SCCS/1647/22), is 5mg.

5.2.2 Dermal / percutaneous absorption**Applicant 1**

According to Applicant 1, the absorption, through all the layers of the eyelid skin, to contact the outer surface of the eye is very likely due to the quantity of DDDE present in the cosmetic product, but also due to the addition of the viscosity-increasing agent. The DDDE stays at the place of the application.

Dermal absorption study

The *in vitro* absorption potential of DDDE present in a representative cosmetic eyelash product formulation was determined in healthy human skin mounted on dynamic cells in an OECD Test Guideline 428-compliant study.

The skin was tape-stripped, and the remaining skin was separated into epidermis and dermis. The stability of the formulation was tested before and after the 24-hour experiment. No reference compound was mentioned.

Results: The absorbed fraction of DDDE after 24 hours was:
6.44% (0.012%); 6.51% (0.018%); 9.12% (0.020%); 10.68% (0.024%)
Total recovery of DDDE was within 85-115%, validating the results.

Result:

The absorbed fraction of the applied test substance was determined to be $6.51 \pm 2.16\%$.

Study design

Reference:	(Athena Cosmetics Inc., 2023)
Date of report:	28.04.2023
Study period:	08.25.2022 – 10.17.2022
Guidelines/Methods:	OECD 428, SCCS Notes of Guidance (12th revision), SCCS basic criteria for the <i>in vitro</i> assessment of dermal absorption of cosmetic ingredient, OECD Guidance 28
Test system:	Human skin obtained from abdominal surgery, dermatomed to a thickness of 310-400 µm.
Group size:	8 donors (Donors 1-4 for 0.020% and 0.024% DDDE, Donors 5-8 for 0.012% and 0.018% DDDE)
Number of cells:	12 cells in total
Test substances:	DDDE and 3H-radiolabelled DDDE in phenoxyethanol
Batch (Purity):	AF-F-0122-01 and 22-0809-93 (Purity: Non-labelled 99.42%, Radiolabelled >98%).
Application area	1cm ²
Active ingredient concentration (w/w)	0.024% (w/w) 0.020% (w/w) 0.018% (w/w) 0.012% (w/w) in a representative cosmetic eyelash product formulation.
Dose	Approximately 1 µCi on each cell (10 µL of test substance/cell).
Amount of reference item applied on skin	2.4 µg 2.0 µg 1.8 µg 1.2 µg
Receptor fluid:	5% w/w Bovine serum albumin, 0.9% NaCl in water
Diffusion cells.	Four donors were included and three diffusion cells per donor were measured for test items, a total of twelve cells was used by each formulation. 48 cells were used for this

	study
Skin temperature:	/
Skin integrity measurement:	Trans-epidermal water loss (1.85 – 10.12 g/m ² /h).
Check of the test item stability at 32°C during 24 hours	Yes
Sampling of Receptor Fluid	At 0.5, 2, 4, 8, 12, and 24 hours.
Stratum corneum tape stripping:	Yes (maximum 20 strips).
GLP:	Yes
Published:	/

Ref.: (Klock, 2023a)

Results and conclusions:**Absorption Results:**

DDDE Dilution 0.024%

Absorption: 10.68% ± 7.18% of applied dose (0.25 ± 0.17 µg/cm²)

DDDE Dilution 0.020%

Absorption: 9.12% ± 7.23% of applied dose (0.19 ± 0.15 µg/cm²)**DDDE Dilution 0.018%**Dermal absorption: 6.51% ± 2.16% of applied dose (0.12 ± 0.04 µg/cm²)

DDDE Dilution 0.012%

Absorption: 6.44% ± 2.14% of applied dose (0.08 ± 0.03 µg/cm²)**Total Recovery:**

Mean recoveries for each dilution condition were within the acceptance criteria of 85-115%, validating the results obtained:

0.024%: 99.11%; 0.020%: 95.53%; 0.018%: 91.02%; 0.012%: 92.00%

According to the Applicant, the study concluded that the dermal absorption of DDDE varies with concentration, with a higher absorption percentage observed in the higher concentration dilutions. The results validate that the experimental conditions were appropriate and that the test items maintained radiochemical purity and stability throughout the experiment. The mean recoveries were consistent across all formulations, confirming the reliability of the data collected during the study. Overall, the findings support the understanding of how different concentrations of DDDE can affect skin absorption, which is crucial for further evaluation in cosmetic formulations.

Applicant 2

(i) Study 1: this study assessed the bioavailability of pure TPEA and TPEA in a cosmetic product, finding bioavailability rates of 42.2% for pure TPEA and 24.4% for the cosmetic product after 24 hours. However, these results were considered unreliable due to improper inclusion of the stratum corneum in bioavailability calculations and other deviations from guideline requirements. The study also lacked key methodological standards such as appropriate skin temperature control and donor compliance.

Detection System: HPLC-MS for detecting Ethyl Tafluprostamide (TPEA) and its metabolite, Tafluprost acid (TPFA).

Study design

Reference:	(Weghuber et al., 2022) (Walberg, 2022) (composition of the cosmetic product)
Date of report:	04.10.2022
Study period:	24.10.2022 – 25.05.2022
Guidelines/Methods:	OECD 428, SCCS Notes of Guidance (11 revision, 2021), SCCS basic criteria for the <i>in vitro</i> assessment of dermal absorption of cosmetic ingredient (2010)
Test system:	Human full thickness skin membranes (fresh, 4 mm, commercially abdominal skin samples)
Group size:	3 female donors (age: 32 - 40 years)
Number of cells:	In total 8 – 11 cells (depending on the experiment)
Test substances:	Ethyl Tafluprostamide (6.0 µg/cm ²) Eyelash product containing 0.015% Ethyl Tafluprostamide (20 µL/cm ²)
Batch (Purity):	Batch: 0442914 (Purity: 100% (HPLC)) Not specified
Vehicle:	Ethanol (50% solution) None (undiluted use)
Dose:	6.0 µg/cm ² 20 µL/cm ²
Receptor fluid:	Hanks' Balanced Salt solution (HBSS) supplemented with 5 g/L albumin (BSA)
Diffusion cells:	Static diffusion cells
Skin temperature:	Skin temperature was not assessed throughout experiments. But prior to use, the skin samples were incubated at 37°C; 5 % CO ₂ , 95 % humidity
Skin integrity:	Transepidermal electrical resistance (TEER)
Exposure:	Topical open to skin surface area of 0.5 cm ²
Exposure time:	2, 4, and 24 h
Sampling:	2, 4, 24 h
Positive control:	Caffeine 50 µg (100 µg/cm ² ; diluted in 50% ethanol)
GLP:	No
Published:	No

Following analysis, the bioavailability of the pure Ethyl Tafluprostamide was considered by the study author to be 12.3 ± 2.2 % after 4 hours and 42.2 ± 23.1 % after 24 hours, while after application of the cosmetic product for 24 hours it was considered to be 24.4 ± 16.5 %. According to an earlier review the presented bioavailabilities of the study author have to be considered as invalid as a key requirement of the respective OECD 428 guideline and the SCCS requirements (References: (SCCS, 2023) (SCCS/1358/10, 2010)) for a reliable study were not met.

Ref.: (Walberg, 2023)

(ii) Study 2: Another study investigated the *in vitro* dermal absorption of Ethyl Tafluprostamide (0.015%) from a cosmetic formulation using [3H]-Ethyl Tafluprostamide as a tracer. Human split-thickness skin from four female donors was mounted on dynamic diffusion cells and exposed to the test substance for 24 hours. The total recovery of the applied dose was 92.62%, with 16.19% of the Ethyl Tafluprostamide considered bioavailable. The receptor fluid, along with the epidermis and dermis, contained the absorbed portion, while the rest remained on the skin surface. Based on the results, the dermal absorption value of 25.77% was used for the Margin of Safety (MoS) calculation according to SCCS guidelines. *Detection System:* Scintillation counting (Tricarb 2910TR) for radiolabeled content analysis.

Study Design:

Reference:	(Eurofins, 2022); Walberg, 2022 (composition of the cosmetic product)
Date of report:	20.12.2022
Study period:	August 2022 – October 2022
Guidelines/Methods:	OECD 428, SCCS Notes of Guidance (11th revision), SCCS basic criteria for the <i>in vitro</i> assessment of dermal absorption of cosmetic ingredient, OECD Guidance No. 28
Test system:	Human split-thickness skin membranes (302 - 400 µm)
Group size:	4 female donors (age: 31 - 59 years)
Number of cells:	12 cells in total (3 replicates from each of the 4 donors)
Test substances:	Ethyl Tafluprostamide (0.015%) [³ H]-Ethyl Tafluprostamide Eyelash serum product without Ethyl Tafluprostamide
Batch (Purity):	0652603 (Purity: 98.5 (HPLC)) 22-0809-93 (Purity: > > 98% (HPLC), Specific activity: 138.971 mCi/mg, radiochemical purity: 99.37% - 100%) LPS-0001
Application area	1 cm ² (flow of receptor fluid = 1.5 mL/h)
Active ingredient concentration (w/w)	0.015%
Dose	5 mg/cm ²
Amount of reference item applied on skin	0.75 µg
Receptor fluid:	6% polyethylene glycol 20 oleyl ether in PBS (0.01 M, pH 7.4, Flow of receptor fluid: 1.5 mL/h)
Diffusion cells.	Dynamic cells (application area = 1 cm ² , flow of receptor fluid = 1.5 mL/h)
Skin temperature:	32 ± 1°C
Skin integrity measurement:	Transepidermal water loss (TEWL)
Check of the test item stability at 32°C during 24 hours	Yes (n=1)
Sampling of Receptor Fluid	30 min, 2h, 4h, 6h, 12h and 24h after application (total volume :36 m)
Stratum corneum tape stripping:	
GLP:	Yes
Published:	No

The summary of the absorption of Ethyl Tafluprostamide through human split thickness skin is presented in following Tables:

Distribution of ³H-Ethyl Tafluprostamide after application of the eyelash serum product to human skin (expressed in % applied dose)

Reference item	Ethyl Tafluprostamide (% applied dose)	
Test item	Eyelash serum product	
No. of cells	n=12	
	Mean	SD
Total stratum corneum strips	14.16	10.71
Skin Excess*	62.27	5.92
Epidermis	10.94	7.70
Dermis	0.83	0.66
Epidermis + Dermis	11.77	7.88
Receptor fluid	4.42	4.00
Epidermis + Dermis + receptor fluid	16.19	9.58
TOTAL RECOVERY	92.62	8.91

*Skin excess corresponds to: Washing + Donor compartment rinsing + Remaining skin.

Distribution of ^3H -Ethyl Tafluprostamide after application of the eyelash serum product to human skin (expressed in $\mu\text{g}_{\text{eq}}/\text{cm}^2$)

Reference item	Ethyl Tafluprostamide (% applied dose)	
Test item	Eyelash serum product	
No. of cells	n=12	
	Mean	SD
Test substance applied ($\mu\text{Ci}/\text{cm}^2$)	0.44	0.04
Total stratum corneum strips	0.11	0.08
Skin Excess*	0.48	0.06
Epidermis	0.09	0.07
Dermis	0.0061	0.0046
Epidermis + Dermis	0.09	0.07
Receptor fluid	0.034	0.031
Epidermis + Dermis + receptor fluid	0.13	0.08
*Skin excess corresponds to: Washing + Donor compartment rinsing + Remaining skin.		

Conclusion:

Under the conditions of this study, the *in vitro* dermal absorption of Ethyl Tafluprostamide (0.015%) from a cosmetic formulation as test item by using the ^3H -Ethyl Tafluprostamide as a tracer applied for 24 hours onto the surface of healthy human skin mounted on dynamic cells showed a mean total recovery was $92.62\% \pm 8.91\%$

The bioavailable portion as measured for skin absorption was $16.19\% \pm 9.58\%$ [$0.034 \pm 0.031 \mu\text{g}/\text{cm}^2$].

Therefore, the value of 25.77 % (mean plus standard deviation) will be used as dermal absorption value for the Margin of safety (MoS) calculation. This value considers already a high variation according to the basic SCCS criteria for the *in vitro* assessment of dermal absorption of cosmetic ingredient

Ref.: (SCCS/1358/10, 2010); (Walberg, 2023)

SCCS comments

The two dermal absorption studies are difficult to compare as they were performed under very different conditions, although both were carried out by the same CRO and both done under GLP and quality control. Applicant 1 used 4 different dilutions of which one was the maximally allowed concentration of 0.018% (2 were lower and 1 was higher); 1.8 μg was applied on skin of the 0.018% dilution; concentration dependency was clearly present for the 4 concentrations tested.

Applicant 2 used one concentration of 0.015% which is lower than the maximal concentration allowed; 0.75 μg was applied on skin (42% of applicant 1).

Different results were obtained, namely Applicant 1 proposed a dermal absorption of $6.51 + 2.16\% = 8.67$, whereas Applicant 2 proposed $19\% + 9.58\% = 25.77\%$ (both mean value +1SD). Given the big differences in proposed DA and also the high SD of the second study, the SCCS considers that a conservative approach is warranted for this highly active substance. Therefore, a default value of 50% for dermal absorption would be more appropriate.

The data from dermal absorption studies also indicate that there is substantial systemic exposure when a formulation containing DDDE is applied on the skin. Considering that the intended application is in close proximity to the eyes, this needs to be taken into special consideration.

5.2.3 Oral absorption/bioavailability**Applicant 1**

According to Applicant 1, the oral absorption or bioavailability of DDDE can be assessed based on its physicochemical properties.

Considering its molecular weight of 452.5 g/mol, the water solubility of 1.05 g/L and log Kow of 2.74, DDDE was calculated to have a moderate oral absorption potential.

Ref.: (Athena Cosmetics Inc., 2023)

SCCS comment

The meaning of 'moderate' oral absorption has not been further specified.

Applicant 2

No oral absorption data was available

SCCS comment

In the absence of an oral bioavailability study, when relevant, the default value of 50% is appropriate according to the SCCS Notes of Guidance (SCCS/1647/22).

5.2.4 Toxicokinetic data**Additional Toxicokinetic data**Previous submission

Applicant 1: Photographic evidence that the cosmetic product X does not contact the eye: ophthalmic slit lamp microscope to assess photographically whether the product migrates from where it is applied to contact the fluid surrounding the eye. the product with an added small amount of ophthalmic solution, was applied to the upper eyelash margin of the adult female test subjects. None of the product + dye migrated into the eye despite normal blinking by the subject and the fact that the presence of the dye slightly decreased the viscosity of the product.

Ref.: (Athena Cosmetics Inc., 2020)

Clinical evidence that DDDE does not have a physiological effect on the eye: a 28-day study was conducted, involving 19 female subjects to assess the ocular safety and tolerability of the product. The examination was done with a slit lamp of the eyelids and measurement of the intraocular pressure (IOP) in each eye. The study showed there was no statistically significant change in IOP in 19 human volunteers.

Ref.: (Athena Cosmetics Inc., 2020)

New submission:**Skin metabolism**

The skin metabolism of the test substance DDDE (purity not specified) was investigated in an OECD Test Guideline 428-compliant *in vitro* dermal penetration study. Methodologies for analytical detection using fresh human skin were established. The test substance was applied at 3.0 µg (6.0 µg/cm² diluted in 50% ethanol) for 2, 4 or 24 hours to human skin on a static transwell system using a physiological buffer as receptor fluid. The identification and quantification of DDDE, its metabolite Tafluprost free acid (TPFA) and the reference substance caffeine were carried out using a Vanquish ultra-high-performance liquid chromatography (UHPLC) system. For the quantification of DDDE and the metabolite TPFA a six-point, weighted (1/x), external calibration in the concentration range from 0.05 µg/mL to 2.5 µg/mL was applied. The analytes were identified according to their specific masses in individual mass

traces. DDDE was found to be extensively metabolised into TPFA ($68.5 \pm 2.7\%$) after 24 hours.

Test substance	DDDE (purity not specified)
Study Type	<i>In vitro</i> dermal penetration study (according to OECD Test Guideline 428)
Application Concentration	3.0 µg (6.0 µg/cm ² diluted in 50% ethanol)
Application Duration	2, 4, or 24 hours
Sample Type	Fresh human skin (4 mm thickness)
Number of Donors	3
Solvent	50% ethanol
Receptor Fluid	Physiological buffer
Solvent	Caffeine
Reference Compound	Tafluprost free acid (TPFA)
Metabolite	Vanquish UHPLC with HPLC-MS
Detection System	Six-point, weighted (1/x), external calibration (range: 0.05 µg/mL to 2.5 µg/mL)
Metabolism Results	DDDE metabolized into TPFA ($68.5 \pm 2.7\%$) after 24 hours

Ref.: (Weghuber et al., 2022)

SCCS comments

This study shows the transformation of DDDE to TPFA, which acts directly as a potent agonist of prostaglandin F receptor (FP receptor) and reduces intraocular pressure by increasing the outflow of aqueous humor in the eye.

Applicant 2

STUDY 1

New submission: skin metabolism (idem as for reviewer 1, above study on TPEA):

See section 2.2 Dermal/percutaneous absorption (Applicant 2 (i) Study 1)

In addition, and as TPEA share the same main metabolite with Tafluprost, key and relevant kinetics studies derived from the pharmacology/toxicology FDA-NDA review and evaluation performed by the US Food and Drug Administration on Tafluprost (FDA, 2011) are presented below.

STUDY 2

Evaluation of the binding ratios of Tafluprost acid (AFP-172) to rat, rabbit, dog, and human serum albumin by an ultrafiltration method:

As Tafluprost acid (AFP-172) is the main metabolite of Tafluprost and the active ingredient, binding ratios to rat, rabbit, dog, and human serum albumin were evaluated by an ultrafiltration method followed by LC-MS/MS detection. Tafluprost acid (AFP-172) was tested at concentrations of 10, 100, and 500 ng/ml for albumin from all test species and additionally at 1 ng/ml for rabbit and human serum albumin.

Only negligible amounts of Tafluprost acid (AFP-172) were shown to bind to the ultrafiltration filter unit. The binding ratios of Tafluprost acid (AFP-172) to rat, rabbit, and dog serum albumin occurred in the range of 92.6 to 97% at Tafluprost acid (AFP-172) concentrations of 10 - 500 ng/ml. The binding ratio of 500 ng/ml Tafluprost acid (AFP-172) to human serum albumin was 99.2%. The binding ratios of Tafluprost acid (AFP-172) to human serum albumin at Tafluprost acid (AFP-172) concentrations of 1 - 100 ng/ml could not be obtained. This was apparently the case because at these lower concentrations, due to the high degree of protein binding, concentrations of free Tafluprost acid (AFP-172) in the filtrate were below the limit

of quantification for the LC-MS/MS detection method.

Reference: FDA, 2011

STUDY 3

***In vitro* metabolism of [³H]-Tafluprost (AFP-168) and Tafluprost acid (AFP-172) by rat, monkey, and human hepatocytes:**

Cryopreserved hepatocytes from Sprague-Dawley rats, Cynomolgus monkeys, and humans were thawed, assessed for viability and standard metabolic activity and incubated with 100 µM concentrations of [³H]-Tafluprost (AFP-168) or Tafluprost acid (AFP-172) for 0, 120, and 240 minutes. Upon termination, Tafluprost (AFP-168), Tafluprost acid (AFP-172) and metabolites were analyzed using an LC/MS method.

As results, in hepatocytes from all species, [³H]-Tafluprost (AFP-168) was completely metabolized to various metabolites after 240 minutes of incubation. In monkey and human hepatocytes, eleven metabolites were detected, and 7 metabolites were detected in rat hepatocyte incubations.

Following incubation with Tafluprost acid (AFP-172), Tafluprost acid (AFP-172) and the same 10 metabolites were detected in monkey and human hepatocytes, and in rat hepatocytes. The metabolites for each species and the putative metabolite identities are summarized in the table below. Five metabolites (1,2-dinor- Tafluprost acid (AFP-172), 1,2,3,4-tetranor-Tafluprost acid (AFP-172), hydroxy-1,2-dinor- Tafluprost acid (AFP-172), hydroxy-1,2,3,4-tetranor- Tafluprost acid (AFP-172) and Tafluprost acid (AFP-172) were detected in hepatocytes from all three species incubated with either [³H]-Tafluprost (AFP-168) or Tafluprost acid (AFP-172). Also, a total of seven potential glucuronide products were detected with one (No. 7) apparently occurring only in rat hepatocytes.

Metabolites Identified in Rat, Monkey, and Human Hepatocytes After Incubation with Either [³H]-Tafluprost (AFP-168) or Tafluprost acid (AFP-172)

Metabolite	Incubation with [³ H]-AFP-168			Incubation with AFP-172		
	Rat	Monkey	Human	Rat	Monkey	Human
AFP-172	x	x	x	x	x	x
1,2-dinor-AFP-172	x	x	x	x	x	x
1,2,3,4-tetranor-AFP-172	x	x	x	x	x	x
Hydroxy-1,2-dinor-AFP-172	x	x	x	x	x	x
Hydroxy-1,2,3,4-tetranor-AFP-172	x	x	x	x	x	x
AFP-172 glucuronide No. 1	x	x	x	x	x	x
AFP-172 glucuronide No. 2		x	x	x	x	x
AFP-172 glucuronide No. 3		x	x	x	x	x
AFP-172 glucuronide No. 4		x	x		x	x
AFP-172 glucuronide No. 5		x	x	x	x	x
AFP-172 glucuronide No. 6		x	x	x	x	x
AFP-172 glucuronide No. 7	x			x		

x Metabolite detected.

SCCS comments

The studies provide an overview of the metabolism of Tafluprost and the resulting metabolite Tafluprost acid. Pharmacokinetic data from monkeys and rats show efficient systemic clearance and minimal tissue retention, while serum albumin binding studies show strong plasma protein binding of Tafluprost acid. Species-specific metabolism studies reveal pathways for Tafluprost and Tafluprost acid across humans and animals, with some human-specific metabolites identified.

Ethyl Tafluprostamide, or Tafluprost Ethyl Amide, is an amide derivative of Tafluprost. Tafluprostamide has been reported to be inactive, which after administration (and absorption through the cornea) is hydrolysed to form Tafluprost acid - the biologically active metabolite

form. This metabolite is of importance when evaluating the (adverse) systemic health effects of Tafluprost Ethyl Amide.

5.2.5 Calculation of SED/LED

SED Calculation

Applicant 1

$SED = E_{product} \times C/100 \times DAp/100$

With

- $E_{product}$ – Considering 50% of the measured ‘average amount’ or ‘maximum amount’ of the cosmetic product per kg bw is 0.04-0.067 mg/kg bw/day for a 60 kg adult (*i.e.*, 50% x 4.8 mg/day ÷ 60 kg = 0.04 mg/kg bw/day; 50% x 8 mg/day ÷ 60 kg bw = 0.067 mg/kg bw/day)
- C –0.018%
- DAp –8.67% (see 2.2; mean value +1SD)

The resulting estimated SEDs from daily application of the cosmetic eyelash product containing 0.018% of DDDE are calculated to be:

- SED (based on average amount of 4.8 mg the cosmetic eyelash product) = 6.24E-07 mg/kg bw/day [*i.e.*, (0.04 mg/kg bw x 0.018% x 8.67%)].
- SED (based on maximum amount of 8 mg the cosmetic eyelash product) = 1.04E-06 mg/kg bw/day [*i.e.*, (0.0667 mg/kg bw x 0.018% x 8.67%)].

Applicant 2

According to the Applicant 2, there are no daily exposure levels provided for eyelash and eyebrow sera in Europe in the most recent SCCS guidance document (SCCS, 2023a). The eyelash serum is applied as a thin layer to the eyelid above the eyelash line. This is comparable to the use of a liquid eyeliner. However, there are clear differences in exposure as eyeliner will generally be applied above and below the eyelash lines of the upper and lower eye lids, respectively. And in contrast to the single use of the eyelash sera, eyeliner will be applied twice a day (Reference: SCCS, 2023a).

To close the existing data gap, the applicant has performed analyses on the exact amount of the daily uses of his eyelash serum and eyebrow serum according to his application instructions (products’ labels and information leaflets) including HPLC analytic on TPEA. The following results were obtained which will be used for the subsequent calculations:

Eyelash serum: 1.5 ± 0.6 mg (corresponding to 0.24 ± 0.10 µg TPEA)

Eyebrow serum: 5.2 ± 1.3 mg (corresponding to 0.80 ± 0.20 µg TPEA)

According to the Applicant, for reasons of consistency, the obtained NOAEL of 0.3 µg/kg bw/d of the pre- and postnatal developmental toxicity study of the structural analogue Tafluprost, which was also used by the FDA-NDA reviewer for the calculation of its safety margin based on body surface area, will be used as PoD for the calculation of the Margin of Safety (MoS) compliant to the guidance of the SCCS and in line with the NOAEL selection of the German Federal Institute for Risk Assessment (BfR) in its health assessment of an eyelash growth serum product containing TPEA (synonym: Dechloro-Dihydroxy-Difluoro-Ethylprostenolamide in 2017 (DDDE) (References: SCCS, 2023; BfR, 2017; FDA, 2011). This will be related to the DA value of 25.77% (mean: 16.19% plus 9.58% SD) for dermal absorption for the calculation of the Systemic Exposure Dose (SED, mg/kg bw/day).

Eyelash serum with 0,015% TPEA		
Amount of product applied daily (qx)	0.0015	g/day
Default human body weight	60	kg
Relative daily exposure ($E_{product} = qx/60 \text{ kg}$)	0.025	mg/kg bw/d

Opinion on Prostaglandin analogues used in cosmetic products

Dermal absorption as percentage (Dap (%))	25.77	%
Concentration in the product (C (%))	0.015	%
Systemic exposure dose (SED, mg/kg bw/d) = $E_{\text{product}} \times C (\%) / 100 \times Dap (\%) / 100$	0.001	µg/kg bw/d
Point of departure (PoD, pre- /postnatal developmental toxicity study, rats, i.v. injections of the structural analogue Tafluprost)	0.3	µg/kg bw/d
Route-to-Route Faktor	1	
PoDsys = PoD x Route-to Route Faktor	0.3	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	310	

Eyebrow serum with 0,015% Ethyl Tafluprostamide		
Amount of product applied daily (qx)	0.0051	g/day
Default human body weight	60	kg
Relative daily exposure ($E_{\text{product}} = qx / 60 \text{ kg}$)	0.09	mg/kg bw/d
Dermal absorption as percentage (Dap (%))	25.77	%
Concentration in the product (C (%))	0.015	%
Systemic exposure dose (SED, mg/kg bw/d) = $E_{\text{product}} \times C (\%) / 100 \times Dap (\%) / 100$	0.003	µg/kg bw/d
Point of departure (PoD, pre- /postnatal developmental toxicity study, rats, i.v. injections of the structural analogue Tafluprost)	0.3	µg/kg bw/d
Route-to-Route Faktor	1	
PoDsys = PoD x Route-to Route Faktor	0.3	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	91	

Aggregate Exposure: Eyelash serum and eyebrow serum each with 0.015% TPEA		
Eyelash serum SED	0.0011	µg/kg bw/d
Eyebrow serum SED	0.003	µg/kg bw/d
PoDsys = PoD x Route-to Route Faktor	0.3	µg/kg bw/d
Margin of Safety (MoS = PoDsys/SED)	75	

According to the Applicant, this approach can be adjusted based on valid scientific data, particularly from toxicokinetics and toxicodynamics, on a case-by-case basis. Tafluprost acid, the main metabolite of Tafluprost, was shown to bind strongly to serum albumin across various species, and its rapid metabolism eliminates species differences, omitting the need for an interspecies variability factor in safety assessments. A margin of safety of 25 is deemed valid and robust, with conservative calculations ensuring worst-case scenarios are covered. Despite the conservative approach, the actual safety margins for exposure to eyelash and eyebrow serums (310, 91, and 75) far exceed the acceptable limit of 25. Thus, Ethyl Tafluprostamide is considered fit for use in cosmetic products at concentrations up to 0.015%.

Ref.: Cayman, 2023b

SCCS comments

The uncertainty for such a pharmacologically-active substance is unknown and needs to be sufficiently large to be conservative and protective for the consumer. There is a high variability in the dermal absorption (DA) values determined by both Applicants,

therefore the use of a DA of 50% is more appropriate.

There is no toxicological point of departure (PoD) available for DDDE, a PoD was only proposed for Tafluprost via read-across. However, the read-across for Tafluprost is not acceptable because it was not carried out in a systematic scientific way.

The PoD of Tafluprost was determined via IV route, which further adds to the overall uncertainty.

The Applicant claims interspecies kinetics are similar, but not enough details were provided. Therefore, omitting the interspecies factor for this very active compound as proposed by Applicant 2 is not acceptable. Applying the interspecies factor leads to MoS lower than 100 – insufficient to regard it safe.

Exposure to both eyelash and eyebrow serum, with a concentration of 0.015 and 0.018% DDDE, respectively, needs to be considered together.

5.3 TOXICOLOGICAL EVALUATION

5.3.1 Skin irritation

Applicant 1

a) *In vitro* tests

- *In vitro* skin irritation assay using the EpiDerm™ RhE model

Method	EpiDerm™ RhE model, OECD Test Guideline 439, following GLP standards
Assay	MTT assay (triplicate)
Exposure Duration	60 minutes
Sample Type & Number	30 µL of DDDE (99.78% purity) applied to human epidermis tissue
Reference Compounds	Negative control: DPBS buffer; Positive control: 5% Sodium Dodecyl Sulfate (SDS)
Incubation Conditions	25 hours at 37 ± 1 °C, 5 ± 1 % CO ₂ , and ≥95% relative humidity
Detection System	Mitochondrial conversion of MTT to formazan blue measured spectrophotometrically
Viability Results	Tissue viability in replicates: 3%, 17.2%, 27.7%
Observed Fluctuations	Attributed to incomplete washing (due to high viscosity of the test substance) or biological variability
Outcome	DDDE was considered at least irritant to the skin in the EpiDerm™ RhE test under test conditions

Ref: (Brandt, 2023b)

SCCS comment

In the EpiDerm RhE assay, tissue viability shows inconsistent results, but they all show skin irritation (3, 17.2 and 27.7%).

-*In vitro* EpiSkin™ RhE test; small model / Reconstructed human Epidermis

Method	<i>In vitro</i> EpiSkin™ RhE test, OECD TG 439
Test Substance	Undiluted DDDE applied to EpiSkin™ tissue samples
Procedure	Done in triplicate
Incubation Conditions	42 hours at 37 ± 1 °C, 5 ± 1 % CO ₂ , and ≥95% relative humidity
Exposure Duration	15 minutes (minimum advised in OECD 439)
Sample Type	Reconstructed human Epidermis (RhE) using the EpiSkin™ small model
Sample Size	Three replicates for each condition (test substance, positive control, and

Reference Compounds	negative control)
Detection System	Negative control: DPBS buffer; Positive control: 5% SDS
Viability Result	MTT assay, measuring mitochondrial conversion of MTT to formazan blue by spectrophotometry
Outcome	Mean cell viability: 96.0% DDDE was considered non-irritant to the skin in the EpiSkin™ RhE test

Ref.: (Budda, 2022)

b) In vivo tests

A Human Repeated Insult Patch Test (HRIPT) with a cosmetic eyelash formulation containing 0.025% DDDE, was not found to be irritating in 51 human volunteers. In this study, the formulation was applied to the skin on the back (intrascapular area) of 51 human subjects (ages 18 – 59 years) for 24 hours, 3 times/week for 3 consecutive weeks (total of 9, 24-hour exposures). Before each reapplication, the test sites were evaluated by trained laboratory personnel. Fourteen days after the removal of the last induction patches, a challenge and re-challenge patch was applied to each of the study participants. Two out of 51 evaluations were scored "Erythema throughout at least ¾ of the patch area", all other evaluations were scored "No". No adverse reactions of any kind were reported during this study, thus 0.025% DDDE did not show any signs or symptoms indicative of a skin sensitisation response in any of 51 panelists completing the study.

Ref.: (Gunt, 2009)

SCCS comments

The SCCS Opinions are based on safety evaluation of cosmetic ingredients, and as such, studies performed with products are not considered. The SCCS considers HRIPT as an unethical test.

Applicant 2**Study Design:**

Reference:	ToxiCoop, 2022c
Date of report:	11.11.2022
Study period:	14.09.2022 – 16.09.2022
Guidelines/Methods:	OECD 439, Commission Regulation (EU) No 2019/1390, SCCS Notes of Guidance, 11 th revision, 2021
Test system:	EpiSkin™ Small Model (EpiSkin™SM), three-dimensional human skin model
Replicates:	Triplicate
Test substance:	Ethyl Tafluprostamide
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	None
Dose:	Neat test item
Route:	Topical application
Exposure:	15 ± 0.5 min.
Post-exposure expression:	42 ± 1-h
Negative control:	Phosphate Buffered Saline (PBS)
Positive control:	Sodium Dodecyl Sulfate (SDS, 5% aqueous solution)
GLP:	Yes
Published:	No

Material and methods:

The possible *in vitro* skin irritation potential of Ethyl Tafluprostamide (batch: 0652603-2, purity: 98.5%) was investigated according to OECD 439 under GLP conditions considering also current SCCS guidance criteria. The EpiSkin™ Small Model (EpiSkin™SM), a three-dimensional human skin model (EPISKIN Laboratories Lyon, France) was used to evaluate the skin irritation potential of the test substance.

Disks of EPISKIN (three units) were treated with test item and incubated for 15 minutes ($\pm$ 0.5 min) at room temperature. Exposure of test material was terminated by rinsing with PBS solution. Epidermis units were then incubated at 37 ± 1 °C for 42 hours ($\pm$ 1 h) in an incubator with 5 ± 1 % CO₂, ≥ 95 % humidified atmosphere. The viability of each disk was assessed by incubating the tissues for 3 hours ($\pm$ 5 min) with MTT solution at 37 ± 1 °C in 5 ± 1 % CO₂, ≥ 95 % humidified atmosphere, protected from light. The precipitated formazan was then extracted using acidified isopropanol and quantified spectrophotometrically. SDS (5 % aqueous solution) and PBS treated epidermis were used as positive and negative controls, respectively. For each treated tissue viability was expressed as a percentage relative to negative control.

Results:

The obtained results of the optical density (OD) measured at 570 nm of each replicate and the calculated % viability of the cells are summarized in the table below:

Optical Density values and viability percentages of the controls and test item

Substance	Optical Density (OD)		Viability (%)
Negative Control: PBS	1	0.921	95
	2	1.017	105
	3	0.963	100
	mean	0.967	100
	standard deviation (SD)		5.01
Positive Control: SDS (5 % aq. solution)	1	0.108	11
	2	0.133	14
	3	0.069	7
	mean	0.103	11
	standard deviation (SD)		3.30
Test Item: Ethyl Tafluprostamide	1	0.965	100
	2	1.024	106
	3	0.792	82
	mean	0.927	96
	standard deviation (SD)		12.48
- Mean blank OD value was 0.0341 - Optical density means the mean value of the duplicate wells for each sample (rounded to three decimal places)			

The test item Ethyl Tafluprostamide did not show significantly reduced cell viability in comparison to the negative control (mean viability: 96 %). All obtained test item viability results were above 50 % when compared to the viability values obtained from the negative control. Therefore, the test item was considered to reveal no-irritant to skin.

Positive and negative controls showed the expected OD and cell viability values within acceptable limits and indicated the suitability and sensitivity of the test system. The mean OD value of the blank sample was below 0.1.

Conclusion:

Under the conditions specified in this study, the test substance reveals no skin irritation potential. According to the current OECD Guideline No. 439, the test item Ethyl Tafluprostamide is considered as non-irritant to skin. There is no need for classification (UN GHS No Category).

Ref.: (Toxicoop, 2022c)

5.3.2 Mucous membrane irritation / eye irritation**Applicant 1**a) *In vitro tests**-In vitro* EpiOcular™ EIT assay (99.78% neat oil DDDE)

Method	<i>In vitro</i> EpiOcular™ Eye Irritation Test (EIT), according to OECD TG 492
Procedure	Done in duplicate
Test Substance	DDDE applied to a three-dimensional human cornea tissue model
Dose Applied	30 µL
Exposure Duration	28 minutes
Incubation Conditions	120 minutes at 37 ± 1 °C, 5 ± 1 % CO ₂ , and ≥95% relative humidity
Detection System	MTT assay; spectrophotometric measurement of mitochondrial conversion of MTT to formazan blue
Reference Compounds	Negative control: Demineralized water; Positive control: Methyl acetate
Viability Results	Tissue viability: 98.4% and 99.3% (replicates)
Mean Viability	98.8%
Outcome	DDDE was considered non-irritant to the eye

Ref: (Brandt, 2023c)

-In vitro eye irritation, Hen's egg test – chorioallantoic membrane assay with cosmetic eyelash formulations containing 0.025% DDDE

Method	Four CAMs rinsed with 5 mL physiological saline after 20 seconds. Observations made before application and at 30 seconds, 2 minutes, and 5 minutes post-exposure.
Exposure Amount	0.3 mL or 0.3 g of test and reference substances
Reference Compounds	- AlmayOne (coat mascara, 50%) - Maybelline waterproof eyeliner (ultra-eyeliner, 50%)
Detection Method	CAM scores
Test Sample Results	CAM score of 0 (zero) at all time points (30 seconds, 2 minutes, and 5 minutes), indicating no irritation potential.
Reference Product Scores	Mean CAM score: 0.5 (mascara), 0.75 (eyeliner)
Outcome	Test sample showed no irritation potential.

Ref.: (Nitka, 2009)

SCCS comments

As mentioned before, studies performed with cosmetic products are not considered in the SCCS Opinions.

The Het-CAM test has no OECD TG and can only be regarded as providing support in a weight of evidence (WoE) approach. Only two replicates were used, which is below the typical standard for robust *in vitro* testing.

b) *In vivo test*

The test substance (0.05% DDE) was applied daily for 28 days to the eyelid margin at the base of the upper eyelashes (in accordance with the directions in the package insert). The study was conducted under the supervision of an ophthalmologist. The Intra ocular pressure (IOP) was measured in each eye of each subject at the beginning and end of the study. Apart from IOP measurements, ocular irritation was assessed by the supervising ophthalmologist who queried each subject about adverse reactions and also conducted an ophthalmic examination on day 0 and day 28. The ophthalmic examination was performed with a slit

lamp and included the subject's eyelids, cornea, conjunctive, anterior chambers, papillary reactions, and visual acuity. Any reaction in the eyes and or the eye contour area was evaluated by the ophthalmologist and observations were recorded. Tolerance to the test substance was evaluated based on the observed reactions and their degree of severity, as well as on the reproducibility from one volunteer to another. No ocular irritation was reported during the entire length of the study by the ophthalmologist. Also, there was no statistically significant reduction in IOP over the 28-day study.

Only minor ocular effects were self-reported by 4 of 19 volunteers including slight dryness sensation around the eye areas (5%), slight itching sensation (21%), slight stinging sensation (16%), slight eye-watering and redness (11%) and moderate to high burning sensation (5%). According to the Applicant, these minor self-reported adverse reactions were consistent with allergic reactions.

Ref.: (Sebesten, 2010)

SCCS comments

The study included 19 volunteers, which is a relatively small sample size for a human study. Minor adverse effects such as dryness, itching, stinging, and redness were self-reported by 4 of the 19 participants (21%), which is high. Objective scoring should have been done using a standardised scoring system or a questionnaire for assessing subjective symptoms. Additionally, the correlating self-reported symptoms with objective clinical findings would have helped to clarify whether these effects were truly related to the test substance and an allergic reaction. If 21% of the volunteers showed an allergic reaction, that would be extremely high.

Applicant 2

STUDY 1

In vitro Eye Irritation in the EpiOcular™ Model

Reference:	ToxiCoop, 2023b
Date of report:	10.01.2023
Study period:	26.10.2022 – 17.11.2022
Guidelines/Methods:	OECD 492, SCCS Notes of Guidance, 11 th revision, 2021
Test system:	EpiOcular™ Human Cell Construct (MatTek Corporation)
Test substance:	Ethyl Tafluprostamide
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	None
Dose:	Neat test item
Route:	Topical application
Exposure:	6 hours (± 15 min)
Post exposure:	18 hours ± 15 minutes
Negative control:	Sterile deionized water:
Positive control:	Methyl acetate (MA)
GLP:	Yes
Published:	No

Material and methods:

The possible *in vitro* eye irritation potential of Ethyl Tafluprostamide (batch: 0652603-2, purity: 98.5%) was investigated according to OECD 492 under GLP conditions considering also current SCCS guidance criteria. The EpiOcular™ Human Cell Construct (MatTek Corporation) was used to evaluate the test substance. Prior to the treatment the tissues were pre-wetted with 20 µL of Ca⁺⁺Mg⁺⁺Free- Dulbecco's phosphate buffered saline (DPBS) and were incubated at standard culture conditions for 30 ± 2 minutes. Disks of EpiOcular™ (two units) were treated with the test item and incubated for 6 hours (± 15 min) at standard

culture conditions (37 ± 2 °C in an incubator with 5 ± 1 % CO₂, ≥ 95 % humidified atmosphere). Exposure of the test item was terminated by rinsing with Ca⁺⁺Mg⁺⁺ Free-DPBS solution. After rinsing, the tissues were incubated for a 25 ± 2 minutes immersion incubation at room temperature. At the end of this period, the test item treated tissues were incubated for 18 hours $\pm$ 15 minutes at standard culture conditions for post exposure. Fresh assay medium was used during these periods. The viability of each disk was assessed by incubating the tissues for 3 hours ($\pm$ 15 min) with MTT solution at 37 ± 2 °C in an incubator with 5 ± 1 % CO₂, protected from light, ≥ 95 % humidified atmosphere. The precipitated formazan was then extracted using isopropanol and quantified spectrophotometrically. Sterile deionized water and methyl acetate treated tissues were used as negative and positive controls, respectively. The disks of EpiOcular™ (two units / control) were treated with positive and negative control and incubated for 6 hours $\pm$ 15 minutes.

Results:

Ethyl Tafluprostamide induced significantly reduced cell viability in comparison to the negative control with a mean viability of 11 %. Thus, all obtained test item viability results were below the 60 % prediction criteria when compared to the viability values obtained from the negative control.

Positive control viability results showing a mean tissue viability of 6 % and fulfilled therefore the acceptance criteria of below 50 % when compared to the viability values obtained from the negative control. Furthermore, the mean optical density (OD) value of the two negative control tissues (mean OD value: 1.910) were between 0.8 and 2.8. So, the positive and negative controls showed the expected values within acceptable limits and indicated the suitability and sensitivity of the test system. The difference of viability between the two relating tissue replicates ranged from 0 % to 2 % which was below the threshold value 20 %. The mean OD value of the blank sample (0.0333) was below the threshold value of 0.1.

Conclusion:

Under the conditions of this *in vitro* study, the results obtained with DDDE indicated that according to the guideline OECD 492 criteria, the test item has to be categorized as "No prediction can be made".

As a consequence, subsequently the *in vitro* eye irritation test using isolated chicken eyes was performed in addition.

Ref.: (Toxicoop, 2023b)

STUDY 2

In vitro Eye Irritation in isolated chicken eyes

Reference:	ToxiCoop, 2023c
Date of report:	14.04.2023
Study period:	20.01.2023 – 20.02.2023
Guidelines/Methods:	OECD 438, Commission Regulation (EU) No 1152/2010, SCCS Notes of Guidance, 11 th revision, 2021
Test system:	Isolated chicken eyes Strain of chicken: ROSS 308 Source: Taravis KFT, Sárvár, Hungary
Replicates:	Triplicate
Test substance:	Ethyl Tafluprostamide
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	None
Dose:	Neat test item (0.03 g)

Opinion on Prostaglandin analogues used in cosmetic products

Route:	Topical application
Exposure:	10 seconds
Observation:	Pre-treatment and at approximately 30, 75, 120, 180 and 240 minutes after the post-treatment rinse
Negative control:	Sodium chloride (NaCl)
Positive control:	Imidazole
GLP:	Yes
Published:	No

Material and methods:

The corneas were evaluated pre-treatment and at approximately 30, 75, 120, 180, and 240 minutes after the post-treatment rinsing. The endpoints evaluated were corneal opacity, swelling, fluorescein retention, and morphological effects. All of the endpoints, with the exception of fluorescein retention (which was determined only at pre-treatment and 30 minutes after test item exposure) were determined at each of the above time points. Based on the results of the first experiment a confirmatory experiment was performed to confirm the negative outcome.

The positive control Imidazole was ground prior to use and was applied in an amount of 0.03 g/eye by powdering the entire surface of the cornea attempting to cover the cornea surface uniformly with the positive control, respectively. As the test item is highly viscous it was applied with a spatula to the entire surface of a plastic film (approximately 2 cm²). This plastic film was applied to the cornea by hand to cover the entire surface of the cornea. Three test item treated eyes, three positive control treated eyes, and one negative control treated eye (which was treated with 30 µL of NaCl 9 g/L saline solution) were used in the first and additional experiment. After an exposure period of 10 seconds from the end of the application, the cornea surface was rinsed thoroughly with about 20 mL saline solution at ambient temperature, and this procedure was repeated for each eye. The positive control Imidazole was stuck onto the cornea surfaces in all eyes (three eyes) for 30 minutes after the post-treatment rinsing in the first and additional experiment. The positive control treated cornea surfaces were not totally cleared at 240 minutes after the post-treatment rinse in the first and the additional experiment.

Results:

The positive control Imidazole revealed a corrosive/severely irritating reaction, while the negative control had no significant effects on the chicken eye in the first and in the additional experiment indicating the validity, sensibility and suitability of the test system.

The exposure to Ethyl Tafluprostamide resulted in three times class I reactions (based on the corneal swelling of 1 % within 240 minutes categorized as "No swelling", the corneal opacity score of 0.5 (very faint opacity) categorized as "No opacity" and the fluorescein retention of 0.5 categorized as "No fluorescein retention") in the first experiment. In the confirmatory second experiment the overall ICE classes of the test item treated corneas were again three times class I (based on corneal swelling of 1 % within 240 minutes categorized as "No swelling", corneal opacity score of 0.3 (very faint opacity) categorized as "No opacity" and the fluorescein retention of 0.2 categorized as "No fluorescein retention") as can be seen in the tables below:

First experiment - Results of Ethyl Tafluprostamide

Observation	Value	ICE Class
Mean maximum corneal swelling at up to 75 min	1%	I
Mean maximum corneal swelling at up to 240 min	1%	I
Mean maximum corneal opacity	0.5	I
Mean fluorescein retention	0.5	I
Other Observations	None	
Overall ICE Class	3 x Class I	

Confirmatory second experiment - Results of Ethyl Tafluprostamide

Observation	Value	ICE Class
Mean maximum corneal swelling at up to 75 min	1%	I
Mean maximum corneal swelling at up to 240 min	1%	I
Mean maximum corneal opacity	0.3	I
Mean fluorescein retention	0.2	I
Other Observations	None	
Overall ICE Class	3 x Class I	

Conclusion:

According to the Applicant, under the conditions of this *in vitro* isolated chicken eyes test, the results obtained with Ethyl Tafluprostamide in the led to an overall categorization of 3 x Class I in the combination of the investigated three endpoints in the first and the additional experiment. According to OECD 438 criteria, Ethyl Tafluprostamide is categorized as "No Category" and thus, as a substance that does not require classification for eye irritation or serious eye damage under the UN GHS classification system.

In addition, the structural analogue Tafluprost did not cause irritation to the ocular structures in a 52-week monkey study after repeated application of concentrations up to 0.05% directly into the eye. Specific investigations were performed by means of a slit lamp and involved the cornea, conjunctiva, anterior chamber, iris, lens, and vitreous humor on sedated monkeys once pre-dose and once during Weeks, 13, 26, 39, and 52 (2 to 6 hours after the first dosing). The different ocular structures were scored for irritation.

During Week 26, two animals of each gender in at 0.05% demonstrated pigment around the lower nasal limbus. However, this change was not evident in Weeks 39 and 52 and did not correlate with histopathology. No Tafluprost (AFP-168) related changes in the irritation score for any of the ocular structures occurred during the treatment period. Even pigment transiently visible at week 26 at the limbus (junction between cornea and mucosa) in one male and one female monkey was completely reversible, as it could not be detected at week 39 or 52 or in histopathology (FDA, 2011).

Ref.: (Toxicoop, 2023c)

SCCS comments

According to the results shown, DDDE seems to be not irritating to the eye.
The test of Tafluprost in monkeys is not relevant for consideration in this evaluation.

5.3.3 Skin sensitisation**Applicant 1****a) *In chemico* test**

Method	Direct Peptide Reactivity Assay (DPRA), OECD Test Guideline 442C
Sample Type & Number	DDDE, Synthetic Cys- and Lys-peptides; triplicate samples
Concentrations	1:10 molar ratio (Cys-peptide) and 1:50 molar ratio (Lys-peptide)
Exposure	22 hours at 25°C
Solvent	Acetonitrile
Reference Compound	Cinnamic aldehyde (positive control)
Detection System	Peptide depletion measured via HPLC-UV
Outcome	Mean Cysteine Peptide Depletion: 5.61% ± 0.44% (Negative DPRA Prediction)

However, the test substance showed turbidity right after mixing the test substance solution with buffer and the Lys-peptide-solution. After the incubation period, precipitation was

observed and therefore only the supernatant was used for measurement. The study found minimal reactivity towards the Cys-peptide and Lys-peptide, resulting according to the Applicant in a negative DPRA prediction.

Ref.: (Brandt, 2023a)

Method	Direct Peptide Reactivity Assay (DPRA), OECD Test Guideline 442C
Formulation	An opalescent formulation was obtained, not suitable for HPLC analysis
Exposure	Incubated with Cysteine peptide for 24 hours
Sample Type & Number	DDDE, synthetic Cys-peptides; triplicate samples
Concentrations	1:10 molar ratio of the test substance with Cys-peptides
Solvent	Acetonitrile
Reference Compound	Cinnamic aldehyde (positive control)
Detection System	Peptide depletion measured via HPLC after 24±2 hours incubation at 25°C
Outcome	No co-elution observed. Mean cysteine peptide depletion: 5.61% ± 0.44% (Negative DPRA Prediction)

Ref.: (Sagi, 2022)

SCCS comments

The studies indicated minimal reactivity; however, the presence of precipitation in the assays may have confounded the results, suggesting potential underestimation of DDDE's reactivity. Precipitation was noted following incubation and therefore a final conclusion on the lack of reactivity cannot be drawn with sufficient confidence as peptide depletion may be underestimated according to the OECD 442C guideline criteria. In case of precipitation, a positive result could still be used, but a negative result is uncertain and should be interpreted with due care. DPRA results are not justified.

b) *In vitro* test: KeratinoSens according to OECD TG 422D

Method	KeratinoSens™ Assay, OECD TG 442D, 2 replicates (I and II)
Sample Type & Number	12 concentrations of DDDE (0.98 µM, 1.95 µM, 3.91 µM, 7.81 µM, 15.63 µM, 31.25 µM, 62.5 µM, 125 µM, 250 µM, 500 µM, 1000 µM, 2000 µM)
Cytotoxicity Threshold	Viability below 70% was considered cytotoxic and not further evaluated for luciferase induction
Concentrations	0.98 µM to 2000 µM (varied across tests)
Exposure	48 hours
Solvent	Dimethyl sulfoxide (DMSO), final concentration 1%
Reference Compound	Cinnamic aldehyde (5 concentrations ranging from 4 to 64 µM) as positive control
Detection System	Luciferase gene induction measured via luminescence; cytotoxicity assessed by cell viability (MTT assay)
Outcome	Cytotoxic effects observed at doses ≥500 µM in repetitions I and II. Cell viability rose to 80% at lower concentrations, with reductions of 50% and 30% in viability.
Repetition I Outcome	Significant increase in luciferase induction at 250 µM, but no dose-dependence detected
Repetition II Outcome	Significant increase in luciferase induction at 250 µM, with maximal fold induction of 1.8-fold
Additional Evaluation	OECD TG 497 indicated results of repetitions I and II were not congruent, leading to a 'borderline' conclusion

Ref.: (Hummel-Kocsi, 2022)

SCCS comment

Inconsistencies between repetitions led to a "borderline" conclusion of the KeratinoSens by

the Applicant, raising questions about the reliability of the findings.

c) Human data

Method	Human Repeat Insult Patch Test (HRIPT)
Procedure	Application, 10-14 day resting period, challenge patches applied, evaluations after 48 & 96 hours
Sample Type & Number	54 healthy volunteers (12 males and 42 females)
Concentrations	7.5% DDDE in phenoxyethanol (diluted to 0.267% for application in deionized water)
Application Details	Samples applied to the back under semi-occlusive patches three times per week for three weeks
Solvent	Phenoxyethanol / Deionized water
Reference Compound	1% sodium lauryl sulfate (positive control)
Detection System	International Contact Dermatitis Research Group scoring scale for skin reactions
Outcome	All 486 skin evaluations scored "0", indicating no reaction.
Positive Control	Eighteen participants showed mild reactions (Grade 1 or 2) to 1% sodium lauryl sulfate
Negative Control	No reactions to distilled water (negative control)
Study Validity	Test deemed valid; no adverse reactions reported
Conclusion	DDDE did not cause skin sensitization or cumulative irritation

Ref.: (Martin, 2022)

SCCS comments

Only 33 % of the volunteers showed mild reactions for the positive control, which raises questions about the validity/sensitivity of the test. The SCCS considers HRIPT as unethical.

Method	Human Repeat Insult Patch Test (HRIPT)
Exposure	
Sample Type & Number	cosmetic product, 52 healthy volunteers (4 males and 48 females).
Concentrations	7.5% DDDE in phenoxyethanol (diluted to 0.267% for application in deionized water)
Application Details	product was applied in the intrascapular area of the skin for 24 hours, three times per week over three consecutive weeks, totaling nine exposures.
Solvent	Not specified
Reference Compound	Not specified
Detection System	International contact dermatitis research group scoring scale
Outcome	Out of 561 evaluations, only two scored "1" (erythema), while the remainder scored "0", indicating no significant adverse reactions. The study concluded that the eyelash product did not show signs of skin sensitisation in any of the participants.

Ref.: (Gunt, 2009)

SCCS comment

As mentioned before, studies performed with cosmetic products (not ingredients) are not evaluated in the SCCS opinions.

Applicant 2**Study 1:***Study Design:* Direct Peptide Reactivity Assay (DPRA)

Reference:	ToxiCoop, 2022b
Date of report:	18.10.2022
Study period:	21.09.2022 – 23.09.2022
Guidelines/Methods:	OECD 442C, OECD 497, SCCS Notes of Guidance, 11 th revision, 2021
Test system:	Direct Peptide Reactivity Assay (DPRA) with high-performance liquid chromatography (HPLC) analysis
Replicates:	Triplicate
Test substance:	Ethyl Tafluprostamide
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	Acetonitrile
Dose:	100 mM test solution
Exposure:	<i>In chemico</i> by combining the test chemical with a solution of the synthetic Cysteine peptide
Exposure duration:	24 ± 2 hours (25 °C)
Controls:	Reference control A: Peptide stock solution with acetonitrile for system suitability check. Reference control B: Peptide stock solution with acetonitrile for peptide stability check. Reference control C: Peptide stock solution with acetonitrile for the test item and the positive control Co-elution controls Test item stock solution and acetonitrile combined with phosphate buffer. Positive control: Peptide stock solution combined with Cinnamaldehyde as positive control stock solution (100 mM) and acetonitrile
GLP:	Yes
Published:	No

Material and methods:

The possible sensitizing potential of Ethyl Tafluprostamide (batch: 0652603-2, purity: 98.5%) was investigated by means of *in chemico* determination with the Direct Peptide Reactivity Assay (DPRA) with high-performance liquid chromatography (HPLC) analysis according to OECD 442C and considering SCCS criteria under GLP conditions. Accordingly, the reactivity of the test item and synthetic Cysteine containing peptides was evaluated by combining the test chemical with a solution of the peptide and monitoring the remaining concentration of the peptide following 24 hours of interaction time at room temperature. Relative concentrations of the peptides following the 24-hour incubation were determined by HPLC with gradient elution and UV detection at 220 nm. The samples were also evaluated with regards to possible occurrence of precipitation. Reaction samples, reference controls A, B and C, co-elution controls and positive controls were prepared and analysed in triplicates. The total HPLC analysis time was less than 30 hours.

Results:

No precipitation was observed as acetonitrile dissolved the test item completely and homogenous and clear solution was achieved.

The mean values of the peak areas versus the concentrations of cysteine peptide showed good linearity with $r^2 = 0.9995$, covering the concentration range from 0.0167 mM to 0.534 mM.

The system suitability was verified as the mean peptide concentration of reference control A

sample replicates was 0.51 mM.

The mean peptide concentration of reference control B sample replicates was 0.49 mM and verified the stability of the peptide over time.

The mean peptide concentration of reference control C sample replicates was 0.49 mM and confirmed that the solvent of the test item did not impact the percent peptide depletion.

Table below showed that the acceptance criteria was met for the positive control with a cysteine peptide depletion value of 70.67 % $\pm$ 0.69 % and that the mean cysteine peptide depletion value obtained with the test item was 5.61 % $\pm$ 0.44 %.

Cysteine peptide depletion values for the positive control and the test item

Sample	Peptide conc. calculated (mM)	Mean peptide conc. (mM)	Peptide depletion (%)	Mean Peptide depletion (%)	Standard deviation (SD) for peptide depletion (%)
Positive control, rep I	0.14	0.14	69.88	70.67	0.69
Positive control, rep II	0.14		70.99		
Positive control, rep III	0.14		71.14		
Test item, rep I	0.47	0.47	5.11	5.61	0.44
Test item, rep II	0.46		5.93		
Test item, rep III	0.47		5.79		

No co-elution was observed with the peptide. The mean cysteine peptide depletion of the test item was 5.61 %, which was clearly below the threshold of 13.89 % indicating a negative DPRA prediction.

Peptide depletion resulting from the positive control cinnamaldehyde was within the expected percentage range and confirmed the suitability and sensitivity of the test system.

According to the Applicant, all validity criteria were met, confirming the validity of the study.

Conclusion:

According to the Applicant, under the conditions of the DPRA for Ethyl Tafluprostamide showed no or minimal reactivity and was therefore concluded to be negative according to the prediction criteria. Thus, revealed no skin sensitizing potential in this *in vitro* system.

Study 2:

Study Design: KeratinoSens™ method (ARE-Nrf2 Luciferase test method)

Reference:	ToxiCoop, 2022e
Date of report:	14.12.2022
Study period:	24.10.2022 – 10.11.2022
Guidelines/Methods:	OECD 442D, OECD 497, SCCS Notes of Guidance, 11 th revision, 2021
Test system:	KeratinoSens™ cell line (immortalized adherent cell line derived from human keratinocytes (HaCaT))
Replicates:	Triplicate
Test substance:	Ethyl Tafluprostamide
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	DMSO
Dose:	1 st test: twelve concentrations: 0.98 μ M - 2000.00 μ M, 2-fold dilution factor

	2 nd test: twelve concentrations: 55.80 µM - 1000.00 µM, 1.3-fold dilution factor
	3 rd test: twelve concentrations: 67.29 µM - 500.00 µM, 1.2-fold dilution
Exposure duration:	48 ± 1 hours (37 ± 1 °C in the presence of 5 % CO ₂)
Negative control:	Vehicle (DMSO)
Positive control:	Trans-Cinnamaldehyde
GLP:	Yes
Published:	No

Material and methods:

The possible sensitizing potential of Ethyl Tafluprostamide (batch: 0652603-2, purity: 98.5%) was studied using the KeratinoSens™ method (ARE-Nrf2 Luciferase test method) according to OECD 442D and considering SCCS criteria under GLP conditions. In total, three independent tests with each three replicates were performed with the test item. In the first test, twelve concentration ranging from 0.98 µM - 2000.00 µM using a 2-fold dilution factor were used. However, due to a borderline effect at a single dose with accompanied cytotoxicity, a second test with twelve concentrations in the range between 55.80 µM - 1000.00 µM and a 1.3-fold dilution factor and third test with twelve concentrations ranging between 67.29 µM - 500.00 µM and using a 1.2-fold dilution were performed. DMSO was used as suitable solvent and negative control, while Trans-Cinnamaldehyde was the positive control. The KeratinoSens™ cells were derived from human keratinocytes (HaCaT) and transfected with selectable plasmids containing luciferase gene under the transcriptional control of the AKR1C2 ARE gene sequence. After 48 hours of incubation, Luciferase gene induction was measured in the cell lysates by luminescence detection using a light producing luciferase substrate. Cytotoxicity and the relative luminescence intensity of luciferase substrate in the lysates were measured and luciferase induction compared to solvent/vehicle control was calculated.

Results:

Precipitation was not observed at any concentration in any of the three tests.

The test item induced cytotoxicity in KeratinoSens™ cells compared to the solvent/vehicle control in all tests at the higher tested concentrations (500.00 µM - 2000.00 µM in the first test, 350.13 µM - 1000.00 µM in the second test and 289.35 µM - 500.00 µM in the third test).

IC₃₀ and IC₅₀ values could be determined for each independent test and the overall IC₃₀ was 278.51 µM, while the overall IC₅₀ was 311.38 µM.

The induction values of the test item did not exceed the 1.5-fold threshold at any tested concentrations in any test compared to the respective negative control and consequently an EC_{1.5} value could not be determined.

With regards to the OECD 497 prediction model criteria, there was only one induction value which exceeded the lower limit of the borderline threshold (1.35-fold) compared to the respective negative control exclusively at the concentration of 250.00 µM and only in the first test. However, as no such reaction was observed in any of the subsequent tests, based on the OECD 442D and OECD 497 prediction model criteria, the test item was concluded negative.

The overall value for maximal fold induction (I_{max}) was determined as 1.25-fold.

The suitability and sensitivity of the test system was confirmed as all validity criteria were met and the negative control reacted as expected without indicating a skin sensitizing potential, while the luciferase activity induction obtained with the positive control (Trans-Cinnamaldehyde) was dose-dependently and statistically significantly above the threshold of 1.5-fold in all tests.

Conclusion:

Overall, under the conditions of this study and based on the obtained results, Ethyl Tafluprostamide showed no *in vitro* skin sensitization potential under the experimental conditions of KeratinoSens™ method (ARE-Nrf2 Luciferase Test Method), when tested up to clear cytotoxic concentrations and considering the OECD 442D and OECD 497 prediction model criteria.

In vivo: in addition, also the sensitizing potential of the structural analogue Tafluprost was investigated (Reference: FDA, 2011). The Adjuvant and Patch test was used to determine the skin sensitization potential of 0.005% and 0.05% Tafluprost (AFP-168) solutions in the guinea pig. The negative control was the vehicle for test item solution and the positive control was 2,4-dinitrochlorobenzene (DNCB). Each group of 8 guinea pigs was injected with emulsified Freund's complete adjuvant intradermally into the shoulder skin in 4 corner areas. At the same time, a closed patch of each test preparation was applied conclusively for 24 hours. The same abrasions and applications were repeated on each of the next two days. Thus, the initial sensitization consisted of 3 days of treatment. The second sensitization was performed at the same area and consisted of application of each test preparation at the same areas for 48 hours. Two weeks later, each test preparation was applied openly to the flank as the challenge procedure.

As results, animals in the vehicle control group, and the 0.005% and 0.05% test item solution did not demonstrate skin reactions in response to the challenge procedure. In contrast, all of the animals in the positive control group demonstrated positive skin reactions after challenge with DNCB. Finally, under the conditions of the study, neither concentration of Tafluprost (AFP-168, 0.005% and 0.05%) demonstrated a potential for skin sensitization.

Ref.: (ToxiCoop, 2022b), (ToxiCoop, 2022e)

SCCS comments

DPRA and KeratinoSens™ tests for DDDE showed negative results.

To support the non-sensitising potential of DDDE, an *in vivo* test was carried out using Tafluprost. As read-across for Tafluprost was not accepted, these results are not considered by the SCCS.

SCCS overall conclusion on skin sensitisation

Applicant 1: no firm conclusion with respect to skin sensitisation could be made. Neither DPRA nor KeratinoSens™ tests for DDDE were reliable. *In vivo* HRIPT tests were performed using DDDE and a cosmetic product containing DDDE. Besides the fact that HRIPT tests are considered unethical by the SCCS, the positive controls in the DDDE test were not convincingly positive. Product results are not considered.

Applicant 2: both DPRA and KeratinoSens™ tests for DDDE were negative, pointing to non-sensitising properties of DDDE. To confirm that skin sensitisation was negative, an *in vivo* human HRIPT study with Tafluprost was carried out. The outcome was negative. HRIPT is considered unethical by the SCCS, and data from Tafluprost is, as explained earlier, not accepted as a surrogate for DDDE.

Taking a collective view of the results, DDDE can be considered as non-sensitising.

5.3.4 Acute toxicity**5.3.4.1 Acute oral toxicity****Applicant 1:**

No acute oral studies could be identified for DDDE. The acute oral toxicity endpoint has therefore been assessed on the basis of an acute toxicity study available for the structural

analogue Tafluprost.

An acute oral toxicity study was conducted on Tafluprost using Sprague-Dawley rats, with 5 animals per sex per group receiving doses of 0, 10, 30, and 100 mg/kg body weight. Test done pre-2009. Following the administration, the rats were monitored daily for signs of toxicity and mortality over a 14-day period. Measurements of body weight, food intake, and water intake were recorded at specified intervals. Throughout the study, no mortalities or significant changes in body weight, food intake, or water intake were noted. However, two animals exhibited a hunched posture and wasted appearance after dosing, and one of these animals also developed chest sores or experienced loss of chest fur, which was resolved by day 14. Overall, the study found that the LD50 for Tafluprost was ≥ 100 mg/kg body weight in rats. The results indicate a low acute toxicity level for Tafluprost within the tested dosage range.

Ref.: (CDER, 2011)

Applicant 2:

There are no acute toxicity studies on Ethyl Tafluprostamide available.

Therefore, data of the structural analogue Tafluprost (Reference: FDA, 2011) were taken as surrogate based on the read-across justification provided in section 'Read Across'.

The acute oral toxicity of Tafluprost (AFP-168) was investigated in male and female Sprague-Dawley rats. Each 5 animals per sex received single oral doses of 0, 10, 30 and 100 mg/kg bw and were observed for 14 days after dosing.

No deaths occurred in any treatment group. The observed clinical findings included hunched posture and wasted appearance beginning one day after dosing in one 10 mg/kg bw and one 100 mg/kg bw animal. One 10 mg/kg bw animal and one 100 mg/kg bw animal exhibited chest sores or loss of chest fur, at 7 days after dosing that resolved by Day 14. There were no noteworthy findings in body weight, food intake, water intake or gross pathology. Therefore, the oral LD50 value of Tafluprost in rats was estimated as > 100 mg/kg bw.

SCCS comments

The data provided by both Applicants is for Tafluprost and not for DDDE, hence it has not been taken into consideration in this evaluation.

Concerning the data provided by Applicant 2, the original study name and reference were missing. Since prostaglandins may cause thrombosis, the observed chest sores should be looked at more carefully as it could be an indication of adverse effect.

5.3.4.2 Acute dermal toxicity

No acute dermal studies could be identified for DDDE or Tafluprost.

5.3.4.3 Acute inhalation toxicity

No acute inhalation studies could be identified for DDDE or Tafluprost.

5.3.4.4 Acute intravenous (IV) toxicity

Applicant 1:

Study1:

Guideline: Not specified

Species/strain:	Rats/Sprague-Dawley
Number of animals:	5/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Vehicle:	Not specified
Batch:	Not specified
Purity:	Not specified
Dose levels:	0, 1, and 3 mg/kg bw
Dose volume:	Not specified
Route:	Intravenous
Observation	14 days
GLP:	Not specified
Study period/year:	Not specified (pre-2009)

Tafluprost (purity not specified) was investigated for acute intravenous toxicity in rats. In this study, Sprague-Dawley rats (5/sex/group rats) were administered single intravenous doses of the test substance at 0, 1 and 3 mg/kg bw. Following exposure, the animals were observed daily for clinical signs of toxicity, and mortality for 14 days. Body weights, food intake and water intake were determined at defined intervals.

Neither mortalities or clinical signs nor marked changes in body weight, food intake, water intake or gross pathology were observed by the study investigators.

Under the conditions of the study, the LD50 of Tafluprost was determined to be >3 mg/kg bw in rats.

Study 2:

Guideline:	Not specified
Species/strain:	Beagle dogs
Number of animals:	2 males/group
Test substance:	Tafluprost
Product name:	AFP-168
Vehicle:	Not specified
Batch:	Not specified
Purity:	Not specified
Dose levels:	0, 0.3, 3, and 30 µg/kg bw (<i>i.e.</i> , 0.0003, 0.003 and 0.03 mg/kg bw)
Dose volume:	Not specified
Route:	Intravenous
Observation	14 days
GLP:	Not specified
Study period/year:	Not specified (pre-2009)

Tafluprost (purity not specified) was investigated for acute intravenous toxicity in dogs. In this study, Beagle dogs (2 males/group) were administered single intravenous doses of the test substance at 0, 0.3, 3, and 30 µg/kg bw. Following exposure, the animals were observed daily for clinical signs of toxicity and mortality for 14 days. Body weights and food intake were determined at defined intervals. Haematology, urinalysis, body temperature, blood pressure or ophthalmologic findings were also evaluated.

No mortalities or marked changes in body weight, haematology parameters, urinalysis, body temperature, or ophthalmologic findings were observed.

Clinical signs including salivation, vomiting, moderate miosis, irregular respiration and

increased heart rate were observed at doses ≥ 0.003 mg/kg bw. The miosis was severe, and elevated blood pressure was also observed at 0.03 mg/kg bw. No adverse effects were noted at 0.0003 mg/kg.

Under the conditions of the study, the LD₅₀ of Tafluprost was determined to be >0.03 mg/kg bw in dogs.

Ref.: (CDER, 2011), (TGA, 2011)

Applicant 2:

Study 1:

The acute intravenous toxicity of Tafluprost (AFP-168) was investigated in male and female Sprague-Dawley rats. Each 5 animals per sex and dose received single intravenous doses 0, 1, and 3 mg/kg bw and were observed for 14 days after dosing.

No deaths occurred in any treatment group. There were no noteworthy findings with regards to clinical findings, body weight, food intake, water intake or gross pathology. Therefore, the intravenous LD₅₀ value of Tafluprost in rats was estimated as > 3 mg/kg bw.

Study 2:

In dogs, the acute intravenous toxicity of Tafluprost (AFP-168) was investigated in male Beagle dogs. Each 2 males per dose received single intravenous doses 0.3, 3.0 and 30 μ g/kg bw and were observed for 14 days after dosing.

No deaths occurred at any Tafluprost (AFP-168) group. At doses ≥ 3 μ g/kg bw the dogs experienced signs of nausea, vomiting, moderate to severe miosis, irregular respiration and increased respiratory rate, increased heart rate, transient prolongation of QTc, and transient increased in serum ALT activity. There were no noteworthy findings with regards to body weight, food intake, hematological parameters, urinalysis, body temperature or ophthalmologic findings. Therefore, the oral LD₅₀ value of Tafluprost in rats was estimated as > 30 μ g/kg bw.

SCCS comments

Both Applicants provided *in vivo* studies for acute IV toxicity for the compound Tafluprost. As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, these results have not been taken into consideration for this Opinion.

5.3.5 Repeated dose toxicity

5.3.5.1 Repeated dose (28 days) oral / dermal / inhalation toxicity

Applicant 1:

Study 1

Guideline: not specified

Species: rats (strain not specified)

Number of animals: 12/sex/group

Test substance: Tafluprost

Product name: AFP-168

Vehicle: Not specified

Batch: Not specified

Purity: Not specified

Route: Intravenous

Dose levels: 0, 10, 30 and 100 µg/kg bw/day
 Duration: 28 days
 Recovery: 14 days
 GLP: Yes
 Study period: Not specified (pre-2013)

The subacute repeated dose toxicity of the analogue Tafluprost (purity not specified) was investigated in rats following intravenous administration of 0, 10, 30 and 100 µg/kg bw/day daily for 28 days followed by a 14-day recovery period. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Haematological parameters, clinical chemistry parameters, urinalysis, ophthalmic and electrocardiographic parameters were examined. At the termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

No treatment-related mortality, clinical signs, or changes in body weight, water consumption, clinical chemistry, urinalysis parameters, ophthalmic parameters, electrocardiograph traces, organ weights, gross pathology or histopathology were noted. Slightly lower haemoglobin concentration, erythrocyte numbers, and packed cell volume occurred in males at 100 µg/kg bw/day and erythrocyte numbers were marginally reduced in males at 30 and 10 µg/kg bw/day. Also, platelet numbers were marginally low in all treated females and males at 30 and 100 µg/kg bw/day. However, significant differences did not occur, and all of these parameters returned to normal levels during the recovery period.

Ref.: (CDER, 2011)

Study 2:

Guideline	Not specified
Species/strain	Dogs/Not specified
Number of animals	4/sex/group
Test substance	Tafluprost
Product name	AFP-168
Vehicle	Not specified
Batch	Not specified
Purity	Not specified
Route	Intravenous
Dose levels	0, 0.1, 1 and 10 µg/kg bw/day
Duration	28 days
Recovery	14 days
GLP	Yes
Study period/year	Not specified (pre-2013)

The subacute repeated dose toxicity of the analogue Tafluprost (purity not specified) was investigated in dogs following intravenous administration of 0, 0.1, 1 and 10 µg/kg bw/day daily for 28 days followed by a 14-day recovery period. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Haematological parameters, clinical chemistry parameters, urinalysis, ophthalmic parameters, and electrocardiograph were assessed. At the termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

No treatment-related mortality, clinical signs, or changes in body weight, food consumption, haematology, body temperature, ophthalmology, gross pathology, organ weights or histopathology were noted. Slight miosis, sporadic salivation and vomiting were observed in both sexes at 1 µg/kg bw/day. Salivation, vomiting, miosis, increased respiratory rate, increased heart rate, and prolonged QTc interval were observed in both sexes at 10 µg/kg bw/day. Increased alanine aminotransferase (ALT) activity, increased urine volume and decreased urinary potassium concentration were observed in both sexes during the dosing

period, but these effects disappeared after the recovery period. Increased diastolic pressure was found only in males and reduced urinary chloride concentration occurred only in females.

Ref.: (CDER, 2011)

Study 3:

Guideline:	Not specified
Species/strain:	Monkey/Not specified
Number of animals:	3/sex/group
Test substance:	Ophthalmic solution containing 0.0005-0.05% Tafluprost
Product name:	AFP-168
Test concentrations:	0, 0.0005%, 0.005%, 0.05%
Batch:	Not specified
Purity:	Not specified
Route:	Topical ocular
Dose levels:	0, 0.15, 1.5 and 15 µg/left eye/time, twice daily (<i>i.e.</i> , 0, 0.3, 3 and 30 µg/eye/day) Equivalent to 0, 0.067–0.1, 0.67–1 and 6.7–10 µg/kg bw/day ¹²
Formulation/Vehicle:	Not specified
Duration:	28 days
GLP:	Yes
Study period/year:	Not specified (pre-2013)

The ocular toxicity of an ophthalmic solution of analogue Tafluprost (purity not specified) was investigated in a subacute toxicity study in monkeys. Monkeys (3/sex/group) were exposed to ophthalmic solutions containing 0, 0.0005%, 0.005% and 0.05% Tafluprost (*i.e.*, equivalent to 0, 0.067–0.1, 0.67–1, 6.7–10 µg/kg bw/day) via the ocular route for 28 days. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Haematological parameters, clinical chemistry parameters, ECG parameters, blood pressure, electroretinogram parameters, macroscopic ocular findings and intraocular pressure were examined. At termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

No treatment-related mortality, clinical signs, changes in body weight, food consumption, macroscopic ocular findings, lens or vitreous body changes, or changes in intraocular pressure, electroretinogram parameters, or eye lash colour, ECG parameters, blood pressure, haematology parameters, clinical chemistry, gross pathology, organ weights or histopathology were noted.

Treatment-related changes were restricted to local changes in iris colour in two animals at 3 µg/eye/day (*i.e.*, 0.67–1 µg/kg bw/day). In addition, transient corneal precipitates, anterior chamber cells, superficial corneal opacity and erosion, positive epithelial topical ocular fluorescein staining, and conjunctiva redness were occasionally observed at the two highest tested doses. The erosion, positive epithelial topical ocular fluorescein staining, and conjunctiva redness were also observed in vehicle control animals and untreated eyes and may have been related to corneal surface drying due to the anaesthesia.

Conclusion:

According to the Applicant, under the conditions of the study, the NOAEL for systemic toxicity was established at 30 µg/eye/day (*i.e.*, 6.7–10 µg/kg bw/day) due to the lack of any systemic effects of Tafluprost in monkeys.

Ref.: (CDER, 2011; TGA, 2012)

SCCS comments

Both Applicants provided *in vivo* sub-chronic repeated dose toxicity studies for the compound Tafluprost. As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, these results have not been taken into consideration for this Opinion.

Further remarks on the Tafluprost studies:

- study 1 in rats (strain not known) shows a NOAEL of 10µg/kg bw/day, contrary to 30µg/kg bw/d as stated by the Applicant.
- study 2 in dogs (strain not known) shows a NOAEL of 1 µg/kg bw/day.
- PoD has been derived by the Applicant from IV route in studies 1 and 2.
- 28-day exposure for studies 1 and 2 for systemic toxicity is too short and an additional factor of 3 will be needed to account for the short time duration.
- study 3 in monkey (strain not known) shows a NOAEL lower than 3 µg/eye/day and not 30µg/kg bw/d as stated by the Applicant.
- eye changes were observed, which are the concerning side effects of this type of compound and cannot be attributed to just local effects without additional evidence.
- dogs seem to be the most sensitive amongst the tested species for Tafluprost.

5.3.5.2 Sub-chronic (90 days) oral / dermal / inhalation toxicity

Applicant 1:**Study 1:**

Guideline:	EMA note for guidance on repeated dose toxicity (CPMP/SWP/1042/99)
Species/strain:	Rats/Crl:CD®(SD)IGSBR
Number of animals:	15/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Batch/Lot:	FP-0002
Purity:	98.6%
Route:	Intravenous
Dose levels:	0, 10, 30 and 100 µg/kg bw/day
Dose volume:	10 ml/kg
Vehicle:	Sterile saline (0.9% NaCl)
Satellite group Duration:	8/sex/group for toxicokinetic analysis 26-weeks
GLP:	Yes
Study deviation:	Yes, study deviations were not considered to have altered the outcome or integrity of the study.
Study year	2001

The subchronic toxicity of Tafluprost (98.6% purity) was investigated for repeated dose toxicity according to the EMA note for guidance on repeated dose toxicity (CPMP/SWP/1042/99) (TGA, 2012). Crl:CD®(SD)IGSBR rats (15/sex/group) were administered intravenously daily at doses of 0, 10, 30 and 100 µg/kg bw/day for 26-weeks. The stability of the dosing solution was confirmed by analysis. During the treatment period, animals were observed for clinical signs, mortality, body weight, food consumption and water consumption at defined intervals. Haematological, biochemical, ophthalmological examination and urinalysis were performed. Electrocardiogram (ECG) measurements were performed pre-

treatment and in Week 26 before dosing. At the termination of treatment, all animals were sacrificed after overnight fasting and macroscopically examined, organs were weighed, and comprehensive histopathology was performed. The study report noted multiple minor deviations from the study protocol. However, these deviations were not considered to have altered the outcome or integrity of the study.

Results:

Fifteen animals (eleven males and four females) died or were sacrificed during the treatment period. The apparent cause of death of one male in the low dose group and two males in the intermediate dose group was due to glomerulonephropathy, but the cause of death for the other animals (all intermediate and high dose) was not apparent.

No treatment-related clinical signs in surviving animals, changes in body weight, water consumption, clinical chemistry, or ophthalmic parameters were noted. Few ECG findings were of uncertain relationship to the test substance administration. In haematological examinations, some effects were evident in males and females. In Week 26, haemoglobin concentration and erythrocyte numbers were slightly decreased in females in all dose groups. Reticulocyte numbers, mean cell volume, mean cell haemoglobin, and red cell distribution widths were slightly higher in females at 30 and 100 µg/kg bw/day. Similar patterns were observed in males at 100 µg/kg bw/day but generally to a lesser extent for most parameters. Platelet numbers and platelet crit were slightly low at all dose levels in males, and the mean platelet volume and platelet distribution width were slightly high in males at 0.03 and 100 µg/kg bw/day. For these parameters, similar patterns were observed in females at 100 µg/kg bw/day but generally, to a lesser extent for most parameters. Neutrophils were also slightly increased in high-dose males, but not in females.

In urinalysis, slightly smaller volumes of urine with slightly greater specific gravity were noted in males. The total output of sodium, potassium, and calcium was similar to controls. These changes were considered to be toxicological insignificant. No changes were observed in females.

Analysis of bone marrow smears revealed reduced numbers of early erythropoietic and myelopoietic cell types and increased late erythropoietic and myelopoietic cell types at 100 µg/kg bw/day. A slight but significant increase in mean spleen weights was recorded in males at 100 µg/kg bw/day and in females at 30 µg/kg bw/day and 100 µg/kg bw/day.

Histopathology findings were noted in bone marrow in femoral and sternum bone, spleen, liver, and kidney in all dose groups. The findings included dose-related hyperostosis and myelofibrosis in the femoral and sternum bone marrow. Also increased haematopoiesis was noted in the spleen, liver, and male femoral bone marrow. Generally, minor liver haematopoiesis occurred as foci in liver parenchyma in all dose groups without a clear dose-dependent trend for severity. A dose-dependent increase in the incidence and severity of femoral bone marrow haematopoiesis was only apparent in males. A dose-dependent increase in the incidence and severity of corticomedullary mineralization was observed only in females. The histopathology was consistent with the known pharmacological activity of prostaglandin F2a analogues. Under the conditions of the study, the LOAEL for the test substance was considered to be 10 µg/kg bw/day in rats.

Ref.: (CDER, 2011)

Study 2:

Guideline:	EMA note for guidance on repeated dose toxicity (CPMP/SWP/1042/99)
Species/strain:	Beagle dogs
Number of animals:	4/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Batch/Lot:	FP-0002
Purity:	98.6%

Route:	Intravenous
Dose levels:	0, 0.1, 1, and 10 µg/kg bw/day
Dose volume:	1 mL/kg
Vehicle:	0.9% sodium chloride
Satellite group Duration:	None 39 weeks
GLP:	Yes
Study deviation:	Yes, study deviations were not considered to have altered the outcome or integrity of the study.
Study year:	2000

The subchronic toxicity of the analogue Tafluprost (98.6% purity) was investigated for repeated dose toxicity according to the EMA note for guidance on repeated dose toxicity (CPMP/SWP/1042/99) (TGA, 2012). Beagle dogs (4/sex/group) were dosed daily via the intravenous route at doses of 0, 0.1, 1, and 10 µg/kg bw/day for 39 weeks. The stability of dosing solution was confirmed by analysis. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Ophthalmological examinations were performed in all animals' pretreatment and in Weeks 13, 26, and 39. An indirect binocular ophthalmoscope was used to examine the eyelids, optic disc, tapedal and non-tapedal fundus, and retinal blood vessels. In addition, a pupillary light response examined in weeks 14, 27, 38/39 of the study for all animals.

Electrocardiograms were performed on all animals. In addition, blood pressure, body temperature, respiratory rate measurements were made pre-treatment and before dosing and 5, 30 and 120 minutes after dosing in weeks 4, 8, 13, 26, and 39. Blood samples for haematological and clinical- chemical examination were collected from all animals before treatment, and in weeks 13, 26, and 39. For urinalysis, urine samples were collected overnight from all animals pre-treatment and in weeks 12, 26, and 38. At the termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

One high-dose male displaying multiple indications of hepatic failure was killed *in extremis* during Week 19 of treatment of the experiment. Several clinical signs were observed on an almost daily basis including salivation, emesis/retching, and pacing at 10 µg/kg bw/day. Emesis/retching and loose faeces were less frequently observed at 1 µg/kg bw/day. Other less frequent clinical signs included panting, subdued mood, vocalization, aggression, loose faeces, poor mobility, and vasodilation were observed at 10 µg/kg bw/day. Several clinical signs were observed in high-dose animals on an almost daily basis including salivation, emesis/retching, and pacing. Emesis/retching and loose faeces were less frequently observed in dogs receiving 1 µg/kg/day. Clinical signs in single high-dose male killed *in extremis* included, inappetence, progressive body weight loss, thin appearance, sluggishness, and yellow coloration of the whole body.

No treatment-related changes in body weight, food consumption, body temperature, or urine parameters were noted.

No ocular changes were considered to be treatment-related other than dose-dependent changes in pupillary light response. Marked miosis was observed shortly after dosing during weeks 14, 27, and 39 at 10 µg/kg bw/day. Miosis was slight to moderate two hours after dosing in weeks 27 and 39 for most high-dose animals. Slight to moderate miosis occurred at 1 µg/kg bw/day after dosing in Weeks 14, 27, and 39. No changes were observed at a low dose level of 0.1 µg/kg bw/day.

In ECG examination, increased heart rate was observed in a dose-dependent manner at 5 and/or 30 minutes after dosing. As a consequence of increased heart rate, RR-intervals were slightly shortened. The QT values corrected for heart rate (QTc) were normal.

Slight but significant increased mean arterial blood pressure (MAP) was observed 30 minutes

after dosing during weeks 4, 26, and 39 in females at 10 µg/kg bw/day. Also, MAP was significantly increased at a single time-point, 5 minutes after dosing in Week 26 in males at 1 and 10 µg/kg bw/day.

Respiratory rates were slightly but significantly elevated in animals at 30 minutes after dosing at 10 µg/kg bw/day. However, the rates were less elevated or similar to control, two hours after dosing indicating a transient effect.

No treatment-related changes in haematology and clinical chemistry parameters were observed in any of the animals except the high-dose male killed in extremis. For this animal, prolonged prothrombin, activated partial thromboplastin times and increased aspartate aminotransferase, alanine aminotransferase, gamma glutamyl transferase and alkaline phosphatase activities and total bilirubin concentration were noted.

The primary gross pathology finding was reddening at injection site but not in a dose-dependent manner. No other treatment-related findings were observed except the high-dose male killed in extremis. For this animal, a small, firm, mottled liver, a small thymus, a slightly enlarged kidney and a discoloured intestinal tract were observed. The relative salivary gland weights were increased by 28 and 37% in males and females respectively compared to controls. No other treatment-related changes were observed.

In histopathology examination, minor adrenal cortical eosinophilia and acinar cell hypertrophy in salivary glands were observed primarily in males and females at 10 µg/kg bw/day. For the affected animals, the salivary gland acini tended to be larger due to an increased amount of normal appearing cytoplasm. No other treatment-related findings were observed except the high-dose male killed in extremis. For this animal, extensive histopathology occurred in the liver. The centrilobular cords were atrophic and consisted mainly of collapsed sinusoids with pigmented cells and few normal hepatocytes. The periportal region consisted of proliferating bile ducts variably surrounded by basophilic hypertrophic hepatocytes interspersed with necrotic cells and mitosis.

Under the conditions of the study, the NOAEL for the test substance Tafluprost was established at 1 µg/kg bw/day in dogs.

Ref.: (CDER, 2011)

Study 3:

Guideline:	Not specified
Species/strain:	Mice/Crl:CD-1(ICR)BR
Number of animals:	12/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Vehicle:	Not specified
Batch/Lot:	F01X010
Purity:	99.5%
Route:	Subcutaneous
Dose levels:	0, 3, 10, 30 and 100 µg/kg bw/day
Duration:	90 days
GLP:	Yes
Study year:	2003

The sub-chronic repeated dose toxicity of the analogue Tafluprost (99.5% purity) was investigated in Crl:CD-1(ICR)BR mice following subcutaneous administration of 0, 3, 10, 30 and 100 µg/kg bw/day daily for 90 days. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Haematological and clinical chemistry parameters were examined. At the termination of

treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

No treatment-related mortality, clinical signs, or changes in body weight, food consumption, haematology parameters, clinical chemistry parameters, gross pathology, organ weight or histopathology were observed at any dose levels.

Under the conditions of the study, the NOAEL for the test substance Tafluprost was 100 µg/kg bw/day in mice.

Ref.: (CDER, 2011; TGA, 2012)

Study 4:

Guideline	Not specified
Species/strain	Rats/Crl CD®(SD)I
Number of animals	GSBR 10/sex/group
Test substance	Tafluprost
Product name	AFP-168
Vehicle	Not specified
Batch/Lot	F01X010
Purity	99.5%
Route	Subcutaneous
Dose levels	0, 3, 10, and 30 µg/kg bw/day
Duration	90 days
GLP	Yes
Study year	2003

The sub-chronic repeated dose toxicity of the analogue Tafluprost (99.5% purity) was investigated in Crl:CD®(SD)IGSBR rats following subcutaneous administration of 0, 3, 10, and 30 µg/kg bw/day daily for 90 days. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Haematological parameters, and clinical chemistry were assessed. At the termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

Only minor and/or dose-independent changes were noted for mortality. No treatment related effects on clinical signs, gross pathology, or changes in body weight, food consumption, haematology, clotting parameters, clinical chemistry, or organ weights were observed.

Histopathology in the spleen and kidney was considered to be treatment related. In the spleen, a minor increase in the incidence and severity of haematopoiesis in high-dose males and females compared to vehicle control animals was noted. In the kidney, a minor increase in the incidence and severity of corticomedullary mineralization in high-dose females was noted. However, the histopathological changes in spleen and kidney were not considered to be statistically significant.

Under the conditions of the study, the NOAEL for the test substance Tafluprost was considered to be 30 µg/kg bw/day in rats.

Ref.: (CDER, 2011)

Study 5:

Guideline:	Not specified
Species/strain:	Monkey/not specified
Number of animals:	3/sex/group
Test substance:	Ophthalmic solution containing 0.0045% Tafluprost
Test concentrations:	0, 0.0045%
Batch/Lot:	100002-04
Purity:	102.4%
Route:	Ocular
Dose levels:	0, 1.35 µg/left eye/time, twice daily (<i>i.e.</i> , 0, 2.7 µg/eye/day) Equivalent to 0, 0.75–0.9 µg/kg bw/day
Dose volume:	30 µL/eye/dose
Formulation/Vehicle:	Vehicle for DE-111 ophthalmic solution contained benzalkonium chloride with a pH of approximately 7.0, and an osmolar ratio of approximately 1)
Satellite group Duration:	None 13 weeks
GLP:	Yes
Study deviation:	Yes, study deviations were not considered to have altered the results or the validity and integrity of the study
Study year:	2010

The ocular toxicity of an ophthalmic solution of analogue Tafluprost (purity: 102.4% purity) was investigated in a subchronic toxicity study in monkeys. Monkeys (3/sex/group) were exposed to ophthalmic solutions containing 0, 0.0045% Tafluprost (*i.e.*, equivalent to 0, 0.67–0.9 µg/kg bw/day) via the ocular route for 13 weeks. The right eyes remained untreated. The stability of the dosing solutions was analytically confirmed. During the treatment period, animals were observed for clinical signs, mortality and body weight at defined intervals. Macroscopic ocular examination as well as fundus examination, slit lamp examination and fluorescein staining of the cornea, and IOP measurements were performed once before dosing, and 2 to 6 hours after dosing in Weeks 4, 8 and 13. Similarly, iridial and eyelash colour examinations were performed before dosing, and 2 hours 26 minutes to 5 hours 13 minutes after dosing in weeks 4, 8, and 13. Electroretinography (ERG) was performed before dosing and during Week 13. Scotopic ERG, oscillatory potentials, 30 Hz flicker, and photopic ERG were measured. At the termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

No treatment-related mortality, changes in body weight, macroscopic ocular findings, fundus, or slit lamp examinations or changes in electroretinogram parameters, gross pathology, organ weights or histopathology were noted. Treatment-related changes in eye colour were observed in animals. A slight tendency for decreased IOP was observed compared to the untreated eyes; however, the reduced IOP in the treated eyes fell within the normal variations. The iridial colour was found to be darkened relative to pre-dose colours in two male animals. The ocular effects were considered to be mainly cosmetic, not associated with loss of function, and not toxicologically significant. These effects were however noted in the warnings and precautions of the product labels.

Conclusion:

Under the conditions of the study, the local (ocular) or systemic NOAEL for the test substance Tafluprost was established at 2.7 µg/eye/day (0.67–0.9 µg/kg bw/day).

Ref.: (CDER, 2011)

SCCS comments for Applicant 1 submission

Applicant 1 provided *in vivo* sub-chronic toxicity studies for the compound Tafluprost. As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, these results have not been taken into consideration for this Opinion.

Furthermore, remarks on the Tafluprost studies:

- study 1 in rats, IV, shows a LOAEL of 10µg/kg bw/day based on histopathological changes in liver, bone marrow and cortico- medulla; IV is not a usual exposure route for NOAEL determination and the use of LOAEL values requires an additional factor of 3 (see SCCS Notes of Guidance (SCCS/1647/22));
- study 2 in dogs, IV, shows a NOAEL of 0.1 µg/kg bw/day for Tafluprost and not 1 µg/kg bw/day as stated by the Applicant, based on occurrence of miosis. Again, IV is not a usual exposure route for NOAEL determination.
- study 3 in mice via the subcutaneous route has proposed a NOAEL of 100 µg/kg bw/day for Tafluprost with too little explanation; subcutaneous administration is not a usual exposure route for NOAEL determination.
- study 4 in rats, NOAEL of Tafluprost of 30 µg/kg bw/day is shown based on histopathologic changes in spleen and kidney; too little explanation is given, SC is not a usual exposure route for NOAEL determination
- study 5 in monkey (strain not known), ocular exposure, shows a NOAEL between 0.7-0.9 µg/kg bw/day for Tafluprost based on eye colour change and change in IOP. As via the ocular route eye changes were observed, which are the concerning side effects of this type of compounds, they cannot be attributed to just local effects without showing additional evidence.
- dogs seem to be the most sensitive amongst the tested species for Tafluprost with respect to systemic toxicity.

Applicant 2**Study 1:**

26-week intravenous administration toxicity study with Tafluprost (AFP-168) in the rat:

Material and methods:

The purpose of study was to investigate the repeated dose toxicity of Tafluprost (AFP-168, Lot FP-0002, Purity: 98.6%) in male and female Sprague-Dawley rats (CrI:CD®(SD)IGSBR) under GLP conditions. Each 15 male and 15 female rats in the main group with complete investigations were dosed intravenously once a day with doses of 0, 10, 30 and 100 µg/kg bw/d for 26 weeks. In addition, a satellite group consisting of each 8 animals per dose and sex were used for toxicokinetic analyses. The dosing volume was 10 mL/kg bw. At the end of the treatment period, the corresponding main study animals were sacrificed and submitted for gross necropsy. Clinical reactions including mortality to treatment were monitored as well as body weight assessment, food and water consumption, ophthalmology, Electrocardiogram (ECG), clinical pathology, organ weights, and gross pathology. Histopathological evaluation was performed on a comprehensive range of organs. Toxicokinetics was analysed in the separate satellite group.

Results:

Mortality - Eleven males and four females in the Main Study, and three male toxicokinetic animals, all from Tafluprost (AFP-168) treatment groups, were sacrificed due to morbidity or died during the treatment period. Some of the animals were sacrificed due to impaired mobility. All of the animals died or were sacrificed between Weeks 6 and 26 of the experiment. The incidence of mortality was Tafluprost (AFP-168)-dose dependent with intermediate and high-dose male animals predominantly affected. For three of the male animals, the cause of death was determined to be kidney glomerulonephropathy; however, no specific cause of death was apparent for the other animals.

Clinical Signs - No clinical signs in surviving animals were attributed to Tafluprost (AFP-168) treatment.

Body Weights - No effects on body weight or body weight gain were attributed to treatment with Tafluprost (AFP-168).

Feed Consumption - Food consumption was similar for all groups

Water Consumption - Water consumption was similar for all groups.

Ophthalmoscopy - No Tafluprost (AFP-168)-related ophthalmoscopy effects were observed.

Electrocardiogram (ECG) - The few ECG findings were of uncertain relationship to Tafluprost (AFP-168) administration. Heart rates for the high-dose males were approximately 12% lower than for controls in Week 26 and this difference was significant, but high-dose males also had a non-significant lower mean heart rate of approximately 10% compared to controls before the start of treatment. Intermediate-dose females also had a significant, approximately 9% lower mean heart rate compared to controls in Week 26, but a similar effect was not observed in high-dose females.

Hematology - Several hematology parameters changed in a dose-dependent manner. In Week 26, hemoglobin concentration and erythrocyte numbers were slightly decreased in females in all Tafluprost (AFP-168) dose groups. Reticulocyte numbers, mean cell volume, mean cell hemoglobin, and red cell distribution widths were slightly higher in females in the 30 and 100 µg/kg bw/day dose groups. Males in the high-dose group demonstrated similar patterns but generally to a lesser extent for most parameters. Platelet numbers and platelet crit were slightly low at all dose levels in males, and the mean platelet volume and platelet distribution width were slightly high in males in the 30 and 100 µg/kg bw/day groups. For these parameters, females in the high-dose group demonstrated similar patterns but generally to a lesser extent for most parameters. Neutrophils were also slightly elevated in high-dose males, but not females

Clinical Chemistry - No Tafluprost (AFP-168)-related changes in any of the hematology parameters were noted.

Urinalysis - Males receiving Tafluprost (AFP-168) produced slightly smaller volumes of urine with slightly greater specific gravity than controls. The concentration of urine sodium, potassium, and calcium was consequently slightly higher than for the controls, but the total output of each electrolyte was similar to controls. Females did not demonstrate a similar pattern, and the changes in males were not thought to have toxicological significance.

Gross Pathology - Not males but females in the Tafluprost (AFP-168) treatment groups demonstrated gross pathology changes in the bone marrow firmness and thickness which correlated with histopathology findings.

Organ Weights - Mean Spleen weights were slightly but significantly higher in males receiving 100 µg/kg bw/d (0.898 g) and in females receiving 30 (0.612 g) and 100 (0.635 g) µg/kg bw/d AFP-168 compared to controls (0.766 g for males and 0.535 g for females).

Histopathology - In animals in the Tafluprost (AFP-168) treatment groups histopathology findings were noted in bone marrow in femoral and sternum bone, spleen, liver, and kidney. In bone marrow in femoral and sternum bone, there was hyperostosis in animals in all

Tafluprost (AFP-168) groups. In the most severe cases, bone marrow cavities were almost completely filled with bony trabeculae. The effect was dose-dependent and females were more affected than males. Also, myelofibrosis was seen in femoral and sternum bone marrow. The myelofibrosis was characterized by lightly eosinophilic fibrous tissue in the marrow cavity, particularly around small blood vessels and the bony trabeculae. This lesion was more prominent and absent in sternum bone marrow in low-dose females. Dose-related severity was evident in femur, but less apparent in the sternum.

Increased hematopoiesis was observed in spleen, liver and male femoral bone marrow. In the spleen, a degree of background hematopoiesis was apparent in vehicle control animals. Tafluprost (AFP-168) treatment produced a dose-dependent increase in severity and females were more affected than males. Control animals did not demonstrate liver or femoral bone marrow hematopoiesis. Generally minor liver hematopoiesis occurred as foci in liver parenchyma in all Tafluprost (AFP-168) treatment groups without a clear dose dependent trend for severity. Increased femoral bone marrow hematopoiesis was only apparent in males and was dose-dependent for severity.

A Tafluprost (AFP-168) dose-dependent increase in the incidence and severity of corticomedullary mineralization was observed only in females.

Toxicokinetics - Plasma concentrations of Tafluprost (AFP-168) were below the limit of quantification for all of the Tafluprost (AFP-168) dose groups with the exception of one pre-dose sample from the 30 µg/kg bw/d dose group in Week 13. Plasma Tafluprost acid (AFP-172) was detected at all dose levels. Also two vehicle control samples tested positive for low levels of Tafluprost acid (AFP-172), but these results were thought to be false positives due to contamination. In the Tafluprost (AFP-168)-treatment groups, C_{max} and AUC_{0-1h} values increased in an approximately dose-proportional manner with the exception noted below. Male values were similar for all three Tafluprost (AFP-168) doses at all three sampling time points, but the female C_{max} values for the high-dose group were approximately 80% lower for the week 13 and 26 samples compared to the Day 1 samples. AUC_{0-1h} values were similar for male and females at all time-points. T_{max} was consistently 1 hour (the first sample timepoint) except for the Day 26 high dose female samples where T_{max} was 5 hours

Conclusion:

The repeated intravenous injection of Tafluprost (AFP-168) to in male and female Sprague-Dawley rats at doses of 0, 10, 30 and 100 µg/kg bw/d and exposure for 26 weeks led to premature deaths in few animals as in total 15 animals (eleven males and four females) in the main study died or were killed during the treatment period. The apparent cause of death for one low dose and two intermediate dose males was glomerulonephropathy, but the cause of death for the rest of the fifteen animals (all intermediate and high dose) was not apparent. Some slight hematological effects including decreased hemoglobin concentration, erythrocyte numbers and packed cell volume were evident in males and females. Analysis of bone marrow smears revealed reduced numbers of early erythropoietic and myelopoietic cell types and increased late erythropoietic and myelopoietic cell types in high-dose animals. Slightly high spleen weights were observed in high-dose males and females and in intermediate-dose females. Histopathology included: dose-related hyperostosis and myelofibrosis in femoral and sternum bone marrow at all treated doses. Also hematopoiesis was increased in spleen, liver, and male femoral bone marrow of Tafluprost (AFP-168) treated animals and increased corticomedullary mineralization in treated females. The histopathology was consistent with known pharmacological activity of prostaglandin F_{2a} analogues. A NOAEL value could not be determined for the 26-week rat i.v. study at the lowest investigated dose of 10 µg/kg bw/day. Thus, this dose represents the LOAEL.

SCCS comments

The study was performed with Tafluprost, not with DDDE. IV route is also not the correct means of exposure for repeated dose toxicity testing of cosmetic ingredients.

Study 2:

39-week intravenous administration toxicity study with Tafluprost (AFP-168) in the dog:

Material and methods:

The purpose of study was to investigate the repeated dose toxicity of Tafluprost (AFP-168, Lot FP-0002, Purity: 98.6%) in male and female Beagle under GLP conditions. Each 4 male and 4 female dogs were dosed intravenously once a day with doses of 0, 0.1, 1 and 10 µg/kg bw/d for 39 weeks. The dosing volume was 1.0 mL/kg bw. At the end of the treatment period, the animals were sacrificed and submitted for gross necropsy. Clinical reactions including mortality to treatment were monitored as well as body weight assessment, food consumption, ophthalmology, Electrocardiogram (ECG), blood pressure, body temperature, respiration rate, clinical pathology, toxicokinetics, organ weights, and gross pathology. Histopathological evaluation was performed on a comprehensive range of organs.

Results:

Mortality - One male in the high dose group was killed in extremis during Week 19 of treatment. A number of clinical signs, gross pathology, coagulation parameter, clinical chemistry, and histopathology findings were observed for this animal as noted below in appropriate sections.

Clinical Signs - Several clinical signs were observed in high-dose animals on an almost daily basis including salivation, emesis/retching, and pacing. Emesis/retching and loose feces were less frequently observed in dogs receiving 1 µg/kg bw/d AFP-168. Other clinical signs that were observed less frequently in high-dose animals included panting, subdued mood, vocalization, aggression, loose feces, poor mobility, and vasodilation. Clinical signs were occasionally observed during the dosing procedure but generally occurred between 5 and 30 minutes after dosing. For the single high-dose male killed in extremis, clinical signs included: inappetence, progressive body weight loss, thin appearance, sluggishness, and yellow coloration of the whole body.

Body Weights - Body weights were not significantly different between groups.

Feed Consumption - Food consumption was similar for the Tafluprost (AFP-168) treatment groups compared to controls.

Ophthalmoscopy - No ocular changes were considered related to Tafluprost (AFP-168) administration other than dose-dependent changes in pupillary light response. High-dose animals demonstrated marked miosis shortly after dosing during Weeks 14, 27, and 39. Miosis was slight to moderate two hours after dosing in Weeks 27 and 39 for most high-dose animals. Slight to moderate miosis occurred in most intermediate-dose (1 µg/kg bw/d) animals after dosing in Weeks 14, 27, and 39. Low dose (0.1 µg/kg bw/d) and control animals exhibited a similar range of pupillary responses including normal pupil size, slight miosis, or moderate mydriasis shortly after dosing and for up to two hours after dosing.

Electrocardiogram (ECG) - Heart rates increased in an Tafluprost (AFP-168) dose-dependent manner 5 and/or 30 minutes after dosing. As a consequence of increased heart rate, RR-intervals were slightly shortened. The QT values corrected for heart rate (QTc) were normal.

Blood pressure - The mean arterial blood pressure (MAP) of high-dose females was slightly but significantly elevated compared to vehicle control animals 30 minutes after dosing during Weeks 4, 26, and 39. Also MAP was significantly elevated in intermediate- and high-dose males at a single time-point, 5 minutes after dosing in Week 26.

Body Temperature - No Tafluprost (AFP-168)-related changes in body temperature were observed.

Respiratory Rate - Respiratory rates were slightly but significantly elevated in high-dose males 5 and 30 minutes after dosing during Weeks 8, 13, 26, and 39. In high-dose females, respiratory rates were significantly increased only in Weeks 26 and 39, 30 minutes after dosing. Respiratory rates were generally less elevated or similar to control values two hours after dosing indicating a transient effect.

Hematology - No Tafluprost (AFP-168)-related changes in hematology or coagulation parameters were observed in any of the study animals except the high-dose male killed in extremis. For this animal, prothrombin and activated partial thromboplastin times were prolonged.

Clinical Chemistry - No consistent or Tafluprost (AFP-168)-dose-dependent changes in any of the plasma chemistry parameters were observed except in the high-dose male killed in extremis. For this animal, aspartate aminotransferase, alanine aminotransferase, gamma glutamyltransferase and alkaline phosphatase activities and total bilirubin concentration were elevated.

Urinalysis - No Tafluprost (AFP-168)-related changes in any of the urinalysis parameters were observed.

Gross Pathology - The primary gross pathology finding was injection site reddening but not in a dose-dependent manner for incidence. No other Tafluprost (AFP-168)-related gross pathology findings were observed. In addition, in the high-dose male killed in extremis, a small, firm, mottled liver, a small thymus, and a slightly enlarged kidney were observed. The intestinal tract was also discolored in this animal.

Organ Weights - The relative salivary gland weights of high-dose males and females were increased 28 and 37% respectively compared to controls. None of the other measured organ weights were changed by treatment with Tafluprost (AFP-168).

Histopathology - Minor adrenal cortical eosinophilia and acinar cell hypertrophy in salivary glands was observed primarily in high-dose males and females. For the affected animals, the salivary gland acini tended to be larger due to an increased amount of normal appearing cytoplasm. No other Tafluprost (AFP-168)-related histopathology findings were noted. For the male killed in extremis, extensive histopathology occurred in the liver. The centrilobular cords were atrophic and consisted mainly of collapsed sinusoids with pigmented cells and few normal hepatocytes. The periportal region consisted of proliferating bile ducts variably surrounded by basophilic hypertrophic hepatocytes interspersed with necrotic cells and mitosis.

Toxicokinetics - Tafluprost (AFP-168) was measurable only at very low levels in a few samples, but this may have been due to sample storage beyond the established limits for AFP-168 in dog plasma (42 days at -70°C). AFP-168 was not detected in blood samples obtained from vehicle control animals.

Tafluprost acid (AFP-172) was detected in plasma at all Tafluprost (AFP-168) dose levels on each sampling occasion with the exception of Group 2 (0.1 µg/kg bw/d) in Week 4 where no plasma Tafluprost acid (AFP-172) was detected. Tafluprost acid (AFP-172) was not detected in blood samples obtained from vehicle control animals. Substantial gender differences were not apparent for Tafluprost acid (AFP-172) for C_{max} and AUC values and both of these parameters increased in an approximately dose-proportional manner. Values were similar on all the sampling occasions indicating an absence of Tafluprost acid (AFP-172) accumulation in plasma. T_{max} values ranged from 1 to 5 minutes.

Conclusion:

The repeated intravenous injection of Tafluprost (AFP-168) to in male and female Beagle dogs at doses of 0, 0.1, 1.0 and 10 µg/kg bw/d for a period of 39 weeks led to specific toxicity primarily in the high dose (10 µL/kg bw/d) animals. The observed findings included transient clinical signs of nausea, transient miosis, transient slight elevation in heart rate, blood pressure as well as respiration rate. Enlarged salivary glands and histopathological findings in the adrenal and salivary glands. In addition, one high dose male was killed in extremis

in the week 19 displaying multiple indication of hepatic failure. The overall NOAEL of Tafluprost for beagle dog after 39-week i.v. treatment was 1 µg/kg bw/day.

Ref.: (CDER, 2011)

SCCS comments for Applicant 2 submission subchronic studies

Applicant 2 provided *in vivo* subchronic toxicity studies for the compound Tafluprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, these results have not been taken into consideration for this Opinion.

Further SCCS remarks on the Tafluprost studies:

- study 1 in rats, IV daily injection of Tafluprost, 26 weeks, shows a LOAEL of 10µg/kg bw/day based on histopathological dose-related changes, hyperostosis, myelofibrosis in bone marrow, hematopoiesis increase in spleen, liver, bone marrow, increased cortico-medullary mineralisation in female animals. Several animals died, and the results indicate dose-dependency. Using a LOAEL for MoS calculation requires an additional factor of 3.
- also, IV is not an adequate exposure route for NOAEL determination
- study 2 in dogs, IV injection daily, 39 weeks, shows for Tafluprost a NOAEL of 0.1 µg/kg bw/day (not 1 µg/kg bw/day as stated by the Applicant) based on occurrence of miosis; IV is not an adequate exposure route for NOAEL determination
- dogs are a more sensitive species for systemic toxicity study of Tafluprost than rats.

5.3.5.3 Chronic (> 12 months) toxicity

Applicant 1

Study 1:

Guideline:	EMA note for guidance on repeated dose toxicity (CPMP/SWP/1042/99) Monkey/not specified
Species/strain:	4/sex/group
Number of animals:	Ophthalmic solution containing 0.0005-0.05% Tafluprost AFP-168
Test substance:	0, 0.0005, 0.005 and 0.05%
Product name:	D01114, D01115, D01116
Test concentrations:	98, 100, 102% respectively
Batch/Lot:	Topical ocular
Purity:	0, 0.15, 1.5, 15 µg/left eye/time, twice daily (<i>i.e.</i> , 0, 0.3, 3 and 30 µg/eye/day)
Route:	Equivalent to 0, 0.075–0.1, 0.75–1 and 7.5–10 µg/kg bw/day ¹⁴ 30 µL/eye/dose
Dose levels:	The vehicle for the 0.05 (30 µg/eye/day), 0.005% (3µg/eye/day) and 0.0005% (0.3 µg/eye/day) AFP-168 ophthalmic solutions contained Tween 80, NaH ₂ PO ₄ X H ₂ O,
Dose volume:	EDTA x 2Na, glycerine, 0.01 % benzalkonium chloride, and NaCl
Formulation/Vehicle:	None 52 weeks Yes Yes, study deviations were not considered to have altered the results and integrity
Satellite group	of the study 2001
Duration:	
GLP:	
Study deviation	Not specified (pre-2013)

Study year:

Monkeys (4/sex/group) were exposed ophthalmic solutions containing 0, 0.0005%, 0.005% and 0.05% Tafluprost (*i.e.*, equivalent to 0, 0.075– 0.1, 0.75–1 and 7.5–10 µg/kg bw/day) via the ocular route for 52 weeks. The right eyes remained untreated. The stability of dosing solutions was analysed but no stability information was provided. During the treatment period, animals were observed for clinical signs, mortality body weight and food consumption at defined intervals. Macroscopic ocular examination as well as fundus examination, slit lamp examination and fluorescein staining of the cornea, and IOP measurements were performed once before dosing, and 2 to 6 hours after first dosing in weeks 13, 26, 39 and 52. Similarly, iridial and eyelash colour examinations were performed once before dosing, and once after dosing in Weeks 13, 26, 39 and 52. ERG was performed once before dosing and during weeks 25 and 26, 29 or 30 , 51 or 52. Scotopic ERG, oscillatory potentials, 30 Hz flicker, and photopic ERG were measured. Electrocardiograms (EKG) were performed on all animals (unanaesthetised, temporarily restrained) once before treatment and in weeks 13, 26 and 52 (prior to and 15 minutes after the first dosing). Heart rates (beats/minute), ECG intervals, RR, P, PR, QRS, and QT intervals (QTc and Qt dispersion), as well as voltage measurement of P, R, S, and T were measured. Blood samples for haematological and clinical-chemical examination were collected from all animals once before treatment, and once in Weeks 13, 26, and 52. For urinalysis, urine samples were collected overnight from all animals once pre-treatment and once in Weeks 13, 26, and 52. At termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Results:

No treatment-related mortality, changes in body weight, food consumption, fundus examination, pupil size, irritation score, corneal examination or changes in electroretinography, blood pressure, haematology, clinical chemistry, or urine parameters,

were noted.

Only eye-related clinical signs were considered to be treatment-related. A sunken eye was noted at the two highest doses and darker iris colour was noted in all doses in at least 3 of 4 animals of each gender in the beginning as early as Day 23, and this change was permanent throughout the experiment. Four animals (three males and one female) demonstrated sunken eyes on single occasions. Many of the same animals demonstrated blue-grey discolouration of the lower eyelid. A statistically significant decrease in IOP occurred in males and females in the high-dose group during Week 26. The IOP of the treated eye remained lower compared to the untreated eye for the rest of the study, but not significant. Slight decreases in IOP also occurred in low and intermediate-dose groups during Week 26. The lower eyelid darkening was observed in the intermediate and high dose group.

In gross pathology, no treatment-related changes were noted other than iris darkening, sunken eye. No treatment-related changes in organ weights were noted. Significant increase in the group mean thymus/brain weight ratios relative to control animals were noted in two males of intermediate and high dose groups. However, for all other animals in these groups, non-statistical increases were noted in similar thymus/brain weight ratios and absolute thymus and thymus weight/body weight ratios.

No systemic organ histopathology was considered to be related to the treatment. Most of the animals in all dose groups demonstrated increased melanocyte pigment in the iris stroma of the left eye which is in line with the observed iris darkening. In addition, the treated eyes showed increased pigment around the hair follicles and minimal focal inflammation in the epithelium predominantly in high-dose males and females.

According to the Applicant, all the ocular effects were considered to be mainly cosmetic, not associated with loss of function, and not toxicologically significant.

Conclusion:

Under the conditions of the study, the Local (ocular) and systemic NOAEL for the test substance Tafluprost in monkeys was established at 30 µg/eye/day (7.5–10 µg/kg bw/day).

Ref.:(CDER, 2011; (TGA, 2012))

Study 2:

Study type and species: 78 weeks, subcutaneous study, mice (CrI:CD- 1(ICR)BR) (51/sex/dose)

Doses:0, 10, 30 and 100 µg/kg bw/day (Tafluprost)

Key findings: no significant adverse effects at any dose level

NOAEL / LOAEL: Systemic and carcinogenicity NOAEL = 100 µg/kg bw/day*

Ref.: (CDER, 2011)

Study 3:

Study type and species: 24 months, subcutaneous study, rats (Crj:CD(SD) IGSBR) (60/sex/dose)

Doses:0, 3, 9 and 30 µg/kg bw/day (Tafluprost)

Key findings: At 3 µg/kg bw/day, significant reduction in body weight in both sexes. Non-neoplastic Histopathology changes including hyperostosis of the sternum and femur in some animals and increased incidence of extramedullary haematopoiesis in the spleen in males at all dose levels

NOAEL / LOAEL: Systemic NOAEL = 3 µg/kg bw/day*

Ref.: (CDER, 2011)

SCCS comments on chronic studies

The provided *in vivo* chronic repeated dose toxicity study is for the compound Tafluprost.

As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, these results have not been taken into consideration for this Opinion.

Additional comments from the Applicant on the Tafluprost study:

- study 1 in monkeys (strain is not known), ocular topical application of Tafluprost, 52 weeks, shows according to the Applicant a NOAEL of 7.5-10 µg/kg bw/d, but even at the lowest dose of 0.075-0.1 µg/kg bw/day some eye treatment-related effects were noted such as iris colour changes, blue-green discoloration of lower eye lid, some IOP decrease, sunken eye in certain animals, melanocyte pigments in iris stroma. As via the ocular topical route eye changes were seen, which are the most delicate side effects of this type of compounds, they cannot be attributed to just cosmetic effects as mentioned by the Applicant. LOAEL = 0.075-0.1 µg/kg bw/day and an additional factor 3 for using a LOAEL must be applied.
- study 2 in mice, 78 wks with Tafluprost, SC exposure, only mentions a NOAEL= 100 µg/kg bw/day and no side effects without any further explanation. This study is not acceptable.
- study 3 in rats, SC exposure to Tafluprost, 24 months provided a LOAEL=3 µg/kg bw/day based on histopathological changes, hyperostosis of sternum and femur, extramedullary haematopoiesis in spleen of males at all doses; no further details are given. A factor of 3 applies for using a LOAEL. The study is not acceptable as no experimental details were provided.

5.3.6 Reproductive and developmental toxicity

5.3.6.1 Fertility and reproductive toxicity

Applicant 1:

Material and methods:

Guideline:	Not specified
Species/strain:	Rats/Crl:CD(SD)IGSBR rats
Number of animals:	24/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Batch/Lot:	FP-0002
Purity:	98.6%
Route:	Intravenous
Dose levels:	0, 10, 30 and 100 µg/kg bw/day
Dose volume:	10 mL/kg
Formulation/Vehicle:	0.9% sodium chloride
Satellite groups:	No
Duration:	Two weeks before mating, throughout the mating period, and until Day 6 of gestation for the females or until necropsy in Week 9 of the treatment period for the males.
GLP:	Yes
Study deviation:	Yes, study deviations were not considered to have altered the results or integrity of the study
Study year:	2001

The reproductive toxicity of the analogue Tafluprost (98.6% purity) was investigated in rats

via the IV route. Crl:CD(SD)IGSBR rats (24/sex/group) were dosed daily at 0, 10, 30 and 100 µg/kg bw/day via the intravenous route for two weeks before mating, throughout the mating period, and until Day 6 of gestation for the females or until necropsy in Week 9 of the treatment period for the males. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals.

The stage of oestrous was recorded for each female from 15 days prior to treatment until mating confirmation. The stages of oestrous were assessed via daily vaginal washings. Rats were paired on a 1:1 basis within each treatment group during the mating period. A vaginal copulatory plug or the presence of sperm in a vaginal washing confirmed positive evidence of mating. The day on which mating was confirmed was designated Day 0 of gestation. On Gestation Day (GD) 13, mated females were sacrificed and examined macroscopically. The ovaries and uteri were removed and examined. The pregnancy status, number of corpora lutea, number and intrauterine position of implantations with recording of live embryos, early intrauterine deaths, and late intrauterine deaths were recorded.

Rats were sacrificed and examined macroscopically for structural or pathological changes. The tissues included ovaries, uterus, cervix, vagina, pituitary, testes, epididymides, seminal vesicles, prostate, coagulation gland, and lesions from all adult animals were retained. The histopathology of reproductive organs from all control and high-dose animals were examined.

Results:

No treatment-related changes in body weight, food consumption, fertility or histopathology were noted.

Mortality was found in 2 animals on Day 10 and 15 respectively at 100 µg/kg bw/day. These animals were found dead within one hour of dosing. The animals appeared unremarkable at necropsy. Treatment-related clinical signs including pale extremities were observed in all dose groups immediately after treatment and appeared to be of approximately equal severity in all groups. This effect was observed daily from the second day of treatment in the 100 µg/kg bw/day and from the third day of treatment for the 10 and 30 µg/kg bw/day groups. The effect was transient and disappeared after one hour.

No treatment-related effects were observed on the oestrous cycle except one control pairing, all males mated within the initial four days of mating. No treatment-related effects were observed on fertility parameters (Mating/Fertility Index, Corpora Lutea, Preimplantation Loss, etc.). The mating index was found to be 100% for all treatment and control groups. The fertility and fecundity indexes were 100, 91.7, 91.7, and 100% for both males and females in the control group, low, intermediate and high dose group respectively. On GD 13, no effects were observed on the outcome of the pregnancy and all pregnant females had live embryos. The mean number of corpora lutea and the mean number of implantations per female were similar to the controls. Similar or fewer losses in the mean pre-and post-implantation loss were observed in the treated animals when compared to the control animals.

In histopathology examination, no treatment-related changes in reproductive organs were observed. No treatment-related abnormalities were observed in any of the cell types present within the different stages of the spermatogenic cycle in testis staging.

Conclusion:

Under the conditions of the study, the NOAEL for reproduction/fertility of the test substance Tafluprost was established at 100 µg/kg bw/day in rats based on the absence of toxicity.

Ref.: (CDER, 2011)

Applicant 2

Intravenous study on fertility and early embryonic development of Tafluprost (AFP-168) in the rat: See study here above of Applicant 1.

SCCS comments

The cause of death in both studies, immediately after dosing, was not further examined, raising concerns about unexplored toxicity effects. Also, the repeated appearance of pale extremities immediately after dosing in all treated groups may indicate a physiological stress

response that is insufficiently explored.

5.3.6.2 Developmental Toxicity

Applicant 1

Study type,	Doses	Key findings	NOAEL	Reference
<i>Dose range finding studies</i>				
Prenatal developmental toxicity, intravenous study, rats (strain not specified) (7 females/dose)	0, 10, 30 and 100 µg/kg bw/day Tafluprost Gestation day (GD): Not specified	Pale extremities were observed immediately after dosing in animals at 30 and 100 µg/kg bw/day A dose-dependent increase in the incidence of post-implantation loss in the intermediate (12.1%) and high-dose (49.6%) groups compared to controls (6.0%) with two high-dose animals having litters with no live fetuses.	NOAEL for maternal toxicity = 10 µg/kg bw/day* NOAEL for developmental toxicity = 10 µg/kg bw/day*	(CDER, 2011)
Prenatal and postnatal development toxicity, intravenous study, rats (strain not specified) (7 females/dose)	0, 0.3, 1, 3 and 10 µg/kg bw/day Tafluprost GD: Not specified.	Decreased maternal body weight gains at ≥3 µg/kg bw/day. The increased number of dead fetuses and reduced birth index and viability index on Day 4 of lactation in the animals at 3 and 10 µg/kg bw/day	NOAEL for maternal toxicity = 1 µg/kg bw/day* NOAEL for developmental toxicity = 1 µg/kg bw/day*	(CDER, 2011)
Prenatal developmental toxicity, intravenous study, rabbits (strain not specified) (7 females/dose)	0, 1, 3 and 10 µg/kg bw/day Tafluprost GD: Not specified.	Clinical signs include tremors, splayed legs and excessive licking immediately after dosing in animals at 10 µg/kg bw/day. Intrauterine deaths with no live fetuses in 2 females at 1 µg/kg bw/day, and no live for higher conc	LOAEL for maternal toxicity = 1 µg/kg bw/day* LOAEL for developmental toxicity = 1 µg/kg bw/day*	(CDER, 2011)

Applicant 2**Study 1:****Intravenous study of embryo-fetal development (*i.e.* prenatal developmental toxicity) of Tafluprost (AFP-168) in the rat:**

Material and methods:

The possible effect of Tafluprost (AFP-168, batch: FP-002, purity: 98.6%) on embryo-fetal development was studied in predated female Sprague-Dawley rats (CrI:CD®(SD)IGSBR) under GLP conditions. Each 24 animals per sex and group received daily intravenous (i.v.) doses of 0, 3, 10, and 30 µg/kg bw/d from gestation day (GD) 6 to 17. The dosing volume was 10 mL/kg bw. All animals were sacrificed on GD 20 and pregnancy as well as fetal outcome parameters were assessed. In addition, blood samples were obtained and processed to plasma for toxicokinetic analysis on GD 6 and GD 17.

Results:

Mortality - One female receiving 10 µg/kg bw/d of Tafluprost (AFP-168) died during blood sampling procedures on GD 17, but this death was considered unrelated to AFP-168 administration.

Clinical signs - No clinical signs considered to be related to the test item administration were observed.

Body weight - Mean body weights did not differ between groups.

Food consumption - There were no apparent effects on food intake.

Toxicokinetics - Neither Tafluprost (AFP-168) nor Tafluprost acid (AFP-172) was detectable in plasma obtained from control females. Also, Tafluprost (AFP-168) plasma levels above the lower limit of quantification were detectable only in a few animals in the Tafluprost (AFP-168)-dose groups. Plasma Tafluprost acid (AFP-172) was generally detectable for Tafluprost (AFP-168)-treated animals, but only above the lower limit of quantification for 1 – 15 minutes after dosing. Consequently, plasma AUC exposure values could not be consistently calculated for the highest dose group of 30 µg/kg bw/d. Plasma C_{max} increased in a dose-dependent manner and appeared to increase with repeated dosing. The T_{max} value for all treatment groups was 1 minute after dosing.

Necropsy:

Cesarean Section Data (Implantation Sites, Pre- and Post-Implantation Loss, etc.)

- The ovaries and uteri of each female were removed and the following parameters were assessed: pregnancy status, gravid uterus weight, and the number of corpora lutea. Also, the number and intrauterine position of implantations were assessed and categorized as live fetuses, early intrauterine deaths, late intrauterine deaths, and dead fetuses. Early intrauterine deaths were classified as those with decidual or placental tissue, and late intrauterine deaths were classified as those with fetal tissue in addition to placental tissue. Fully formed fetuses that appeared to have died shortly before necropsy were classified as dead fetuses.

The 20% mean post-implantation loss was greater in the high-dose group compared to the 8.6% loss in the vehicle control group. Two high-dose females had total litter loss as did one female in the vehicle control group due mainly to late intrauterine deaths for all three females. High-dose females also demonstrated an increased number of early and late intrauterine deaths, increased post-implantation loss, and decreased mean numbers of fetuses compared to control females. However, the differences were not statistically significant except for the number of high-dose dams having late intrauterine deaths. The mean number of corpora lutea

and the mean number of implantations were similar to those of the controls for all the Tafluprost (AFP-168)-treatment groups.

Offspring (Malformations, Variations, etc.) - Fetal weights were significantly decreased in the intermediate and high-dose groups compared to controls with mean fetal weights of 3.70, 3.59 and 3.89 g respectively. The mean placental weight was slightly but not significantly higher for the high-dose group (0.61 g) compared to the control group (0.55 g). This difference was largely driven by a very high placental weight for one high-dose animal.

The visceral malformations were not affected by the test item in terms of the number of fetuses or litters affected. Visceral malformations included but were not limited to additional structure in the eye, severe renal pelvic cavitation, severely distended left ureter, umbilical hernia, absent kidneys and ureters, and abnormal lung lobulation. However, specific visceral malformations occurred in only one litter per dose group, different visceral malformations occurred with the different test item doses and the least number of malformations occurred in the high dose group at 30 µg/kg bw/d.

Skeletal malformations occurred in the intermediate (10 µg/kg bw/d) group and high dose group at 30 µg/kg bw/d. In the high-dose group there was a greater number of variations of the lumbar centra, thoracic arcs and thoracic centra. There was also a dose-related significant increase in the numbers of intermediate and high-dose litters with unossified 5th sternebra.

Conclusion:

When Tafluprost (AFP-168) was investigated for its possible effect on embryo-fetal development in female Sprague-Dawley rats at daily intravenous (i.v.) doses of 0, 3, 10 and 100 µg/kg bw/d from GD 7 to 17, no test item related effect regarding maternal toxicity occurred. The test item led to clear signs of prenatal developmental toxicity including teratogenicity. The highest dose of 30 µg/kg bw/d led to an increased number of intrauterine deaths, reduced fetal maturity and a low incidence of defects of the vertebral column. Similar findings, but at a lower frequency and severity occurred in the intermediate dose of 10 µg/kg bw/d. None of such effects were noted at the low dose of 3 µg/kg bw/d.

According to the Applicant, the NOAEL for maternal toxicity was 30 µg/kg bw/d, the highest dose investigated, while the NOAEL for prenatal developmental toxicity including teratogenicity was 3 µg/kg bw/d in the rat.

Study 2:

Intravenous study of embryo-fetal development (i.e. prenatal developmental toxicity) of Tafluprost (AFP-168) in the rabbit – Initial study

Material and methods:

The possible effect of Tafluprost (AFP-168, batch: FP-002, purity: 98.6%) on embryo-fetal development was studied in predated female rabbits (CrI.NZW/Kbl BR) under GLP conditions. Each 24 animals per sex and group received daily intravenous (i.v.) doses of 0, 0.03, 0.1 and 0.3 µg/kg bw/d from gestation day (GD) 1 to 19. The dosing volume was 1 mL/kg bw for the control, intermediate and high dose groups and 0.3 mL/kg bw for the intermediate dose group. All animals were sacrificed on GD 29 and pregnancy as well as fetal outcome parameters were assessed. In addition, blood samples were obtained and processed to plasma for toxicokinetic analysis on GD 7 and GD 19.

Results:

Mortality - No animals were found dead during the experiment. However, single animals in the vehicle control and the low dose (0.03 µg/kg bw/d) groups, and eleven and three mothers in the intermediate dose group (0.1 µg/kg bw/d) and high dose group (0.3 µg/kg bw/d)

groups **aborted their pregnancies** between GD 17 and GD 23 and were euthanized prior to GD 29.

Clinical signs - No clinical signs occurred after dosing. The only clinical signs observed were red tissue and fluid under the cages of the mothers that aborted their pregnancies.

Body weight - A slight weight loss relative to the control animals occurred in the intermediate and high dose animals between GD 7 and GD 10 and between GD 10 and GD 15. The difference was significant only for the intermediate dose between GD 12 and GD 15. Thereafter the growth rate was similar to those of the controls.

The mean gravid uterus weight in the two intermediate-dose females with viable fetuses was significantly less than that of the controls.

Food consumption - Food intake was not reduced by test item treatment.

Toxicokinetics – Tafluprost (AFP-168) was not detected in plasma from the high dose and control animals and the other two groups were not examined for plasma Tafluprost (AFP-168) concentrations. Tafluprost acid (AFP-172) was detectable in the plasma of females in the 0.1 µg/kg bw/d and 0.3 µg/kg bw/d groups on GD 7 and GD 19, but only in the 1 and 5 minute samples. Plasma Tafluprost acid (AFP-172) concentrations were only detectable in the low dose group (0.03 µg/kg bw/d) in the 1 minute sample on GD 7. The Tafluprost acid (AFP-172) toxicokinetic parameters, C_{max} , T_{max} , and $AUC_{(0-120 \text{ min})}$ could not be calculated due to insufficient data.

Necropsy:

Cesarean Section Data (Implantation Sites, Pre- and Post-Implantation Loss, etc.) – Dose-related effects on the pregnancy outcome parameters were recorded in all three treatment groups. In 10 intermediate dose group females and in 13 high dose group females that had evidence of pregnancy on GD 29, the mean number of implantations were significantly less than in the control females. The mean implantation values for the control, intermediate and high dose groups were 10.5, 7.9 and 7.8, respectively. In addition, in 8/10 intermediate dose group females and in 13/13 high-dose females, all implantations died early in gestation leaving no live fetuses. The respective values of 91.6% and 100% for the mean incidences of post-implantation loss for the intermediate dose group and high dose group were significantly higher than the mean incidence of post-implantation loss for control females (13.1%). Also, in the intermediate and high-dose groups the mean number of corpora lutea (3.6 for both groups) was significantly lower compared to control animals (11.5).

In the low dose group, the mean post-implantation loss was 29.2% compared to 13.1% in control females with three low dose females and one control female having total intra-uterine death with no live fetuses. In low-dose females that had some live fetuses, the percentage of post-implantation loss (18.0%) was slightly higher than that of control females (8.7%) with the majority of intrauterine deaths occurring late in gestation.

Offspring (Malformations, Variations, etc.) - In the high-dose group there were no live fetuses. In the two litters with surviving fetuses in the intermediate dose group, the mean fetal weight was similar to that of the controls. Placental weight was slightly but not significantly higher than for the controls and both values fell within the historical control range. Also, the mean litter weight was reduced compared to controls due to the reduced number of live births and one litter contained only females and one litter was approximately equal for each gender. In the low dose group, the mean fetal, and placental weights, and the litter gender proportionality were similar to those of the controls. The mean litter weight reflected the reduced number of live fetuses.

One control fetus had a severe brain malformation, exposed meninges through a skull opening. In the low-dose group, fetal malformations were observed in nine fetuses from six mothers. Six of the affected fetuses had defects of the skull, brain, and/or spine and three had abdominal wall defects. The overall number of fetal variations in the low-dose group was similar to that of the controls, but the skeletal variations in the low-dose group were more varied than in the control group and often associated with the malformations. No fetal malformations were observed in the two intermediate dose litters with surviving fetuses, and

visceral and/or skeletal variations were similar in incidence and type compared to the control litters.

Conclusion:

According to the Applicant, when Tafluprost (AFP-168) was investigated for its possible effect on embryo-fetal development in female rabbits at daily intravenous (i.v.) doses of 0, 0.03, 0.1, and 0.3 µg/kg bw/d from GD 1 to 19, In the early post-implantation loss occurred to a significantly elevated degree in the intermediate dose group and high dose group. high-dose group there were no live fetuses, and in the intermediate-dose group only two litters contained live fetuses. Abortions occurred in eleven and three of the intermediate dose females and high dose females compared to one control female. In the low dose group, the majority of litters were viable, but three fetuses in separate litters had abdominal wall malformations, and six fetuses in five litters had cranial and/or spinal malformations. A NOAEL was not established.

However, from the reviewer perspective there is a clear limitation due to a deviation in the study design compared of the other prenatal developmental toxicity studies in the rat and in the second rabbit study. In this initial study in rabbits, the dosing was already initiated at GD 1. This means treatment started already shortly after fertilization, transport of the zygote up to the implantation of the embryo in the endometrium of the uterus, which is generally finished at GD 6 or 7 in rats and rabbits, respectively. This is a very sensitive period and any disturbance by handling, dose application or toxic effects by substances should be avoided. Otherwise, a very high rate of pre-implantation loss up to total loss is very likely. As a consequence, no litters or dose-dependently too few litters and/or fetuses will be available for a reliable assessment of fetal development and morphology. This was exactly observed in this initial rabbit study. With regards to these consequences, the respective OECD and ICH guidelines generally recommend to start on GD 6 or 7 in rats and in rabbits as this is the timepoint where the implantation process is generally completed and the placenta development is initiated.

Study 3:

Intravenous study of embryo-fetal development (i.e. prenatal developmental toxicity) of Tafluprost (AFP-168) in the rabbit - Second study

Material and methods:

Due to the observed effects in the first study with severe effects on pregnancy outcome and prenatal developmental toxicity preventing a reliable assessment, a second study in predated female rabbits (CrI.NZW/Kbl BR) was performed. In this study, Tafluprost (AFP-168, batch: FP-002, purity: 98.6%) was studied again under GLP conditions. Each 24 main group animals per sex and group received daily intravenous (i.v.) doses of 0, 0.001, 0.003 and 0.01 µg/kg bw/d from gestation day (GD) 7 to 19. The dosing volume was 0.1 mL/kg bw for the control and high dose groups, 0.01 mL/kg bw for the low dose group and 0.03 mL/kg bw for the intermediate dose group. All animals were sacrificed on GD 29 and pregnancy as well as fetal outcome parameters were assessed. In addition, blood samples were obtained and processed to plasma for toxicokinetic analysis on GD 7 and GD 19 in additional three pregnant satellite females per group in the low, intermediate and high dose groups.

Results:

Mortality -No morbidity or deaths were attributed to treatment with the test item occurred. Three control females and four intermediate-dose females were killed on the advice of the veterinary surgeon due to very low food consumption at the start of the study.

Clinical signs - All of the animals that were euthanized due to very low food consumption appeared thin, and of these animals one control female and three intermediate-dose females exhibited distended urinary bladders upon euthanasia and necropsy.

In the animals surviving until GD 29, one high-dose animal appeared thin between GD 25 and GD 29 coinciding with weight loss and reduced food intake during this period. Also, at necropsy for this animal, the liver was pale and mottled.

Body weight - There were no changes in mean body weight that were considered related to test item administration. Also, there were no Tafluprost (AFP-168)-related changes in mean gravid uterus weight.

Food consumption - As noted above, the food intake of three control animals and four intermediate-dose animals was very low from GD 4 and these animals were euthanized. Also, low food intakes were observed for two high-dose females and two vehicle control females between GD 23 and 29, but these changes were not considered related to the test item administration.

Toxicokinetics - No quantifiable levels of Tafluprost (AFP-168) or Tafluprost acid (AFP-172) were observed in any of the treatment samples. All plasma concentrations were below the lower limit of quantification (LLOQ).

Necropsy:

Cesarean Section Data (Implantation Sites, Pre- and Post-Implantation Loss, etc.) - The mean number of fetuses, and the group mean pre- and post-implantation loss was not affected by the test item treatment.

Offspring (Malformations, Variations, etc.) - Mean fetal, litter, and placental weights were unaffected by treatment. External and/or visceral malformations were seen in 3, 4, 4, and 6 fetuses from the control, low, intermediate, and high-dose groups respectively. In addition, skeletal malformations were seen in 2, 8, 6, and 8 fetuses from the same respective groups. However, for each individual malformation the incidence was 0 or 1 per Tafluprost (AFP-168) treatment group with a mean percent $\leq 2\%$ of the fetuses in each group. No individual malformation was significantly increased in any of the test item-treatment groups compared to the vehicle control group.

The number of fetuses with external, visceral, and skeletal variations in all the test item-treatment groups was generally similar to that of the controls. The number of fetuses in the treatment groups with abnormally pale contents in the gall bladder and/or non-eruption of the incisors were higher than in the controls; however, the differences were not statistically significant.

Conclusion:

When Tafluprost (AFP-168) was investigated again for its possible effect on embryo-fetal development in female rabbits a second study at daily intravenous (i.v.) doses of 0, 0.001, 0.003 and 0.01 $\mu\text{g/kg bw/d}$ from gestation day (GD) 7 to 19, there were no dose-related deaths or abortions and there were no effects on clinical signs, body weights and food intake. At caesarean section, no effects on reproductive parameters including post-implantation loss were observed.

Mean fetal, litter and placental weights were unaffected by treatment. No individual visceral or skeletal malformation was statistically significantly increased in the Tafluprost (AFP-168) treatment groups.

Thus, the NOAEL for maternal and prenatal developmental toxicity was 0.01 $\mu\text{g/kg bw/d}$, the highest dose investigated in this second rabbit study. Importantly, at this dose, plasma concentrations of Tafluprost acid (AFP-172) were consistently below the limit of detection (20 pg/ml).

Study 4:**Study for effects of Tafluprost (AFP-168) on the pre-and postnatal development including maternal function in rats via intravenous administration:****Material and methods:**

The possible effect of Tafluprost (AFP-168, lot # 037010, purity: 101.5%%) on pre- and postnatal development including maternal function was investigated in female Sprague-Dawley rats (CrI:CD®(SD)IGSBR) under GLP conditions. Each 22 pregnant females per group received daily intravenous (i.v.) doses of 0, 0.3, 1, 3, and 10 µg/kg bw/d from gestation day (GD) 6 until lactation day (LD) 20, which covered the period from implantation up to weaning. The dosing volume was 3 mL/kw bw. F0 dams, F1 and F2 generation animals were comprehensively investigated for clinical effects, developmental landmarks including neurological assessment and were subjected to gross pathology.

Results:**F0 Dams**

Survival - One high-dose female died during the course of the study due to unknown causes.

Clinical signs - Only loss of fur was noted in one animal. No Tafluprost (AFP-168)-related clinical signs were noted.

Body weight - No test item-related changes in body weight were noted.

Feed consumption - The females receiving 3 µg/kg bw/d demonstrated a significant, transient decrease in food consumption from LD 4 to LD 7, but due to a lack of a similar effect in the high-dose group, this decrease did not appear to be treatment-related.

Uterine content - There were no significant differences between the control and test item treatment animals with regard to the number of implantation sites, total number of newborns, delivery index, or proportion of male live newborns.

Necropsy observation - Dams that were necropsied on Day 21 did not exhibit any test item-related or abnormal gross pathology findings.

Delivery and Gestation Parameters - One 3 µg/kg bw/d female delivered her newborn late on GD 25. However, no abnormal delivery was evident in any dam, and there were no significant differences between any of the Tafluprost (AFP-168)-treatment groups in the duration of gestation, gestation index, or delivery index.

Nursing Parameters - One dam in each of the 0.3, 3, and 10 µg/kg bw/d groups and 3 dams in the 1 µg/kg bw/d group exhibited poor nursing behavior on LD 0, 1, or 2. Poor nursing behavior resulted in the deaths of all offspring from 2 dams in each of the 1, 3, and 10 µg/kg bw/d groups within 2 days after birth.

F1 Generation:**Survival:**

At birth - The high-dose (10 µg/kg bw/d) group revealed a greater number of dead newborns, and a lower birth index compared to controls. This effect was not statistically significant, but it was considered test item related. There were no significant differences between any of the Tafluprost (AFP-168)-treatment groups and the control group in the number of implantation sites, total number of newborns, delivery index, or male proportion of live newborns.

During Lactation - The viability index on LD 4 was lower in the 1, 3, and 10 µg/kg bw/d groups compared to the control group with a significant reduction for the high-dose group. This effect

was considered to be test item related. The weaning index was comparable between all the Tafluprost (AFP-168)-treatment groups and the control group.

Clinical signs:

Pre-weaning - Some of the newborns in the 1, 3, and 10 µg/kg bw/d groups exhibited no milk in the stomach at 0, 1, or 2 days of age. All of the newborns from 2 dams in each of these groups died within 2 days of birth. Some of the offspring from 2 or 3 dams in the control and 0.3, 1, and 3 µg/kg bw/d groups died during the lactation period as did some of the offspring from 10 dams in the 10 µg/kg bw/d group.

Post-weaning - No abnormal clinical signs were evident in any offspring after weaning.

Body weight:

Pre-weaning: Body weights of male and female newborns in the 10 µg/kg/day group were significantly lower than control offspring and this difference was considered AFP-168-related. However, the offspring in all groups demonstrated similar body weights after ≥ 4 days of age.

Post-weaning: Body weights of female offspring in the 10 µg/kg group were significantly but transiently lower than control group animals at 28 days of age. Because of its transient nature, this effect was not considered to be AFP-168-related.

Physical development:

Pre-weaning - The incidence of pinna unfolding at 3 days of age was statistically lower in the high-dose group (56.5%) compared to the control group (95.5%). This effect was considered related to growth retardation associated with the low birth weights in the high-dose group (10 µg/kg bw/d). At ≥ 4 days of age, all of the groups demonstrated approximately 100% incidence of pinna unfolding.

Post-weaning - All F1 offspring exhibited positive visual placing responses, pupillary reflexes, Preyer's reflexes and pain responses.

Neurological assessment:

Pre-weaning - There was no significant difference between the control and the test item-treated groups for the incidence of back righting or negative geotaxis before weaning.

Post-weaning - All F1 offspring exhibited positive visual responses, pupillary reflexes, Preyer's reflexes, and pain responses.

Gross Pathology

Post-weaning - No gross pathology considered related to the test item treatment was observed in any male or female offspring culled at four days of age or in offspring at 21 days of age. A few Tafluprost (AFP-168)-treated animals at both ages demonstrated dilation of the renal pelvis, but the incidence was low in each group and the effect was not dose-related. No gross pathology was evident at 10 weeks of age.

Reproduction - No Tafluprost (AFP-168)-related changes in preputial separation or vaginal opening were observed. Also, there were no significant differences between any of the AFP168-treatment groups and the control group for mating or the fertility index.

Other - No test item-related gross pathology was noted in males after mating. Also, no abnormal clinical signs or gross pathology were evident in any female, including non-pregnant females. No significant differences in F1 female body weights were noted during gestation for the AFP-168 treatment groups versus the control group.

F2 Generation:

Survival - There were no significant differences between the control and Tafluprost (AFP-168) treatment groups for embryonic mortality.

Other - There were no significant differences between the control and test item treated groups in the number of corpora lutea, implantations, live F2 embryos, or pre-implantation loss.

Conclusion:

According to the Applicant, when Tafluprost (AFP-168) was investigated for its possible effect on pre- and postnatal development including maternal function in female Sprague-Dawley at daily rats intravenous (i.v.) doses of 0, 0.3, 1, 3, and 10 µg/kg bw/d from GD 6 until LD 20, no treatment related maternal deaths and no effects on body weight, food consumption, nor findings at necropsy. Some dams receiving 1 µg/kg bw/d Tafluprost (AFP-168) and higher doses exhibited poor nursing behavior, which resulted in decreased viability indices on day 4 of lactation to offspring. Increased mortality of newborns, decreased body weights, and delayed pinna unfolding at Day 3 of lactation were noted in offspring in the 10 µg/kg/day group. No other treatment-related effects were evident in physical development, sensory functions, genital development and fertility of F1 offspring in any treatment group. Thus, the NOAEL values were considered to be 10 µg/kg bw/d for maternal toxicity in F0 dams and 0.3 µg/kg bw/d for pre- and postnatal development toxicity in the F1 generation animals. For the F2 generation animals, there were no significant differences between the control and all Tafluprost (AFP-168) treatment groups in the number of corpora lutea, implantations, live F2 embryos, pre-implantation loss or embryonic mortality and therefore the respective NOAEL for pre- and postnatal developmental toxicity was 10 µg/kg bw for the F2 animals.

SCCS comments

Both Applicants provided *in vivo* developmental/reproductive studies for the compound Tafluprost. As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, these results have not been taken into consideration for this Opinion.

Further remarks for the Tafluprost studies:

From the studies on Tafluprost by both Applicants, it appears that of the species examined (rat, rabbit), the rabbit is a more sensitive species than the rat:

- Applicant 1 determined a LOAEL for developmental toxicity of 1µg/kg bw/ for rabbit versus a NOAEL of the same value for rat.
- Applicant 2 observed no prenatal toxicity in the rat when the compound was given directly after mating at 3µg/kg bw/d, whereas in the rabbit even 0.03µg/kg bw/d was detrimental and caused spontaneous abortion.
- When the protocol was adapted in later dosing, the NOAEL for the rat was found to be 10µg/kg bw/for pre-and postnatal developmental toxicity. For the rabbit the NOAEL was 0.001µg/kg bw/d for maternal toxicity and prenatal development since still visceral changes and skeletal malformations were observed, although not significant according to the Applicant

The question about species difference for maternal toxicity and prenatal toxicity is important and cannot be ignored since human exposure will include women of child-bearing age. Data for humans are essential, certainly in the light of prostaglandins in general being known for their effects on uterine contractions. In that respect, it was shown in the literature via *in vitro* tests that prostaglandin F2 alpha analogues can cause contractions of uterine smooth muscle which could theoretically cause miscarriages (Sakai et al. 2022). Of course, exposure concentration plays a pivotal role.

Furthermore, the studies by Applicant 2 show that the effects of Tafluprost on uterine contractions (abortions in the animals) are very dependent on the protocols used, which creates a high uncertainty for the human situation.

Seen that there is a lack of human data for reprotoxicity/developmental toxicity and a high uncertainty about species differences in terms of species sensitivity, methodology and occurrence of side effects, Tafluprost can only be considered a risk for the unborn child at any concentration.

For DDDE, no developmental/reproductive toxicity data are available, which are considered to be crucial for its safe use.

Ref.: (Sakai et al., 2022)

5.3.7 Mutagenicity / genotoxicity**5.3.7.1 Mutagenicity / genotoxicity *in vitro*****Applicant 1****Study 1:**

Guideline:	OECD Guideline 471(2020)
Test system:	<i>Salmonella typhimurium</i> strains TA 1535, TA 1537, TA 98 and TA 100, <i>Escherichia coli</i> WP2 uvrA (pKM101)
Replicates:	Triplicate
Test substance:	DDDE (neat oil) CAS 1185851-52-8
Vehicle:	DMSO
Batch:	TAF-10-1122-01
Purity:	99.78%
Test concentrations:	Experiment I: plate incorporation method 3.16, 10.0, 31.6, 100, 316, 1000, 2500 and 5000 µg/plate (TA100) 31.6, 100, 316, 1000, 2500 and 5000 µg/plate (TA98, TA1535, TA1537, E. coli WP2 uvrA (pKM101)) Experiment II: pre-incubation method 3.16, 10.0, 31.6, 100, 316, 1000, 2500 and 5000 µg/plate (TA100, TA1535 [without S9-mix]) 31.6, 100, 316, 1000, 2500 and 5000 µg/plate (TA98, TA1535 [with S9-mix], TA1537, E. coli WP2 uvrA (pKM101))
Negative control:	Purified water and DMSO
Positive control:	4-nitro-o-phenylene-diamine, methylmethanesulfonate sodium azide (without S9-mix); 2-aminoanthracene (with S9-mix)
Metabolic activation:	S9 liver microsomal fraction prepared from rats in house or obtained from Trinova Biochem (also from rats) GmbH, Gießen, Germany
GLP:	Yes
Study period:	2023

The mutagenicity potential was evaluated for the test substance DDDE in an *in vitro* bacterial reverse mutation assay, according to OECD Guideline 471 in compliance with GLP. In this assay, the test substance DDDE was tested according to the plate incorporation method (experiment I) and the preincubation method (experiment II) using *Salmonella typhimurium* strains TA98, TA100, TA1535, TA1537 and tester strain E. coli WP2 uvrA (pKM101). The test concentrations ranged from 3.16 to 5000 µg/plate in both experiment I and II.

Results:

No precipitation of the test substance was observed in any tester strain used in experiment I and II (with and without S9-mix).

In experiment I, toxic effects of the test substance were observed in tester strains TA98, TA100 and TA1535 (with and without metabolic activation) and in tester strain E. coli WP2 uvrA (pKM101) (without metabolic activation) at concentrations of 2500 µg/plate and higher. In experiment II, toxic effects of the test substance were noted at concentrations of ≥316 µg/plate (with and without S9- mix), depending on the particular tester strain (*i.e.*, at ≥1000 µg/plate in the strain TA 98; ≥316 µg/plate in the strain TA 100 and TA1535; ≥2500 µg/plate in the strains TA 1537 and E. coli WP2 uvrA)

No biologically relevant increases in revertant colony numbers of any of the five tester strains

were observed following treatment with the substance at any concentration level, neither in the presence nor absence of metabolic activation in experiments I and II.

Overall, it can be stated that during the described mutagenicity test and under the experimental conditions reported, the test substance did not cause gene mutations by base pair changes or frameshifts in the genome of the tester strains used.

Conclusion:

Under the conditions of the study, DDDE was not mutagenic in the bacterial reverse mutation assay (Ames test), neither in the presence nor absence of metabolic activation.

Ref.: (Klock, 2023b)

Study 2:

Guideline:	OECD Guideline 487 (2016)
Test system:	Human lymphocytes
Replicates:	Duplicate
Test substance:	DDDE (neat oil) CAS 1185851-52-8
Vehicle:	DMSO
Batch:	TAF-10-1122-01
Purity:	99.78% (1.22 % NMT Impurity)
Test concentrations:	Pre-Experiment: 7.8, 15.6, 31.3, 62.5, 125, 250, 500, 1000, 1500, 2000 µg/mL (with and without S9-mix) Experiment I: Short term exposure 250, 325 and 350 µg/mL (without S9-mix) 100, 250 and 300 µg/mL (with S9-mix) Experiment II: Long term exposure 25, 50, 100µg/mL (without S9-mix)
Negative control:	Cell culture medium
Positive control:	Methylmethanesulfonate (MMS) and colchicine (without S9-mix) Cyclophosphamide (CPA) (with S9-mix)
Metabolic activation:	S9 liver microsomal fraction prepared from rats in house
GLP:	Yes
Study period:	2023

The genotoxicity of the test substance DDDE was evaluated in an OECD Guideline 487 and GLP- compliant *in vitro* MNT using human lymphocytes. In this assay, the test substance, dissolved in DMSO, was tested in the presence and absence of metabolic activation (*i.e.*, phenobarbital/β-naphthoflavone-induced rats' liver S9-mix). The selection of the concentrations was based on data from the pre-experiment. In the main experiment, the test concentrations in experiment I with 4 h short-term exposure were 250, 325 and 350 µg/mL, without S9 mix and 100, 250 and 300 µg/mL, with S9 mix. In experiment II, with 44 h long-term exposure, the test concentrations were 25, 50 and 100 µg/mL, without S9 mix.

Methylmethanesulfonate (50 and 65 µg/mL) and cyclophosphamide (15 µg/mL) were used as clastogenic controls. Colchicine (0.02 and 0.4 µg/mL) was used as aneugenic control. A solvent control (DMSO) was also included in the test.

Results:

No precipitation of the test substance was observed up to the highest concentration used in experiments I (with and without S9-mix) and II (without S9-mix).

In experiment I (without S9-mix), no increase of the cytostasis above 30% was noted up to 250 µg/mL. At 325 and 350 µg/mL, a cytostasis of 37 and 55%, respectively, was observed. In experiment I (with S9-mix), no increase of the cytostasis above 30% was noted up to 100 µg/mL. At 250 and 300 µg/mL, a cytostasis of 44 and 59%, respectively, was observed.

In experiment II (without S9-mix), no increase of the cytostasis above 30% was noted up to 50 µg/mL. At 100 µg/mL, a cytostasis of 66% was observed.

The numbers of micronucleated cells were within the historical control limits of the solvent control, and they did not show a biologically relevant increase compared to the concurrent solvent control in both experiments I and II.

The micronucleated cell frequency of the negative and solvent control was also within the historical control limits. Clastogenic (methylmethanesulfonate, cyclophosphamide) and aneugenic positive controls (colchicine) induced distinct and statistically significant increases in the micronucleus frequency, demonstrating the validity of the assay.

Overall, treatment with test substance DDDE resulted did not result in a statistically significant increase in the number of cells with micronuclei compared to concurrent solvent controls in experiments I and II, both in the presence and absence of an S9-mix.

Conclusion:

Under the conditions of the study, the test substance DDDE did not induce structural and/or numerical chromosomal damage in human lymphocytes.

Ref.: (Klock, 2023c)

Applicant 2

Study 1: Gene mutation assays using bacteria (Ames)

Study Design:

Reference:	ToxiCoop, 2023a
Date of report:	11.01.2023
Study period:	08.11.2022 – 01.12.2022
Guidelines/Methods:	OECD 471, Council Regulation (EU) No 440/2008 B13/14 Mutagenicity, SCCS Notes of Guidance, 11th revision, 2021
Test system:	<i>Salmonella typhimurium</i> strains TA98, TA100, TA1535, TA1537, <i>Escherichia coli</i> WP2 uvrA
Replicates:	Triplicates
Test substance:	Ethyl Tafluprostamide, CAS 1185851-52-8
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	DMSO
Concentrations:	Plate incorporation and preincubation tests: 0, 5, 16, 50, 160, 500, 1600, 5000 µg/plate The same concentrations were tested with and without S9
Positive controls:	-S9-mix: Sodium azide (SAZ): 2 µg/plate for TA100, TA1535 4-Nitro-o-phenylenediamine (NPD): 4 µg/plate for TA98 9-Aminoacridine (9AA): 50 µg/plate for TA1537 Methyl methanesulfonate (MMS): 2 µL/plate for WP2 uvr A +S9-mix: 2-aminoanthracene (2-AMA): 2 µg/plate for TA98, TA100, TA1535, TA1537, 50 µg/plate for WP2 uvr A
Metabolic activation:	Phenobarbital (PB) and β -naphthoflavone (BNF)-induced rat liver S9 was provided by Trinova Biochem

Negative control:	Vehicle and DMSO
GLP:	Yes
Published:	No
Study period:	2022

Material and methods:

Ethyl Tafluprostamide (batch: 0652603-2, purity: 98.5%) was investigated for its possible mutagenic potential in *Salmonella typhimurium* (*S. typhimurium*) strains TA98, TA100, TA1535, TA1537 and *Escherichia coli* (*E. coli*) strain WP2 uvrA according to OECD 471 under GLP conditions. The tested concentrations of the test item dissolved in DMSO were 0, 5, 16, 50, 160, 500, 1600, 5000 µg/plate for all strains with and without the addition of a rat liver homogenate metabolizing system (S9 fraction of Phenobarbital (PB) and β-naphthoflavone (BNF)-induced rat liver). Three replicates for each concentration level and controls were used. The study included preliminary experiments (solubility trial, solubility test and a concentration range finding test (informatory toxicity test)), an initial mutation test (plate incorporation test) and a confirmatory mutation test (pre-incubation test).

Results:

The test item was adequately soluble in dimethyl sulfoxide (DMSO) at 50 mg/mL and a clear, transparent and homogeneous solution was gained. No precipitation of the test item was observed on plates in the tested bacterial strains at any examined concentration levels with and without metabolic activation.

Bacteriotoxicity in the initial test was observed in all *Salmonella typhimurium* strains at the highest tested concentration (5000 µg/plate) in the absence and/or presence of S9 and also in *Salmonella typhimurium* TA98 at 1600 µg/plate (-S9). In the confirmatory mutation test, bacteriotoxicity was observed in all *Salmonella typhimurium* strains at the highest two tested concentrations (5000 and 1600 µg/plate) in the absence and presence of exogenous metabolic activation system (±S9) as well as in *Escherichia coli* WP2 uvrA at the highest tested concentration which was 5000 µg/plate (±S9).

Regarding mutagenicity, (±S9) no substantial increases in revertant colony numbers were observed in any of the five tester strains following treatment with Ethyl Tafluprostamide at any concentration level, neither in the presence nor absence of metabolic activation system (S9). Sporadic increases in revertant colony numbers compared to the vehicle control values were observed in all independently performed experiments. However, there was no tendency of higher mutation rates with increasing concentrations beyond the generally acknowledged border of biological relevance in the performed experiments.

The revertant colony numbers of vehicle control (DMSO) plates with and without S9 demonstrated the characteristic mean numbers of spontaneous revertants that were in line with the corresponding historical control data ranges in all cases. The positive controls showed the expected biological relevant increases in induced revertant colonies and all validity criteria were met indicating the suitability and sensitivity of the test system.

Conclusion:

Under the conditions of this test, Ethyl Tafluprostamide did not induce gene mutations by frameshift or base-pair substitution in the genome of the tester strains, when tested with and without metabolic activation up to cytotoxic and/or regulatory compliant concentrations. Thus, Ethyl Tafluprostamide is considered to reveal no mutagenic potential in this bacterial reverse mutation assay.

Ref.: (ToxiCoop, 2023a)

Study 2: Mammalian Cell Micronucleus Test (MNT)**Study Design:**

Reference:	ToxiCoop, 2022d
Date of report:	05.12.2022
Study period:	04.10.2022 – 05.10.2022
Guidelines/Methods:	OECD 487, Council Regulation (EC) No 2017/735, Annex Part B, B.49, SCCS Notes of Guidance, 11th revision, 2021
Test system:	L5178Y tk+/- 3.7.2c mouse lymphoma cell line
Replicates:	Duplicates
Test substance:	Ethyl Tafluprostamide, CAS 1185851-52-8
Batch (Purity):	0652603-2 (purity: 98.5% (HPLC))
Vehicle:	DMSO
Concentrations and exposure:	4/24 h exposure/sampling time without S9-mix: 4.4, 8.8, 17.6, 35, 70, 140, 150* and 160* µg/mL 4/24 h exposure/sampling time with S9-mix: 17.6, 35, 70, 130, 140 and 150 µg/mL 24/24 h exposure/sampling time without S9-mix: 2.2, 4.4, 8.8, 17.6 and 35* µg/mL test item (*Due to high cytotoxicity level at indicated concentrations, evaluation was not performed)
Positive controls:	-S9-mix: 4-Nitroquinoline N-Oxide (NQO): 0.025 µg/mL Colchicine (COL): 0.04 µg/mL (4-h exposure), 0.011 µg/mL (24 h exposure) +S9-mix: Cyclophosphamide monohydrate (CP): 3.13 µg/mL
Metabolic activation:	Phenobarbital (PB) and β-naphthoflavone (BNF)-induced rat liver S9 was provided by Trinova Biochem
Negative control:	Vehicles
GLP:	Yes
Published:	No

Material and methods:

The possible clastogenic and aneugenic potential of the test Ethyl Tafluprostamide (0652603-2 (purity: 98.5%)) dissolved in DMSO was evaluated according to OECD 487 under GLP conditions by flow cytometric investigating its effects on the frequency of micronuclei in cultured L5178Y cell line cells. The cells were exposed to the test item at concentrations of 0, 4.4, 8.8, 17.6, 35, 70, 140, 150 and 160 µg/mL for 4 hours without S9, at 17.6, 35, 70, 130, 140 and 150 µg/mL for 4 hours with S9 and at 2.2, 4.4, 8.8, 17.6 and 35 µg/mL for 24 hours without S9 in duplicates. The concentrations were selected on the basis of cytotoxic investigations in a preliminary study with and without metabolic activation using S9 mix of phenobarbital and β-naphthoflavone induced rat liver. During the exposure period, the cells was covered with medium containing the test item. The exposure period was 4 and 24-hours in the absence and 4-hours in the presence of S9 mix at 37°C in a humidified atmosphere of 5% CO₂. The exposure period was followed by washing the cells with RPMI 5 and 10 mL of growth medium (RPMI 10) was added for the further incubation until the cell harvest. Sampling was made after 24 hours to enable approximately 2 normal cell cycles from the beginning of the exposure. Toxicity was determined by cell counting. Based on the cell counts relative increase in cell counts (RICC) was calculated. The pH and osmolality of each treatment solution was measured at the start of the treatment. In total 20000 SYTOX positive and EMA negative cells were analyzed for micronucleus formation. Appropriate positive control substances (-S9-mix: 4-Nitroquinoline N-Oxide (NQO), Colchicine (COL) and +S9-mix Cyclophosphamide monohydrate (CP)) and the vehicle as negative control were examined in parallel.

Results:

There was no precipitation of the test item at any dose level tested. No biologically relevant

changes in pH or osmolality of the test system were noted at the different dose levels tested. After 4-hour treatments, clear cytotoxicity was observed at 140 µg/mL in the absence of metabolic activation and at 150 µg/mL in the presence of metabolic activation. Clear cytotoxicity was also observed in the 24-hour treatment at 17.6 µg/mL.

Therefore, doses of 150 and 160 µg/mL in the absence of the metabolic activation system at the 4- hours treatment and dose of 35 µg/mL in the absence of the metabolic activation system at the 24-hours treatment were not evaluated due to high cytotoxicity level.

Beside the observed clear cytotoxic response, in this *in vitro* mammalian cell micronucleus test, number of micronuclei showed occasionally statistically significant increases compared to the concurrent and historical controls after short-term treatment only. However, when Tafluprost Ethyl Amide was examined both in the absence and in the presence of metabolic activation (at 4-hours treatments), these increases were considered as not biologically relevant. Even though the test item showed a statistically significant increase compared to the concurrent solvent at the clearly and required 55+/-5% cytotoxic concentration at 140 µg/mL in the absence of the metabolic activation system at the 4-hours treatment. Noteworthy that the response was not consistent as there were clear differences between replicate A and B. At concentrations of 35 and 130 µg/mL in the presence of metabolic activation system at 4-hours treatment statistically significant increases were evident but none could be noted below (at 17.6 µg/mL), in between (70 µg/mL) or above (140 and 150 µg/mL). Thus, clearly no concentration-dependency was given under these conditions. Finally, all of these occasionally recorded increases were clearly within the historical control ranges. Therefore, these were considered as biologically not relevant and as not indicative for a clastogenic/genotoxic potential.

At 24-hours treatments in the absence of metabolic activation Tafluprost Ethyl Amide led to no statistically significant or biologically relevant increases in the number of micronuclei in comparison to the concurrent or historical controls.

The number of MN of the concurrent negative control groups were in the historical control range. However, the mean values of the concurrent controls were generally lower than the historical control. The difference attaining statistical significance for the 24- hour treatment/sampling without S9-mix.

The positive controls 4-Nitroquinoline N-Oxide (0.025 µg/mL/ both 4-, and 24-hour treatments), Colchicine (0.040 µg/mL / 4-hour and 0.011 µg/mL/ 24-hour treatment) and Cyclophosphamide (3.13 µg/mL/ 4-hour treatment) caused expected biologically relevant increases of cells with micronuclei compared to the solvent. Even though, 4-Nitroquinoline N-Oxide at 4-hour treatment and Colchicine at 24-hour treatment showed lower genotoxicity it is acceptable due to lower cytotoxicity (approx. 30%). Thus, the suitability and sensitivity of the test system was demonstrated and the study is considered valid.

Conclusion by Applicant:

Under the conditions of this study, Ethyl Tafluprostamide induced clear cytotoxicity at the high concentrations and led occasionally to statistically significant increases in breakage and /or chromosomal loss in L5178Y cells after 4-hours treatments in the absence and in the presence of metabolic activation. However, these increases were neither consistent nor concentration-dependent and in any case not different from the historical control ranges. Therefore, these observations were finally considered as biologically not relevant.

After 24-h treatments clearly no increases in micronucleus formation in L5178Y cells in the absence of metabolic activation was noted up to the highest tested concentration, which was proven to induce clear cytotoxicity.

Finally, the test item was considered to have no selective genotoxic/clastogenic potential in this micronucleus test in L5178Y cells, when tested up to cytotoxic concentrations.

Ref.: (ToxiCoop, 2022d)

Opinion on Prostaglandin analogues used in cosmetic products

4-hour treatment without S9-mix / 24-hour sampling time

Test group	Dose	S9-mix	Treatment/ sampling time (h)	Replicate	Total Number of Nucleated Events	Total Number of MN Events	MN Frequency
Solvent control (DMSO)	10 µL/mL	–	4/24	A	10000	37	0.4
	10 µL/mL	–	4/24	B	10000	60	0.6
	Mean				10000	49	0.5
Tafluprost ethyl amide	4.4 µg/mL	–	4/24	A	10000	34	0.3
	4.4 µg/mL	–	4/24	B	10000	44	0.4
	Mean				10000	39	0.4
	8.8 µg/mL	–	4/24	A	10000	36	0.4
	8.8 µg/mL	–	4/24	B	10000	88	0.9
	Mean				10000	62	0.5
	17.6 µg/mL	–	4/24	A	10000	41	0.4
	17.6 µg/mL	–	4/24	B	10000	51	0.5
	Mean				10000	46	0.5
	35 µg/mL	–	4/24	A	10000	52	0.5
	35 µg/mL	–	4/24	B	10000	40	0.4
	Mean				10000	46	0.5
	70 µg/mL	–	4/24	A	10000	39	0.4
	70 µg/mL	–	4/24	B	10000	38	0.4
	Mean				10000	39	0.4
	140 µg/mL	–	4/24	A	10000	104	1.0
	140 µg/mL	–	4/24	B	10000	58	0.6
	Mean				10000	81 ¹	0.8
4-Nitroquinoline N- Oxide	0.025 µg/mL	–	4/24	A	10000	102	1.0
	0.025 µg/mL	–	4/24	B	10000	86	0.9
	Mean				10000	94**	0.9
Colchicine	0.040 µg/mL	–	4/24	A	10000	188	1.9
	0.040 µg/mL	–	4/24	B	10000	194	1.9
	Mean				10000	191**	1.9

¹= p < 0.01 to the concurrent negative control and p < 0.05 to the historical negative control

**=p < 0.01 to the concurrent negative control and to the historical negative control

4-hour treatment with S9-mix / 24-hour sampling time

Test group	Dose	S9-mix	Treatment/ sampling time (h)	Replicate	Total Number of Nucleated Events	Total Number of MN Events	MN Frequency
Solvent control (DMSO)	10 µL/mL	+	4/24	A	10000	45	0.5
	10 µL/mL	+	4/24	B	10000	50	0.5
	Mean				10000	48	0.5
Tafluprost ethyl amide	17.6 µg/mL	+	4/24	A	10000	45	0.5
	17.6 µg/mL	+	4/24	B	10000	47	0.5
	Mean				10000	46	0.5
	35 µg/mL	+	4/24	A	10000	100	1.0
	35 µg/mL	+	4/24	B	10000	59	0.6
	Mean				10000	80**	0.8
	70 µg/mL	+	4/24	A	10000	46	0.5
	70 µg/mL	+	4/24	B	10000	31	0.3
	Mean				10000	39	0.4
	130 µg/mL	+	4/24	A	10000	100	1.0
	130 µg/mL	+	4/24	B	10000	60	0.6
	Mean				10000	80**	0.8
	140 µg/mL	+	4/24	A	10000	45	0.5
	140 µg/mL	+	4/24	B	10000	51	0.5
	Mean				10000	48	0.5
	150 µg/mL	+	4/24	A	10000	61	0.6
	150 µg/mL	+	4/24	B	10000	32	0.3
	Mean				10000	47	0.5
Cyclophosphamide	3.13 µg/mL	+	4/24	A	10000	646	6.5
	3.13 µg/mL	+	4/24	B	10000	602	6.0
	Mean				10000	624**	6.2

**p < 0.01 to the concurrent negative control and to the historical negative control

Study 3: Ames test (Tafluprost)

Tafluprost (APF-168; purity 99%) at concentrations as high as 5000 µg/plate did not induce mutations in *Salmonella typhimurium* strains TA98, TA100, TA1535 and TA1537 or in the *Escherichia coli* (E. coli) strain WP2 uvrA in the presence and absence of metabolic activation (S9-mix).

(Ref: CDER 2011)

Study 4: In vitro assay in mammalian cells (Tafluprost)

Chinese hamster lung cells (CHL) treated with Tafluprost (AFP-168; purity 99%) in the absence and presence of metabolic activation under all treatment conditions exhibited frequencies of cells with structural aberrations, which were similar to those observed in concurrent negative control cultures, which in all but one case fell within the historical ranges for negative controls. Cultures treated with the test item for 3 hours in the absence of S9 mix resulted in cultures with increased frequencies of cells with polyploidy. However, for unclear reasons, the concurrent untreated control culture showed also polyploidy. Experiments with exposure of the cells for 20 hours in the absence of metabolic activation revealed clearly no effects regarding structural or numerical Aberrations (polyploidy). Therefore, the obtained results suggest that a factor other than Tafluprost (AFP-168) may have induced the observed polyploidy and Tafluprost was finally considered as negative when tested with/without metabolic activation up to cytotoxic concentration in CHL cells.

(Ref: CDER 2011)

SCCS comments

Full reports of studies 3 and 4 were not available. As they were conducted on Tafluprost, they cannot be considered as representative for DDDE.

Considering that the 2 Ames tests on the test substance DDDE and 2 *in vitro* micronucleus

tests were negative, there is no concern for genotoxicity of DDDE under the *in vitro* conditions applied.

5.3.7.2 Mutagenicity / genotoxicity *in vivo*

Applicant 2

In vivo clastogenicity assay in mice (Micronucleus assay):

When Tafluprost (AFP-168) was investigated for its potential to induce micronuclei *in vivo* in the bone marrow of treated male and female mice (outbred CD-1) at intraperitoneally injected doses of 0, 175, 350 and 700 mg/kg bw, the groups of mice receiving the test item exhibited polychromatic erythrocyte (PCE) to nonchromatic erythrocyte (NCE) ratios that were similar to or lower than values for the vehicle control group. Bone marrow from animals receiving the test item at 350 mg/kg bw (24-hour sample time) and 700 mg/kg bw (24- and 48-hour sample times) exhibited reduced PCE/NCE ratios indicative of bone marrow cytotoxicity. Group mean frequencies of micronucleated PCE for all the Tafluprost (AFP-168) treatment groups were similar to those of the vehicle control group at both sampling times. No significant differences were apparent. Thus, Tafluprost (AFP-168) it did clearly not induce micronucleated polychromatic erythrocytes indicative of clastogenicity in the bone marrow of mice at 24 hours and at 48 hours after single i.p. dosing.

(Ref: CDER 2011)

SCCS comments

The data available *in vivo* on the substance Tafluprost is not informative for the genotoxicity of DDDE without adequate justification for read-across. Also, the full study report was not available and the intraperitoneally route of exposure used is not relevant for cosmetic exposure.

Conclusion on Mutagenicity/ genotoxicity by the Applicant:

According to the Applicant, the genotoxicity of neat DDDE has been assessed in an Ames test as well as in an *in vitro* MNT using human lymphocytes. No mutagenic or clastogenic response was observed in these recent OECD guideline-compliant assays. This is further supported by the absence of genotoxicity observed for the analogue Tafluprost in an Ames test, *in vitro* chromosomal aberration assay and *in vivo* MNT in mice. Based on this information, DDDE is not assessed to be genotoxic.

Overall SCCS comment

Taking all available data into consideration, the SCCS is of the opinion that there is no concern regarding genotoxicity of DDDE.

5.3.8 Carcinogenicity

Applicant 1

Study 1: SAR analysis

The SAR analysis was conducted using the OECD QSAR Toolbox v.4.6 (OECD, 2023) and the expert rule-based tool, Derek Nexus tool from Lhasa (Barber et al., 2015) v.6.2.1. DDDE does not present any structural alert indicating a potential for carcinogenicity.

SAR analysis of DDDE

SAR tool	Profiler/Endpoint	Results
OECD QSAR Toolbox v.4.6	Carcinogenicity (genotox and non-genotox) alerts by ISS (v.2.7)	No alert found
	Oncologic Primary Classification (v.4.3)	Not classified
Derek Nexus v.6.2.1	Bladder urothelial hyperplasia	No alert
	Carcinogenicity	No alert
	Photocarcinogenicity	No alert

Study 2: QSAR analysis

Statistically-based QSAR analysis was conducted using the carcinogenicity models of the VEGA platform (Benfenati et al., 2013) v. 1.2 and the 'liver specific cancer (rat/mouse *in vivo*)' of the Danish (Q)SAR models platform (Danish (Q)SAR Databas, 2023).

As shown in the table below, predictions by the VEGA v.1.2 and the Danish (Q)SAR models are not considered reliable as all predictions are outside the applicability domain for most of the criteria set by each model, and the expert analysis of the nearest neighbours of the training set does not indicate an adequate similarity to DDDE.

QSAR analysis of DDDE

QSAR platform	Model	Results
VEGA v.1.2	Carcinogenicity model CAESAR v.2.1.10	Positive - Outside the applicability domain
	Carcinogenicity model (ISS) 1.0.3	Negative - Outside the applicability domain
	Carcinogenicity model (IRFMN-ISSCAN-CGX) 1.0.1	Positive - Outside the applicability domain
	Carcinogenicity model (IRFMN-Antares) 1.0.1	Positive - Outside the applicability domain*
	Carcinogenicity oral classification model (IRFMN) 1.0.1	Negative - Outside the applicability domain*
	Carcinogenicity inhalation classification model (IRFMN) 1.0.1	Negative - Outside the applicability domain*
Danish (Q)SAR models	Liver specific cancer (rat/mouse <i>in vivo</i>)	Inconclusive - Outside the applicability domain

* Based on expert judgement

Based on the experience acquired so far, the current QSAR models for carcinogenicity are not powered enough to provide reliable results.

Therefore, the carcinogenicity data available on the analogue Tafluprost was additionally used by the Applicant for assessing this endpoint.

Study 3: 18-month carcinogenicity study in mouse via the SC route (Tafluprost)

Guideline:	EMA note for guidance on carcinogenic potential (CPMP/SWP/2877/00) (TGA, 2012)
Species/strain:	Mice/CrI:CD-1(ICR)BR
Number of animals:	51/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Batch/Lot:	F01X010
Purity:	100.7%
Route:	Subcutaneous
Dose levels:	0, 10, 30 and 100 µg/kg bw/day
Dose volume:	10 mL/kg
Formulation/Vehicle:	The stock AFP-168 solution was 0.0015% AFP-168 dissolved in the vehicle (0.9% saline) 13-week range-finding study
Basis of dose selection	
Satellite groups:	Yes; 18 animals/sex/group for low dose group and 26 animals/sex/group for intermediate and high dose group respectively for toxicokinetic analysis:
Dual control:	Yes, two saline groups
Interim sacrifice:	No
Duration:	18 months
GLP:	Yes
Study deviation:	Yes, study deviations were not considered to have altered results.
Study year:	or integrity of the study 2004

The carcinogenicity study of Tafluprost (100.7% purity) was investigated in mice according to the EMA note for guidance on carcinogenic potential (CPMP/SWP/2877/00) (TGA, 2012). Based on the findings in a 13-week dose range finding study, CrI:CD-1(ICR)BR mice (51/sex/group) were dosed daily via the subcutaneous route at 0, 10, 30 and 100 µg/kg bw/day for 18 months. The dual control groups were used. The stability of dosing solutions was analytically confirmed. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. The moribund animals were sacrificed, and full necropsy were performed. Blood samples for haematological examination were collected from unfasted animals at the terminal sacrifice. At termination of the treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

No treatment-related mortality, clinical signs, haematology parameters, gross pathology or nonneoplastic histopathology were noted. In neoplastic histopathological evaluation, a non-significant increased incidence rates for neoplastic lesions were observed in the high-dose group when compared to the vehicle control groups. Under the study conditions, the NOAEL for systemic toxicity and carcinogenicity of the test substance was set at the highest tested dose of 100 µg/kg bw/day.

Ref.: (CDER, 2011)

Study 4: 2-years carcinogenicity study in rats (Tafluprost)

Guideline:	EMA note for guidance on carcinogenic potential (CPMP/SWP/2877/00)
Species/strain:	Rats/Crj:CD(SD)IGSBR
Number of animals:	60/sex/group
Test substance:	Tafluprost
Product name:	AFP-168
Batch/Lot:	037010
Purity:	101.5%
Route:	Subcutaneous
Dose levels:	0, 3, 9 and 30 µg/kg bw/day
Dose volume:	3 mL/kg
Formulation/Vehicle:	The AFP 168 stock solution was 0.0015% AFP-168 dissolved in the vehicle (isotonic sodium chloride solution).
Basis of dose selection	13-week range-finding study
Satellite group	Yes; 12 animals/sex/group for toxicokinetic analysis:
Dual control	Yes, two saline groups
Interim sacrifice	No
Duration:	2 years
GLP:	Yes
Study deviation	Yes, study deviations were not considered to have altered the results or integrity of the study
Study year:	2003

The carcinogenicity of the analogue Tafluprost (101.5% purity) was investigated in a study according to the EMA note for guidance on carcinogenic potential (CPMP/SWP/2877/00) (TGA, 2012). On the basis of the findings in a 13-week dose range finding study, Crj:CD(SD)IGSBR rats (60/sex/group) were dosed daily via subcutaneous route at 0, 3, 9 and 30 µg/kg bw/day for 2 years. The dual control groups were used. The stability of dosing solutions was analysed and proven to be stable. During the treatment period, animals were observed for clinical signs, mortality, body weight and food consumption at defined intervals. Animals that died during the study were necropsied and examined for gross pathology as soon as possible. Blood samples were obtained, when possible, from moribund animals, before the animals were euthanized. For both the animals that died and the moribund animals, following necropsy the organs were weighed, and histopathology sections were prepared. Blood samples for haematological examination were collected from animals one day after the termination of dosing. At termination of treatment, all animals were sacrificed and macroscopically examined, organs were weighed, and comprehensive histopathology was performed.

Only minor and/or dose-independent changes in mortality, clinical signs, and food consumption were noted. Significant decreases in body weight were noted in male and female animals at 30 µg/kg bw/day. Treatment-related significant changes in haematology parameters included decreased red blood cells and increased white blood cells in male animals at 30 µg/kg bw/day. However, the change in RBCs was considered to be related to hyperostosis and an associated decrease in bone marrow cavity volume. Increased white blood cell counts were noted in males in the 30 µg/kg bw/day group. This change was driven by extremely high values in two males suffering from leukaemia. However, overall, the incidence of leukaemia determined through histopathology was not increased in the 30 µg/kg group compared to the vehicle control groups.

Gross pathology did not show presence of any statistically significant or treatment related lesions in any organs.

Absolute and relative spleen weights were increased significantly in males at 30 µg/kg bw/day.

The increased spleen weights were influenced due to leukaemia in one male and more severe extramedullary haematopoiesis in males at 30 µg/kg bw/day. Also, absolute, and relative adrenal weights were increased significantly in males at 30 µg/kg bw/day. However, no correlation was observed with histopathological evidence of adrenal tumours in any dose groups.

The relative but not absolute weights of kidney liver, lungs, heart, and brain were significantly increased in males and/or females at 30 µg/kg bw/day. However, these relative organ weight changes were considered attributable to decreased body weight in the high-dose group and not considered to be a direct treatment-related effect. Other weight changes included decreased absolute brain weight and increased relative brain weight high-dose group animals were also considered to be attributed to decreased body weight.

In histopathological evaluation, none of the neoplastic lesions in any of the dose groups was considered to be treatment related. Some significant differences were noted for the incidence of adenocarcinoma and adenoma in the pars distalis of the pituitary between male animals in one or both of the vehicle control groups and one or more of the dose groups. Also, a significant increase in the specific mammary gland and uterine tumours in female rats in one or more dose groups compared to one of the vehicle control groups were noted. In general, no dose-related pattern was observed in tumour incidence changes and in some instances, lower tumour incidences were observed in the treatment groups compared to the controls. These comparisons suggest that the pituitary gland, mammary gland, or uterine tumour incidence changes were toxicologically insignificant. Overall, no treatment-related biologically significant changes in the incidence of neoplastic lesions were observed.

Treatment-related non-neoplastic histopathological changes were observed which included hyperostosis of the sternum and femur in some animals and increased incidence of extramedullary haematopoiesis in the spleen in males of all dose groups. This change was attributable to a decreased volume of the bone marrow cavity due to hyperostosis.

Under the study conditions, the NOAEL for carcinogenicity was set at 30 µg/kg bw/day. Taking into consideration the non-neoplastic histopathology changes in sternum and femur in some animals and increased incidence of extramedullary haematopoiesis in the spleen in males at all dose levels, the LOAEL for systemic toxicity of the test substance Tafluprost can be considered at 3 µg/kg bw/day.

Ref.: (CDER, 2011)

Applicant 2

Study 1: Carcinogenicity study of Tafluprost (AFP-168) in rats via subcutaneous (s.c.) administration:

After repeated S.C. injection of Tafluprost (AFP-168) to in male and female Sprague-Dawley **rats** at doses of 0, 3, 9 and 30 µg/kg bw/d for up to 24 months only minor and/or dose-independent changes were noted for mortality, clinical signs and food consumption for the test item treatment groups compared to the vehicle control groups. Body weights were significantly decreased in the high dose group (30 µg/kg bw/d) male and female rats. Selective Tafluprost (AFP-168) changes in hematology (decreased red blood cells, increased white blood cells), and organ weights (increased spleen and adrenal weights) were observed. Non-neoplastic findings included hyperostosis of the sternum and femur in some animals and increased incidence of extramedullary hematopoiesis in the spleen in males at all Tafluprost (AFP-168)-treated groups. Tafluprost (AFP-168) did not produce biologically significant changes in the incidence of neoplastic lesions. Thus, Tafluprost (AFP-168) was considered to reveal no carcinogenic potential in rats after invasive long-term s.c. treatment.

Study 2: 78-week subcutaneous (S.C.) administration oncogenicity study in the mouse:**Key Study Findings**

Daily subcutaneous administration of Tafluprost AFP-168 (10, 30, and 100 µg/kg/day) for 78 weeks to mice did not result in appreciable compound-related toxicity. No AFP-168- related changes were noted for mortality, body weights, clinical signs, hematology parameters, gross pathology, and non-neoplastic histopathology. In addition, incidence rates for neoplastic lesions were not significantly increased for the high-dose AFP-168 treatment group compared to the vehicle control groups.

Conclusion on carcinogenicity by the Applicant:

No carcinogenicity studies could be identified for DDDE. However, the absence of genotoxicity and structural alerts for carcinogenicity using *in silico* tools together with the absence of significant treatment-related tumorigenic potential of the analogue Tafluprost in two chronic carcinogenicity assays in mice and rats via subcutaneous routes at doses up to 100 µg/kg bw/day, indicate that there is only a low carcinogenicity concern for DDDE at the intended use concentration of 0.018%.

SCCS comments

As expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. However, since DDDE is not a genotoxic compound, the SCCS has no concern about genotoxic carcinogenicity.

Furthermore, Applicant 1 suggested a LOAEL for carcinogenicity for Tafluprost of 3µg/kg bw/d (much lower than 100µg/kg/d).

A summary of mutagenicity and carcinogenicity toxicity studies submitted by Applicants 1 and 2 is provided in this table:

	Studies	Test substance:	Result	Ref
GENOTOX				
Applicant 1	Study 1: Ames Guideline: OECD Guideline 471 Test system: Salmonella typhimurium strains TA 1535, TA 1537, TA 98 and TA 100, Escherichia coli WP2 uvrA (pKM101)	DDDE (neat oil) CAS 1185851-52-8	Negative Reliable	Klock, 2023b
	Study 2: MN <i>in vitro</i> Guideline: OECD Guideline 487 Test system: Human lymphocytes	DDDE (neat oil) CAS 1185851-52-8	Negative Reliable	Klock, 2023c
Applicant 2	Study 1: Gene mutation assays using bacteria (Ames)	Ethyl Tafluprostamide CAS 1185851-52-8	Negative Reliable	ToxiCoop, 2023a
	Study 2: Mammalian Cell Micronucleus Test (MNT)	Ethyl Tafluprostamide CAS 1185851-52-8	Negative	ToxiCoop, 2022d
	Study 3: Ames test	Tafluprost Cas 209860-87-7	Negative	CDER 2011
	Study 4: <i>In vitro</i> chromosomal aberration assay in mammalian cells	Tafluprost CAS 209860-87-7	Negative Not reliable due to polyploidy	

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			observed in negative controls and treated cells
	<i>In vivo</i> clastogenicity assay in mice (Micronucleus assay)	Tafluprost (AFP-168)	Negative
Carcinogenicity			
Applicant 1	QSAR	DDDE	Not reliable
	18-month carcinogenicity study in mouse via the SC route	Tafluprost	Non-significant increased incidence rates for neoplastic lesions the high-dose group after s.c.; NOAEL for systemic toxicity and carcinogenicity of the test substance was set at the highest tested dose of 100 µg/kg bw/day. (CDER, 2011; TGA, 2012)
	2-years carcinogenicity study in rats	Tafluprost	NOAEL for carcinogenicity was set at 30 µg/kg bw/day. The LOAEL for systemic toxicity of the test substance considered at 3 µg/kg bw/day. (CDER, 2011; TGA, 2012)
Applicant 2	Carcinogenicity study of Tafluprost (AFP-168) in rats via subcutaneous (s.c.) administration	Tafluprost	Negative FDA, 2011
	78-week subcutaneous (s.c.) administration oncogenicity study in the mouse	Tafluprost	Negative

DDDE is the molecule of concern. The data for Tafluprost cannot be accepted as read-across has not been done in a scientifically acceptable way.

Full reports are not available of the 'CDER' studies, and the studies that are available are concerned with a different substance, namely Tafluprost.

5.3.9 Photo-induced toxicity

5.3.9.1 Phototoxicity / photo-irritation and photosensitisation

Applicant 1

No publicly available photo-induced toxicity studies on DDDE could be identified. However, a recently conducted OECD Guideline 101 UV/VIS study with DDDE (neat oil) revealed an absorption band in the range 210 – 240 nm with maximum absorption at 226 nm and an absorption band in the range 250 – 285 nm with three maxima at 265 nm, 258 nm and 276 nm. The molar extinction coefficients (MCEs) were in the range 1046.2 to 1306.1 L*Mol⁻¹cm⁻¹.

1 for the three maxima.

As the MCE results are above the cut-off of $>1000 \text{ L/mol}\cdot\text{cm}^{-1}$, a photo-reactivity potential cannot be entirely ruled out. However, considering the maximum absorbance wavelength cut-off, which is below 313 nm, additional *in vitro* phototoxicity testing is not required as per the SCCS NoG (SCCS, 2023).

Ref.: (Johannes, 2023)

Applicant 2

No trigger to investigate this endpoint as Ethyl Tafluprostamide showed consistently three absorption maxima in the UV/VIS spectrum at 220 nm, at 270 nm and at 276 nm under acidic (pH 1.0), neutral (pH 7.4 – 7.7.) and alkalic (pH 13.8 – 14.1) conditions, respectively. At the wavelength of 300 nm and above no absorption was noted. According to EFSA and SCCS no phototoxicity testing should be performed in cases where a test material only absorbs at wavelengths lower than 313 nm, which is clearly the case with Ethyl Tafluprostamide.

Ref.: (Consilab, 2023a);(SCCS, 2023)

(Moreover, there is no indication to assume that Ethyl Tafluprostamide may exhibit phototoxic properties as no extended light absorbing chromophore is evident in the chemical structure.

5.3.9.2 Photomutagenicity / photoclastogenicity

/

5.3.10 Human data

Applicant 1

Clinical studies

The skin irritation and sensitisation potential of DDDE was evaluated in clinical studies. Under the test conditions, the eyelash cosmetic **products** containing up to 0.025% DDDE, were neither irritating nor sensitising to skin.

The eye irritation potential of DDDE was evaluated in clinical studies. Under the test conditions, eyelash cosmetic products containing up to 0.025% DDDE were not irritating to eyes.

Clinical trials data on tafluprost – Ophthalmic solution

The safety of test substance Tafluprost (0.001%-0.005% ophthalmic solution) was assessed in phases I, II and III clinical trials, in compliance with GCP and regulatory requirements. The phase II and phase III clinical trials performed in a masked way are the most important studies for assessing safety. The subjects received at least one dose of Tafluprost eyedrop at a concentration of 0.0015% concentration, directly to the surface of the eye. The studies assessed the preservative-containing and/or preservative-free formulation. Long-term data is derived from Study 15-003 over 12 months and Study 74458 over 24 months.

Subjects were observed for adverse events (AE), laboratory evaluations (blood chemistry, haematology, and urinalysis) and vital signs assessment. In addition, specific ocular safety assessments were conducted. The status of the cornea, conjunctiva, iris, lens, vitreous and the retina were investigated routinely by recording visual acuity, visual fields, slit-lamp microscopy and ophthalmoscopy. Conjunctival hyperaemia was examined using standard photos as reference. Iris, eyelash and eyelid changes were assessed from photos by an independent, masked assessor. Microscopy in selected centres was used to assess corneal endothelial cells in Study 74458 and 15-003. Aqueous flare was measured in selected centres in Study 74458.

Results:

In Phase I clinical trials (Study 15005 and 77551), the most prevalent adverse events (AEs) were ocular redness and ocular hyperaemia. There were neither serious adverse events nor premature discontinuations due to adverse events. No unexpected findings were detected in the ocular safety variables or systemic safety variables. Formulations were well tolerated and safe .

In Phase II/III trials (15002, 74458, 15003, 74460 and 77550), AEs occurred in 484/724 (66.9%) patients treated with Tafluprost and ocular AEs in 343/724 (47.4%) subjects. The most common AE in patients treated with 0.0015% Tafluprost was ocular/conjunctival hyperaemia, reported in 16% of subjects, followed by eye pruritus (7.5%), eye irritation (6.8%), eye pain (5.9%), growth of eye lashes (4.1%), visual field defects (4.1%), dry eye (3.9%) and blurred vision (3.3%). Most ocular AEs (77%) were considered by the investigators to be related to study medication.

Systemic adverse events included headache was the most commonly reported at 6.9%, followed by nasopharyngitis (6.6%), cough (4.0%), and hypertension (4.0%). No marked effects were observed on blood pressure, heart rate or laboratory examinations.

Conclusion by Applicant:

According to Applicant 1, the results of the clinical trials support the clinical safety and efficacy of Tafluprost eyedrops for the treatment of elevated intraocular pressure in patients with open-angle glaucoma or ocular hypertension.

In addition, the data from the clinical trials did not reveal any critical adverse reactions with regards to safety of the product.

Overall, the administration of Tafluprost eyedrops directly onto the surface of the eye, appeared to be well tolerated despite the high incidence of ocular AEs. The study subject's discontinuation due to an AE and an ocular AE was low (*i.e.*, 3% and 2% of subjects respectively).

Ref.: (CDER, 2012)

Post-marketing experience

In post-marketing experience with Tafluprost, the most common adverse effects were ocular hyperaemia and eye redness. The events were consistent with Phase I-III clinical trials. No marked systemic effects were observed. The spontaneous post-marketing data of Tafluprost formulation were consistent with the safety profile from the clinical studies and provides reassurance that no new safety issues have emerged with prolonged use.

SCCS comments

The SCCS only evaluates the substances used as cosmetic ingredients for safety to the consumer, **not formulations or final products** as such. However, although Tafluprost is not intended for use in cosmetic products, it is clear from the clinical trials that a very high ratio of ocular adverse effects was evident with the use of pharmaceutical eye drops containing Tafluprost at concentrations that were much lower than what could be anticipated if Tafluprost was used in cosmetic products. Also, ocular adverse effects were observed during post-marketing surveillance.

Applicant 2

Eyelash serum and eyebrow serum **product** demonstrated clearly very good tolerability after prolonged daily application to the eyebrow serum of up to 56 weeks in 30 or 32 human volunteers and under supervision of medical specialists (dermatologist, ophthalmologist, reference (Dermatest, 2023b) (Dermatest, 2023c).

In the clinical application study with the **eye serum product**, all volunteers tolerated the product very well and the usage did not lead to ophthalmologically relevant ocular mucosal and other ocular changes in any test subject when daily applied for 56 weeks. Although there was an incidental misapplication into the eye at very few occasions. Visual acuity and especially the Intraocular Pressure (IOP) as most sensitive parameter were also not affected during the testing period as verified by the ophthalmologist. The dermatological assessment of skin compatibility after 56 days of application was that from a dermatologist point of view that no relevant skin reactions occurred in the test area. The product was very well tolerated. Neither intolerance reactions in the sense of irritation nor allergic reactions were observed. Accordingly, from a dermatological point of view, there is no high risk of irritation or sensitization for the tested product. In this respect it has to be emphasized again that all protective mechanisms of the eye for protection against and removal of foreign substances are intact, when using an eyelash serum. These mechanisms are the eyelid reflex and an open lacrimal duct.

Also, the clinical study with the **eyebrow serum product** showed clearly that the product was very well tolerated after daily application for 56 weeks. All tested volunteers showed healthy eye mucosa before, during and after the application study. Within the use period there were no findings indicating or suggesting side effects in the ocular tissues or periocular tissues of the volunteers. This was reasonable as there was no misapplication due to the clear instructions and in addition, none of the test subjects (0%) had any of the product run into their eyes during application. Finally, from the dermatologist and ophthalmologist point of view, there were no relevant skin and eye mucous membrane reactions in the test area and the product was tolerated very well. Neither intolerance reactions in the sense of irritation nor allergic reactions were observed.

SCCS comments

As mentioned above, **cosmetic formulations and final products** are not evaluated by the SCCS. There are no human data regarding DDDE available *per se*. Also, data from Tafluprost trials (with low concentrations) are not acceptable for DDDE. However, taking all observations into consideration, a high incidence of ocular adverse events has been observed that includes redness and hyperaemia. The frequent occurrence of these events suggests a need for a more stringent safety assessment.

5.3.11 Special investigations

Assessment of intraocular pressure and ocular effects in humans

Applicant 1

The eye irritation potential of a cosmetic eyelash product containing 0.025% of DDDE was investigated in 19 human volunteers. The test substance was applied daily in accordance with the directions in the package insert for 28 days. The study was conducted under the supervision of an ophthalmologist. The IOP and ocular effects were evaluated. Study is discussed in section 'Irritation and corrosivity'.

The within-eye differences in IOP from the beginning to the end of the study were not statistically significant ($t > 0.05$). No ocular adverse effects were noted by the study ophthalmologist. Only minor ocular effects were self-reported by 4 of 19 volunteers including slight dryness sensation around the eye areas (5%), slight itching sensation (21%), slight stinging sensation (16%), slight eye watering and redness (11%) and moderate to high burning sensation (5%). Overall, the test substance was moderately tolerated by the majority of the volunteers.

According to the Applicant, under the conditions of the study, the test substance 0.025% DDDE did not show a significant effect on IOP, but minor ocular effects were observed in the human volunteers.

Assessment of endocrine toxicity**Applicant 1***(i) Existing data and non-test information*

QSAR model		Results
Oestrogen	OECD QSAR Toolbox v.4.6	Strong binder, OH group
	Derek Nexus version 6.2.1	No alerts
	Danish QSAR models	ER α binding: Positive, within the applicability domain (ID) ER α activation: Negative, out of the applicability domain (OD) ER activation: Negative, ID
	VEGA v.1.2.3	ER relative binding affinity: Inactive, OD ER mediated effects: Inactive, ID
	Endocrine disruptome	Medium binding to ER α as agonist/antagonist
Androgen	Derek Nexus version 6.2.1	No alerts
	Danish QSAR models	AR binding: Negative, OD AR activation: Negative, ID AR inhibition: Negative, ID
	VEGA v.1.2.3	AR binding activity: Active, OD
	Endocrine disruptome	Potential binding to AR as antagonist
Thyroid modalities	Derek Nexus version 6.2.1	No alerts
	Danish QSAR models	Thyroid peroxidase (TPO) inhibition: Negative, ID Sodium/iodide symporter (NIS): Negative, ID
	VEGA v.1.2.3	TR α and TR β binding: Inactive, ID
	Endocrine disruptome	Medium probability to bind TR α
Steroidogenesis	Derek Nexus version 6.2.1	No alerts
	Endocrine disruptome	Medium probability to bind to glucocorticoid receptor (GR) as agonist

The Applicant is of the opinion that overall, the (Q)SAR results indicate that DDDE may have some ED activity. However, this will need to be evaluated further based on the *in vivo* studies available for the analogue Tafluprost to determine whether there is a plausible link between the potential ED activity and the ED-mediated adversity (if any).

(ii) In vitro assays providing data about selected ED mechanisms / pathways (OECD

Conceptual Framework Level 2)

No ED relevant *in vitro* assays as defined in the OECD Conceptual Framework Level 2 could be identified for DDDE or its analogue Tafluprost.

(iii) In vivo assays providing data about selected ED mechanisms/ pathways (OECD Conceptual Framework Level 3)

No *in vivo* assays investigating ED selective mechanisms or mode of actions (e.g., Uterotrophic assay - OECD TG 440, Hershberger assay - OECD TG 441) could be identified for DDDE or its analogue Tafluprost. However, safety pharmacology studies for the common hydrolytic metabolite Tafluprost acid, showed an increase in the force of uterine contractions in uteri of non-pregnant rats and rabbits at $\geq 10^{-9}$ g/mL (CDER, 2011).

(iv) In vivo assays providing data on adverse effects on ED related endpoints (OECD Conceptual Framework Level 4)

The potential for endocrine adversity of DDDE can be assessed based on the endocrine, androgen, thyroid and steroidogenesis-mediated ('EATS-mediated') and 'Sensitive to, but not diagnostic of EATS' (EATS-sensitive) parameters (see the below list) evaluated in repeated dose, reproductive, and developmental toxicity studies.

As per the ECHA EFSA guidance document for the identification of endocrine disruptors (EFSA/ECHA, 2018), the EATS-mediated and sensitive parameters typically investigated in these studies include:

- EATS-mediated: Weights and histopathology of thyroid, ovaries, uterus, cervix, vagina, testes, epididymis, prostate and seminal vesicles and sperm parameters (morphology, motility, and count), Oestrus cyclicity.
- EATS-sensitive: Weights and histopathology of adrenals, reproductive and developmental parameters such as fertility, gestation length, litter size and weight, litter viability, and mortality.

The available repeated dose, reproductive and developmental toxicity studies for the analogue Tafluprost do not show any toxicologically significant effects on EATS-mediated parameters (see previous sections). The effects on the EATS-sensitive, but not diagnostic parameters such as litter viability, litter size and weight was taken into consideration while establishing the PoD (see previous sections).

(v) In vivo assays providing more comprehensive data on adverse effects on ED related endpoints over more extensive parts of the life cycle of the organism (OECD Conceptual Framework Level 5)

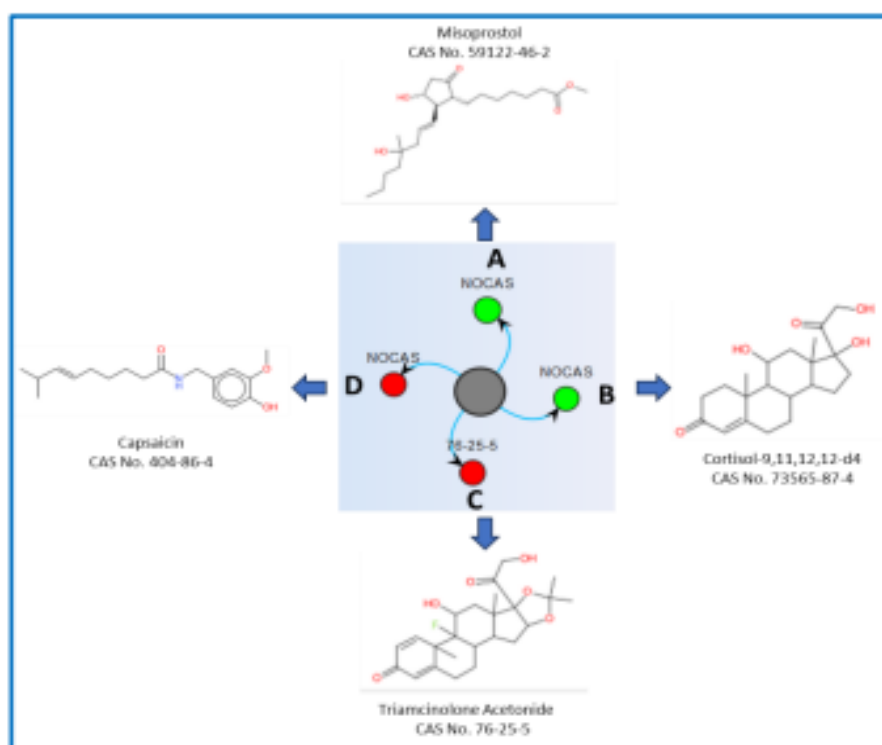
No OECD Level 5 *in vivo* assays defined in the OECD Conceptual Framework Level 5 could be identified for DDDE or its analogue Tafluprost.

Overall conclusion by Applicant on endocrine disrupting properties of DDDE:

Based on the available information from *in silico* modelling for DDDE paired with the evidence from *in vivo* studies for the analogue Tafluprost, DDDE may have some potential for ED activity. Available data from *in vivo* testing of the analogue Tafluprost (*i.e.*, repeated dose toxicity; developmental and reproductive toxicity) did not reveal any evidence for EATS-mediated adversity but showed adverse effects on 'EATS-sensitive' parameters. Due to lack of specificity of these parameters, no conclusive evidence of adverse effects due to an endocrine-related mechanism is available. Furthermore, the proposed point of departure (PoD) for risk assessment is based on a NOAEL derived from a developmental toxicity study via the intravenous route and is therefore considered to also adequately cover for any potential EATS-sensitive effects.

Applicant 2

The models used comprised expert-rules based SAR tools (OECD QSAR Toolbox v.4.6, Derek Nexus v.6.2.1) and statistical based QSAR models (VEGA v.1.2, Danish (Q)SAR models) were used for the *in silico*-based analysis of the carcinogenicity concerns raised. In addition, the ToxRead tool v. 0.25 was evaluated to determine whether this model could provide an answer 'with a reasonable model certainty'. The analogues identified by ToxRead as suitable analogues are shown in the figure below:



Finally, the expert review of the *in silico* data predominantly based on SAR analysis did not support a conclusion for carcinogenicity concerns for Ethyl Tafluprostamide. Conclusions based on QSAR predictions were considered as not reliable. The concern on the non-genotoxic carcinogenicity potential of Ethyl Tafluprostamide can be ruled out for the following reasons:

- 1) The SAR analysis did not indicate any carcinogenicity concern related to DDDE;
- 2) The predictions of the QSAR analysis are not reliable, as they are always outside the applicability domain. Also, it should be pointed out that the current QSAR tools are not powered enough to address the carcinogenicity endpoint;
- 3) The *in silico* analysis using the ToxRead tool did not identify suitable analogues for the carcinogenicity endpoint;
- 4) On the basis of the lack on carcinogenicity in two species shown for the close analogue Tafluprost, DDDE is not considered to be carcinogenic, neither regarding a direct nor a non-genotoxic mechanism.

Based on a scientifically recognized structure-analogy analysis for Tafluprost, which was also carried out by the BfR (Reference: BfR, 2017) and which is also repeatedly used by the applicant, there is no plausible evidence that DDDE possesses the property of a non-genotoxic carcinogen.

As already mentioned in other sections, DDDE revealed no mutagenic/genotoxic potential, when tested in the OECD/SCCS compliant Ames test and *in vitro* micronucleus test and no carcinogenic potential is likely considering the fact that the structural analogue Tafluprost revealed no such potential in rats and mice after (invasive) subcutaneous injection for 24

months or 78 weeks, respectively. As a consequence, the concern that DDDE may be a non-genotoxic carcinogen can be excluded.

SCCS comments

The lack of specific *in vivo* and *in vitro* tests for DDDE (and Tafluprost) on endocrine activity limits the ability to conclusively evaluate potential endocrine risks. The results of *in silico* predictions show a positive receptor binding. *In silico* analysis suggests the potential for endocrine activity.

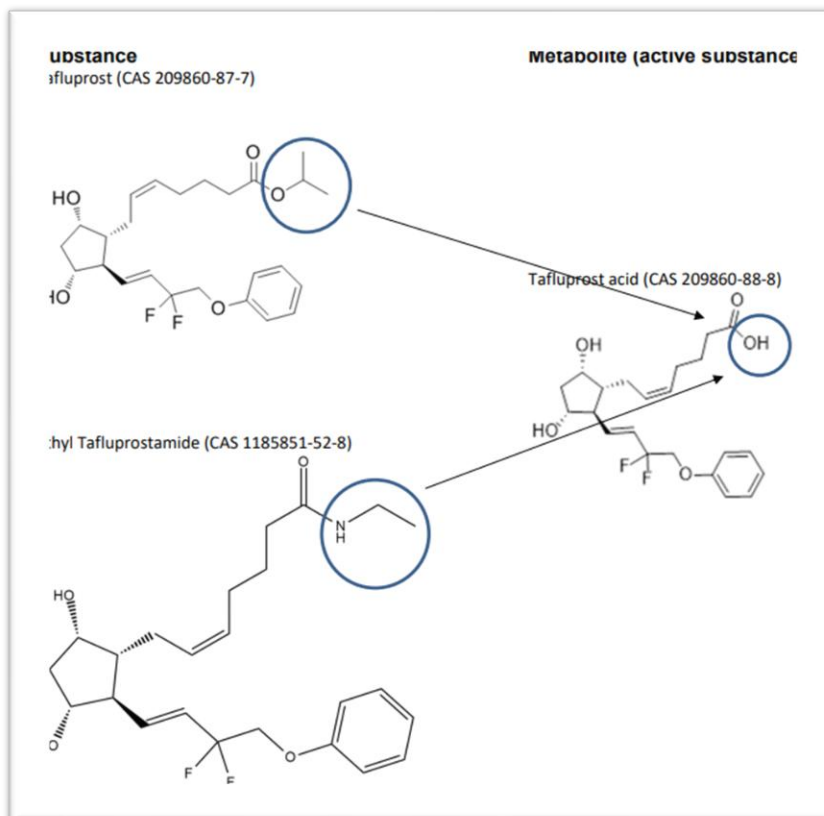
Read across justification by Applicant

Applicant 2

The strongest justification for the use of the read across approach was the obtained evidence that DDDE is metabolized to Tafluprost acid as it was demonstrated in the scientifically valid *in vitro* dermal metabolism, distribution and penetration study presented in section 5.3 Other studies on toxicokinetics. This study was performed using viable human skin showing that the metabolite Tafluprost acid was formed of up to $68.5 \pm 2.7\%$ after 24 hours.

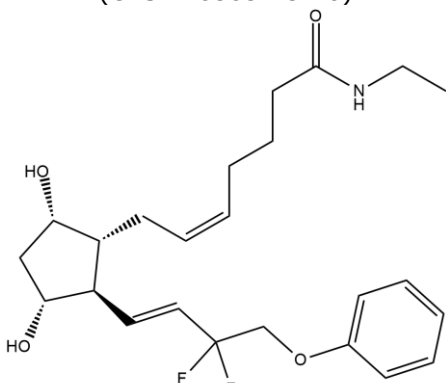
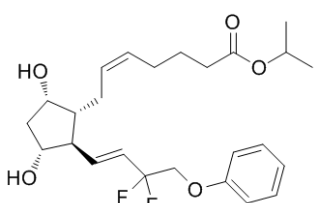
As Tafluprost acid was also demonstrated to be the main metabolite originating from Tafluprost (FDA, 2011,) it is justified to take *in vivo* data from Tafluprost as surrogate for missing *in vivo* data of Ethyl Tafluprostamide.

Furthermore, all three substances share the common structure, namely the Carboxylic acid structure. Ethyl Tafluprostamide is a Carboxylic acid amide, Tafluprost a Carboxylic acid ester and Tafluprost acid a Carboxylic acid as is demonstrated in the figure below.



In addition, it is demonstrated that Ethyl Tafluprostamide and Tafluprost revealed physico-

chemical properties in the same range and Tafluprost showed also a high Dice index of 0.86, which is considered to be indicative of a structural similarity. Details, according to the Applicant, are provided in the table below.

Substance	Tafluprost Ethyl Amide (CAS 1185851-52-8)	Tafluprost (CAS 209860-87-7)
Structure		
SMILES	<chem>CCNC(=O)CCCC=CCC1C(CC(C1C=C(C(COc2ccccc2)(F)F)O)O</chem>	<chem>CC(C)OC(=O)CCCC=CCC1C(CC(C1C=CC(COc2ccccc2)(F)F)O)O</chem>
Physico-chemical properties		
Molecular weight (g/mol)	437.53	452.54
Melting Point (°C)	247.58	204.56
(Predicted:EPI Suite™ v.4.11)		
Boiling Point (°C)	574.90	509.44
(Predicted:EPI Suite™ v.4.11)		
Log Kow	5.03	6.51
(Predicted:EPI Suite™ v.4.11)		
Water solubility (mg/L)	0.091	0.0039
(Predicted:EPI Suite™ v.4.11)		
Dice index		0.86
(similarity indication coefficient for Tafluprost compared to DDDE)		
Toxicological properties		
Eye irritation	No eye irritation potential <i>in vitro</i> (Isolated Chicken Eye Test))	No eye irritation potential <i>in vivo</i> (52-week monkey study, up to 0.05% direct eye application)
Skin sensitization	No skin sensitizing potential <i>in vitro</i> (Direct Peptide Reactivity Assay, KeratinoSens™ method)	No skin sensitizing potential <i>in vivo</i> (Adjuvant and Patch test in Guinea pigs)
Mutagenicity	No mutagenic potential <i>in vitro</i> (negative: Ames test, micronucleus assay)	No mutagenic potential <i>in vitro</i> and <i>in vivo</i> (negative: Ames test, clastogenicity in CHL cells, mice micronucleus test)

Finally, according to the Applicant, considering all of the above presented data, the profile of Tafluprost is considered as sufficiently acceptable for the structural analogue read across approach for DDDE.

Overall SCCS comments

As expressed in the Preamble, the SCCS is of the view that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence.

It is not explained how structural similarity between Tafluprost, DDDE and other analogues

was assessed, since several molecular features can be used as input for the algorithm. Also, as mentioned above, the experimental value for solubility is very different from the predicted one, and therefore it is incorrect to use the predicted value for comparison between the analogues.

5.4 SAFETY EVALUATION (INCLUDING CALCULATION OF THE MoS)

Applicant 1

Selection of the PoD

Considering all available information and in line with the approach chosen by the German BfR for the risk assessment of DDDE in cosmetic eye product formulations (BfR, 2017), the NOAEL of 0.3 µg/kg bw/day derived from the prenatal developmental toxicity study in rats following intravenous administration of Tafluprost is considered as the most appropriate PoD. As the NOAEL is based on critical effects *i.e.*, developmental/teratogenic effects following intravenous dosing, it represents a very conservative and worst-case PoD when compared with the exposure route (*i.e.*, dermal) for the intended application of DDDE present in cosmetic eye product formulations. Further, the study is considered to be of good quality with appropriate dose spacing and it corresponds to the lowest NOAEL established in the available set of studies for Tafluprost.

No bioavailability correction (considering its molecular weight of 452.5g/mol, water solubility of 1.05 g/L and log K_{ow} of 2.74) is required for deriving a systemic dose (PoD_{sys}) as the PoD has been derived from an intravenous study, resulting in a systemic PoD (PoD_{sys}) of 0.3 µg/kg bw/day or 0.0003 mg/kg bw/day.

Exposure assessment

DDDE is used in the cosmetic eyelash product at 0.018% concentration. It is intended to be used as an eyelash conditioner that “helps to strengthen eyelashes while protecting against breakage and brittleness” while “improving flexibility, moisture and shine for bold, beautiful, more dramatic looking lashes” (Athena Cosmetics Inc., 2020). The concentration of DDDE in product X is 0.018%.

Instructions for use of the eyelash cosmetic formulation containing DDDE

The cosmetic eyelash product should be applied with a multi-use fine brush applicator as a thin line directly to eyelashes (primarily the upper eyelashes) above the lash line. It is formulated with a thickener (cellulose gum) to ensure that the product stays on the eyelashes where it is applied and does not migrate to contact the fluid or membranes surrounding the eye. Therefore, the cosmetic eyelash product is designed to affect the appearance of eyelashes and not to come in contact with eyes.

Further, the below ‘Directions for Use’ and ‘Caution Statements’ are provided in the packaging for cosmetic eyelash product in Europe.

Directions For Use	Caution Statements
Once a day, apply a thin line of the cosmetic eyelash product directly to eyelashes, above the lash line. Let dry completely before applying additional beauty products.	Do not get in eye. Rinse immediately with water if eye contact occurs. If irritation develops, reduce frequency of use until irritation resolves. If irritation persists or is excessive, discontinue use and consult a physician. Some users have reported a faint darkening of the eyelash base (primarily with excessive use); if this is of concern, do not use. Keep out of reach of children.

Amount of product applied per application

A very small amount of DDDE is applied per application to the eyelashes. The amount of cosmetic eyelash product applied per brushstroke was determined by weighing the brush applicator, which was part of the container, before and after applying the product to the upper eyelashes. Ten different applicator brushes were used in the study, with an average of 2.4 mg applied to the upper eyelashes with each brush stroke.

Another in-house test was conducted using commercially available mink hair samples. The results showed that the range of the product applied per brush stroke was 0.64 – 3.72 mg. The highest 'average amount' from both procedures (average 2.42 mg) is almost identical to the average amount of 2.4 mg applied (per brushstroke) to human upper eyelashes. However, the highest 'maximum amount' is lower (*i.e.*, 3.72 mg) than the maximum amount applied (4 mg) per brushstroke to human upper eyelashes.

Calculations of the average and maximum amounts of DDDE used per application to the upper eyelashes are presented in Annex V. For risk assessment, both the 'average amount' and 'maximum amount' per brush stroke were considered. For the MoS calculations, the 'average amount' of 4.8 mg/day and the 'maximum amount' of 8 mg/day were used.

Margin of Safety calculation (MoS)

$$\text{MoS} = \text{PoDsys}/\text{SED}$$

With:

PoDsys = Systemic Point of Departure (mg/kg bw/day)

the MoS for systemic toxicity should be ≥ 100 .

SED and MoS calculations

Product daily amount (mg)	SED (mg/kg bw/d)	PoDsys (mg/kg bw/d)	MoS
Based on average daily amount of 4.8 mg the cosmetic eyelash product	6.24E-07	0.0003	481
Based on maximum daily amount of 8 mg the cosmetic eyelash product	1.04E-06	0.0003	288

Conclusion by Applicant

According to the Applicant, the calculated MoS is greater than 100 and thereby supports the safe use of DDDE at a concentration up to 0.018% in cosmetic eyelash products under the conditions presented in this evaluation.

Applicant 2

See section 5.2.5

SCCS comments

The MoS calculation of DDDE by the Applicant was based on a NOAEL derived from a prenatal developmental toxicity study in rats following intravenous administration of Tafluprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a NOAEL for DDDE the SCCS cannot derive a MoS for DDDE.

Furthermore, the SCCS is also of the opinion that SED and MoS cannot be calculated, because human reproductive toxicity and developmental toxicity cannot be excluded for DDDE (as for Tafluprost) and there are in fact strong indications that it causes both (see SCCS comments under 5.3.6.2). Non-genotoxic carcinogenicity can also not be excluded (see SCCS comments under 5.3.7)

5.5 DISCUSSION

A general remark is that nearly no data are present on the substance under consideration, Ethyl Tafluprostamide (DDDE). Instead, several studies have been carried out with Tafluprost, which was considered by both Applicants to be the best read-across surrogate compound. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. Therefore, study results obtained using Tafluprost have not been considered in this opinion.

Only the results obtained by using the compound DDDE are taken into consideration. As DDDE is present in eye lash and eyebrow serum up to a concentration of 0.018% (cf. terms of reference for this Opinion), both products could be used simultaneously and therefore aggregate exposure has to be considered.

Physicochemical properties

DDDE, an amide derivative of Tafluprost, is inactive. It is hydrolysed into Tafluprost acid (TPFA), the biologically active metabolite form.

Impurities have not been identified and quantified.

Exposure assessment and toxicokinetics

-Function and uses

Human exposure to DDDE was determined using eyeliner product. An average of 2.4mg product used with each brush stroke corresponded to 0.432 µg DDDE. The amount of eyeliner according to the Notes of Guidance (SCCS/1647/22), is about the double, namely 5mg.

DDDE is not only present in eye lash (0.015%), but also in eyebrow serum up to a concentration of 0.018%. Both products could be used simultaneously and therefore aggregate exposure to DDDE has to be considered.

-Dermal absorption

A default value of 50% for human dermal absorption is considered as the dermal absorption studies reported show a large variation (8.67% and 25.77%).

-Oral absorption

n As no data on oral absorption are present, a default value of 50% is considered (where applicable).

-Toxicokinetics

It was shown that DDDE is metabolised to TPFA, which acts directly as a potent agonist of prostaglandin F receptor (FP receptor), which reduces intraocular pressure by increasing the outflow of aqueous humor in the eye.

Tafluprost and TPFA were shown to highly bind to serum albumin of different species, including humans. They are both completely metabolised *in vitro*, in isolated hepatocytes of different species. Further toxicokinetic information is not available.

-Calculation of SED/LED and MoS

The MoS calculation of DDDE by the Applicant was based on a NOAEL derived from a study with Tafluprost. As expressed in the Preamble, the SCCS is of the opinion that the proposed

read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a NOAEL for DDDE the SCCS cannot derive a MoS for DDDE.

Toxicological evaluation

Irritation and corrosivity

DDDE is not irritative to the skin as shown *in vitro* and *in vivo*. Also, DDDE is not irritative to the eye.

Skin sensitisation

No firm conclusion for the studies presented by Applicant 1 with respect to skin sensitisation can be made. The results of the *in vitro* studies were doubtful as precipitation occurred in different assays that confounded the results, suggesting potential underestimation of DDDE's reactivity. In case of precipitation, a positive result could still be used, but a negative result is uncertain and should be interpreted with due care. The DPRA and KeratinoSense tests for DDE by Applicant 2 were negative. For confirming that skin sensitisation was negative, HRIPT tests were added by both Applicants. For DDE, either the validity of the test was critical or DDE was present in a product. HRIPT with Tafluprost was negative, but the results were not considered as Tafluprost is not accepted as a suitable analogue for DDDE. In Europe, supporting human HRIPT studies are considered to be unethical.

Taking all results together, DDDE can be considered as non-sensitising.

Acute toxicity

Results provided were only for Tafluprost and not for DDDE, hence not accepted.

Repeated dose toxicity

No repeated dose toxicity study and no PoD were available for DDDE.

On the contrary, in total 7 repeated dose toxicity studies were carried out with Tafluprost, which are not relevant for DDDE:

- as expressed in the Preamble, the SCCS is of the opinion that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence, therefore Tafluprost is not accepted as a surrogate for DDDE;
- different exposure routes were used (IV, SC, oral) for PoD determination which makes comparisons of results difficult;
- clear interspecies differences (rat, mice, dog, monkey) are present, making extrapolation to humans uncertain;
- the dog seems to be the most sensitive species when long-term systemic effects of Tafluprost are considered; for DDDE no data are available. For developmental toxicity of Tafluprost, rabbits are more sensitive than rats; for DDDE no data are available;
- 28-d repeated dose toxicity studies are too short and an additional uncertainty factor of 3 is usually introduced;
- In a study of Tafluprost on monkeys, the ocular route was used and important eye changes occurred for which an unacceptable explanation was given as being just local effects without any further additional evidence. Since eye changes, different from eye irritation, occurred and these side effects are well-known when Tafluprost is used as a drug in eye drops against glaucoma, extra caution is warranted. Although Tafluprost is not intended for use in cosmetic products, it is clear from the clinical trials that a very high ratio of ocular adverse effects was evident with the use of pharmaceutical eye drops containing Tafluprost at concentrations that were much lower than what could be anticipated if Tafluprost was used in cosmetic products.

Reproductive and developmental toxicity

Neither *in vivo*, nor *in vitro* studies are available for DDDE.

All reproductive/developmental toxicity studies were performed with Tafluprost, which is not accepted as a suitable analogue for DDDE since the read-across study is not performed according to scientific standards.

When Tafluprost was used, important species differences were observed for the species examined (rat, rabbit). Species differences for maternal toxicity and prenatal toxicity are important and cannot be ignored since in case of human exposure to eyebrow and eye lash cosmetic products, women of child bearing age are involved and crucial developmental toxicity data relevant for humans are lacking.

Prostaglandins in general are known for their effects on uterine contractions. In that respect, it was shown via *in vitro* tests reported in the open literature that exposure to prostaglandin F2 alpha analogues may cause contractions of uterine smooth muscle which theoretically could be at the origin of miscarriages (Sakai et al. 2022). Exposure concentrations play a pivotal role. The concentrations of Tafluprost used in eye drops are about 10-times lower than if it was used in eye lash and eyebrow cosmetic products. Although Tafluprost is not intended for use in cosmetic products, it is clear from the clinical trials that a very high ratio of ocular adverse effects was evident with the use of pharmaceutical eye drops containing Tafluprost at concentrations that were much lower than what could be anticipated if Tafluprost was used in cosmetic products.

Furthermore, it was shown that uterine contractions and/or abortions occurring in the animals under investigation, were dependent on the protocols used. A difference of a few days in administration of Tafluprost to fertilised animals had detrimental consequences, which were, of the two species examined, most obvious in rabbits. Potential effects in women are unknown and raise concern.

Seen the lack of data with human relevance for reprotoxicity/developmental toxicity and the high uncertainty about species differences in terms of species sensitivity, methodology and occurrence of side effects, Tafluprost is considered not safe for the unborn child at any concentration. For DDDE, no reprotoxicity/developmental toxicity data are available in any species.

Mutagenicity/ genotoxicity

There is no concern with respect to the *in vitro* genotoxicity data of DDDE. Without adequate justification for read-across, the *in vivo* data available for Tafluprost, were not considered informative for the genotoxicity of DDDE. Full study reports were not available, and the intraperitoneal exposure route used was considered to be not relevant for cosmetic exposure.

Carcinogenicity

There were no carcinogenicity studies available for DDDE.

As DDDE is not a genotoxic compound, the SCCS has no concern for genotoxic carcinogenicity. However, the possibility for non-genotoxic carcinogenicity has not been ruled out.

Photo-induced toxicity

No relevant experimental data on photo-induced toxicity of DDDE have been provided. There is no indication to assume that DDDE may exhibit phototoxic properties as no extended light-absorbing chromophore is evident in the chemical structure.

Human data

Some information was available on cosmetic products containing DDDE, but products are not taken into consideration by the SCCS.

Clinical trials of pharmaceutical eye drops with Tafluprost have been done at much lower concentrations than those proposed for DDDE by the Applicant for use in cosmetic products. Still high percentages of ocular adverse effects were observed with these preparations. Also, ocular adverse effects were observed during post-marketing surveillance. In addition, Tafluprost trials are not acceptable for DDDE. Furthermore, a high incidence of ocular adverse events was observed, including redness and hyperaemia. The frequent occurrence of these events suggests a need for cautious use.

Safety evaluation

The MoS calculation of DDDE by the Applicant was based on a NOAEL derived from a prenatal developmental toxicity study in rats following intravenous administration of Tafluprost. As expressed in the Preamble, the SCCS is of the view that the proposed read-across between DDDE and Tafluprost does not fulfil the critical requirements for an unbiased and transparent read-across and is not supported by other line(s) of evidence. In the absence of a NOAEL for DDDE the SCCS cannot derive a MoS for DDDE.

Furthermore, the SCCS is also of the opinion that SED and MoS cannot be calculated, because human reproductive toxicity and developmental toxicity cannot be excluded for DDDE (as for Tafluprost) and there are in fact strong indications that it causes both (see SCCS comments under 5.3.6.2). Non-genotoxic carcinogenicity can also not be excluded (see SCCS comments under 5.3.7).

6. CONCLUSION

1. *In light of the data provided and taking under consideration the concerns raised by the SCCS in Opinion SCCS/1635/21,*
 - a) *does the SCCS consider Ethyl Tafluprostamide safe when used in eyelash and eyebrow products up to a maximum concentration of 0.018 %?*
 - b) *does the SCCS consider Methylamido-Dihydro-Noralfaprostal safe when used in eyelash and eyebrow products up to a maximum concentration of 0.03 %?*
 - c) *does the SCCS consider Isopropyl Cloprostenate safe when used in eyelash and eyebrow products up to a maximum concentration of 0.005 %?*

Having evaluated all the evidence provided by the Applicants to support the safe use of the three prostaglandin analogues (PGAs) - Isopropyl Cloprostenate, *Methylamido-Dihydro-Noralfaprostal* and Dechloro Dihydroxy Difluoro Ethylcloprostenolamide - the SCCS is of the opinion that none of them can be considered safe for use in cosmetic products intended for promoting the growth of eyelashes and eyebrows.

The SCCS conclusion is based on the potent pharmacological activity of the PGAs, even at low concentrations, and the lack of acceptable data that would make it possible to exclude any potential adverse effects resulting from reproductive/developmental toxicity. The latter is crucially important for PGAs in view that users of the PGA-containing cosmetic products will most likely be women of child-bearing age.

More details on the SCCS concerns regarding safety of the PGAs are given in the section 8.

2. *Alternatively, what is according to the SCCS the maximum concentration considered safe for use of Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal and Isopropyl Cloprostenate, respectively, in eyelash and eyebrow products?*

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3. *Does the SCCS have any further scientific concerns regarding the use of Ethyl Tafluprostamide, Methylamido-Dihydro-Noralfaprostal and Isopropyl Cloprostenate in cosmetic products?*

The available evidence has shown that neither of the 3 PGAs assessed in this Opinion has a potential for genotoxicity, and hence no concern over genotoxic carcinogenicity. However, as indicated in the previous SCCS Opinion (SCCS/1635/21), in the absence of experimental data on carcinogenicity, the possibility of non-genotoxic carcinogenicity cannot be excluded.

7. MINORITY OPINION

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8. SCCS CONCERNS

SCCS Concerns over the Safety of the Use of Prostaglandin Analogues (PGAs) in Cosmetic Products

The SCCS has evaluated all submitted information on the three prostaglandin analogues (PGAs) - Isopropyl Cloprostenate (IPCP), 0.005%, Methylamido-Dihydro-Noralfaprostal (MDN) 0.03% and Dechloro Dihydroxy Difluoro Ethylcloprostenolamide (DDDE) 0.015%/0.018%, intended to be used in cosmetic products for promoting the growth of eyelashes and eyebrows. Such products are becoming increasingly available on the international market.

PGAs are well known pharmaceutical compounds that are used for the treatment of certain eye conditions (such as glaucoma, ocular hypertension, and dry eye syndrome). However, such uses are allowed subject to prescription by qualified medical staff to a patient needing treatment and are supervised/monitored for any side effects. As such, they are neither available in Europe as OTC drugs, nor considered appropriate to be extended to unsupervised use by a (non-patient) consumer/ general public. In this regard, the SCCS has noted that the Applicant has suggested specific ways for safe application of the PGA-containing products. It can also be anticipated that detailed instructions for the application method will be supplied with the product. However, such measures cannot provide a safeguard against inappropriate use in a scenario where such products are widely available for use without any medical prescription and oversight to general consumers, who cannot be expected to always follow the instructions for a specific way of application.

The SCCS has also noted that a large part of the submitted information in the provided dossiers is based on data/information from studies relevant to clinical medicine and not *per se* to cosmetic safety assessment. The Applicants conducted new studies on IPCP, MDN and DDDE to provide experimental data on skin and eye irritation, skin sensitisation and genotoxicity. However, no data were provided for any of the PGAs under evaluation for the toxicological endpoints of key importance that are needed for the derivation of a toxicological point of departure (PoD). To fill this gap, the Applicants have proposed the use of read-across based on extrapolation of data from structural analogues. After careful evaluation, the SCCS is of the opinion that the way read-across has been carried out does not fulfil the critical requirements for an unbiased and transparent read-across. No further information from other line(s) of evidence was also provided to support the read-across outcomes. The SCCS has therefore not accepted the use of the conclusions drawn from read-across in the safety assessment of the PGAs under evaluation. Thus, critical information for deriving suitable points of departure is missing and therefore calculation of the margins of safety has not been possible for any of the three PG analogues under evaluation. Furthermore, the SCCS assessment of the available clinical and toxicological data points to serious health concerns for the use of PGAs in eyelash and eyebrow cosmetic products. The key aspects of the SCCS concerns in this regard are discussed below:

Developmental and reproduction toxicity

Major concerns in regard to developmental and reproductive toxicity have already been expressed by BfR in their report on the health assessment of prostaglandins, prostaglandin analogues and prostamides in eyelash growth treatments (BfR, 2018). BfR referred to developmental and reproductive toxicity studies in rodents exposed to Bimatoprost, one of the structural analogues proposed by the Applicant to justify the safety of MDN. In these studies, miscarriages were reported in rodents. In other animal studies cited in the dossier on MDN, and in the BfR (2018) report, reproductive toxicity effects, including shortened gestation periods, foetal deaths and reduced body weight of young animals, were observed. Although (for Bimatoprost) these effects in animal studies occurred at doses higher than the expected human exposure, the advice of the company bringing the eye drops with

Bimatoprost on the market was that the medication should only be used by pregnant women if the potential benefits justified the potential risk to the foetus.

Tafluprost, the analogue proposed by the Applicant to justify the safety of DDDE in cosmetic eyelash products, belongs to the FDA pregnancy category-C compounds (FDA, 2011). Although these FDA categories were replaced in 2015 by narrative descriptions to allow better patient-specific counselling and informed decision, it is important to note that the definition of FDA pregnancy category C is: '*Animal reproductive/developmental studies have shown adverse effects on the fetus and there are no adequate and well-controlled studies in humans*'². Due to these findings, FDA recommends to adequately protect female patients of reproductive age.

Also, a very high ratio of ocular adverse effects has been reported from the use of pharmaceutical eye drops containing Tafluprost at concentrations that were much lower than what could be anticipated if Tafluprost was used in cosmetic products. When seen together with the evidence for potential adverse effects of Tafluprost and other PGAs, on development and reproduction, this raises a serious safety concern because the consumers who are most likely to be using such eyelash and eyebrow cosmetics will be women of child-bearing age. This is also important in view of that PGAs are known to be potent pharmacologically active substances that can exert systemic effects in humans at very low concentrations. Information on the exposure doses at which developmental and reproductive toxicity can be excluded for the PGAs is therefore a prerequisite to establishing that their use in cosmetic products is safe for the consumer. This information is currently not available. In consideration of all these aspects, the SCCS is of the opinion that major concerns over consumer safety exist in regard to the 3 PGAs (IPCP, MDN and DDDE) assessed in this Opinion.

Adverse effects on eyes/ periorbital tissue

From the submitted data and the information retrieved from the open literature, it is clear that there are several reports of undesired effects in the eyes and the skin surrounding the eyes from the medicinal use of prostaglandins and PGAs. The leaflets accompanying the medicines containing PGAs also indicate potential side effects that may occur in the eyes; such as conjunctival hyperemia, stinging/irritation, cataract, dry eye, ocular pain, eye lids darkening, blurred vision and even permanent change/increase of brown pigmentation of the iris. The medicinal use of PGAs in eye drops has been clearly linked to the so-called prostaglandin-associated periorbital syndrome (PAPS), which includes undesirable side effects such as periorbital hyperpigmentation, deepening of the upper eyelid sulcus, and thickening and alterations of the interpupillary distance and orbital structure. Although eyelash and eyebrow cosmetic products are not intended to be applied directly in the eyes, they will be used in close proximity to the eye, and therefore unintentional /accidental spillage into the eyes cannot be excluded. Most of these local side effects will not be detected in the standard local toxicity tests (e.g. eye and skin irritation tests) carried out on cosmetic products and their ingredients.

It is also important to highlight that whilst some undesirable side effects to the eyes and the skin may be acceptable for a pharmaceutical compound in the context of a risk-benefit analysis, such a principle is not applied for cosmetics in Europe (Regulation (EU) N° 1223/2009, preamble 9) where the regulation requires these to be only assessed for safety to the consumer (not risk-benefit analysis). The use of the prostaglandine analogues is, however, associated with undesirable effects on and around the eyes. At least 3 serious adverse effects (SUEs) have so far been officially reported.

Furthermore, the PGAs under consideration bind primarily, although not exclusively, to the Prostaglandin F2 α (PGF2 α) receptor. PG receptors are constitutively present in several human tissues. In this regard, it is notable that:

- Since PGAs have a high affinity and high potency towards PG receptors, they will affect the activity of endogenous prostaglandins.

² Mosby's Dental Drug Reference, 11th Edition, 2014, ISBN 978-0-323-16916-5, Editor: Arthur H. Jeske

- Effective concentrations of PGAs are in the nanomolar range. Therefore, even very small amounts becoming systemically available may lead to pharmacological and toxicological consequences.
- As highlighted in the SCCS/1635/21, *“from in silico analysis of the 15 PG analogues suggests that they may not be mutagenic/genotoxic, but there is a strong indication for their potential carcinogenicity via non-genotoxic mechanisms.”*
- Although the systemic exposure of the different analogues under consideration here is likely to be low, there is, in view of the above-mentioned high potency and widespread presence of specific receptors in human tissues, a high level of uncertainty in regard to the identification of a safe internal exposure dose. Therefore, a safe concentration for use of PGAs in the intended cosmetic applications cannot be established.

Considering all the aspects discussed above, the SCCS is of the opinion that the use of any of the three prostaglandin analogues (DDDE, MDN, IPCP) in eyelash and eyebrow cosmetic products cannot be considered as safe for the consumer.

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10. GLOSSARY OF TERMS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – Appendix 15 - from page 158.

11. LIST OF ABBREVIATIONS

See SCCS/1647/22, 12th Revision of the SCCS Notes of Guidance for the Testing of Cosmetic Ingredients and their Safety Evaluation – Appendix 15 - from page 158.

Abbreviations for Methylamido-Dihydro-Noralfaprostal (MDN)

2AA:	2-Aminoanthracene
4-NOPD:	4-Nitro-o-phenylenediamine
AGN 191522:	Bimatoprost acid
AGN 192024:	Bimatoprost
CAS:	Chemical Abstracts Service
CPA:	Cyclophosphamide
Cpm:	Counts per minute
DMSO:	Dimethyl sulfoxide
DPBS:	Dulbecco's Phosphate Buffered Saline
DPRA:	Direct Peptide Reactivity Assay
EC:	European Commission
EDTA:	Ethylenediaminetetraacetic acid trisodium salt hydrate
EU:	European Union
FBS:	Fetal Bovine Serum
GLP:	Good Laboratory Practice
GPMT:	Guinea pig maximization test
h:	hour
HPLC:	High Performance Liquid Chromatography
<i>i.e.</i> :	id est (that is)
m or M:	male or molar, molarity
MA:	Methyl Acetate
MDN:	Methylamido-Dihydro-Noralfaprostal
MMS:	Methyl methanesulfonate
MNT:	Micronucleus test
MoS:	Margin of Safety
MR:	Mutation Rate
MTT:	Methylthiazole tetrazolium (3-[4,5-Dimethylthiazol-2-yl]-2,5-diphenyl-tetrazolium bromide)
MW:	Molecular Weight
n:	number of determinations, strips or replicates
na or NA:	not applicable
NaN ₃ :	Sodium azide
NC:	Not Calculated
NOAEL:	No observed adverse effect level
ns:	not significant
OD:	optical density
OECD:	Organization for Economic Co-operation and Development
PBS:	Phosphate Buffered Saline
PHA:	Phytohaemagglutinin
PoD:	Point of departure

QAU:	Quality Assurance unit
RCD:	Cleaning of donor compartment
RCR:	Rinsed receptor compartment
RF:	Receptor fluid
RhCE:	Reconstituted Human Cornea Epithelium
RhE:	Reconstructed human Epidermis
RSD:	Relative standard deviation
SC:	<i>Stratum corneum</i>
SD:	Standard deviation or Sprague-Dawley rat
SED:	Systemic Exposure Dose
TEWL:	Transepidermal Water Loss
TK:	Thymidine kinase

Abbreviations for Isopropyl Cloprostenate (IPCP)

2-AA:	2-Aminoanthracene
4-NOPD:	4-Nitro-o-phenylenediamine
BfR:	German Federal Institute for Risk Assessment
CAS:	Chemical Abstracts Service
CPA:	Cyclophosphamide
Cpm:	Counts per minute
DA:	Defined Approaches
DAP:	Dermal Absorption
DMSO:	Dimethyl sulfoxide
DPBS:	Dulbecco's Phosphate Buffered Saline
dpm:	disintegrations per minute
DPRA:	Direct Peptide Reactivity Assay
EC:	European Commission
ECx:	Effective Concentration
EDTA:	Ethylenediaminetetraacetic acid trisodium salt hydrate
EFSA:	European Food Safety Authority
EMA:	European Medicines Agency
EU:	European Union
FBS:	Fetal Bovine Serum
GLP:	Good Laboratory Practice
GPMT:	Guinea pig maximization test
h:	hour
h-CLAT:	human Cell Line Activation Test
HPLC:	High Performance Liquid Chromatography
ICx:	Inhibitory Concentration
<i>i.e.</i> :	id est (that is)
ICH:	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
IPCP:	Isopropyl Cloprostenate
m or M:	male or molar, molarity
MA:	Methyl Acetate
MDN:	Methylamido-Dihydro-Noralfaprostal
MMC:	Mitomycin C
MMS:	Methyl methanesulfonate
MNT:	Micronucleus test
MoS:	Margin of Safety
MR:	Mutation Rate

MTT:	Methylthiazole tetrazolium (3-[4,5-Dimethylthiazol-2-yl]-2,5-diphenyl36 tetrazolium bromide)
MW:	Molecular Weight
n:	number of determinations, strips or replicates
na or NA:	not applicable
NaN ₃ :	Sodium azide
NC:	Not Calculated
NOAEL:	No observed adverse effect level
ns:	not significant
OD:	optical density
OECD:	Organization for Economic Co-operation and Development
OPPTS:	The Office of Prevention, Pesticides and Toxic Substances
PBS:	Phosphate Buffered Saline
PGAs:	Prostaglandin analogues
PHA:	Phytohaemagglutinin
pK _a :	acid dissociation constant
PoD:	Point of Departure
QAU:	Quality Assurance unit
QSAR:	Quantitative Structure-Activity Relationship
RCD:	Cleaning of donor compartment
RCR:	Rinsed receptor compartment
RF:	Receptor fluid
RhCE:	Reconstituted Human Cornea Epithelium
RhE:	Reconstructed human Epidermis
RICC:	Relative Increase in Cell Count
RSD:	Relative standard deviation
SAR:	Structure-Activity Relationship
SC:	Stratum corneum
SD:	Standard deviation or Sprague-Dawley rat
SDS:	Sodium Dodecyl Sulfate
SED:	Systemic Exposure Dose
TEWL:	Transepidermal Water Loss
TK:	Thymidine kinase
UV/VIS:	Ultraviolet-visible spectroscopy

Abbreviations for Ethyl Tafluprostamide (DDDE)

AE:	adverse events
AFP-168:	[³ H]-Tafluprost
AFP-172:	Tafluprost Acid
ALT:	alanine aminotransferase
BfR:	German Federal Institute for Risk Assessment
BNF:	β-naphthoflavone
BSA:	Bovine Serum Albumin
CAS:	Chemical Abstracts Service
CHL:	Chinese hamster lung cells
COL:	Cochicine
CPA:	Cyclophosphamide
CP:	Cyclophosphamide monohydrate
DA :	Dermal absorption
DDDE :	Dechloro Dihydroxy Difluoro Ethylcloprostenolamide = Ethyl Tafluprostamide
DMSO:	Dimethyl sulfoxide
DPRA:	Direct Peptide Reactivity Assay

EATS:	estrogeny, androgen, thyroid and steroidogenesis,
HBBS:	Hanks' Balanced Salt solution
EC:	European Commission
ECG:	Electrocardiogram
EIT:	Eye Irritation Test mean
ERG:	Electroretinography
FPreceptor:	prostaglandin F receptor
GD:	Gestation Day
GHS:	Globally Harmonised System
HaCaT:	immortalized adherent cell line derived from human keratinocytes
HPLC:	High Pressure Liquid Chromatography
HRIPT:	Human Repeat Insult Patch Test
ICE:	Integrated Chemical Environment
ICH:	International Conference on Harmonisation
IOP:	intraocular pressure
LCMS:	Liquid Chromatography-Mass Spectrometry
LLOQ:	lower limit of quantification
MCE:	molar extinction coefficients
MMS:	Methylmethanesulfonate
MNT:	Micronucleus Test
MoS:	Margin of Safety
MTT:	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide
MW:	Molecular Weight
NDA:	Non Disclosure Agreement
NOAEL:	No observed adverse effect level
NQO:	Nitroquinoline N-Oxide
OD:	optical density
OECD:	Organization for Economic Co-operation and Development
PB:	Phenobarbital
PBS:	Phosphate-buffered Saline
pKa:	acid dissociation constant
PoD:	Point of Departure
RhE:	Reconstructed human Epidermis
RICC:	relative increase in cell counts
SC:	subcutaneous
SDS:	Sodium Dodecyl Sulfate
SED:	Systemic Exposure Dose
TEA:	Tafluprost Ethyl Amide
TEER:	Transepidermal electrical resistance
TEWL:	Transepidermal water loss
TPEA :	Ethyl Tafluprostamide
TPFA :	Tafluprost Acid
UV/VIS :	Ultraviolet/ Visible

12. ANNEXES

ANNEXES for Methylamido-Dihydro-Noralfaprostal (MDN)

ANNEX 1 Summary of Toxicokinetic studies performed with Bimatoprost

This Annex provides an overview provided by the Applicant, of kinetics studies derived from the pharmacology/toxicology review and evaluation performed by the Center for Drug Evaluation and Research (CDER) of the US Food and Drug Administration (FDA) on Bimatoprost (Reference: CDER, 2001).

Pharmacokinetics of Bimatoprost (AGN 192024) in Sprague-Dawley rats following single intravenous or single oral administration:

The purpose of the study was to determine the pharmacokinetics of Bimatoprost (AGN 192024) following a single iv or oral administration in rats. Sprague Dawley rats (6/sex/group) were dosed at 1 mg/kg bw intravenously (iv) or at 4 mg/kg bw orally (po). Blood samples were collected at pre-dose and 2, 5, 15, 30, 60, 90, 120, 240, 360, 480 and 1440 min after iv dosing and at pre-dose and 15, 30, 60, 90, 120, 240, 360, 480, 720 and 1440 min after po dosing. Blood concentrations of Bimatoprost (AGN 192024) and its metabolite, Bimatoprost acid (AGN 191522) were measured.

Bimatoprost (AGN 192024) was eliminated rapidly in rat blood following iv administration. The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a moderate oral bioavailability in rats (28%) following a single oral administration.

Pharmacokinetics of Bimatoprost (AGN 192024) in CD rats following single intravenous or single oral administration:

The purpose of this study was to determine the pharmacokinetics of Bimatoprost (AGN 192024) following a single iv or oral administration in rats. Sprague Dawley rats (6/sex/group) were dosed at 1 mg/kg bw iv or 4 mg/kg bw po. Blood samples were collected at pre-dose and 2, 5, 15, 30, 60, 90, 120, 240, 360, 480 and 1440 min after iv dosing and at pre-dose and 15, 30, 60, 90, 120, 240, 360, 480, 720 and 1440 min after po dosing. Blood concentrations of Bimatoprost (AGN 192024) and its metabolite, Bimatoprost acid (AGN 191522) were measured.

Bimatoprost (AGN 192024) was eliminated rapidly in rat blood following iv administration. Bimatoprost (AGN 192024) had a moderate oral bioavailability in rats (28%) following a single oral administration.

Intravenous pharmacokinetics and bioavailability of Bimatoprost (AGN 192024) in cynomolgus monkeys:

The purpose of this study was to determine the pharmacokinetics of Bimatoprost (AGN 192024) following a single iv or oral administration in monkeys. The animals (3/sex/group) were dosed at 1 mg/kg bw iv or 4 mg/kg bw po. Blood samples were collected at pre-dose and 2, 5, 15, 30, 60, 90, 120, 240, 360, 480 and 1440 min after iv dosing and at pre-dose and 10, 20, 40, 60, 90, 120, 240, 360, 480, 720 and 1440 min after po dosing. Blood concentrations of Bimatoprost (AGN 192024) and its metabolite, Bimatoprost acid (AGN 191522) were measured.

The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a very low oral bioavailability in monkeys (2.6%) following a single oral administration.

Pharmacokinetics of Bimatoprost (AGN 192024) in Swiss Webster mice following single intravenous or single oral administration:

The purpose of this study was to determine the pharmacokinetics of Bimatoprost (AGN 192024) following a single iv or oral administration in mice. Swiss Webster mice (4/sex/group) were dosed at 1 mg/kg bw iv or 4 mg/kg bw po. Blood samples were collected at pre-dose

and 2, 5, 10, 15, 30, 60, 90, 120, 240, 360, 480 and 1440 min after iv dosing and at pre-dose and 20, 40, 60, 90, 120, 240, 360, 480, 720 and 1440 min after po dosing. Blood concentrations of Bimatoprost (AGN 192024) and its metabolite, Bimatoprost acid (AGN 191522) were measured.

Bimatoprost (AGN 192024) was eliminated rapidly in mouse blood following iv administration. The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a moderate oral bioavailability in mice (20.6% in females, 58.7% in males) following a single oral administration.

Pharmacokinetics of Bimatoprost (AGN 192024) in CD-1 mice following single intravenous or single oral administration:

The purpose of this study was to determine the pharmacokinetics of Bimatoprost (AGN 192024) following a single iv or oral administration in mice. CD-1 mice (6/sex/group) were dosed at 1 mg/kg bw iv or po. Blood samples were collected at pre-dose and 5, 10, 20, 40, 60, 120, 240, and 360 min after dosing. Blood concentrations of Bimatoprost (AGN 192024) and its metabolite, Bimatoprost acid (AGN 191522) were measured.

Bimatoprost (AGN 192024) was eliminated rapidly in mouse blood following iv administration. The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a moderate oral bioavailability in mice (15.2% in females, 72.1% in males) following a single oral administration.

[3H]-Bimatoprost (AGN 192024): Placental transfer and milk secretion in the rat following a single intravenous administration:

The purpose of this study was to determine the placental transfer of radioactivity in pregnant rats, and to determine the secretion of radioactivity into the milk of lactating rats following a single intravenous dose (1 mg/kg bw of [3H]-Bimatoprost (AGN 192024) in pregnant (on day 18 of pregnancy) and lactating (1 week after parturition) Sprague Dawley rats. Tissue and plasma samples from 3 animals per time point were collected at 0.5, 2, 5 and 24 h (pregnant rats) and 0.5, 1, 3, 6 and 24 h (lactating rats) after dosing. Tissue and plasma samples were analysed for total radioactivity by liquid scintillation counting.

In the part conducted in the pregnant rats, radioactivity was distributed rapidly, and maximum concentrations was reached in most tissues within 30 min with the highest levels observed in the liver (2.44 µg-eq/g) and kidney (1.52 µg-eq/g). Radioactivity was noted in amniotic fluid (C_{max}: 0.304 µg-eq/mL) and fetus (C_{max}: 0.286 µg-eq/g), indicating placental transfer of radioactivity. The results from the lactating rat part showed that the secretion of radioactivity from [3H]-Bimatoprost (AGN 192024) into milk occurred during 24 h after dosing.

***In vitro* protein binding of [3H]-Bimatoprost (AGN 192024) in mouse, rat, rabbit, monkey and human plasma and bovine and human serum albumin:**

The purpose of this study was to determine the extent of *in vitro* plasma protein binding of [3H]-Bimatoprost (AGN 192024) in mice, rats, rabbits, monkeys, humans. The *in vitro* binding to bovine and human serum albumin was also assessed. The results indicated that [3H]-Bimatoprost (AGN 192024) was modestly free in animal plasma proteins (28 - 37%) and slightly unbound to human plasma. The extent of binding did not change over plasma [3H]-Bimatoprost (AGN 192024) concentrations tested (1 -250 ng/mL). [3H]-Bimatoprost (AGN 192024) was a poor ligand to bovine and human serum albumin.

In vitro metabolism of [3H]-Bimatoprost (AGN 192024) in rat, monkey and human liver slices: The purpose of this study was to examine the metabolism of [3H]-Bimatoprost (AGN 192024) in rat, monkey and human liver slices after incubation for 18 h at 37 °C. The results showed that approximately 89%, 84%, 58% and 77% [3H]-Bimatoprost (AGN 192024) were metabolized after 18 h incubation with rat, male monkey, female monkey and human liver, respectively. The liver metabolic profiles of [3H]-Bimatoprost (AGN 192024) were comparable among the species tested. [3H]-Bimatoprost (AGN 192024) was deaminated (conversion to Bimatoprost acid (AGN 191522)) and glucuronidated in liver slices of all the species tested.

***In vitro* metabolism of [3H]-Bimatoprost (AGN 192024) in liver microsomes of mouse, rat, rabbit, monkey and human:**

The purpose of this study was to compare the metabolite profile of [3H]-Bimatoprost (AGN 192024) in human liver microsomes with those of mouse, rat, monkey and rabbit. [3H]-Bimatoprost (AGN 192024) was incubated with the liver microsomes from different species at 37 °C for 10 - 60 minutes. The results showed that Bimatoprost (AGN 192024) was extensively metabolized by human liver microsomes to 3 major and several minor metabolites. The 3 major metabolites were tentatively identified as hydroxylated analogs of Bimatoprost (AGN 192024). The metabolite profiles of Bimatoprost (AGN 192024) microsomes from different species were very similar. The 3 major human metabolites were also noted in each of the animal species tested. The mouse liver microsomes metabolized Bimatoprost (AGN 192024) more rapidly and several additional metabolites were formed. Only rabbit liver microsomes metabolized Bimatoprost (AGN 192024) to form significant amount of Bimatoprost acid (AGN 191522). In conclusion, the metabolite profiles were similar across all 5 species examined.

Identification of human hepatic cytochromes P-450 involved in the metabolism of Bimatoprost (AGN 192024):

The purpose of this study was to determine the specific cytochrome P450 isoforms responsible for the metabolism of Bimatoprost (AGN 192024) *in vitro*. [3H]-Bimatoprost (AGN 192024) was incubated with human liver microsomes at 37 °C for 5 - 10 minutes. It was shown that Bimatoprost (AGN 192024) was extensively metabolized by human liver microsomes to 3 major and 3 minor metabolites. The 3 major metabolites were tentatively identified as the hydroxylated analogs of Bimatoprost (AGN 192024). The formation rate of the 3 major metabolites were correlated best with the activity of CYP3A4/5 isozyme. The involvement of CYP3A4/5 in Bimatoprost (AGN 192024) metabolism was further confirmed by inhibition of metabolite formation by troleandomycin, a specific mechanism-based inhibitor of CYP3A4/5, and by ketoconazole, a specific competitive inhibitor of CYP3A4/5. An assay with cDNA expressed CYP3A4 also confirmed the involvement of CYP3A4/5 in Bimatoprost (AGN 192024) metabolism. In conclusion, the metabolism of Bimatoprost (AGN 192024) to its 3 major oxidative metabolites in human liver microsomes appeared to be catalyzed by CYP3A4/5.

Metabolite profiles in maternal and fetal tissues following a single intravenous administration of [3H]-Bimatoprost (AGN 192024) to pregnant rats:

The purpose of this study was to determine the metabolite profiles in maternal and fetal tissues following a single iv administration of [3H]-Bimatoprost (AGN 192024) (2 mg/kg bw) to pregnant rats (Day 18 of pregnancy). Animals were sacrificed at 0.5, 2 and 5 h after dosing and blood and tissue (liver, kidney, lung, uterus, placenta, ovary and fetus) samples were collected.

Bimatoprost (AGN 192024) was extensively metabolized in the pregnant rats. As many as 22 individual metabolite peaks were seen in the profiles of the various tissues. Bimatoprost (AGN 192024) and Bimatoprost acid (AGN 191522) were in every tissue analysed. Bimatoprost acid (AGN 191522) was the major metabolite in the ovary and uterus (46.7% and 39.7%, respectively). while Bimatoprost (AGN 192024) was the major component seen in the fetal tissues (50.2%). The fraction of Bimatoprost acid (AGN 191522) in the fetus was quite small (6.25%).

Metabolite profiles in blood, urine, feces of rat, monkey and human following a single intravenous administration of [3H]-Bimatoprost (AGN 192024):

The purpose of this study was to determine the metabolite profiles in blood, urine and feces following a single iv administration of [3H]-Bimatoprost (AGN 192024) to rats (2 mg/kg bw), monkeys (1 mg/kg b) and healthy humans. Metabolite profiles were also investigated in blood of mice and rabbits following a single dose at 1 mg/kg bw. Blood samples were collected at 5 min (monkeys), 10 min (humans) and 30 min (mice, rats and rabbits) after dosing. Urine samples were collected up to 4 h (monkeys), 6 h (humans) and 24 h (rats) after dosing. Feces samples were collected up to 48 h (monkey), 120 h (humans) and 24 h (rats) after dosing.

The results indicated that the blood, urinary and fecal metabolite profiles of rats, monkeys and humans were comparable. Bimatoprost (AGN 192024) was extensively metabolized in all species tested via deamination, hydroxylation, N-deethylation and glucuronidation. As many as 14 individual metabolite peaks were seen in the profiles of the various species. Bimatoprost (AGN 192024) and Bimatoprost acid (AGN 191522) were detected in the blood and feces of all species. Bimatoprost (AGN 192024) was the major circulating drug-related component in the rat, monkey, mouse and human, while Bimatoprost acid (AGN 191522) was the major component in rabbit blood. Bimatoprost (AGN 192024) was the largest fraction of radioactivity excreted in rat (8.5 - 14.2%) and monkey (19.4%) urine. Glucuronide conjugate of Bimatoprost (AGN 192024) was the major urinary metabolite in human (26.4%) and monkey (11.2%) urine.

Overall summary of toxicokinetics

The dermal absorption *in vitro* is currently studied in a SCCS and OECD guidance/guideline compliant study using MDN from a cosmetic formulation by using the [3H]- MDN as a tracer and applied for 24 hours onto the surface of healthy human skin.

Therefore, in accordance with the SCCS guidance principles (SCCS/1647/22, Reference: SCCS, 2023a) the default SCCS standard value of 50% will preliminary be used as dermal absorption value for the MOS calculation.

This calculation will immediately be revised when the report of the currently performed dermal penetration study with radiolabeled MDN will be available. This will provide a realistic and cosmetic product specific value and, based on experience with other PGAs, the value for dermal penetration is assumed to be much lower than the default value of 50%.

From the read across justification and with regards to structural-chemical considerations it can be considered as highly likely that MDN shares the same main metabolite with Bimatoprost, namely Bimatoprost acid. Therefore, the information on key and relevant kinetics studies available on Bimatoprost and reliable for MDN can be summarized as follows:

- Pharmacokinetics of Bimatoprost (AGN 192024) in Sprague-Dawley rats following single intravenous or single oral administrations showed that Bimatoprost (AGN 192024) was eliminated rapidly in rat blood following iv administration. The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a moderate oral bioavailability in rats (28%) following a single oral administration.
- Also pharmacokinetics of Bimatoprost (AGN 192024) in CD rats following single intravenous or single oral administration demonstrated that Bimatoprost (AGN 192024) was eliminated rapidly in rat blood following iv administration. Bimatoprost (AGN 192024) had a moderate oral bioavailability in rats (28%) following a single oral administration.
- The intravenous investigation of pharmacokinetics and bioavailability of Bimatoprost (AGN 192024) in cynomolgus monkeys indicated that the blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a very low oral bioavailability in monkeys (2.6%) following a single oral administration.
- In Swiss Webster mice the study on pharmacokinetics of Bimatoprost (AGN 192024) following single intravenous or single oral administration showed that Bimatoprost (AGN 192024) was eliminated rapidly in mouse blood following iv administration. The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a moderate oral bioavailability in mice (20.6% in females, 58.7% in males) following a single oral administration.
- Also in CD-1 mice following single intravenous or single oral administration Bimatoprost (AGN 192024) was eliminated rapidly in mouse blood following iv administration. The blood concentrations of Bimatoprost acid (AGN 191522) were much lower than those of Bimatoprost (AGN 192024). Bimatoprost (AGN 192024) had a moderate oral

bioavailability in mice (15.2% in females, 72.1% in males) following a single oral administration.

- The investigation of [3H]-Bimatoprost (AGN 192024) with regards to placental transfer and milk secretion in the rat following a single intravenous administration showed that in the part conducted in the pregnant rats, radioactivity was distributed rapidly and maximum concentrations were reached in most tissues within 30 min with the highest levels observed in the liver (2.44 µg-eq/g) and kidney (1.52 µg-eq/g). Radioactivity was noted in amniotic fluid (C_{max}: 0.304 µg-eq/mL) and fetus (C_{max}: 0.286 µg-eq/g), indicating placental transfer of radioactivity. The results from the lactating rat part showed that the secretion of radioactivity from [3H]-Bimatoprost (AGN 192024) into milk occurred during 24 h after dosing.
- The results of the *in vitro* protein binding of [3H]-Bimatoprost (AGN 192024) study in mouse, rat, rabbit, monkey and human plasma and bovine and human serum albumin indicated that [3H]-Bimatoprost (AGN 192024) was modestly free in animal plasma proteins (28 - 37%) and slightly unbound to human plasma. The extent of binding did not change over plasma [3H]-Bimatoprost (AGN 192024) concentrations tested (1 -250 ng/mL). [3H]-Bimatoprost (AGN 192024) was a poor ligand to bovine and human serum albumin.
- In the study of *in vitro* metabolism of [3H]-Bimatoprost (AGN 192024) in rat, monkey and human liver slices it was demonstrated that approximately 89%, 84%, 58% and 77% [3H]-Bimatoprost (AGN 192024) were comparably metabolized after 18 h incubation with rat, male monkey, female monkey and human liver, respectively. Bimatoprost (AGN 192024) was deaminated (conversion to Bimatoprost acid (AGN 191522)) and glucuronidated in liver slices of all the species tested.
- *In vitro* metabolism study of [3H]-Bimatoprost (AGN 192024) in liver microsomes of mouse, rat, rabbit, monkey and human showed that Bimatoprost (AGN 192024) was extensively metabolized by human liver microsomes to 3 major and several minor metabolites and that the metabolite profiles were similar across all 5 species examined.
- The *in vitro* study with human liver microsomes with regards to involvement of human hepatic cytochromes P-450 in the metabolism of Bimatoprost (AGN 192024) showed that the test item was extensively metabolized by human liver microsomes to 3 major and 3 minor metabolites. The metabolism to the 3 major oxidative metabolites appeared to be catalyzed by CYP3A4/5.
- The investigation in the metabolite profiles in maternal and fetal tissues following a single intravenous administration of [3H]-Bimatoprost (AGN 192024) to pregnant rats revealed that the test item was extensively metabolized in the pregnant rats. Bimatoprost (AGN 192024) and Bimatoprost acid (AGN 191522) were in every tissue analysed. Bimatoprost acid (AGN 191522) was the major metabolite in the ovary and uterus, while the parent compound was the major component seen in the fetal tissues. The fraction of Bimatoprost acid (AGN 191522) in the fetus was quite small.
- The metabolite profiles in blood, urine, feces of rat, monkey and human following a single intravenous administration of [3H]-Bimatoprost (AGN 192024) indicated that the blood, urinary and fecal metabolite profiles of rats, monkeys and humans were comparable. Bimatoprost (AGN 192024) was extensively metabolized in all species tested via deamination, hydroxylation, N-deethylation and glucuronidation. Bimatoprost (AGN 192024) and Bimatoprost acid (AGN 191522) were detected in the blood and feces of all species. Bimatoprost (AGN 192024) was the major circulating drug-related component in the rat, monkey, mouse, and human, while Bimatoprost acid (AGN 191522) was the major component in rabbit blood.

Finally, it was clearly demonstrated that Bimatoprost was rapidly metabolized to its main metabolite Bimatoprost acid.

Comparative *in vitro* studies indicated clearly that there were no relevant species differences in the metabolic profile. In addition, there was clear evidence that Bimatoprost acid was bound to serum albumin in rats, rabbits, dogs and humans.

Thus, there are clear indications that there are no relevant species difference and consequently the usual factor 4 for interspecies variability in kinetics can be omitted in the calculation of the safety margin (for details see section 5.6. below).

ANNEX 2: Summary of acute toxicity studies with Bimatoprost

Bimatoprost (AGN 192024, 0.1% ophthalmic solution) was investigated for potential systemic toxic effects following a single iv injection in rats under standard conditions and GLP. Each 6 male and 6 female Sprague Dawley rats were dosed once intravenously with the test item at 0.03, 0.3 and 3.0 mg/kg bw. The day of dosing was designated as Day 1. The animals were observed for 14 days following treatment.

No mortality was observed, and no signs of toxicity were noted during the observation period. Body weight was not impaired, and no gross pathological changes were observed in any animal of any groups. Thus, when Bimatoprost was administered in rats as a single iv injection in doses up to 3.0 mg/kg bw no systemic adverse effects were observed. Therefore, the intravenous LD50 value in rats was estimated as > 3.0 mg/kg bw.

In mice, Bimatoprost (AGN 192024, 0.1% ophthalmic solution) was investigated for potential systemic toxic effects following a single iv injection under standard conditions and GLP. Each 3 male and 6 female Swiss-Webster mice were dosed once intraperitoneally (ip) with the test item at 96 mg/kg bw. The day of dosing was designated as Day 1. The animals were observed for 7 days following treatment.

No mortality was observed, and no signs of toxicity were noted during the observation period. Body weight was not impaired, and no gross pathological changes were observed in any animal of any groups. Thus, when Bimatoprost was administered in mice as a single ip injection at 96 mg/kg bw no systemic adverse effects were recorded. Therefore, the intravenous LD50 value in mice was estimated as > 96 mg/kg bw.

ANNEX 3: Summary of repeat-dose toxicity studies with Bimatoprost**Toxicity study with Bimatoprost (AGN 192024) by oral gavage administration to CD-1 mice for 13 weeks followed by a 4-week recovery period:****Material and methods:**

The purpose of study was to investigate the repeated dose toxicity of Bimatoprost (AGN 192024) in male and female mice (CrI:CD-1 (ICR) BR) under GLP conditions and to select dose levels for a subsequent carcinogenicity study. Each 10 male and 10 female mice in the main group with complete investigations were dosed orally by gavage once a day with doses of 0, 4, 8 and 16 mg/kg bw/d for 13 weeks followed by a 4-week recovery period. In addition, a satellite group was used for toxicokinetic analyses. At the end of the treatment period, the corresponding main study animals were sacrificed and submitted for gross necropsy. Clinical reactions including mortality to treatment were monitored as well as body weight assessment, food consumption, ophthalmology, clinical pathology, urinalysis, organ weights and gross pathology. Histopathological evaluation was performed on a comprehensive range of organs.

Results:

Mortality - No treatment-related death was noted. One female animal treated with 16 mg/kg bw/d was found dead on Day 4, and one male animal treated with 4 mg/kg/day of Bimatoprost (AGN 192024) was sacrificed on Day 35. The reasons for these deaths were unknown.

Clinical Signs - No clinical signs were attributed to Bimatoprost (AGN 192024) treatment.

Body Weights - No effects on body weight or body weight gain were attributed to treatment with Bimatoprost (AGN 192024).

Food Consumption - There were no statistically significant differences in food consumption.

Ophthalmoscopy - No Bimatoprost (AGN 192024)-related ophthalmoscopy effects were observed.

Clinical Chemistry - No Bimatoprost (AGN 192024)-related changes in any of the parameters were noted.

Hematology - The total number of WBC was decreased in several treated groups. Since no corresponding pathological findings were noted, and the degree of the decrease was not large the changes may not be toxicologically significant.

Urinalysis - No toxicologically meaningful findings were noted.

Gross Pathology - No treatment-related findings were noted.

Organ Weights - No toxicologically significant differences were noted.

Histopathology - An increased incidence of medullary lymphoid proliferation in the thymus for females at 8 or 16 mg/kg bw/d, and increased incidence of acute inflammatory cells in the superficial layers of the vagina for females at 16 mg/kg bw/d were observed. The thymus changes were fully recovered. The changes in the vagina were partially recovered during the 4-week recovery period. One female animal from high dose group showed extensive deposits of lymphoma throughout many organs. A pulmonary adenoma was detected in one male from 16 mg/kg bw/d after recovery period. No other toxicologically significant findings were observed.

Conclusion:

The repeated oral (gavage) application of Bimatoprost (AGN 192024) to male and female mice at doses of 0, 4, 8 and 16 mg/kg bw/d for 13 weeks followed by a 4-week recovery period led to no treatment-related mortality, clinical signs, body weight and food consumption changes, and ophthalmoscopic findings. Histopathology examinations showed increased incidence of medullary lymphoid proliferation in the thymus for females at 8 or 16 mg/kg bw/d and increased incidence of acute inflammatory cells in the superficial- layers of the vagina for females at 16 mg/kg bw/d. One female animal from the high dose group showed extensive deposits of lymphoma throughout many organs. A pulmonary adenoma was detected in one male from 16 mg/kg bw/d after recovery period. No other toxicologically

significant findings were noted. In all treated groups the exposure was confirmed by available plasma values.

First subchronic toxicity study with Bimatoprost (AGN 192024) by oral gavage administration to CD rats for 13 weeks followed by a 4-week recovery period:

Material and methods:

The purpose of study was to investigate the repeated dose toxicity of Bimatoprost (AGN 192024) in male and female CD rats under GLP conditions. Each 10 male and 10 female rats in the main group with complete investigations were dosed orally by gavage once a day with doses of 0, 0.1, 0.3 and 1.0 mg/kg bw/d for 13 weeks followed by a 4-week recovery period. In addition, a satellite group was used for toxicokinetic analyses. At the end of the treatment period, the corresponding main study animals were sacrificed and submitted for gross necropsy. Clinical reactions including mortality to treatment were monitored as well as body weight assessment, food consumption, ophthalmology, clinical pathology, urinalysis, organ weights and gross pathology. Histopathological evaluation was performed on a comprehensive range of organs. Toxicokinetics was analysed in the separate satellite group.

Results:

Mortality - No treatment-related death was noted.

Clinical Signs - No clinical signs were attributed to Bimatoprost (AGN 192024) treatment.

Body Weights - No toxicologically significant differences in either body weight or body weight gain were noted.

Food Consumption - There were no treatment-related effects on food consumption.

Ophthalmoscopy - No Bimatoprost (AGN 192024)-related changes were observed.

Clinical Chemistry - No toxicologically significant changes were noted.

Hematology - No toxicologically significant changes were noted.

Urinalysis - No toxicologically meaningful findings were noted.

Gross Pathology - No treatment-related findings were noted.

Organ Weights - Ovary weight in treated animals was increased. No other toxicologically significant differences were noted.

Histopathology - Prominent corpora lutea (total of more than 70 for both ovaries) was found in one female at 0.3 mg/kg bw/d. In addition, an increase in numbers of vacuolated corpora lutea was noted in females receiving 0.3 mg/kg bw/d of Bimatoprost (AGN 192024), which were completely reversible after recovery period. No other toxicologically relevant findings were observed.

Conclusion:

The repeated oral (gavage) application of Bimatoprost (AGN 192024) to male and female rats at doses of 0, 0.1, 0.3 and 1.0 mg/kg bw/d for 13 weeks followed by a 4-week recovery period led to no treatment-related mortality, clinical signs, body weight and food consumption changes, and ophthalmoscopic findings. Treatment-related changes were found in the ovaries, characterized by vacuolated corpora lutea in animals treated with 0.3 and 1.0 mg/kg bw/d of Bimatoprost (AGN 192024), which was reversible after recovery period. Total number of ovary corpora lutea was also slightly increased in these animals. Increased ovary weights were noted in treated females. No other toxicologically significant findings were noted.

Second subchronic toxicity study with Bimatoprost (AGN 192024) by oral gavage administration to CD rats for 13 weeks followed by a 4-week recovery period:

Material and methods:

The purpose of study was to investigate the repeated dose toxicity of Bimatoprost (AGN 192024) in male and female CD rats under GLP conditions. Each 10 male and 10 female rats in the main group with complete investigations were dosed orally by gavage once a day with doses of 0, 4, 8 and 16 mg/kg bw/d for 13 weeks followed by a 4-week recovery period. In addition, a satellite group was used for toxicokinetic analyses. At the end of the treatment

period, the corresponding main study animals were sacrificed and submitted for gross necropsy. Clinical reactions including mortality to treatment were monitored as well as body weight assessment, food consumption, ophthalmology, clinical pathology, urinalysis, organ weights and gross pathology. Histopathological evaluation was performed on a comprehensive range of organs. Toxicokinetics was analysed in the separate satellite group.

Results:

Mortality - No treatment-related death was noted.

Clinical Signs - No clinical signs were attributed to Bimatoprost (AGN 192024) treatment.

Body Weights - All treated animal groups demonstrated a decrease in body weight gain. During the recovery period, there were no differences in body weight gains among the different groups.

Food Consumption - Animals receiving the high dose showed a slight decrease in food consumption, which was reversed during the recovery period.

Ophthalmoscopy - No Bimatoprost (AGN 192024)-related changes were observed.

Clinical Chemistry - Treatment-related increases in aspartate amino transferase (AST, formerly GOT) and alanine amino transferase (ALT, formerly GPT) levels were observed in male treated rats, which returned to the control level after 4 weeks recovery period. These changes were not corroborated by histopathological changes, No other toxicologically significant differences were noted.

Hematology - No toxicologically significant changes were noted.

Urinalysis - No toxicologically meaningful findings were noted.

Gross Pathology - No treatment-related findings were noted.

Organ Weights - Ovary weight in treated animals was increased. No other toxicologically significant differences were noted.

Histopathology - Prominent corpora lutea (total of more than 70 for both ovaries) as well as an increase in numbers of vacuolated corpora lutea was found in all treated groups. The increase in treated animals demonstrated a dose-dependent manner. These changes were partially reversible after 4 weeks recovery period. No other toxicologically relevant findings were observed.

Conclusion:

The repeated oral (gavage) application of Bimatoprost (AGN 192024) to male and female rats at doses of 0, 4, 8 and 16 mg/kg bw/d for 13 weeks followed by a 4-week recovery period led to decreases in terminal body weights and body weight gain. ALT and AST activities were increased (2- to 5-fold) in male rats. In the ovaries, prominent corpora lutea (total of more than 70 in both ovaries), vacuolated corpora lutea and increased ovary weights were noted in females at all doses. It was claimed that the mechanisms of the ovarian changes were not known, but these changes were likely related to pharmacologic effects of luteolysis by PGF_{2a} and were considered species-specific. All treatment-related changes showed reversibility during the recovery period.

Chronic toxicity study with Bimatoprost (AGN 192024) by oral gavage administration to CD rats for 52 weeks followed by an 8-week recovery period:**Material and methods:**

The purpose of study was to investigate the repeated dose toxicity of Bimatoprost (AGN 192024) in male and female CD rats under GLP conditions. Each 15 male and 15 female rats in the main group with complete investigations were dosed orally by gavage once a day with doses of 0, 0.1, 0.3 and 2.0 mg/kg bw/d for 52 weeks followed by an 8-week recovery period. In addition, a satellite group was used for toxicokinetic analyses. At the end of the treatment period, the corresponding main study animals were sacrificed and submitted for gross necropsy. Clinical reactions including mortality to treatment were monitored as well as body weight assessment, food consumption, ophthalmology, clinical pathology, urinalysis, organ weights and gross pathology. Histopathological evaluation was performed on a comprehensive range of organs. Toxicokinetics was analysed in the separate satellite group.

Results:

Clinical Observations - No treatment-related mortality and clinical signs were observed.

Body Weights - No treatment-related changes in body weights or body weight gain were noted in males. However, reduced body weight gain was seen in females at 0.3 and 2.0 mg/kg bw/d.

Food Consumption - In female animals, a slight decrease in food consumption (4 - 5%) was noted at 0.3 and 2.0 mg/kg bw/d. This has to be considered to not be toxicologically significant as the decrease was too small to be relevant. No increase in the efficiency of food utilization (food consumed/body weight gain) was seen in these animals (6.3) relative to the control animals (6.7).

Ophthalmoscopy - No Bimatoprost (AGN 192024)-related changes were observed.

Clinical Chemistry - An increase in ALT and AST activities was noted in most treated groups. Since the increase was small, reversible and was not associated with histopathological changes, these changes can be considered to be not toxicologically significant. No other toxicologically significant differences were noted.

Hematology - No toxicologically significant changes were noted.

Urinalysis - No toxicologically meaningful findings were noted.

Gross Pathology - No treatment-related findings were noted.

Organ Weights - No toxicologically significant differences were noted.

Histopathology - Cellular vacuolation in a proportion of corpora lutea was found in one rat at 0.3 mg/kg and 3 rats at 2.0 mg/kg bw/d. In recovery animals, vacuolated corpora lutea was only seen in one female at 2.0 mg/kg bw/d and the proportion of corpora lutea with cellular vacuolation in this rat was less than that seen in the main study animals, indicating that these changes were reversible after 8 weeks recovery. No other toxicologically relevant findings were observed.

Conclusion:

The repeated oral (gavage) application of Bimatoprost (AGN 192024) to male and female rats at doses of 0, 0.1, 0.3 and 2.0 mg/kg bw/d for 52 weeks followed by an 8-week recovery period led to ovary findings in the form of cellular vacuolation in corpora lutea in several rats at 0.3 and 2.0 mg/kg bw/d. Therefore, the ovaries were identified as the target organ for Bimatoprost (AGN 192024). The decreases in terminal body weights and body weight gain were also noted at the same doses in females. A small increase in ALT and AST activities was noted in the treated male and female rats with no corresponding histopathological findings. All treatment-related changes showed reversibility during the recovery period. No other toxicologically significant findings were noted.

The no observed adverse effect level (NOAEL) was considered to be 2.0 mg/kg bw/d for males and 0.1 mg/kg bw/d for females.

ANNEX 4: Summary of reprotoxicity and developmental toxicity studies on Bimatoprost

The reproductive and developmental toxicology of the structural analogue Bimatoprost was assessed in a fertility and general reproduction toxicity study in rats, in prenatal developmental toxicity studies in rats and mice including range finding studies and in a developmental and perinatal/postnatal reproduction toxicity study in rats including a postnatal behavioral/functional evaluation. The key information is provided below, while details can be found in the CDER document (Reference: CDER, 2001).

Oral (gavage) fertility and general reproduction study of Bimatoprost (AGN 192024) in rats**Material and methods:**

The toxicity potential of Bimatoprost (AGN 192024) on fertility and general reproductive performance was studied in male and female Sprague-Dawley rats under GLP conditions. Four groups of male rats (25/group) were treated with 0; 0.1; 0.3 and 0.6 mg/kg bw/d Bimatoprost (AGN 192024) by oral gavage for 10 weeks, then cohabitated with female rats (25/group) given similar doses for 15 days. Cohabitation was on a 1:1 ratio and continued until evidence of mating (copulatory plug or sperm in the estrus smear) was observed. The day on which evidence of copulation was observed was designated as Day 0 of gestation. Dosing was continued in males until evidence of copulation was observed, and in female rats until gestation Day 7. Males were sacrificed following cohabitation and females were sacrificed on gestation Day 10 for caesarean-section.

Results:

Mortality and clinical signs - During the study period, no test item-related mortality was noted, and no remarkable clinical signs attributable to the treatment were recorded.

Body weight - There were no test item-related effects on body weight.

Food consumption - Food intake was not affected by Bimatoprost (AGN 192024) treatment.

Cohabitation Data - With regards to mating, no toxicologically significant, treatment-related changes were noted

Estrous Cycle Examination - No treatment-related changes were noted.

Uterine and Ovarian Examinations - No toxicologically significant changes were noted. At 0.6 mg/kg bw/d, one animal had a litter of 3 non-viable embryos. However, this was a single case, and the total incidence of non-viable embryos was still within the historical range. Therefore, it was not considered treatment-related.

Necropsy examinations - No treatment-related changes were noted in male and female treated rats.

Organ weights and histopathology - No treatment-related differences were noted.

Sperm analysis - No test item-related abnormal findings were observed.

Conclusion:

When Bimatoprost (AGN 192024) was investigated for its possible effect on fertility and general reproductive performance in male and female Sprague-Dawley rats at daily oral (gavage) doses of 0; 0.1; 0.3 and 0.6 mg/kg bw/d for 10 weeks, then cohabitated with female rats given similar doses for 15 days, no parental toxicity or adverse effects on fertility or general reproduction were observed in rats treated at the doses up to 0.6 mg/kg bw/d. Thus, the NOAEL for male and female fertility and general reproductive performance was 0.6 mg/kg bw/d, the highest dose investigated.

Oral (gavage) prenatal developmental toxicity study of Bimatoprost (AGN 192024) in mice:**Material and methods:**

The possible effect of Bimatoprost (AGN 192024) on prenatal embryo-fetal development was studied in mice (CrI:CD-1) under GLP conditions. Pregnant mice (25/group) received 0, 0.1, 0.3 and 0.6 mg/kg bw/d or vehicle control via daily oral (gavage) administration from Day 6 of gestation to Day 15 of gestation. The application volume was 5 mL/kg bw. The day on which evidence of copulation was observed was designated as Day 0 of gestation. On Day 18 of gestation each female was euthanatized and caesarean sectioned. A gross necropsy of the thoracic, abdominal and pelvic viscera was performed. The number of implantation sites and corpora lutea, early and late resorptions, and live and dead fetuses were recorded. The fetuses were weighted, sexed and examined for external malformations and variations. One-half of the fetuses were examined for soft tissue defects. The other half of the fetuses were examined for skeletal anomalies.

Results:

Clinical Observations - Abortion occurred in one mid dose mouse (gestation Day 17) and 3 high dose mice (gestation Days 16 and 17). In addition, premature deliveries were noted in 3 high dose mice on gestation Day 18. The abortion and premature deliveries were considered treatment-related. Post mortem examinations showed no abnormal findings in dams or fetuses.

Body weight - No treatment-related body weight changes were noted.

Necropsy: No treatment-related changes were recorded.

Uterine and Ovarian Examinations - There were one and two dead fetuses in the 0.3 and 0.6 mg/kg bw/d groups, which was within the historical control ranges and was not considered treatment-related. Beside these incidental findings there were no effects regarding corpora lutea, implantations, litter size, resorption, live fetuses/dam, live fetal body weights or live male fetuses per litter.

Fetal examinations - The external, visceral and skeletal findings noted were common in this species and strain, were within the historical control ranges, were not dose-dependent, and were not considered test item-related.

Conclusion:

When Bimatoprost (AGN 192024) was investigated for its possible effect on embryo-fetal development in pregnant mice (CrI:CD-1) at 0, 0.1, 0.3 and 0.6 mg/kg bw/d via daily oral (gavage) administration from Day 6 of gestation to Day 15 of gestation, dose-related abortion and early delivery occurred in the 0.3 and 0.6 mg/kg bw/d groups. Necropsy examination revealed no test item-related abnormalities. No developmental toxicities were noted at any dose groups. In conclusion, Bimatoprost (AGN 192024) was not considered to be teratogenic under the conditions of this study. The NOAEL for maternal toxicity was considered as 0.1 mg/kg bw/d, while the NOAEL for prenatal developmental toxicity including teratogenicity was 0.6 mg/kg bw/d, the highest dose investigated.

Oral (gavage) prenatal developmental toxicity study of Bimatoprost (AGN 192024) in rats:**Material and methods:**

The possible effect of Bimatoprost (AGN 192024) on prenatal embryo-fetal development was studied in female Sprague-Dawley rats under GLP conditions. Pregnant rats (25/group) were given the test item at doses of 0.1, 0.3 and 0.6 mg/kg bw/d or the vehicle control via daily oral (gavage) administration from Day 7 of gestation to Day 17 of gestation. The application volume was 1.5 mL/kg bw. The day on which evidence of copulation was observed was designated as Day 0 of gestation. On Day 20 of gestation each female was euthanatized and caesarean sectioned. A gross necropsy of the thoracic, abdominal and pelvic viscera was performed. The number of implantation sites and corpora lutea, early and late resorptions,

and live and dead fetuses were recorded. The fetuses were weighted, sexed and examined for external malformations and variations. One-half of the fetuses were examined for soft tissue defects. The other half of the fetuses were examined for skeletal anomalies.

Results:

Clinical Observations - Abortion occurred in 2 rats at 0.6 mg/kg bw/d on gestation Day 19, Necropsy of these rats revealed no gross lesions. Localized alopecia was noted in 8 dams in the 0.6 mg/kg bw/d group vs. 4 in the control group. Five of the 8 dams had alopecia on the ventral surface while none in the control group. This was considered treatment-related. No other toxicologically significant abnormal findings were noted in clinical observations

Body Weights - Body weights were not affected by the treatment.

Food Consumption - Food consumption was not affected by the drug during the treatment period.

Necropsy examinations — No treatment-related changes were noted

Uterine and Ovarian Examinations - No toxicologically significant, biologically relevant findings were observed, regarding corpora lutea, implantations, litter size, resorption, live fetuses/dam, live fetal body weights or live male fetuses per litter.

Fetal examinations – The external, visceral and skeletal findings noted were common in this species and strain, were within the historical control ranges, were not dose-dependent, and were not considered test item-related.

Conclusion:

When Bimatoprost (AGN 192024) was investigated again for its possible effect on prenatal embryo-fetal development in female rats at daily oral (gavage) doses of 0, 0.1, 0.3 and 0.6 mg/kg bw/d from gestation day (GD) 7 to 17, two abortions occurred in the 0.6 mg/kg bw/d group. At the same dose, increased incidence of alopecia was noted. Necropsy examination revealed no test item-related abnormalities. No prenatal developmental toxicities were noted at any dose groups. In conclusion, Bimatoprost (AGN 192024) was not considered to be teratogenic under the conditions of this study. The NOAEL for maternal toxicity was considered as 0.3 mg/kg bw/d, while the NOAEL for prenatal developmental toxicity including teratogenicity was 0.6 mg/kg bw/d, the highest dose investigated.

Oral (gavage) developmental and perinatal/postnatal reproduction toxicity study of Bimatoprost (AGN 192024) in rats, including a postnatal behavioral/functional evaluation:

Material and methods:

The possible effect of Bimatoprost (AGN 192024%) in the pregnant/lactating females and in the development of the offspring following exposure of the female from implantation through weaning was investigated in female Sprague-Dawley rats under GLP conditions.

Pregnant rats (25/group) received doses of 0.05, 0.1, 0.3 and 0.6 mg/kg bw/d or vehicle control via daily oral (gavage) administration from Day 7 of gestation to Day 20 of lactation. The day on which evidence of mating was observed was designated as Day 0 of gestation. The day of birth was defined as Day 1 of lactation.

F0 dams, F1 and F2 generation animals were comprehensively investigated as follows:

F0 parturition and F1 litter observations - The pregnant animals were allowed to deliver and the duration of gestation, litter size and viability were examined. All pups in a litter were individually weighted. During pre-weaning observations, each litter was evaluated for viability twice daily. Clinical observations were recorded once daily and body weights were measured weekly.

F1 postweaning observations — Selected F1 generation rats from 0, 0.05, 0.1 and 0.3 mg/kg bw/d groups were observed for viability (twice daily), clinical observations (weekly), body weights and food consumption (weekly). Beginning at 24 days postpartum, one male and one female from each litter were evaluated in a passive avoidance test for learning, short. and long-term retention.

Beginning at day 28 postpartum* females were evaluated for the age of vaginal patency.

Beginning at day 39 postpartum, males were evaluated for the age of preputial separation.

Beginning at 70 days postpartum, one male and one female from each litter were evaluated in a water-filled M-maze for overt coordination, swimming ability, learning and memory.

F1 Reproduction Assessment - At approximately 90 days of age, the F1 generation rats within each dosage group were to cohabitation, one male rat per female rat, with the exclusion of sibling mating. Female rats with sperm observed in a smear of the vaginal contents and a copulatory plug observed were considered to be at Day 0 of gestation and assigned to individual housing.

F0 Necropsy - After the 21-day postpartum period, female rats were euthanized and a gross necropsy of the thoracic, abdominal and pelvic viscera was performed. The number and distribution of implantation sites were recorded.

F1 Necropsy - Pups found dead were examined for gross lesions. All pups culled on Day 21 of lactation were examined for gross lesions. Male F1 rats selected for reproduction assessment were terminated after cohabitation period and a gross necropsy of the thoracic, abdominal and pelvic viscera was performed. On gestation Day 21; female rats were sacrificed and caesarean section, and a gross necropsy of the thoracic, abdominal and pelvic viscera was performed. The female rats were examined for the number of corpora lutea, implantation sites, live and dead fetuses and early and late resorptions.

F2 fetuses - Each fetus was weighted and examined for sex and internal alternations.

Results:

F0 Dams

F0 Maternal Observations - No treatment-related mortality and clinical signs were noted.

F0: Maternal Body Weight — Prior to gestation Day 18, maternal body weight gain was comparable among different groups. The average body weights from gestation Days 18 to 20 were reduced in the 0.6 mg/kg bw/d group, which was 91.7% of the body weights in control animals. The decrease was attributed to the litter loss (dead or resorbed conceptuses) in this group. During lactation period, maternal body weights were comparable among the groups.

F0. Maternal Food Consumption - Food consumption was comparable among different groups during gestation and lactation periods.

F0: Maternal Necropsy Findings - No abnormal findings were noted.

F0: Reproductive Parameters - In the 0.6 mg/kg bw/d group, 3 dams were not observed to have delivered a litter by gestation Day 25. Post mortem examination showed that 2 of these rats had early resorptions and the other one delivered the litter without observations. In 0.3 mg/kg bw/d and 0.6 mg/kg bw/d groups, the duration of gestation was reduced, peri- and postnatal pup mortality was increased.

F1 Generation:

F1: Offspring Viability and Survival –Treatment-related changes included decreases in the number of pups per litter at birth, the number of live born pups per fitter, the percentage of pups surviving after 1 and 4 days, and pup weights during the preweaning period at 0.3 and 0.6 mg/kg bw/d.

F1 Observations - No treatment-related mortality and clinical signs were noted during prepartum period and during gestation period.

F1: Body Weights - Postweaning and gestation body weights and body weight gain for the F1 generation male and female rats were not affected by administration of the test item to the F0 generation dams.

F1 Food Consumption - No toxicologically significant changes in food consumption were observed.

F1 Sexual Maturation - The average day on which preputial separation was evident in males and vaginal patency was evident in females were comparable between control and treated groups.

F1 Behavioral Evaluation - No biologically relevant toxicologically significant differences in the values for learning, short-term retention, long-term retention or response inhibition were noted in the F1 generation male and female rats.

F1 Necropsy Observations - No treatment-related, toxicologically significant findings were noted. In the males, the terminal body weights, testes and epididymides weights were comparable among different groups.

F1 Cohabitation Data and Reproductive Parameter – In the 0.3 mg/kg bw/d group, apparently the number of days in cohabitation was increased and the fertility index was slightly decreased.

F2 Generation:

F2 Fetal External Observations - No treatment-related abnormal findings were noted.

Conclusion:

When Bimatoprost (AGN 192024) was investigated for its possible effect on pre- and postnatal development including maternal function in female Sprague-Dawley rats at daily oral (gavage) doses of 0, 0.05, 0.1, 0.3 and 0.6 mg/kg bw/d from GD 6 until LD 20, no treatment-related mortality, clinical signs of toxicity, body weight and food consumption changes were noted in F0 rats. At 0.3 mg/kg bw/d, the duration of gestation was reduced and peri- and postnatal pup mortality was increased. Test item related effects included decreases in the number of pups per litter at birth, the number of liveborn pups per litter, the percentage of pups surviving after 1 and 4 days, and pup weights during the preweaning period at 0.3 and 0.6 mg/kg bw/d. Sexual maturation and behavioural evaluations showed no abnormal findings in F1 generation rats in any dosage groups. In mating and fertility assessment in F1 generation, rats from 0.3 mg/kg bw/d group showed a slight increase in the number of days required for mating, and a slight decrease in the fertility index. The NOAEL was determined as 0.1 mg/kg/day in this study.

ANNEX 5: Summaries of mutagenicity studies on Bimatoprost

***In vitro* studies**

Ames test:

Bimatoprost (AGN 192024) at concentrations as high as 5000 µg/plate did not induce mutations in *Salmonella typhimurium* strains TA98, TA100, TA1535 and TA1537 or in the *Escherichia coli* (*E. coli*) strain WP2 uvrA in the presence and absence of metabolic activation (S9-mix).

***In vitro* assay in mammalian cells:**

Bimatoprost was investigated to evaluate the mutagenic potential based on quantitation of forward mutations at the thymidine kinase locus of L5178Y mouse lymphoma cells. The study included a preliminary toxicity assay and a mutagenicity assay.

In the preliminary assay, suspension growth relative to the solvent controls was 0% at 1000 and 5000 µg/ml with and without S9 activation. Based on the results of the toxicity assay, the doses chosen for the mutagenicity assay ranged from 50 to 900 µg/ml for both the non-activated and S9-activated cultures. In the mutagenicity assay, cytotoxicity was evidenced by the inhibition of the total growths to 28% for the non-activated cultures at concentration of 452 µg/ml and 10% for S9-activated cultures at concentration of 472 µg/ml. No concentration-related increase in mutant frequency was observed. Under the conditions of this study, Bimatoprost did not produce a positive response in the non-activated and S9 activated systems in L5178Y mouse lymphoma cells.

***In vivo* study**

Bimatoprost (AGN 192024) was investigated for its potential to induce micronuclei *in vivo* in the bone marrow of treated male mice (CD-1) at intravenously (I. v.) injected doses of 0, 5, 10 and 20 mg/kg bw. The bone marrow was harvested 24 and 48 h after dosing. The frequency of micronucleated cells was expressed as percent micronucleated polychromatic erythrocytes (MNPCE). In dose range finding assay, 3 mice/sex/dose were intravenously treated with a single dose of 0, 5, 10 and 20 mg/kg bw and were observed for toxicity for 2 days. All animals appeared normal throughout the 2-day observation period. Based on this assay, the high dose of 20 mg/kg, the maximum dose possible with the concentration limit of 0.2%, was determined. In the main test, Bimatoprost induced no signs of clinical toxicity and was not cytotoxic to the bone marrow. It did not induce a significant increase in micronucleated PCEs over the levels observed in the vehicle controls at any harvest timepoints. Thus, Bimatoprost (AGN 192024) did clearly not induce micronucleated polychromatic erythrocytes indicative of clastogenicity in the bone marrow of mice at 24 hours and at 48 hours after single iv dosing.

Ref: Schilling 2024

ANNEX for ISOPROPYL CLOPROSTENATE (IPCP)

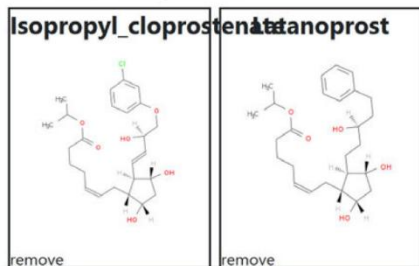
ANNEX 6: Results of similarity assessment between Isopropyl Cloprostenate, Latanoprost, Bimatoprost, and Travoprost prepared by Applicant 2 (IPCP Opinion)

APPENDIX B by Applicant: CHEMMINE STRUCTURAL SIMILARITY RESULTS

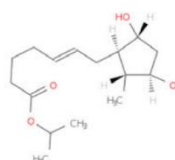
Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



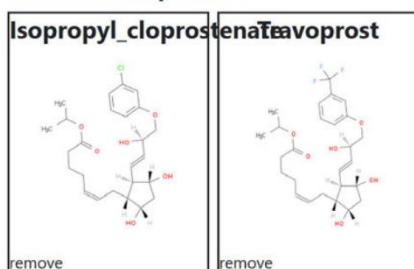
AP Tanimoto: 0.623311
MCS Tanimoto: 0.4651
MCS Size: 20
MCS Min: 0.6452
MCS Max: 0.6250
SMILES:
C(OC(=O)CCC/C=C\C[C@H]1[C@H](C[C@H]1C(=O)O)O)C(C)C
Isopropyl_cloprostenate



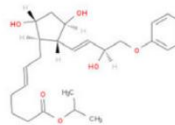
Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



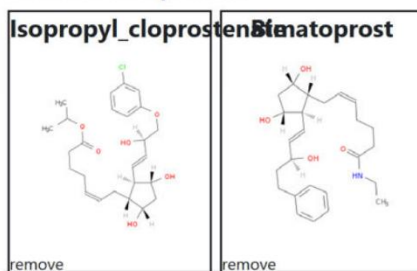
AP Tanimoto: 0.742812
MCS Tanimoto: 0.8611
MCS Size: 31
MCS Min: 0.9688
MCS Max: 0.8857
SMILES:
C(OC(=O)CCC/C=C\C[C@H]1[C@H](C[C@H]1C(=O)O)O)C(C)C
Isopropyl_cloprostenate



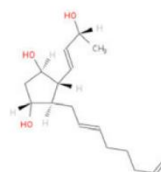
Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



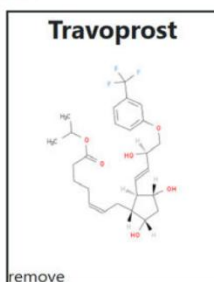
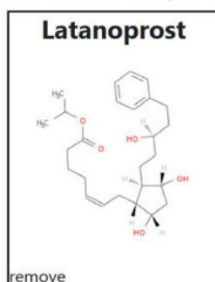
AP Tanimoto: 0.549085
MCS Tanimoto: 0.4762
MCS Size: 20
MCS Min: 0.6667
MCS Max: 0.6250
SMILES:
C(=O)CCC/C=C\C[C@H]1[C@H](C[C@H]1C(=O)O)O)C(C)C
Isopropyl_cloprostenate



Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



AP Tanimoto: 0.534009

MCS Tanimoto: 0.4348

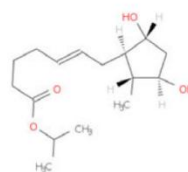
MCS Size: 20

MCS Min: 0.6452

MCS Max: 0.5714

SMILES:

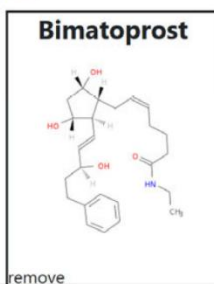
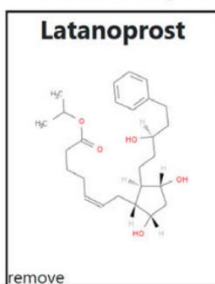
C(OC(=O)CCC/C=C\C[C@H]1[C@](C[C@H])(C(CO)O)O)O
Latanoprost



Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



AP Tanimoto: 0.595745

MCS Tanimoto: 0.3556

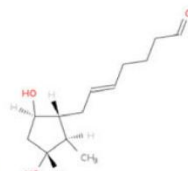
MCS Size: 16

MCS Min: 0.5333

MCS Max: 0.5161

SMILES:

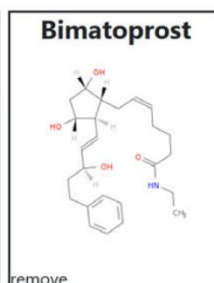
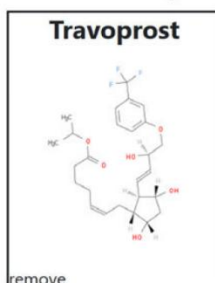
C(=O)CCC/C=C\C[C@H]1[C@@H](C[C@H]([C@@H]1C)O)O
Latanoprost



Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



AP Tanimoto: 0.471429

MCS Tanimoto: 0.4444

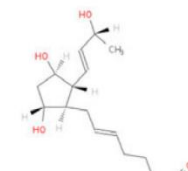
MCS Size: 20

MCS Min: 0.6667

MCS Max: 0.5714

SMILES:

C(=O)CCC/C=C\C[C@H]1[C@@H](C[C@H]([C@@H]1C)O)O
Travoprost



APPENDIX C by Applicant: CHEMMINE DETAILED SIMILARITY COMPARISON

Isopropyl Cloprostenate to Cloprostenol Comparison

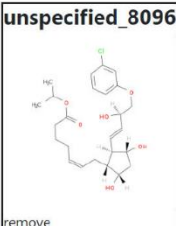
chemminetools.ucr.edu/similarity/

ChemMine Tools About Help Downloads Optional: Login or register to save data.

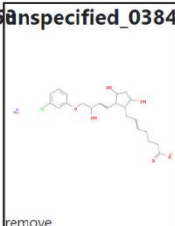
Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



remove

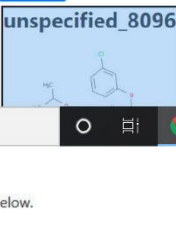


remove

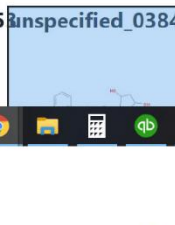
AP Tanimoto: 0.724665
MCS Tanimoto: 0.8768
MCS Size: 29
MCS Min: 0.9667
MCS Max: 0.9062
SMILES: C1(C(C(C1)O)/C=C/C(C(C1O)O)O)O
unspecified_809653

All x

Update



remove

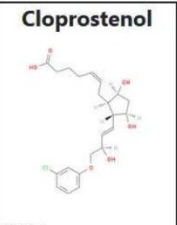


remove

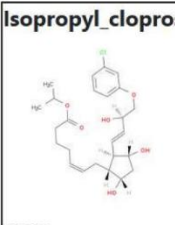
Compound Similarity

Select two compounds to compare from the grid below.

Selected Compounds



remove



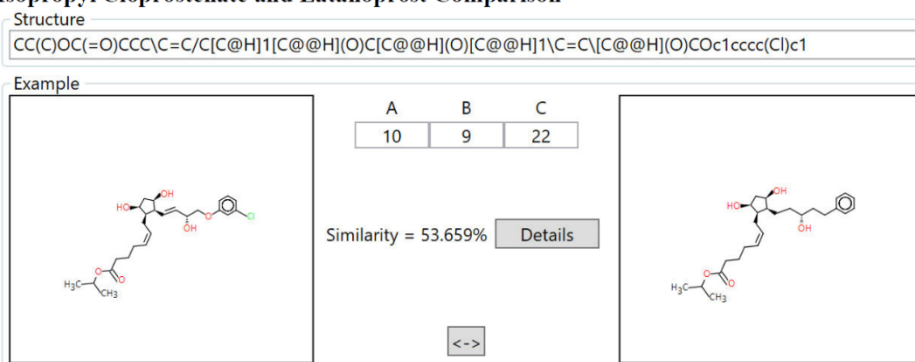
remove

AP Tanimoto: 0.724665
MCS Tanimoto: 0.9062
MCS Size: 29
MCS Min: 1.0000
MCS Max: 0.9062
SMILES: C(COc1cccc(c1)Cl)/C=C/C(C(C1O)O)O
Cloprostenol

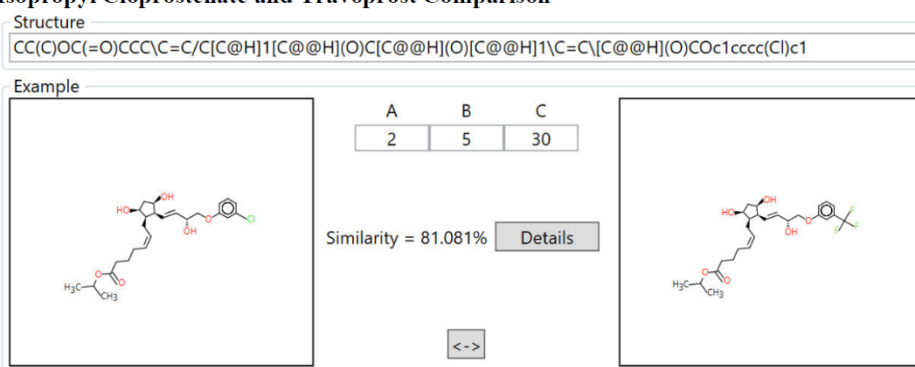
8:26 PM
6/5/2020

APPENDIX D by Applicant: QSAR TOOLBOX STRUCTURAL SIMILARITY RESULTS

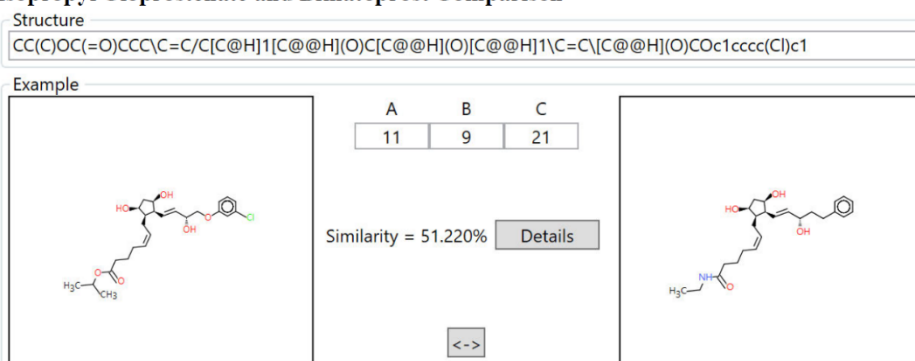
Isopropyl Cloprostenate and Latanoprost Comparison



Isopropyl Cloprostenate and Travoprost Comparison



Isopropyl Cloprostenate and Bimatoprost Comparison

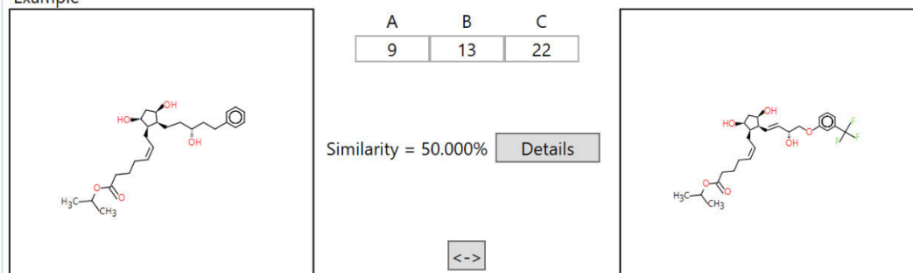


Latanoprost and Travoprost Comparison

Structure

CC(C)OC(=O)CCCC=CCC1C(O)CC(O)C1CCC(O)CCc1cccc1

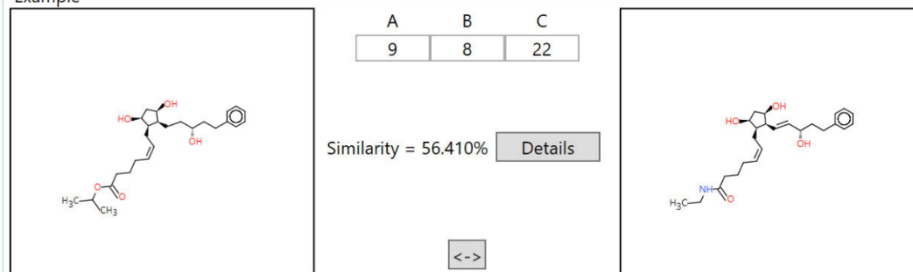
Example

**Latanoprost and Bimatoprost Comparison**

Structure

CC(C)OC(=O)CCCC=CCC1C(O)CC(O)C1CCC(O)CCc1cccc1

Example

**Travoprost and Bimatoprost Comparison**

Structure

CC(C)OC(=O)CCCC=CCC1C(O)CC(O)C1C=CC(O)COc1cccc(c1)C(F)(F)F

Example

